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CHARACTERIZATION OF SOME PROCOAGULANT AND ANTICOAGULANT PROTEIN(S) AND PEPTIDE(S) PURIFIED FROM INDIAN RUSSELL'S VIPER (*Daboia russelii russelii*) VENOM

A thesis submitted

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Doctor of Philosophy

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Dedicated to my loving parents……..

Mrs. Cecilia C. Thakur

&

Late Mr. Sunil Thakur

For their endless support, unconditional love and untold sacrifices…

ABSTRACT

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Snakebite is an important but a neglected health hazard in the tropical and sub-tropical countries of Africa, South-east Asia and South America. Approximately 3,000 species of snakes have been documented of which about 600 are reported to be venomous and responsible for a global incidence of 2,500,000 snakebites annually. India is one of the countries of the South-east Asia that has recorded the highest incidence of envenomation (81,000 per year) and the maximum number of deaths (11,000 per year) due to snakebite; albeit this does not reflect the true magnitude of the problem because most of the snakebites occurring in the rural areas of the country remain unreported.

Russell's Viper (*Daboia russelii*), an important member of the Viperideae family of snakes is widely distributed over large areas of Asia from Pakistan to Taiwan and the Lesser Sunda islands. Russell's Viper is one of the most frequently encountered venomous snake responsible for majority of snakebite morbidity and mortality in many Southeast Asian countries including Thailand, Myanmar, China, Indonesia as well as the Indian subcontinent. Russell's Viper envenomation is an occupational health hazard and a grave problem particularly for the farmers and rice cultivators in Southeast Asia. This species of snake is particularly abundant in these areas because of the easy availability of its primary dietary preference which includes rodents (rats/mice) which thrive near cultivated areas such as rice fields.

Russell's Viper venom (RVV) is hemotoxic in nature which leads to extensive cell and tissue damage. The venom of Russell's Viper is a mixture of many enzymatic and non-enzymatic proteins which often complement each other to induce enhanced toxicity. Most of these proteins affect the normal process of hemostasis either by virtue of their ability to prolong (anticoagulant proteins) or shorten (procoagulant proteins) the coagulation time. Moeover, studies related to clinical manifestations of Russell's Viper envenomation in a region and composition of RVV of that particular geographical location have suggested that due to

differences in geographical origin, RVV composition varies that results in differences in the clinical manifestations following RVV envenomation. Therefore, understanding the structure-function properties of the various procoagulant and anticoagulant components present in RVV of a particular geographical location can help us in designing appropriate antivenom strategies for effective hospital management of Russell's Viper envenomations.

The normal process of hemostasis involves the controlled and regulated activation of components of the coagulation cascade to maintain a delicate balance between clot formation and clot dissolution. These two opposite processes are in equilibrium, keeping the hemodynamic properties of the blood constant. Many snake venom proteins and peptides are reported to specifically target critical molecules of the homeostatic systems of the prey/victim via enzymatic or non-enzymatic mechanisms leading to hemostatic disturbances. Such potent and highly specific snake venom proteins/peptides have thus been indispensable in elucidating the complex physiology of mammalian blood coagulation and fibrinolysis. Additionally, understanding the role of individual RVV components will not only provide an insight into the contribution of the isolated protein/peptide in the overall pathological effects of Russell's Viper envenomation, but this study may also be helpful in identification of potential therapeutic molecules from RVV.

For the ease of understanding, this thesis has been divided into seven chapters.

Chapter I- *Introduction:* The first chapter of the thesis provides a brief appraisal of snakebite problems throughout the world, India included. This chapter also provides an insight into the functional attributes of snake venom protein(s)/peptide(s) with particular reference to RVV in terms of their pathological significance and biomedical applications.

Chapter II- Review of literature: The second chapter reviews the recent data on procoagulant and anticoagulant protein(s)/peptide(s) isolated from RVV. This

chapter also focuses on the pathophysiological significance of the isolated procoagulant and anticoagulant protein(s)/peptide(s) from RVV and provides a brief detail on the biomedical application of such venom-derived protein(s)/peptide(s).

Chapter III- Materials and methods: This chapter includes the materials and methods employed for performing the various experiments which include the isolation and purification processes of the different anticoagulant and procoagulant protein(s)/peptide(s) from RVV. The techniques used for the biochemical and pharmacological characterization of the isolated protein(s)/peptide(s) as well as for understanding the mechanism(s) of action of the isolated protein(s)/peptide(s) have been described.

Chapter IV- Chapter VI- Results and discussion: These three chapters analyses the data of the experiments and discusses the results of this study.

Chapter IV- This chapter describes the characterization and anticoagulant mechanism of a low-molecular mass peptide named Ruviprase purified from RVV. The chapter also explores the anticancer potential of this peptide purified from RVV and advocates its potential therapeutic application(s) in future.

Chapter V- This chapter describes the isolation, characterization and procoagulant mechanism of an oligomeric protease named Rusviprotease purified from RVV.

Chapter VI- This chapter describes the isolation, characterization and pathophysiological significance of an anticoagulant toxic fraction, GF-VI DEAE-II and its cytotoxin-like component named Rusvitoxin purified from RVV.

Chapter VII- *Conclusion:* This chapter describes the conclusion on the major findings and visualizes the future prospects of the research findings of this study.

The first major finding of the present study was the characterization and anticoagulant mechanism of a low-molecular mass peptide purified from Russell's Viper (*Daboia russelii russelii*) venom along with the elucidation of its anticancer potential. Fractionation of RVV on a cation exchanger CM-Cellulose column resulted in the elution of three basic anticoagulant peaks (CMAC-I, CMAC-II and CMAC-III). Further fractionation of CMACV-III on a Sephadex G-50 gel-filtration column resulted in separation of four protein peaks (GF-I, GF-II, GF-III, and GF-IV). The fraction GF-IV was subjected to RP-HPLC and the active anticoagulant fraction was termed as Ruviprase which contributed to 0.1% of the total RVV proteins. Ruviprase exhibited a molecular mass of 4,423.6 Da and was found to be composed of α-helices (61.9 %) and random coils (38.1%). The partial N-terminal sequence (E^1-V^2-X^3-W^4-W^5-W^6-A^7-Q^8-L^9-S^{10}) of Ruviprase demonstrated significant similarity (80.0%) with an internal sequence of apoptosis-stimulating protein reported from the venom gland transcriptome of *Ophiophagus hannah* and *Python bivittatus;* albeit Ruviprase did not show sequence similarity with existing thrombin/factor Xa inhibitors. Ruviprase demonstrated a potent *in vitro* anticoagulant property via non-enzymatic mechanism of inhibition of thrombin and factor Xa by slow binding kinetics. Ruviprase inhibited thrombin by binding to its active site via an uncompetitive mechanism with a *Ki* value and dissociation constant (K_D) of 0.42 μM and 0.46 μM, respectively. Ruviprase inhibited thrombin's amidolytic activity towards its preferred substrate T1637 as well as its fibrinogen clotting ability with a IC_{50} value of 0.23 μM and 1.52 μM, respectively. The corresponding Hill co-efficient was determined as 1.08 and 5.21, respectively suggesting strong positive-cooperative binding between thrombin and Ruviprase. Ruviprase also inhibited the platelet aggregating property of thrombin by virtue of its ability of inhibiting the enzymatic activity of thrombin Though Ruviprase could not inhibit the amidolytic activity of factor Xa towards its preferred substrate chromogenic substrate F3301, Ruviprase demonstrated mixed inhibition (Ki = 0.16 μM) of factor Xa towards its physiological substrate prothrombin. The IC_{50} value and Hill co-efficient for the

inhibition of prothrombin activation by factor Xa were determined as 0.46 μM and 2.08, respectively. Furthermore, the biological properties of Ruviprase could not be neutralized by commercial polyvalent (PAV) or monovalent antivenom (MAV) against RVV suggesting the low immunogenicity of this low molecular mass snake venom peptide. Ruviprase at the therapeutic dose of plasmin (2.0 mg kg^{-1}) was non-toxic and showed potent *in vivo* anticoagulant activity after 6 h of intraperitoneal treatment in Balb/c mice. Because of the potent anticoagulant property as well as non-toxic nature of Ruviprase, the possible therapeutic application of this peptide as an antithrombotic agent for combating thrombosis-associated ailments appears promising. In addition to the strong anticoagulant property, Ruviprase also demonstrated potent anticancer potential towards mammalian breast cancer cells. The IC_{50} value of Ruviprase towards MCF-7, EMT6/AR1, HeLa cells under identical experimental conditions was determined as 4 ± 0.3 ng μl^{-1}, 23 ± 1.5 ng μl^{-1} and 5.8 ± 0.7 ng μl^{-1}, respectively after 24 h of treatment. This indicates that Ruviprase has highest cytotoxcity towards MCF-7 cells whereas it exhibited least inhibitory activity against EMT6/AR1 cell. Our studies suggest that Ruviprase demonstrated pro-apoptotic property by virtue of its ability to cause p53-dependent G2 arrest in MCF-7 cells. Ruviprase also induced mitochondria dependent apoptosis via excess ROS generation as well as alteration of mitochondrial transmembrane potential.

The second major finding of the present study was the purification, characterization and procoagulant mechanism of an oligomeric protease isolated from RVV. Fractionation of crude RVV on Sephadex G-50 gel-filtration column resolved the venom into 11 peaks. The first peak (GF-I) was found to demonstrate protease activity and strong procoagulant effects on platelet poor plasma (PPP). This protein peak was subjected to HiPrep CM FF 16/10 cation exchange chromatography that resulted in the separation of GF-I into 6 peaks (GF-I-CM-I, GF-I-CM-II, GF-I-CM-III, GF-I-CM-IV, GF-I-CM-V and GF-I-CM-VI). The fourth peak, GF-I-CM-IV (Rusviprotease) demonstrated strong fibrinogenolytic as

well as procoagulant property and contributed to 0.8% of the total RVV. Rusviprotease is characterized to be a basic multimeric protein which predominantly exists in RVV in the form of a 27 kDa dimer of 15.1 kDa monomers. Rusviprotease also exists in other multimeric forms (tertameric and hexameric forms) apart from its dimeric form. Rusviprotease is a basic glycoprotein, but it has lower carbohydrate content than previously reported proteases. Rusviprotease was eluted as a bound fraction in cation-exchange column, suggesting the basic glycoprotein nature of Rusviprotease. Additionally, chemical modification studies as well as LC MS/MS analysis of Rusviprotease clearly indicated the metalloprotein nature of Rusviprotease. Rusviprotease decreased the plasma recalcification time, prothrombin time (PT) as well as activated prothrombin time (aPTT) of PPP suggesting that it affects the common pathway of the coagulation cascade. Elucidation of the procoagulant mechanism of Rusviprotease revealed that it demonstrated prothrombin activation property independent of the requirement of any co-factors similar to the prothrombinase complex. Like most group A prothrombin activators, Rusviprotease cleaved prothrombin independent of any co-factor requirement generating meizothrombin which is further cleaved to form thrombin. The *Km* and *Vmax* value of Rusviprotease towards prothrombin was determined as 1.73 μM, and 153.5 nM thrombin generated min^{-1} μmol^{-1}. Spectrofluorometric study provided evidence of interaction of Rusviprotease with factor Xa. Rusviprotease augmented the prothrombin activating property of the factor Xa-prothrombinase complex by 2.5 fold thus enhancing thrombin formation by the factor Xa-prothrombinase complex probably due to its additive effect on factor Xa induced prothrombin activation. In addition to the prothrombin-activating property, Rusviprotease also demonstrated the ability to degrade the α-chain of fibrinogen similar to many snake venom prothrombin activators. The present study indicates that MAV moderately surpasses PAV in neutralizing the catalytic and pharmacological properties of Rusviprotease suggesting the therapeutic superiority of MAV over PAV in neutralizing such proteases in RVV. Unlike many of the SVMPs, Rusviprotease

lacked *in vitro* basement membrane hydrolytic effects clearly suggesting the *in vitro* non-toxic or non-hemorrrhagic nature of Rusviprotease. Intravenous injection of Rusviprotese to BALB/c mice (0.07 mg kg^{-1}) resulted in *in vivo* defibrinogenation and prolonged the tail-bleeding time suggesting that its treatment rendered the blood incoagulable without any toxic effect in mice up to 48 h of treatment. In conclusion, Rusviprotease is the first example of a prothrombin activator purified from *Daboia russelii russelii* venom.

The third and final major finding of the present study involved the isolation and characterization of a toxic fraction from RVV. This toxic RVV fraction (GF-VI DEAE-II) was isolated from crude venom by a combination of gel-filtration and anion-exchange chromatography. The components of the isolated fraction were further separated by RP-HPLC. GF-VI DEAE-II represented 1.6% of the total RVV proteins and was found to comprise a 27.6 kDa protease named RP-I (0.04%, w/w) and a 6.6 kDa peptide named Rusvitoxin (1.11%, w/w). The peptide mass fingerprinting (PMF) analysis of RP-I showed its identity with snake venom serine proteases. PMF as well as secondary structure analyses of Rusvitoxin, which was devoid of detectable enzymatic activity, demonstrated its close identity with snake venom three finger toxins, cytotoxins and cardiotoxins particularly from *Naja* sp. The fact that the fraction separated into two distinct protein components under reducing conditions, a ~35 kDa protein (RP-I) and a ~9 kDa peptide (Rusvitoxin), unequivocally indicated that these proteins co-eluted in the single fraction. Since the components of the toxic fraction could not be separated under various non-denaturing chromatographic conditions except RP-HPLC, it may be suggested that the two molecules (RP-I and Rusvitoxin) remain in close association with each other. The co-elution of a high molecular mass enzyme (~35 kDa) along with a low molecular mass peptide (~9 kDa) upon size exclusion chromatography advocates the above explanation. Present study also suggests that GF-VI DEAE-II preferentially degrades fibrin over fibrinogen unlike most of the previously identified snake venom fibrinolytic enzymes. The preferred fibrin specificity of the

protease component of GF-VI DEAE-II is probably due to its better binding ability with cross-linked fibrin rather than with fibrinogen. Furthermore, a comparison of the fibrino(geno)lytic activity of human plasmin with GF-VI DEAE-II suggests that these two direct-acting fibrinolytic proteases have different specificities and cleavage preference for fibrin(ogen). More than 80% of the *in vitro* proteolytic (fibrinolytic) and anticoagulant property of GF-VI DEAE-II was neutralized by both PAV and MAV against RVV (at 1: 15 antigen:antivenom, w/w) suggesting that these antivenoms could possibly neutralize the effects of this fraction in whole RVV. GF-VI DEAE-II demonstrated mild anticoagulant effect on PPP as it prolonged plasma recalcification time and prothrombin time (PT) but did not affect the activated partial thromboplastin time (aPTT). GF-VI DEAE-II lacked proteolytic effects on extracellular matrix proteins suggesting the absence of hemmorhagic effect. Additionally, GF-VI DEAE-II was found to be non-cytotoxic and non-hemolytic; but demonstrated *in vivo* toxicity in BALB/c mice with an LD_{50} value (*i.p.*) of 2.3 mg kg^{-1}. As suggested by our studies, the three-finger cytotoxin-like molecule (Rusvitoxin) was non-toxic in mice at the tested dose of 5 mg kg^{-1} unlike most of the snake venom cytotoxins. Furthermore, RVV serine proteases are in general found to be non-lethal in mice at the above dose although they affect the coagulation cascade. Due to the low yield and non-toxic nature demonstrated by serine proteases isolated from snake venom, the toxicity study of protease fraction (RP-I) was not done. GF-VI DEAE-II induced lethargy and hind-leg paralysis in mice within 10 min of *i.p.* injection (LD_{50} dose) suggesting its contribution in incapacitating the agile mammalian prey. The *in vivo* pharmacological properties of the isolated anticoagulant fraction included hyperfibrinogenomia, altered plasma levels of coagulation factors (factors X and IX) and significantly altered ($p < 0.05$) levels of pro- and anti-inflammatory cytokines viz. TNF-α, IL-6 and IL-10 in treated mice. Histological observations of tissues and biochemical properties of serum obtained from mice treated with GF-VI DEAE-II suggested multiple organ dysfunctions which are the typical clinical

manifestation of Russell's Viper bite. Present study suggests the significant role of this venom fraction in RVV-induced pathogenesis in prey/victims.

In a nutshell, the thesis deals with (i) the isolation, characterization and elucidation of anticoagulant and pro-apoptotic mechanism of a 4.4 kDa peptide. This is the first report of a natural snake venom inhibitor with the dual ability of inhibiting both thrombin and factor Xa (ii) isolation, characterization and procoagulant mechanism of a oligomeric metalloprotease which exist predominantly in the form of a 26.8 kDa dimer. This is the first report on the isolation and characterization of prothrombin activating protease from RVV, and (iii) isolation, characterization and pathophysiological significance of a toxic anticoagulant venom fraction comprising of a 27.6 kDa fibrinolytic protease and a 6.6 kDa three finger-cytotoxin like peptide from RVV of eastern India origin. This is the first report on the identification of a three-finger toxin like peptide from RVV which remains in close association with a fibrinolytic protease to demonstrate enhanced toxicity.

DECLARATION

I hereby declare that the thesis entitled "**BIOCHEMICAL AND PHARMACOLOGICAL CHARACTERIZATION OF SOME PROCOAGULANT AND ANTICOAGULANT PROTEIN(S)/PEPTIDE(S) FROM RUSSELL'S VIPER (*Daboia russelii russelii*) VENOM**" is an authentic work carried out under the supervision of Dr. Ashis K. Mukherjee, Professor, Department of Molecular Biology and Biotechnology, Tezpur University, Assam – 784 028. No part of this work has been presented for any other degree or diploma earlier. Due to unavailability of proper facilities in Tezpur University, the following experiments/sample analyses were carried out at other institutes.

1. MALDI-TOF-MS, circular dichroism and cell culture (cytotoxicity) studies of the protein/peptide samples were performed at Indian Institute of Technology, Guwahati, Assam, India.
2. N-terminal sequencing and anticancer studies were performed at Indian Institute of Technology, Mumbai, Maharashtra, India.
3. N-terminal sequencing of one of the protein was done at INTAS Biopharmaceutical, Ahmedabad, India.
4. Peptide mass fingerprinting analysis was performed at National Institute of Plant Genomic Research, New Delhi or Centre for Cellular and Molecular Platforms, NCBS-TIFR, Bangalore, India.
5. Animal experiments were done at Defence Research Laboratory, Tezpur, Assam, India.

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CERTIFICATE OF THE SUPERVISOR

This is to certify that the thesis entitled "**CHARACTERIZATION OF SOME PROCOAGULANT AND ANTICOAGULANT PROTEIN(S) AND PEPTIDE(S) PURIFIED FROM INDIAN RUSSELL'S VIPER (*Daboia russelii russelii*) VENOM"** submitted to the School of Sciences, Tezpur University in partial fulfillment for the award of the degree of Doctor of Philosophy in Molecular Biology and Biotechnology is a record of research work carried out by Ms. RUPAMONI THAKUR, Research Scholar, Department of Molecular Biology and Biotechnology, Tezpur University, Assam, under my supervision and guidance.

All help received by her from various sources have been duly acknowledged.

No part of this thesis has been submitted elsewhere for award of any other degree.

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CERTIFICATE OF THE EXTERNAL EXAMINER AND ODEC

This is to certify that the thesis entitled "**CHARACTERIZATION OF SOME PROCOAGULANT AND ANTICOAGULANT PROTEIN(S) AND PEPTIDE(S) PURIFIED FROM INDIAN RUSSELL'S VIPER (*Daboia russelii russelii*) VENOM"** submitted by Ms. RUPAMONI THAKUR to Tezpur University in the Department of Molecular Biology and Biotechnology under the School of Sciences in partial fulfillment of the requirement for the award of the degree of Doctor of Philosophy in Molecular Biology and Biotechnology has been examined by us and found to be satisfactory.

The Oral Defence Evaluation (ODEC) recommends for the award of the degree of Doctor of Philosophy

Signature

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Date: 01 June, 2015

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External Supervisor

Date: 01 June, 2015

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Rupamoni Thakur

Place: Tezpur **(Rupamoni Thakur)**

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ABBREVIATIONS

Abbreviation	Full form
ALP	Alkaline phosphatase
AO	Acridine orange
AT-III	Antithrombin-III
BSA	Bovine serum albumin
CM	Carboxymethyl
CPK-MB	Creatine phosphokinase
DEAE	Diethylaminoethyl
DIC	Disseminated intravascular coagulopathy
DMEM	Dulbecco's modified eagle medium
DMSO	Dimethyl sulfoxide
DTT	Dithiothreitol
EB	Ethidium bromide
ECM	Extracellular matrix
EDTA	Ethylenediaminetetraacetic acid
EMEM	Eagle's minimal essential medium
FPA	Fibrinopeptide A
FPB	Fibrinopeptide B
FPLC	Fast protein liquid chromatography
Hb	Hemoglobin
IAA	Iodoacetic acid
IC_{50}	Half maximal inhibitory concentration
IL	Interleukin
i.p.	intraperitoneal
i.v.	intravenous
LC MS-MS	Liquid chromatography–mass spectrometry
LD_{50}	median lethal dose
LAAO	L-amino oxidase

Abbreviation	Full form
MALDI-TOF-TOF-MS	Matrix-assisted laser desorption/ionization –Time of flight- mass spectrometry
MAV	Monovalent antivenom
MCH	Mean corpuscular hemoglobin
MCHC	Mean corpuscular hemoglobin concentration
MCV	Mean corpuscular volume
MTT	(3-(4,5-Dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide)
PAV	Polyvalent antivenom
PBS	Phosphate buffered saline
pBPB	Para-bromophenacryl bromide
PC	Phosphatidylcholine
PLA_2	Phospholipase A_2
PI	Propidium iodide
PMSF	Phenyl methyl sulphonyl floride
PPP	Platelet poor plasma
PRP	Platelet rich plasma
PS	Phosphatidylserine
RBC	Red blood corpuscles
ROS	Reactive oxygen species
RP-HPLC	Reversed phase-high performance liquid chromatography
RVV	Russell's Viper venom
SD	Standard deviation
SDS-PAGE	Sodium dodycyl sulphate-polyacrylamide electrophoresis
SGOT	Serum glutamic oxaloacetic transaminase
SGPT	Serum glutamic pyruvic transaminase

Abbreviation	Full form
Snaclec	Snake venom C-type lectin protein
SPR	Surface plasmon resonance
SVMP	Snake venom metalloprotease
SVSP	Snake venom serine protease
SVTLE	Snake venom thrombin-like enzyme
TEMED	Tetramethylethylenediamine
TFA	Trifloroacetic acid
TLCK	N-tosyl-L-lysine chloromethyl ketone
TNF-α	Tumour necrosis factor-α
TPCK	Tosyl phenylalanyl chloromethyl ketone
WBC	White blood corpuscles
WHO	World health organization
1X PBS	10 mM phosphate buffered saline
3FTx	Three finger toxin

Chromogenic substrates

Abbreviation	Full form
B2133	N-Benzoyl-Pro-Phe-Arg-p-nitroanilide hydrochloride
B3133	Nα-Benzoyl-L-arginine 4-nitroanilide hydrochloride
F3301	CH_3OCO-D-CHA-Gly-Arg-pNA
I0898	D-Ile-Pro-Arg p-nitroanilide dihydrochloride
T1637	N-(p-Tosyl)-Gly-Pro-Arg p-nitroanilide acetate salt
V0882	D-Val-Leu-Lys 4-nitroanilide dihydrochloride

CHAPTER I

INTRODUCTION

1 INTRODUCTION

1.1 Introduction

With more than 3,000 documented species [1-3], snakes are often considered as one of the most fascinating yet dreaded and creepy reptile which inhabits planet Earth. Since the dawn of time, snakes have enthralled the human mind and have often been a part of our elite culture, religion and mythology. Though most of the snakes aren't as deadly as their looks, however, this species of reptile has gained a reputation as one of the most fearsome creature. By and large, snakes are divided into two categories based on the presence or absence of the venom gland: venomous and non-venomous snakes. About 600 species of the living snakes are reported to be venomous in nature [3-4]. These elongated, limbless, carnivorous reptiles can survive in a wide gamut of habitats and can successfully thrive on land or in water [2, 5].

All venomous creatures, including venomous snakes, possess a refined mechanism for capturing, immobilizing and digesting their prey by employing one of the most interesting facets possessed by them – 'Venom'. Venom is generally composed of numerous biologically active compounds which predominantly include enzymatic and non-enzymatic proteins, peptides and low molecular weight compounds in addition to other important components [6-11]. Every individual component of the venom has a special role to play with characteristic properties of high affinity, selectivity and potency towards its substrate which can be explored to serve as potential drugs and scaffolds of drug design [9, 12-14].

Russell's Viper is one of the deadliest snakes found across the countries of South-East Asia including India [15-16]. This brief prologue introduces the major concerns governing snakebite cases and the contribution of Russell's Viper to the global and national epidemiology of snake envenomations. The chapter also focuses on the contributions of snake venom proteins/peptides which affect the process of hemostasis via its procoagulant or anticoagulant effects.

1.2 Taxonomy and global distribution of venomous snakes

The suborder Serpentes comprises approximately 3000 species of snakes [1-3]. All venomous snakes reportedly belong to the Colubroidea superfamily [2, 17] which has been subsequently subdivided into four families; Colubridae, Atractaspididae, Elapidae and Viperidae [18]. The Colubridae are the largest and most assorted family of snakes comprising of approximately 1,800 species distributed worldwide [19]. The Atractaspididae contain around 30 species which are distributed across Africa and Arabia [18]. The Elapidae consists of about 350 species which are widely distributed across the Americas, Africa, the Middle East, Asia and Australasia [18]. There are two sub-families of the Elapidae family; the Elapinae (coral snakes, cobras, mambas and kraits) and the Hydrophiinae (sea snakes). The elapids are front-fanged snakes, the majority of which produce extremely potent neurotoxic venom, and are of significant medical importance to humans [2, 18]. The Viperidae contains around 310 species and have a worldwide distribution (except Australasia), occupying a range of terrestrial habitats across the Americas, Africa, Europe and Asia [2, 18]. The Viperidae contains two main sub-families; the Viperinae (pit-less Vipers e.g. *Bitis*, *Cerastes*, *Echis* and *Daboia* genera) and the Crotalinae (pit Vipers e.g. *Agkistrodon*, *Bothrops* and *Crotalus* genera) [2, 18]. Viperids which includes the Russell's Viper are front-fanged snakes proficient of producing highly potent hemotoxic venom and are responsible for a substantial number of human mortality and morbidity globally [2, 15].

1.3 Distribution of venomous snakes in India

More than 250 species and subspecies of snakes have been documented in India, out of which about 50 are highly venomous [5, 20]. The snake species posing major health risks in India are classified under the "big four" which includes members of the elapidae family such as the Indian spectacled cobra (*Naja naja*) and common krait (*Bungarus caeruleus*), as well as members of the Viperidae family such as Russell's Viper (*Daboia russelii*) and Saw-scaled Viper (*Echis carinatus*) [21]. These four species are widely distributed throughout the country

although areas such as the North-east region of India, the Himalayan region and the Andaman and Nicobar Islands have their own distinctive snake fauna [21]. Other venomous snake species frequently encountered in India include the central Asian cobra (*Naja oxiana*), Indian monocellate cobra (*N. kaouthia*), greater black krait (*Bungarus niger*), banded krait (*B. fasciatus*), Wall's krait (*B. walli*), Sind krait (*B. sindanus*) and hump nosed pit Viper (*Hypnale hypnale*) [20, 21]. A few other species of snakes prevalent in India are coral snakes (*Calliophis*, *Sinomicrurus*), sea snakes (*Laticauda*, *Kerilia*, *Lapemis*, *Hydrophis*, *Enhydrina*, *Astrotia*, *Pelamis*), pit Vipers (*Trimeresurus*, *Protobothrops*, *Gloydius*, *Ovophis*) and the king cobra (*Ophiophagus hannah*) [21].

1.4 Snake envenomation: A major health concern

Snake envenoming is an important and neglected public health problem in many tropical and sub-tropical countries. In spite of the rigorous and focused investigations on the global incidence of snakebite mortality and morbidity, the true universal burden of snakebite and its associated mortality are often obscure [3, 22-23]. This is because, in most of the countries, hospital-based data on snakebite incidence and mortality is often incomplete since many victims do not seek hospital treatment and prefer traditional medications [3, 24-25]. Very few countries especially those pertaining to the Central and South American region possess a dependable catalog of snakebite incidence which is probably due to better snakebite management strategies, including the development of effective antivenin therapy [3]. Because of the frequent neglect in public and private sector expenditure and attention at local, national, and international levels, and their concentration among the poor, snakebite has been declared as a neglected tropical disease [3, 26]. It is suggested that majority of snakebites occur in rural areas of resource poor countries in Asia and Africa in particular [3, 23-24, 26]. Additionally, snakebite varies seasonally and geographically within countries; i.e., high incidences are reported during agricultural activity, particularly in South-east

Asian countries like India, Bangladesh, Sri Lanka and sub-tropical African countries [3].

1.4.1 Snake envenomations as a neglected tropical disease: The global scenario

Snakebite is an important but neglected health hazard in the tropical and sub-tropical countries of Africa, South-east Asia and South America [3, 24, 26]. Snakebite was recognised for the first time by WHO as a neglected tropical disease in the year 2009 [23]. The first assessment of global snakebite incidence and mortality was undertaken by Swaroop and Grab in 1954 [27]. Their study suggested a global total of 30,000–40,000 deaths from snakebite per year which was an underestimation of the true mortality rate because they relied on hospital and dispensary admissions and excluded central Europe and North Asian countries [23]. As per the recent reports provided by Kasturiratne et al., at least 421,000 envenomings and 20,000 deaths occur each year due to snakebite globally. These figures could be as high as 1,841,000 envenomings and 94,000 deaths [3]. The vast majority of the estimated burden of snakebite is in South and Southeast Asia, sub-Saharan Africa, and Central and South America [3] (Fig. 1.1). However, mortality due to snakebite was reported lower in Central and South America when compared to other high incidence regions because of better snakebite management systems [3]. Estimates of snakebite incidence in sub-Saharan Africa, especially for East Africa are under-reported for many parts of this region [3].

Snakebite cases have been a massive cause of mortality particularly in the rural regions of West African savanna, where approximately 500 snakebites and 40 deaths occur in every 100,000 population [23, 28]. In Malumfashi, Nigeria, 19% of survivors had persistent sequelae and in Kilifi, Kenya, per 100 000 population per year, there were 151 bites and seven deaths [23].

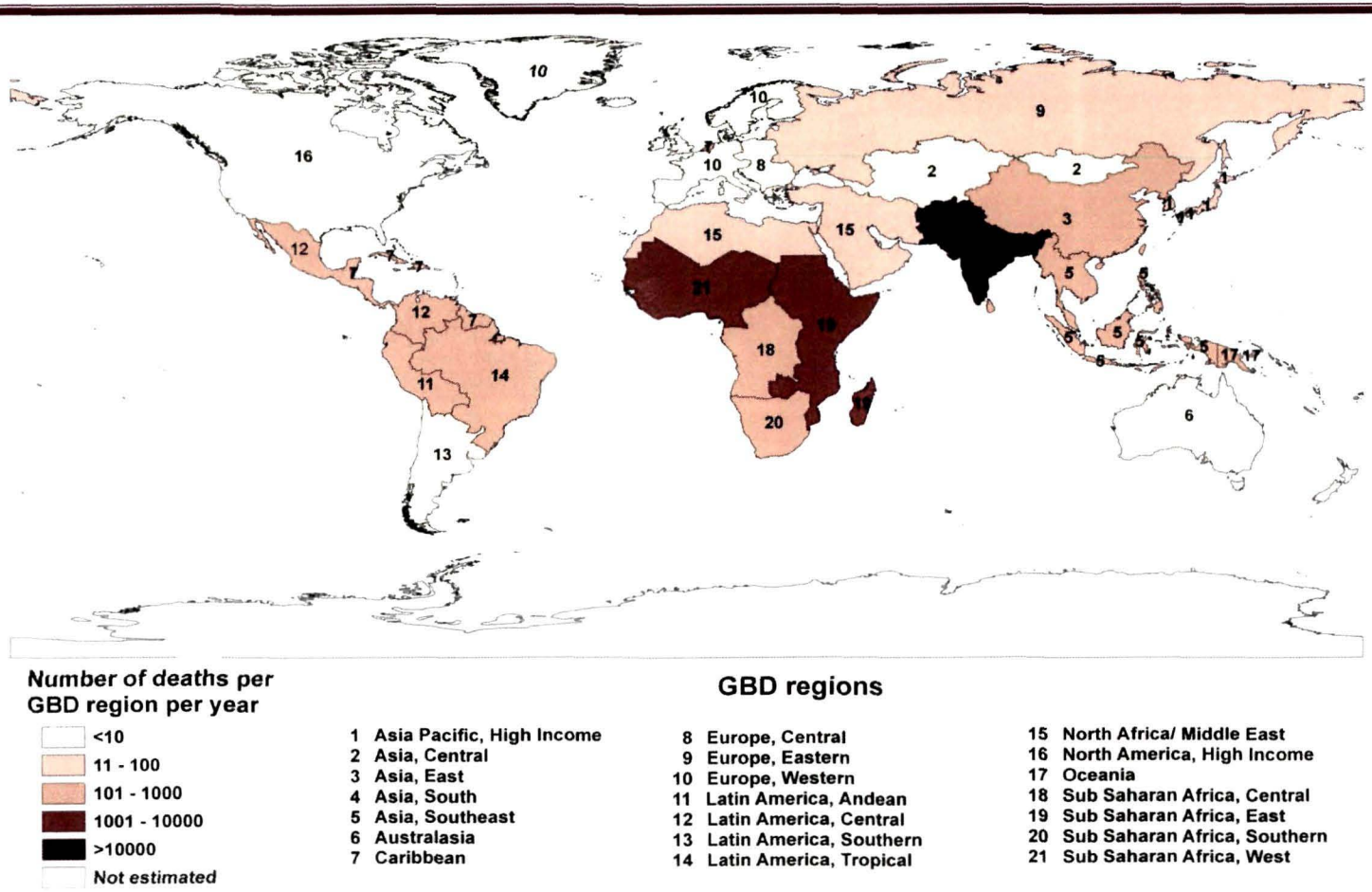


Fig. 1.1 Regional estimates of deaths due to snakebite (Low estimate) [3].

Globally, South Asia has the highest incidence and mortality rates of snakebite [29]. Bites by venomous snakes in this region is reported to cause symptoms such as local tissue damage, neuroparalysis, systemic haemorrhages, generalized myotoxicity, acute renal failure, or complex combinations of these [29]. Apart from India, the other countries of South Asia which harbor majority of snakebite fatality cases include Pakistan, Sri Lanka, Bangladesh, Nepal and Myanmar.

In Pakistan, there are reports of more than 20,000 snakebite deaths annually [26]. In Nepal, there is an estimate of 20,000 snakebites and more than 200 deaths in hospitals annually [25]. In Vietnam, an estimated 300,000 bites per year, and a case fatality ratio of 22% was reported among plantation workers bitten by Malayan pit Vipers in the year 1992-98 [26]. In Mynamar, there were reports of more than 14,000 bites and 1,000 deaths in 1991 [26]. A survey conducted in Bangladesh estimated an annual incidence of 4.3 snakebite cases per 100,000 population and a case fatality of 20% [30]. As per a survey in rural Sri Lanka, hospital data records showed less than half of the deaths due to snakebite [29].

In order to deal with this grave global concern; antivenom is the only specific treatment for snakebite [23, 26]. However, the existing antivenom market covers a very limited number of medically significant species [23, 26]. Hence, the requirement for the development of better and profuse antivenom strategy is a welcome measure for dealing with this dreaded tropical disease.

1.4.2 Snake envenomations as a neglected tropical disease: The Indian scenario

Snakebite is an important cause of mortality and morbidity in modern India. In fact, the importance of snakebite cases has been systematically underestimated and under-reported in India. A recent research on snakebite suggested that it is responsible for an estimated 45,900 cases which constitutes 5% of all injury deaths and nearly 0.5% of all deaths in India in the year 2001-03 [20]. This report was

generated as a part of the India's first national survey of the causes of death, the Million Death Study, undertaken in 2001-03 by the Registrar General of India and the Centre for Global Health Research [31]. On the contrary, the Government of India's Central Bureau of Health Intelligence has estimated as low as 61,507 bites and 1124 deaths in 2006; 76,948 bites and 1359 deaths in 2007 and as high as 50,000 average deaths each year in India due to snakebite [32]. The statistical disparity and the under-reporting of snakebite cases have been attributed to the lack of data in the public health sectors particularly in the rural areas [20, 33].

Snakebite deaths in India occurs mostly in rural areas (97%), are more common in males (59%) than females (41%), and peaks during the monsoon months (from June to September) [20]. Annual snakebite deaths are reported to be greatest in the states of Uttar Pradesh, Orissa, Andhra Pradesh and Bihar (4,500) [20] (Fig. 1.2). In another report of hospital-based studies, mortality rates ranged from 3% to 20% in Northern India [34]. In Maharashtra, an estimated 10,000 annual venomous snakebites account for 2000 deaths [23].

Other states with a large number of snakebite cases include West Bengal, Tamil Nadu, Chhattisgarh and Kerala [20, 33, 35]. Unfortunately, there is no record of snakebite mortality and morbidity cases in the north-eastern states of India which harbor a vast variety of venomous snakes.

1.5 Indian Russell's Viper (*Daboia russelii russelii*): A brief overview

The Russell's Viper, commonly known as Daboia, Chain Viper or Indian Russell's Viper, is a venomous Viper snake, found predominantly in Asia [36]. The species was named in honour of Dr. Patrick Russell, a Scottish surgeon and naturalist, who had described this reptile, and named the genus on its Hindi term "Daboia", which means "the lurker."

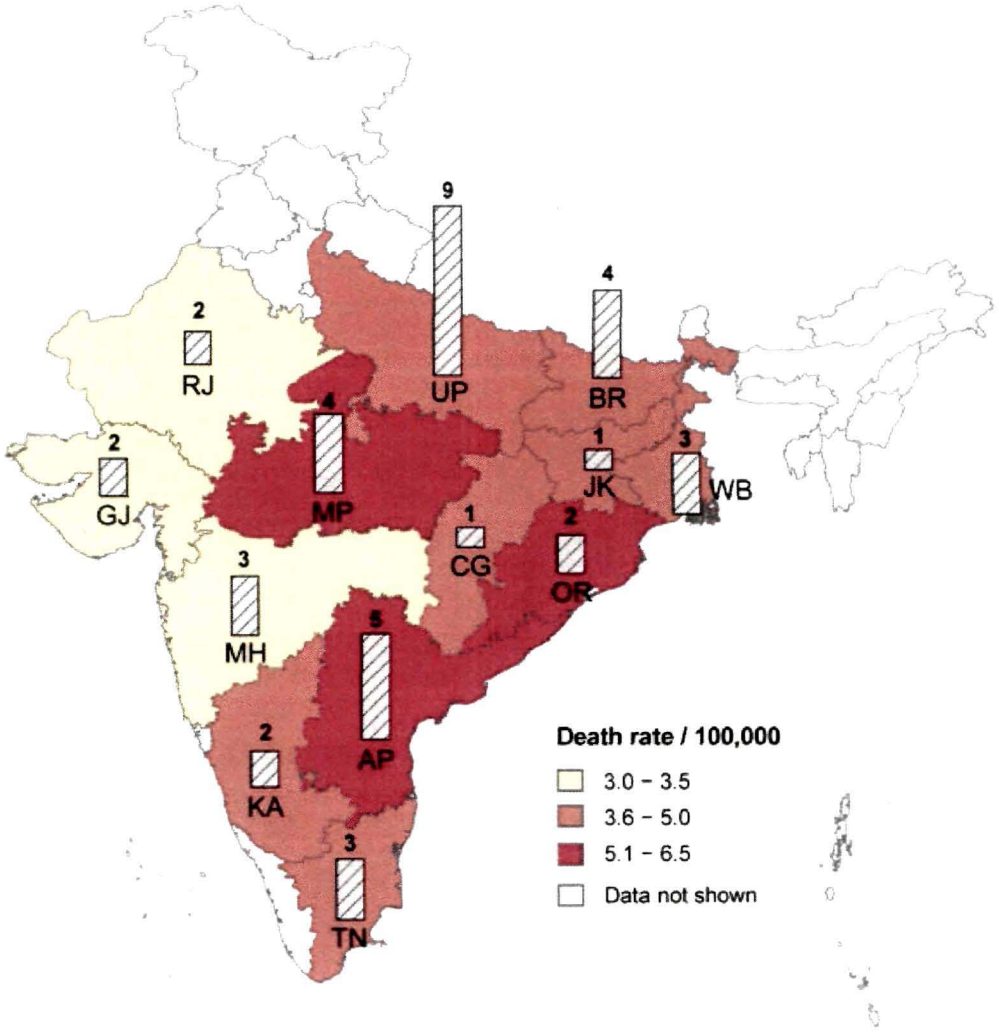


Fig. 1.2 Estimated snakebite deaths in India. AP-Andhra Pradesh, BR-Bihar, CG-Chhattisgarh, GJ-Gujarat, JH-Jharkhand, KA-Karnataka, MP-Madhya Pradesh, MH-Maharashtra, OR-Orissa, RJ- Rajasthan, TN-Tamil Nadu, UP-Uttar Pradesh, WB-West Bengal [20].

1.5.1 Systemic classification

Kingdom: Animalia

Phylum: Chordata

Group: Vertebrata

Subphylum: Gnathostomata

Class: Reptilia

Subclass: Diapsida

Order: Squamata

Suborder: Serpentes

Infraorder: Alethinophidia

Superfamily: Colubroidea

Family: Viperidae

Subfamily: Viperinae

Genus: *Daboia*

Species: *russelii*

Sub species: *russelii*

1.5.2 Distinctive features

Russell's Viper is a medium to large-sized reddish-brown snake. The head of a Russell's Viper is flattened, triangular and distinct from the neck. The dorsal scales of a Russell's Viper are strongly keeled and the dorsal color pattern is of a deep yellow or brown ground color with series of symmetrical dark brown spots that run the length of the body in a distinctive chain-like pattern [5, 37]. (Fig. 1.3).

Fig.1.3 Indian Russell's Viper (*Daboia russelii russelii*) [38].

1.5.3 Description

Adult Russell's Vipers are reported to be slow and sluggish in behaviour whereas the juveniles are generally more active. The Russell's Viper is a rough scaled snake with a flattened and triangular head distinct from the neck marked by the presence of vertical eye pupils [39]. The crown of the head of the Russell's Viper is covered with irregular but strongly fragmented scales. Russell's Viper is recognized by its stout body and the presence of strongly keeled dorsal scales [39]. The tail of a Russell's Viper is relatively shorter in comparison to the body [39]. The dorsal body of Russell's Viper consists of a deep yellow or brown color, with a regular chain-like pattern running along the length of its body [39]. The ventral side of a Russell's Viper is white, yellow or pink, often with an irregular scattering of dark spots. Russell's Viper has an average length of 100 cm but can attain a maximum length of 180 cm [39].

1.5.4 The venom delivery apparatus

The venom glands are modified parotid glands which are connected to an elongated and specialized structure for venom delivery called fangs. Opisthoglyphous snakes (Colubridae) are rear-fanged snakes possessing rear-grooved elongated fangs at the back of the mouth [40-41]. Proteroglyphous snakes (Elapidae) are front-fanged snakes possessing deeply grooved fangs forming a channel through which venom is directed. Proteroglyphous fangs are relatively short, and therefore elapids are required to maintain contact with their prey during injection of venom [40-41]. Solenoglyphous snakes (Viperidae) are front-fanged snakes with long hollow, tubular hyper-extendable fangs resembling hypodermic needles [40-41]. The snakes belonging to the Viperidae family are characterized by the presence of moveable maxilla and tubular front fang unit [40-41]. The fangs of Viperids such as the Russell's Viper are designed so as to ensure efficient venom delivery upon strike enabling the infusion of a large quantity of venom into the victim/prey to exert rapid pharmacological effects (Fig. 1.4).

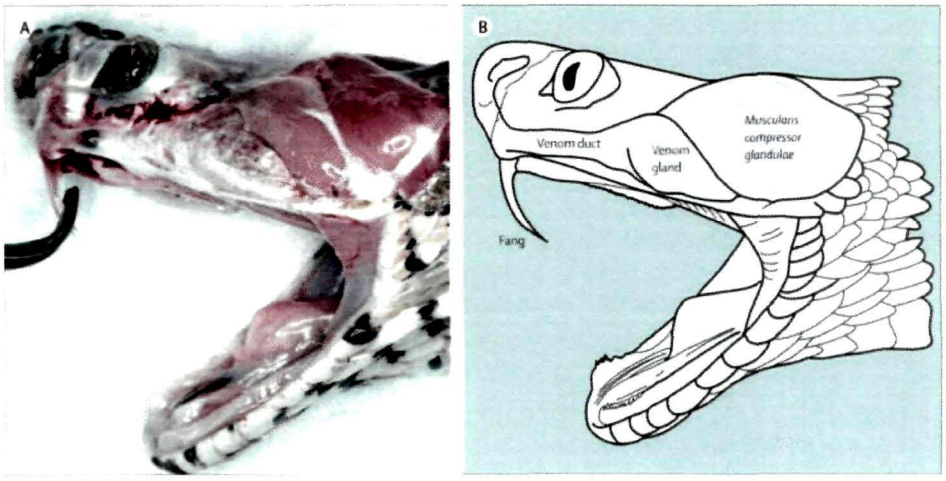


Fig. 1.4 Venom apparatus of Russell's Viper (*Daboia siamensis*) (A) Dissected specimen. (B) Annotated diagram of dissected specimen. Figure adapted from Mackessey and Baxter [43].

1.5.5 Habitat & Distribution

Russell's Viper is distributed with varying densities and occurrence throughout the Indian subcontinent which includes India along with its neighboring countries such as Pakistan, Bangladesh, Bhutan, Nepal and Sri Lanka [36-37]. Its distribution is rather discontinuous in other parts of South-east Asia such as Southern China, Taiwan and the Lesser Sunda islands [42].

On the basis of its geographical distribution, *Daboia russelii* has been categorized into a number of subspecies. The species inhabiting the Indian subcontinent (India, Pakistan and Bangladesh) has been classified as *Daboia russelii russelii* [42]. The other subspecies of Russell's Viper include *Daboia russelii nordicus* pertaining to Northern Indian region, *Daboia russelii pulchella* found in the Sri Lankan region, whereas *Daboia russelii formosensis* and *Daboia russelii limitis* are abundant in Taiwan and in Indonasia, respectively [15, 42]. Recently, on the basis of morphological characteristics and mitochondrial DNA analysis, Russell's Viper has been categorized into two subspecies, namely *Daboia russelii russelii* which is abundant in the Indian subcontinent and *Daboia russelii siamensis* reported in regions other than the Indian subcontinent [42] (Fig. 1.5).

In India, Russell's Viper is profusely distributed in the states of Northern India such as Punjab and is very common along the Western Coast and its hills in Southern India and up to Bengal [36-37]. It is rare in the Ganga valley, Northern Bengal and North-eastern states of India [36].

Russell's Viper is mostly found in open fields but may also be found in forested plantations and farmlands near human settlements [15]. Russell's Viper is preferably encountered in plains and coastal lowlands. Though Russell's Viper is generally not found at high altitudes, but they are reported to survive in altitudes as high as 2300–3000 m [5].

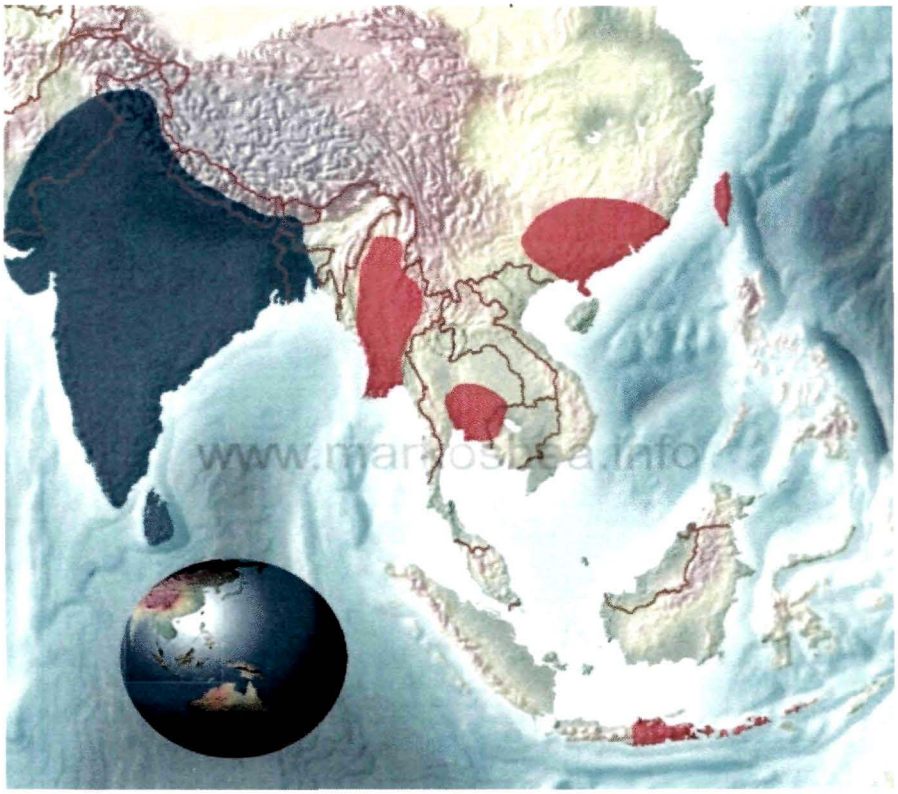


Fig.1.5 Geographical distribution of Russell's Viper. *Daboia russelii russelii*, (blue) and Southeast-Asian Russell's Viper, *Daboia russelii siamensis*, (red) in Southeast Asia [44].

1.5.6 Russell's Viper envenomation: pathophysiology and clinical manifestations

Like most Viperids, the venom of Russell's Viper is hemotoxic in nature leading to profuse cell and tissue damage [36]. On an average, an adult Russell's Viper can inject 225-250 mg of venom to its prey/victim [45]. For most humans, the lethal dose of RVV is approximately 40–70 mg [39]. In general, Russell's Viper envenomation symptoms commence with extreme pain at the site of the bite and regional lymph nodes, immediately followed by swelling of the affected extremity, vomiting, epigastric pain and drowsiness [15, 46]. Besides local swelling and bleeding, blistering and necrosis are also observed at the bite site in about 2% of the cases [15, 46]. The major clinical manifestations of Russell's Viper

envenomation are systemic coagulopathy and tissue hemorrhage [15]. Systemic bleeding is a common symptom resulting in bleeding from the gums, hematuria, bloody sputum and conjunctival edema [15, 46]. Renal failure and severe disseminated intravascular coagulation (DIC) occurs in approximately 25-30% of untreated bites [15]. Other clinical manifestations of Russell's Viper include intravascular hemolysis, pituitary infarction, neuromyotoxicity, generalized capillary permeability and hypotension [15]. Death from septicemia, kidney, respiratory or cardiac failure may occur within 1 to 14 days post-bite or even later [36].

Compositional variation in snake venom due to dietary and geographical differences can have severe outcomes for snakebite victims by rendering the specific antibodies found in antivenoms ineffective against heterologous toxins found in different venoms [23, 42, 46]. Studies related to clinical manifestations of Russell's Viper envenomation and composition of RVV of a particular geographical location have suggested that due to differences in geographical origin, RVV varies in its composition and illustrates differences in the clinical manifestations following envenomation [46-48]. Symptoms such as neurotoxicity and intravascular haemolysis were predominantly noticed followed by Russell's Viper bite in victims from Sri Lankan regions, whereas complicacies such as pituitary hemorrhage and increased capillary permeability were observed in envenomation by the same species from the geographical regions of Burma and South India [46, 48].

The composition and lethal potency of RVV is also reported to show region-ywise variation within India [47]. In the eastern region of India, RVV exhibit procoagulant effects (at low doses) and anticoagulant effects (at higher doses) and correspondingly Russell's Viper bite is characterized by spontaneous haemorrhage [48-49]. On the contrary, RVV of all the other regions exhibit procoagulant effects [48]. Russell's Viper bite in South India is characterized by neurotoxic and myonecrotic symptoms [48, 50]. RVV from Western India is mainly hemorrhagic

[48]. RVV from the eastern region of India was reported to lack neurotoxicity [46] but contributed to intravascular hemolysis and hemorrhage [46].

Because of the variations in RVV due to geographical disparities, it is important to study the pathophysiological contribution of the individual venom components. Such studies can help researchers as well as clinicians in developing better antivenin strategies, pertaining to a local area.

1.6 Snake venom composition: A brief overview

Snake venoms are the most multifaceted and complicated forms of all naturally produced venoms and poisons. Snake venom is a highly modified salivary secretion which has evolved eventually to aid this limbless creature for three natural purposes. Firstly, the venom serves as an offensive weapon in incapacitating and immobilizing their prey [51]. Secondly, it is used as a defensive tool against their predator and thirdly, the venom abets in digestion of their prey [52]. Snake venom has evolved into a complex mixture of many biologically active compounds which include various organic and inorganic compounds. Approximately, 90-95% of the snake venom components are proteins and peptides which are biologically active components of the venom contributing to toxicity [53-54]. The non-protein components which constitute the rest of the venom (5%) consist of low-molecular substances such as amino acids, amines, lipids, nucleosides and nucleotides, carbohydrates and metal ions [55]. Most of these components are biologically less active or assist the venom proteins and peptides to exert their toxicity [55].

The proteins and peptides of snake venoms are either enzymatic or non-enzymatic in nature [9, 11-12]. A large portion of these proteins are enzymes (molecular weight 13,000-150,000) which are mainly hydrolases and are generally composed of exopeptidases, endopeptidases, phosphodiesterases and phospholipases which assist in prey digestion [56-58]. Non-enzymatic proteins, on the other hand contribute towards prey immobilization [11, 59]. Snake venom contains at least 26

types of enzymes, 12 of which are common in all venom although their levels vary markedly [56]. Viperid and crotalid venoms are strongly proteolytic in nature due to the presence of about 80 – 95% enzymes in the total dry matter, whereas elapid and hydrophid venoms are weakly proteolytic with only 25 – 70% enzymes [56-58]. Some of the important enzymatic and non-enzymatic snake venom components which affect the process of blood coagulation and fibinolysis are discussed in sections 1.8 and 1.9, respectively.

1.7 Brief overview of the coagulation system

Blood coagulation is a complex cascade of events that results in the formation of stable fibrin clot. The various coagulation factors and plasma proteins, their molecular masses and their plasma concentrations (in mammals) have been listed in Table 1.1 [60].

Under normal physiological conditions, the process of hemostasis maintains a subtle balance between clot formation and clot dissolution [58]. The process of blood coagulation is triggered by vascular injury, leading to vasoconstriction in the vicinity of the injured vessel, changes in the vascular endothelial surface and the release of procoagulant factors [58]. The coagulation system is responsible for the conversion of soluble fibrinogen into fibrin [58]. The fibrin clot reinforces the platelet plug formed during primary hemostasis.

Initially during blood coagulation, the coagulation factor VII from the plasma comes in contact with tissue factor (TF) on cells outside the vasculature, as a result of blood vessel injury, and undergoes activation to form factor VIIa [61]. The factor VIIa/TF complex eventually activates the zymogens, factor IX and factor X to their activated forms, factor IXa and factor Xa, respectively. Factor Xa along with factor Va, membrane phospholipids and Ca^{2+} ions form a prothrombinase complex on the surface of TF-bearing cells and activate prothrombin to thrombin on the surface of platelets [61]. In the following step, the formation of the fibrin clot results from the activation of fibrinogen by thrombin. Therefore, thrombin

formation is a critical step in coagulation because thrombin not only initiates fibrin formation, but also activates major coagulation factors such as factor V, factor VIII, factor XIII as well as platelets [58, 61].

Table 1.1 Plasma concentration of blood coagulation factors [60].

Component	Molecular Weight (daltons)	Plasma Concentration (μg/ml)	Plasma Concentration (μM)
Fibrinogen (I)	330,000	3000	9.09
Prothrombin (II)	72,000	100	1.388
Factor V	330,000	10	0.03
Factor VII	50,000	0.5	0.01
Factor VIII	330,000	0.1	0.0003
Factor IX	56,000	5	0.08928
Factor X	58,800	8	0.13605
Factor XI	160,000	5	0.031
Factor XII	80,000	30	0.375
Factor XIII	320,000	10	0.03124
Protein C	62,000	4	0.0645
Protein S	69,000	10(free)	0.1449
Protein Z	62,000	2.2	0.0355
Prekallikrein	86,000	50	0.5814
HK	110,000	70	0.6363
Fibronectin	450,000	300	0.6667
Antithrombin III	58,000	290	5
Plasminogen	90,000	216	2.4
Urokinase	53,000	0.1	0.001887
Heparin Cofactor II	66,000	90	1.3636
$Alpha_2$-Antiplasmin	63,000	60	0.9524
Protein C Inhibitor	57,000	4	0.0702
$Alpha_2$-Macroglobulin	725,000	2100	2.8966

Following the healing of the wound, the fibrinolytic system is activated which gradually dissolves the clot mainly due to the catalytic action of the serine protease, plasmin, which degrades the fibrin network [58]. Plasmin is produced by plasminogen activation catalyzed by activators such as tissue-type plasminogen activator (t-PA) and/or urokinase type plasminogen activator (u-PA) [58].

1.8 Snake venom components affecting blood coagulation and fibrinolysis: A classification based on their structural aspects

1.8.1 Snake venom proteases

Proteases or proteolytic enzymes function as molecular knives incising long amino acid sequences into fragments, a process that is essential for the synthesis, biochemical identification and ultimate destruction of all proteins [62]. Proteases are highly selective and show substrate specificity. According to their catalytic mechanisms, proteases are classified into the following six types: aspartic, cysteine, glutamic, metallo, serine, and threonine [63]. Among various lethal and non-lethal enzymatic components, proteases contribute significantly to the overall composition of most of the snake venoms including RVV. The venom proteases that have been studied so far exhibit vivid pharmacological properties and mostly belong to either metallo- or serine protease family.

1.8.1.1 Snake venom metalloproteases

Snake venom metalloproteases (SVMPs) play a significant role in venom-induced pathogenesis by damaging the local tissues and causing tissue necrosis [49, 64-67]. Structurally, SVMPs are monozinc endopeptidases that are classified under the subfamily of 'reprolysins' along with the ADAMs ('a disintegrin and metalloprotease' proteins), [68-71], as they share a common overall domain organization [69, 72]. These reprolysins are classified as part of the 'metzincin' superfamily of zinc-dependent metalloproteases which exhibit zinc binding environments besides possessing the consensus sequence 'HEXXHXXGXXH' and a methionine turn [68, 70-71]. After numerous speculations, SVMPs have

finally been classified into three main groups based on the differences in their domain structure per se [72] (Fig. 1.6):

i. P-I class of SVMPs are the simplest class of proteases comprising only the metalloprotease domain.

ii. P-II class of SVMPs possesses a disintegrin-like domain in addition to the metalloprotease domain. This class of SVMPs is further classified into 6 different subclasses.

iii. P-III SVMPs in addition to the metalloprotease and disintegrin-like domains also contain cysteine-rich domains. The P-III SVMPs are further divided into four subclasses based on their distinct post-translational modifications such as homo-dimerization (P-IIIc), or proteolysis between the metalloprotease and disintegrin-like domains (P-IIIb) [73]. It is important to note that previously classified heterotrimeric P-IV SVMPs that contains an additional snake C-type lectin-like (snaclec) domain is now included in the P-III group as a subclass (P-IIId), since no P-IV gland trancriptome has been established to date [73]. Consequently, the P-IIId class of SVMPs is considered as a post-translationally modified derivative of the P-IIIa subclass of the P-III SVMP [72-73].

SVMPs exhibit a large variation in molecular weights ranging from 15 kDa to over 100 kDa [65, 67, 69, 74]. Crystallographic studies of many of the SVMPs have helped the researchers to understand the structure-function properties of this class of proteases in a much better way. In addition to their pathological significance, SVMPs also remain relevant targets for the development of novel therapeutic agents as they have been shown to interact at several stages of the blood coagulation cascade causing changes in the coagulopathy through either the stimulation or prevention of fibrin formation [75-76].

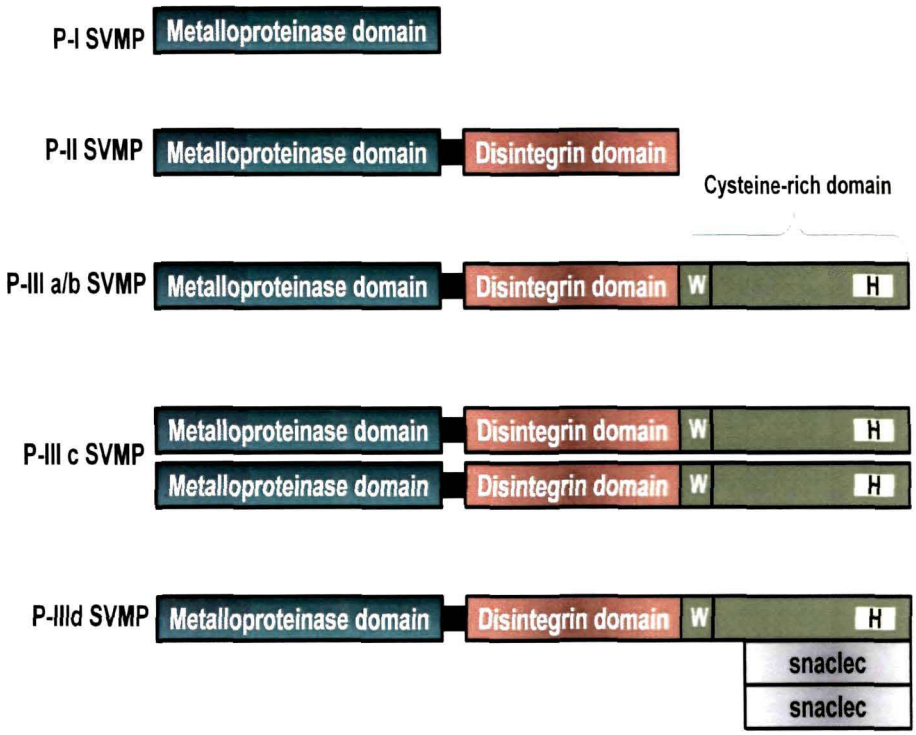


Fig. 1.6 Schematic diagram of the domain structure of SVMPs. Each domain or subdomain is represented by a different color. The metalloprotease domain; the disintegrin (or disintegrin-like) domain; the cysteine rich "wrist" subdomain (W); the cysteine-rich "hand" subdomain (H); snaclec, snake venom C-type lectin-like domain [73].

1.8.1.2 Snake venom serine proteases

Serine proteases are characterized by the presence of a typical and highly reactive "catalytic triad" in their active site, which comprises of serine[195], histidine[57] and aspartic acid[102] [77] (Fig. 1.7). The catalytic triad is a highly conserved and coordinated structure. In the event of enzyme catalysis each of the key amino acids of the catalytic triad has a specific role to play. The catalysis of a peptide cleavage is characterized as a 'ping-pong' catalysis which involves many intermediates [77]. Briefly, the hydroxyl group of serine acts as a nucleophile and attacks the carbonyl carbon of the scissile peptide bond of the target substrate [77]. The histidine nitrogen co-ordinates with serine by accepting the hydrogen from the

hydroxyl group of serine. Finally, the carboxyl group of aspartic acid forms hydrogen bonds with histidine leading to the release of the peptide molecule [77].

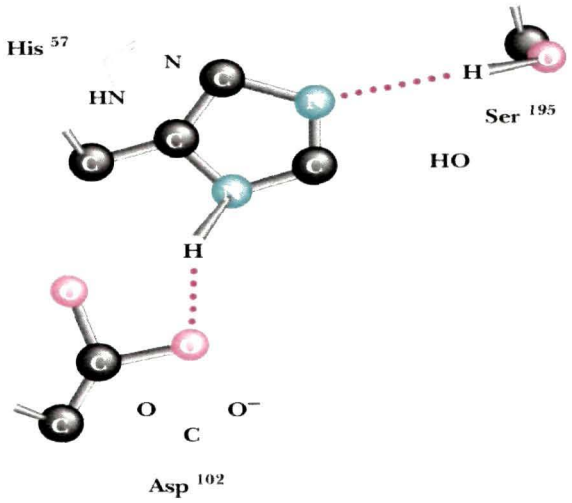


Fig. 1.7 The catalytic triad of a typical serine protease (chymotrypsin) [78].

Snake venoms serine proteases (SVSPs) are reported to contribute significantly in the pharmacological attributes of venoms of Viperidae, Crotalidae, Elapidae and Colubridae families of snakes [79]. The SVSPs belong to the MEROPS peptidase family S1, subfamily S1A (chymotrypsin-A subfamily) [63] and are reported to originate from the glandular kallikrein that later progress to obtain coagulation and fibrino(geno)lytic activities [63]. The enzymatic properties of these proteases are influenced significantly by the peptide moiety adjacent to the scissile bond in their substrates. It has also been elaborated that SVSPs illustrate rigid macromolecular substrate specificity in contrast to the lesser specific activity of trypsin [79].

Despite possessing a high degree of similarity in their amino acid sequences and three dimensional structures, SVSPs exhibit different substrate specificity and biological functions [79-83]. It has been suggested that a relatively minor surface

residue changes can result in significant differences in the preference for hydrolyzing a particular macromolecular substrate by SVSPs [79-83]. Single venom may contain several protease isoenzymes which are encoded by multigene family and during the process of evolution; duplication followed by variation of a single ancestral gene resulted in formation of several protease isoenzymes in single venom [84]. Majority of the SVSPs have been reported to affect the hemostasis of victim/prey owing to their procoagulant or anticoagulant nature in addition to demonstrating proteolytic cleavage of fibrin(ogen), kininogen, factor V, factor X, prothrombin, protein-C or plasminogen as well as platelet aggregation [79-83, 85-88].

1.8.2 Phospholipase A_2

Phospholipase A_2 (PLA_2) are a specialized group of enzymes that catalyzes the hydrolysis of the *sn*-2 position of glycerophospholipids, leading to production of free fatty acids and lysophospholipids [89]. PLA_2 enzymes are found to be ubiquitously present as both extracellular and intracellular forms and hydrolyze a diverse array of phospholipids [89]. On the basis of their structure, the PLA_2 superfamily is classified into 16 groups (GI to GXVI) [90]. Unlike naturally occurring mammalian PLA_2s, snake venom PLA_2 enzymes are secretory PLA_2s ($sPLA_2$) which are often toxic and in addition to their potential role in the process of prey digestion, also exhibit an extensive variety of pharmacological effects by interfering in normal physiological processes leading symptoms such as neurotoxicity, cardiotoxicity, myotoxicity, platelet aggregation, oedema, haemolysis, anticoagulation, convulsion, and hypotension [45, 83, 91-98].

Venoms of snakes belonging to Viperidae family are rich sources of PLA_2 enzymes [45, 92-93, 95-100]. These enzymes have originated through a process of 'accelerated evolution' that has lead to the incorporation of multiple amino acid substitutions of solvent exposed charged residues in these molecules [101]. Interestingly, there seems to be no clear relationship between the strength of enzymatic activity (extent of phospholipid hydrolysis) and pharmacological action

of PLA_2s (such as its anticoagulant effects) *in vivo* [102]. However, studies related to the membrane domain hypothesis of snake venom PLA_2s have suggested that the pharmacological properties of these enzymes may be attributed to their differential mode of membrane damaging effects [45, 91, 96-98]. Additionally, many of the snake venom PLA_2s also affect the coagulation cascade due to its non-enzymatic interaction with members of the coagulation cascade [94, 96, 98].

1.8.3 Protease inhibitors

Snake venoms contain a wide variety of inhibitors which interacts or binds with different proteases of the coagulation cascade, thereby leading to their inactivation. Many of these protease inhibitors belong to the family of Kuntitz-type serine protease inhibitors with a few exceptions [59, 103-106]. The Kunitz-type serine protease inhibitors are ubiquitous and consist of approximately 60 amino acid residues including three disulfide bridges, and a P1 site responsible for the specificity of these inhibitors to their cognate enzymes [107]. Consequently, these inhibitors have been divided into five classes: body trypsin inhibitors, venom chymotrypsin inhibitors, venom trypsin inhibitors, double-functional toxins, and potassium channel blockers [108].

The Kunitz-type serine protease inhibitors are known to be involved in various physiological processes including blood coagulation and fibrinolysis [59, 103-106]. The physiological significance of these snake venom protease inhibitors and their role in the process of hemostasis has been discussed in details in section 1.9.2.1.

1.8.4 Snaclecs

Snaclecs are a group of lectin-proteins which resemble the typical folding in classical C-type lectins, such as selectins and mannose binding proteins [109-111]. Typically, snaclecs have a basic structure of αβ-heterodimers covalently linked by a disulfide bridge. Some snaclecs contain a single αβ-heterodimers, whereas the

others are multimeric αβ-heterodimers linked by either non-covalent or covalent interactions [109-111].

Snaclecs are one of the profuse components of snake venom with various biological activities including anticoagulant and platelet-modulating activities [109-113]. Their action on platelets has often served as useful tools to investigate platelet membrane receptors and signaling pathways [109, 112, 114]. Many of these snaclecs bind to members of the coagulation cascade which play a critical role in maintaining hemostasis such as prothrombin and factor IX and factor X, thereby leading to their activation or inactivation [109-113]. In addition, snaclecs also bind to membrane receptors on platelets and other cells [109-1113]. Most of the snaclecs have more than one binding site and/or may interact with more than one protein/receptor [109, 111, 115]. Moreover, snaclecs are also known to demonstrate haemorrhagic effects [109-111].

1.9 Snake venom components affecting blood coagulation and fibrinolysis: A classification based on their functional aspects

The process of blood coagulation and fibrinolysis maintains a physiological balance under normal conditions [12, 58, 61]. Some exogenous proteins/peptides from snake venoms target the blood coagulation system [12, 58, 61; 86-88]. These compounds specifically attack critical and key members of the coagulation cascade and disrupt the homeostatic system. Such potent and highly specific snake venom proteins/peptides have been obligatory in elucidating the complex physiology of mammalian blood coagulation and fibrinolysis. On the basis of their overall effect on the coagulation cascade, these snake venom proteins having enzymatic and non-enzymatic properties have been classified into procoagulant or anticoagulant proteins (Fig. 1.8).

1.9.1 Enzymatic proteins from snake venom affecting coagulation and fibrinolysis

1.9.1.1 Procoagulant enzymatic proteins from snake venom

Snake venoms harbor a potential number of components that interact with proteins of the coagulation cascade and the fibrinolytic pathway. Procoagulant venoms such as that of Viperideae and Crotalideae family contain components that hasten clot formation.

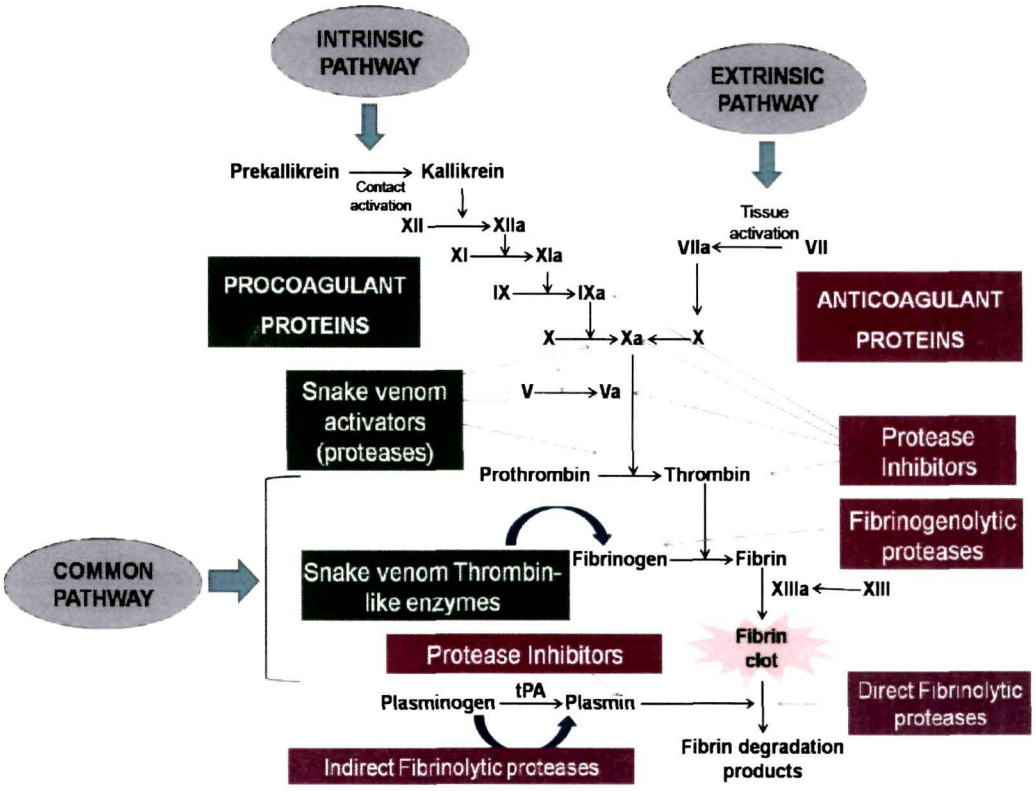


Fig. 1.8 Effect of snake venom proteins on the coagulation system.

All the snake venom procoagulant proteins characterized to date are either serine- or metallo- proteases responsible for the activation of the zymogenic forms of specific coagulation factors in the coagulation cascade [83, 116-119]. Given below

are the various types of snake venom procoagulant proteins identified till date, which impede hemostatic balance.

1.9.1.1.1 Factor VII activators

Factor VII belongs to the serine protease family and the main role this 50 kDa protein is to initiate the process of coagulation in conjunction with tissue factor [120]. Cleavage of its Arg^{152}–Ile^{153} peptide bond converts the zymogen into the physiologically active, two-chain factor VIIa [120]. Upon vessel injury, tissue factor is exposed factor VIIa circulating in the blood [120-121]. The complex of factor VIIa with tissue factor further catalyzes the conversion of factor IX and factor X into the active proteases, factor IXa and factor Xa, respectively [121].

A few snake venom proteins are known to demonstrate procoagulant effects by virtue of ability to activate factor VII [122]. However, to date, snake venom proteins that activate only factor VII have not been isolated [61, 86, 122]. Notably, most of snake venom factor VII activators are also responsible for the activation of other coagulation factors as well [122]. Very often, factor VII activation by snake venom proteins is potentiated by the presence of phospholipids and Ca^{2+}-ions [61].

1.9.1.1.2 Factor V activators

Blood coagulation factor V is a single-chain glycoprotein with a molecular mass of 330 kDa [123]. It plays an important role in the process of blood coagulation. Under physiological conditions, the zymogen, factor V is activated to its active form, factor Va, upon the enzymatic action of thrombin via the enzymatic cleavage at three positions, after Arg^{709}, Arg^{1018} and Arg^{1545} of factor V [123]. Factor Va, a two-chain molecule which is composed of a heavy (105 kDa) and a light chain (71 kDa), in turn, accelerates factor Xa-catalysed prothrombin activation by more than 1,000-fold, leading to procoagulant effects [123].

Factor V can also be activated by a wide range of snake venoms proteins isolated from venoms of snakes of the Viperideae, Crotaldeae and Elapideae family [124].

It has been proposed that factor V activators owe their specificity for the Arg^{1545} bond, which is crucial for the activation of factor V [83, 125-128].

1.9.1.1.3 Prothrombin activators

Prothrombin is a single chain vitamin K-dependent glycoprotein (72 kDa) consisting of an N-terminal Gla domain, two kringle domains and a serine protease domain, which exposes its catalytic site upon activation to thrombin [87, 129-130]. Prothrombin activation occurs via the proteolytic action of the prothrombinase complex, at two internal peptide bonds: Arg^{274}–Thr^{275} and Arg^{323}–Ile^{324} (Fig. 1.9) [131-134]. Both the activation pathways occur during factor Xa-catalyzed prothrombin activation, as both prethrombin 2 (primary cleavage after Arg^{271}) and meizothrombin (primary cleavage after Arg^{320}) are detected as intermediary products [131-132, 135]. Activation of prothrombin releases thrombin from the C-terminal end of the precursor protein, while the Gla and kringle domain-containing region from the amino terminal portion of prothrombin remain attached to the phospholipid membrane [130-132, 135]. Thrombin consists of a light chain (6 kDa) and a heavy chain (31 kDa) covalently associated through a disulfide bond, with the serine protease active site situated in the heavy chain [136-137].

One of the central reactions of the coagulation cascade which is often a target of snake venom proteases is the process of prothrombin activation. This is a vital reaction in the blood coagulation cascade resulting in the formation of thrombin which catalyzes the final step of the coagulation process [138]. Under physiological conditions, factor Xa catalyzed prothrombin activation is substantially enhanced (~300,000 fold) in the presence of the prothrombinase complex formed by the assemblage of Ca^{2+}, negatively charged phospholipids and factor Va [138]. On the basis of their structural properties and co-factor requirements during prothrombin activation, snake venom prothrombin activators have been classified into four groups [87, 130, 139-143].

a. Group A prothrombin activators are metalloproteases which do not require any co-factors such as Ca^{2+}, phospholipids or factor Va for prothrombin activation [139].

b. Group B prothrombin activators are also metalloproteases but require the presence of Ca^{2+} for promoting their prothrombin activating property [141].

c. Group C prothrombin activators are serine proteases which require Ca^{2+} and negatively charged phospholipids, but not factor Va for optimum activity [142] while,

d. Group D prothrombin activators are serine proteases which require factor Va in addition to the other co-factors [143].

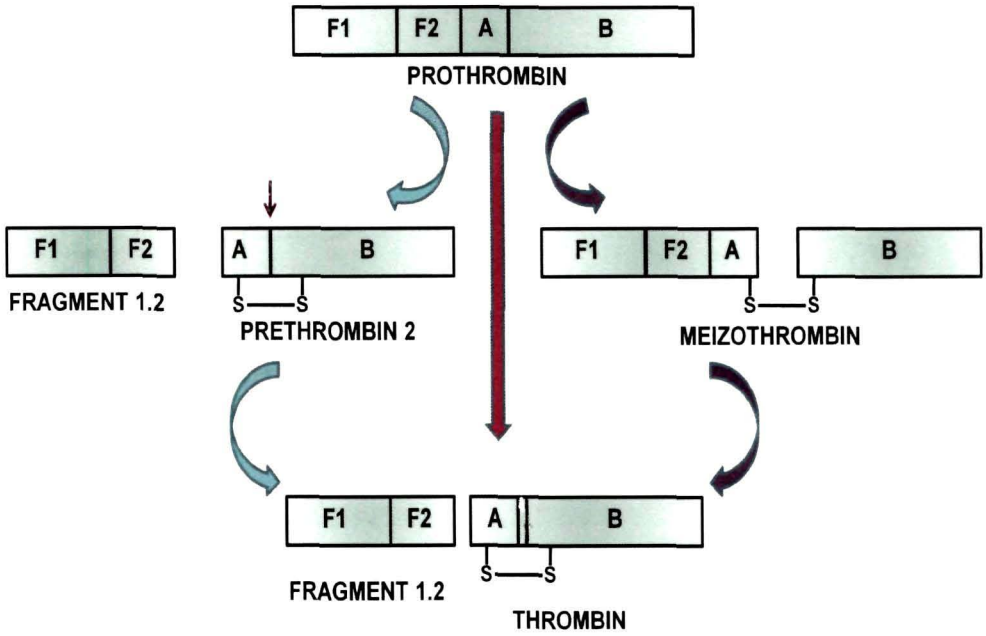


Fig. 1.9 Schematic representation of human prothrombin activation. Prothrombin consists of a membrane-binding N-terminal domain (fragment 1.2) and a C-terminal catalytic domain (prethrombin 2). There are two possible pathways of prothrombin activation leading to the formation of the intermediates: prethrombin 2 and meizothrombin. A third possible pathway of activation is also

predicted which involves direct activation of prothrombin to thrombin without the release of any intermediate (shown with a red arrow).

1.9.1.1.4 Factor X activators

Blood coagulation factor X is one of the chief components of the haemostatic system which is synthesized as a single-chain glycoprotein of 488 amino acids [144]. Upon secretion, factor X undergoes post-translational modifications and the mature protein circulates in the blood as a two-chain protein- the light chain and a heavy chain [144]. Under physiological conditions, the activation of factor X is a membrane bound process which is brought about by cleavage of a single peptide bond at Arg^{194}-Ile^{195} which removes the heavily glycosylated first 52 amino-terminal residues of the heavy chain and upon which the active site becomes fully exposed [144]. The zymogen, factor X is cleaved by the factor VIIa-tissue factor complex via the extrinsic pathway [121, 144].

Blood coagulation factor X activators have been described in the venom of many snake species belonging to the genus Viperidae, Crotalidae and Elapidae [144-146]. So far, only a few SVSPs and SVMPs have been identified to activate factor X [144-145]. SVMPs are known to activate factor X in a Ca^{2+}-dependent manner and belong to P-IIId class of SVMPs [144-145, 147].

1.9.1.1.5 Thrombin-like enzymes

Under physiological conditions, prothrombin is proteolytically cleaved to form thrombin in the coagulation cascade, which ultimately results in the prevention of blood loss during injury as already described in section 1.9.1.1.3. Thrombin is a trypsin-like serine protease that converts soluble fibrinogen into insoluble fibrin strands which leads to the generation of a hemostatic plug or a pathologic thrombus [148-149]. In addition, thrombin potentiates the procoagulant process by activating other coagulation factors such as factors V, VIII, XI and XIII [150]. Conversely, thrombin also aids the anticoagulant process by activation of protein C [150].

Venom of snakes belonging to the Viperideae and Crotalideae family are rich in serine proteases that are functionally and structurally related to thrombin 75, 80-82, 151]. Snake venom thrombin like enzymes (SVTLEs) are divided into three groups, depending on which fibrinopeptide they cleave off, fibrinopeptide A (FPA) from fibrinogen Aα-chain, fibrinopeptide B (FPB) from fibrinogen Bβ-chain or both [151-152]. SVTLEs usually cleave off either FPA (SVTLE-A) or FPB (SVTLE-B), but under rare circumstances they can cleave both fibrinopeptides (SVTLE-AB) in a manner similar to thrombin [75, 80-82, 151-152]. These SVTLEs show less than 40% sequence identity with human thrombin and share approximately 67% sequence identity amongst themselves [80, 86].

Unlike thrombin, SVTLEs are highly specific for thrombin and do not usually activate factor XIII or other coagulation factors, and cleave fibrinopeptide only at a single position [80, 86, 151]. This results in formation of abnormal, "loose" fibrin clots composed of short polymers not stabilized by transglutaminase-catalyzed crosslinking [61, 80-82, 151-152]. Such clots can be easily degraded by the fibrinolytic system. As such, SVTLEs have been considered as defibrinogenating agents for managing unwanted clots in patients suffering from certain thromboembolic disorders [80-82, 151-153]. In addition to activating thrombin, some SVTLEs also are also known to affect the process of haemostasis by leading to induction of platelet aggregation [151, 154].

1.9.1.2 Anticoagulant enzymatic proteins from snake venom

1.9.1.2.1 Anticoagulant PLA_2 enzymes

Apart from the membrane damaging ability, a number of snake venom PLA_2 enzymes are characterized as inhibitors of the coagulation cascade [45, 91, 93, 96-98, 155]. These anticoagulant PLA_2s are classified into strong or weak anticoagulant enzymes on the basis of their degree of anticoagulant property which might be attributed to the non-enzymatic functions of these catalytic proteins [96, 98, 155-158]. The PLA_2s demonstrate anticoagulant activity by virtue of its ability

to inhibit the prothrombinase complex or the extrinsic tenase (tissue factor–Factor VIIa) complex [96, 98, 155, 157-158].

1.9.1.2.2 Fibrinogenolytic proteases

A majority of the snake venom fibrinogenolytic enzymes belong to SVMPs [75, 159-161]. These enzymes preferentially cleave the Aα-chain of fibrinogen, with a lower specificity for the Bβ-chain of fibrinogen [75, 159-161]. Some of these fibrin(ogen)olytic SVMPs can also act directly on fibrin leading to its dissolution and are not inhibited by blood protease inhibitors (the serpins), in general [75, 159-161]. Very often, α-fibrinogenases inhibit or delay the process of blood coagulation, because truncated fibrinogen formed as a result of the protease does not form a strong fibrin clot, thereby leading to the easy physical destruction of fibrinogen. However, many of the fibrin(ogen)olytic SVMPs, which lack any hemmorhagic effects and can break down fibrin-rich clots and avert the progression of clot formation suggesting thrombolytic effects [75, 159-161]. Fibrin(ogen)olytic SVSPs, on the other hand usually cleave the Bβ-chain of fibrinogen preferentially and have lower activity towards the Aα-chain of fibrinogen [160]. Many of these anticoagulant fibrinogenolytic serine proteases are not thrombin-like proteins and are responsible for direct degradation of fibrinogen or fibrin [75, 159-161].

1.9.1.2.3 Protein C activators

Protein C is a vitamin K-dependent, two-chain zymogen activated by thrombin [162]. Under physiological conditions, the zymogen of protein C circulating in the blood is activated by thrombin [162]. The activated protein C in turn degrades factor Va and factor VIIIa and is therefore anticoagulant in nature [162-163]. Most venom protein C activators show sequence similarity to venom serine proteases [83, 164]. Most of the protein C activators have been purified from *Agkistrodon* venoms while some have been isolated from *Bothrops*, *Trimeresurus*, or *Cerastes* venoms [83, 164]. Unlike thrombin-catalysed protein C activation, which requires

thrombomodulin as a cofactor, venom activators directly convert protein C into the active form [162]. Snake venom seine proteases that are protein C activators are characterized not only to prolong *in vivo* clotting time but also to prolong thrombus formation in the arteriovenous shunt *in vivo* [162, 164]. However, so far, no significant data are available on the structure–function relationships of this class of enzymes.

1.9.2 Non-enzymatic proteins from snake venom affecting blood coagulation and fibrinolysis

Numerous non-enzymatic proteins have been characterized from snake venoms over the past few decades. These includes the three-finger toxins [165-168], protease inhibitors [59, 103-106], snaclecs [109-113], nerve growth factors [169], bradykinin-potentiating peptides [170], natriuretic peptides [171], cobra venom factors [172], vascular endothelial growth factors [173], sarafotoxins [174], waprins [175], vespryns [176] and veficolins [177].

Many of these proteins and peptides not only play an important role in prey immobilization but also contribute to the overall understanding of venom toxicity [11]. The proteins and peptides within each family share notable similarities in their primary, secondary and tertiary structures, but they may differ from each other in their pharmacological effects [11].

A few of these non-enzymatic proteins and peptides are known to affect the process of hemostasis. These include the Kunitz-type serine protease inhibitors, snaclecs and a few anticoagulant three finger toxins, in general. Most of these non-enzymatic proteins/peptides possess anticoagulant property by virtue of their ability to bind and/or inhibit different coagulation factors [59, 103, 112-113, 178-189].

It is to be noted that, none of these non-enzymatic proteins are known to have a procoagulant effect on the process of blood coagulation.

1.9.2.1 Non-enzymatic anticoagulant snake venom proteins

1.9.2.1.1 Factor IX/X inhibitors

A number of anticoagulant snaclecs or protease inhibitors that bind with factor IX and/or factor X, have been isolated from various snake species [59, 106, 113, 179, 182, 188]. On the basis of their ligand recognition differences, these proteins are classified as:

- Factor IX/factor X binding proteins, which interacts with both factor IX or factor X in a 1:1 molar ratio [179];
- Factor IX-binding proteins, which do not interact with factor X [181];
- Factor X-binding proteins, which bind predominantly to factor X [59, 106, 113, 182].

Most of these inhibitors are heterodimers with highly similar structures. The factor Xa inhibitors are basically classified under two groups- direct and indirect factor Xa inhibitors [190]. Direct factor Xa inhibitors are those which inhibit factor Xa directly without the requirement of any intermediary components, whereas indirect factor Xa inhibitors demonstrate an AT-III mediated factor Xa inhibition [190].

1.9.2.1.2 Thrombin inhibitors

Thrombin, a 36 kDa trypsin-like serine protease is synthesized as a zymogen (prothrombin) in the liver and is a central and multifunctional component of the hemostatic system [136]. As described in section 1.9.1.1.3, prothrombin is proteolytically cleaved by the prothrombinase complex to form thrombin which is essentially responsible for all major thrombotic processes, including physiologic homeostasis and pathologic thrombosis. The haemostatic system also comprises of endogenous inhibitors of thrombin such as AT-III, which maintains a balance of active thrombin in the blood circulation thus avoiding the formation of unwanted clots [136]. Any defect in the delicate balance between coagulation factors, most importantly thrombin and their endogenous inhibitors can lead to a serious

haemostatic disorder such as the formation of thrombi in the blood vessels. This is likely to lead to complications such as myocardial infarction and other cardiovascular disorders [136].

Besides its central role in hemostasis, it is conventionally accredited that thrombin promotes cancer growth by supporting cellular proliferation, tumor adhesion to subendothelial matrix, metastasis and tumor-associated angiogenesis in cancer [191-192]. Thrombin is therefore, an important pharmaceutical target for the treatment and prevention of thrombosis and cancer progression [193-194]. Many anticoagulant agents, particularly thrombin inhibitors may be used as prophylaxis in atrial fibrillation to avoid thromboembolism as well as for delaying cancer progression [191, 193-197].

It has been well established that thrombin's catalytic activity is enhanced by some recognition domains which are responsible for regulation of the specificity of the enzyme's interaction with different substrates and inhibitors [137, 198-199]. Among the recognition domains, exosite I is responsible for fibrinogen recognition and exosite II is responsible for heparin binding [137, 198-199] (Fig. 1.10A). Both sites are distant from the catalytic pocket but are involved in the specific binding of thrombin to several macromolecular substrates, inhibitors and modulators [137, 198-199]. As such, exogenous venom thrombin inhibitors have been classified into different groups on the basis of their ability to bind to the different sites of thrombin (Fig. 1.10B). These exogenous snake venom thrombin-inhibitors belong to the family of snake venom PLA_2 enzymes, snaclecs, snake venom proteases or Kunitz-type serine protease inhibitor family [59, 94, 96, 98, 106, 180, 184-185, 187].

As discussed above and in section 1.9.1.2.1, it is important to note that some of the snake venom PLA_2 enzymes demonstrate anticoagulant property by virtue of their ability to non-enzymatically inhibit blood coagulation proteins such as thrombin and/or factor Xa in addition to their catalytic ability of membrane phospholipids hydrolysis [94, 96, 98, 115].

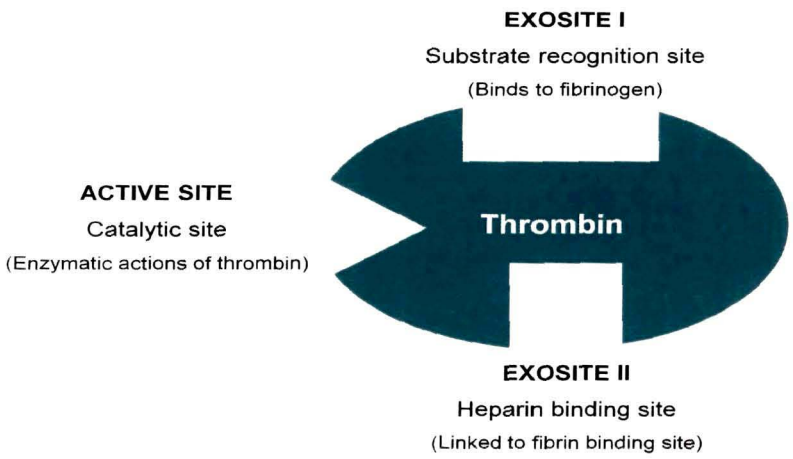


Fig. 1.10A A schematic representation of the thrombin molecule.

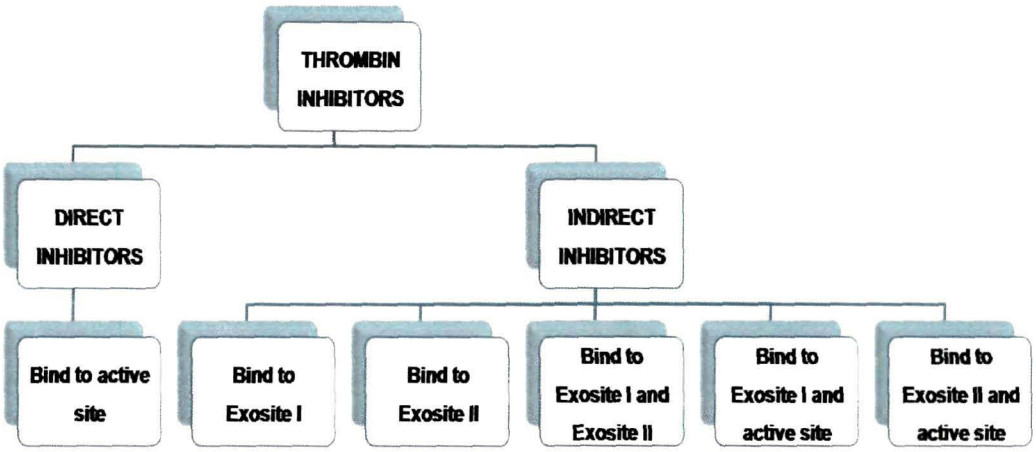


Fig. 1.10B Classification of exogenous thrombin inhibitors on the basis of their ability to bind to the different sites of thrombin.

1.9.2.1.3 Anticoagulant three finger toxins

This family of non-enzymatic polypeptides contains 60–74 amino acid residues and are found in the venoms of elapids, hydrophiids, and colubrids as well as reported in the transcriptomes of viperids and crotalids [165-168]. The three finger toxins (3FTxs) are rich in disulphide bonds contain four to five disulphide bridges, of which four are conserved in all members [200]. However, despite their structural similarity, these 3FTxs differ widely in their activities, being mostly

cardiotoxic and neurotoxic but some of them also acting on the haemostatic system [165-168].

The anticoagulant and antiplatelet effects of 3FTXs were first identified in cardiotoxins isolated from *Naja nigricollis crawshawii* (spitting cobra) venom [201]. It has also been reported that some of the non-enzymatic 3FTxs can form stables complexes with other 3FTxs or venom proteins and synergistically enhance their anticoagulant effects [88] via non-enzymatic inhibition of factor VIIa and factor VIIa/TF complex [186].

1.10 Biomedical applications of snake venom proteins: A brief overview

Snake venom components particularly those affecting the process of hemostasis, have been consistently explored for the development of novel and better therapeutic agents to deal with various life-threatening diseases such as thrombosis associated ailments, hypertension and cancer [7, 9, 12, 14, 202]. Most of the procoagulant or anticoagulant snake venom proteins are highly specific for an individual component of the coagulation cascade [7, 9, 12, 14, 202]. This specificity has been used for practical applications and for the development of several routine laboratory tests. For example, venom factor V activators are excellent tools for studying the structure-function relationship of factor Va and are often used in diagnostic tests for quantification of plasma factor V levels and for the screening of defects in the protein C pathway [12].

Like snake venom factor V activators, factor X activators are also used extensively in the diagnosis of quantitative conversion of the zymogen factor X into factor Xa (eg. RVV-X) [12]. Such enzymes are used for determination of factor X as well as for distinguishing between factor VII and factor X deficiency in plasma [203] and for lupus anticoagulant assay [204]. Most of the procoagulant venom proteins are used in the diagnosis of coagulation factor deficiencies by the determination of coagulation times of blood plasma treated with these proteins [7, 9, 12, 14, 202]. Snake venom prothrombin activators are used to selectively detect fully

carboxylated prothrombin and lupus anticoagulant and even factor Va deficiency in human plasma [12, 139, 205].

Thrombosis-associated disorders such as cardiovascular diseases are a leading cause of mortality in both developing and developed countries [206]. Unfortunately, many natural anticoagulants used as antithrombotic drugs are associated with adverse side effects [207-208]. Notably, the natural anticoagulants used as antithrombotic drugs such as heparin and warfarin are associated with side effects such as extensive bleeding complications [207-208]. As such, snake venom derived potent anticoagulants such as inhibitors of thrombin, factor Xa etc have been considered as an alternative for the development of better antithrombotic agents [180, 207-208]. Additionally, low toxicity anticoagulants are in great demand for therapeutic use in the prevention and treatment of occlusive thrombosis [206].

Other than the non-enzymatic anticoagulant proteins and peptides, SVTLEs as well as fibrinogenolytic enzymes have been studied extensively to develop agents that lead to defibrinogenation or digestion of fibrin clots directly; thereby suggesting potential application of such enzymes for the treatment of thrombotic ailments such as strokes, heart attacks and other diseases associated with the formation of thrombus [75, 80-83, 151, 209-210].

SVTLEs are consistently used for clinical defibrinogenation of blood plasma [75, 80, 151]. SVTLEs such as Ancrod, marketed as Viprinex, have been successfully used to treat ischemic heart diseases, deep vein thrombosis and peripheral occlusive disorders [211]. Snake venom fibrinolytic proteases such as fibrolase [212] have been explored for their antithrombotic and thrombolytic potential. Unfortunately, the recombinant form of fibrolase (Alfimeprase) failed as a potential drug in the phase II clinical trial [213].

A few snake venom proteins affecting the process of hemostasis that have been successfully developed as diagnostic/therapuetic agents are listed in Table 1.2.

Table 1.2 Diagnostic applications of snake venom proteins.

Snake venom component	Example	Mol mass (kDa)	Source	Biological functions	Applications	Reference
Prothrombin activator	Ecarin	55	*Echis carinatus*	Activation of prothrombin	Determination of prothrombin levels in patients undergoing anticoagulant therapy	[214]
SVTLE	Batroxobin	43	*Bothrops moojeni*	release of fibrinopeptide A and the formation of fibrin I monomer	Determination of fibrinogen in plasma, Measurement of the batroxobin clotting time (Reptilase® time) as a heparin-insensitive parallel to the thrombin time, to investigate dysfibrinogenemia, and to test the contractile system of platelets	[215]
Factor X activator	RVV-X	92.8	*Daboia russelii*	Activation of Factor X	Measurement of Factor X	[216]
Factor V activator	RVV-V	29	*Daboia russelii*	Activation of Factor V	Measurement of Factor V	[125]
Protein C activator	ACC-C	37	*Agkistrodon contortrix contortrix*	Activation of Protein C	Determination of protein C and protein S levels.	[162]
SVTLE	Ancrod	35	*Agkistrodon rhodostoma*	Defibrination	Treatment of ischemic stroke, deep vein thrombosis and peripheral occlusive diseases, Used as an alternative to heparin in cardiopulmonary bypass	[211]

1.10.1 Snake venom proteins as potential anticancer agents

Cancer, a life threatening disease is characterized by uncontrolled cell growth and spread of abnormal cells [217, 218]. It is one of the leading causes of death worldwide [217, 218]. There are several measures undertaken to treat this grave medical concern which involves surgery, radiation therapy, chemotherapy, hormone therapy, immunotherapy, etc [217, 218]. However all these therapeutic procedures have their own drawbacks and severe side effects such as bone marrow suppression, neuropathy, jaundice, low blood pressure, increased risk of thromboembolism [217, 218]. As such, search for natural compounds that can serve as potential anticancer agents have intrigued researchers across the globe.

Many snake venom proteins and toxins have been reported to demonstrate anticancer potential towards various cancerous cell lines [218-222]. The snake venom proteins, like other naturally occurring anticancer molecules lead to *in vivo* and/or *in vitro* anti-tumour effects by triggering one or more of the actions listed below [218]-

i. Inhibiting cell proliferation
ii. Damaging cell membranes
iii. Disorganization of actin filaments
iv. Bax-Bcl_2 ratio
v. Inhibition of p50 and p65 translocation into nucleus
vi. Increasing Ca^{2+} influx
vii. Induction of cytochrome c release
viii. Inhibition of NF-κB
ix. Inhibition of nucleic acid synthesis
x. Increases the expression of pro- apoptotic protein such as p53 and p21
xi. ROS generation
xii. Changes the expression of proteins controlling of cell cycle
xiii. Inhibition of STAT3 signal

The first report on the anticancer potential of snake venom was reported by Calmette and his colleagues' in 1933 [223]. Since then, the search for potential anticancer agents from snake venom has been a subject of interest amongst researchers. Amongst the various snake venom components, disintegrins snake venom C-type lectins, cytotoxins, L-amino oxidases and PLA_2s, are found to demonstrate anticancer effects [218-222].

A few snake venom proteins with anticancer potential that have been well characterized include RGD-containing disintegrins. Triflavin (7.6 kDa), an RGD-containing disintegrin isolated from *Trimerisurus flavoviridis*, was first shown to effectively inhibit cell adhesion, migration and angiogenesis [224]. Another disintegrin, an antiplatelet snake venom peptide Rhodostomin (7.1 kDa) isolated from the venom of *Crotalus atrox* could inhibit thrombin-enhanced metastasis [225]. Rhodostomin prevented adhesion, migration and invasion of tumour cells and also tumour growth of human breast cancer cells in bone as well [225]. Another dintegrin showing potential anticancer properties was Salmosin (7.4 kDa) isolated from the venom of the *Agkistrodon halys brevicaudus* [226]. Salmosin could suppress both solid and metastatic tumor growth [226]. A few other examples of snake venom disintegrins with anticancer potential are Contortrostatin (13.5 kDa), purified from *Agkistrodon contortrix contortrix*, which is one of the best studied disintegrin with anticancer potential, Viperistatin (4.4 kDa) isolated from *Vipera palaestinae*, and Lebestatin (4.4 kDa) isolated from *Macrovipera lebetina*, all of which target integrin $\alpha_1\beta_1$ [227-229].

A few C-type lectin proteins with anticancer potential are Lebectin (30 kDa), isolated from *Macrovipera lebectina* which inhibited integrin-mediated adhesion, migration and invasion of human tumour cells [230] and Vixapatin (9.8 kDa) purified from *Vipera paleastinae* venom [231].

Snake venom L- amino oxidases (LAAOs) are reported to be capable of promoting cytotoxicity in different cell lines. Although the cytotoxicity mechanisms of snake

venom-LAAOs have not been fully clarified, it is postulated that the lipids present in cell membranes can be damaged by the reactive oxygen species (ROS) generated by the action of these snake venom proteins [232]. A few SV- LAAO's with anticancer potential are ACTX-6 and ACTX-8 from the venom of *Agkistrodon acutus* [233-234], BatroxLAAO from *Bothrops atrox* [235], and Apoxin-I from *Crotalus atrox* [236].

Amongst the snake venom toxins, Taipoxin (35 kDa), a neurotoxin isolated from the venom of the *Oxyuranus scutellatus scutellatus* was found to demonstrate anticancer potential by virtue of its ability to inhibit SCLC cell lines [237]. INN-toxin (6.9 kDa), is yet another eample of a neurotoxin from *Naja naja* venom showing cytotoxic acticity on Ehrlich Ascites tumour cells [238]. Cardiotoxin III (6.7 kDa), from *Naja naja atra* venom, is one of the extensively studied venom toxin with anticancer activity [239]. Another cardiotoxin, NK-CT1 (6.7 kDa), from Indian *Naja kaouthia* venom has been found to be cytotoxic to human leukemic cell lines U937, K562 and adherent glioma cells C6 and human ovarian cancer cells SKOV3 [220].

It has also been established that thrombin mediated fibrin clot formation plays a vital role in camouflaging tumor cells in addition to promoting their attachment to the vascular endothelium, thus aiding cancer progression and metastasis [193-194]. Though, it has been postulated that thrombin inhibitors can be a potential anticancer agent [240], nevertheless, none of the snake venom thrombin inhibitors have been tested for anticancer property. As such the search for snake venom anticoagulants with potential pro-apoptotic properties is of potential interest.

1.11 Aims and objectives of the present study

The present study aims to identify and characterize new or previously uncharacterized procoagulant and anticoagulant proteins or peptides from Russell's Viper (*Daboia russelii russelii*) venom, their pathophysiological affects and their mechanisms of action. Another important objective was to explore the possibility of development of potential therapeutic agents from RVV that can be used to treat and/or prevent life threatening diseases such as thrombosis and cancer.

Therefore, the objectives of the present study were:

1. Isolation of some procoagulant or anticoagulant protein(s) and peptide(s) from Indian Russell's Viper (*Daboia russelii russelii*) venom.
2. Biochemical and pharmacological characterization of the isolated protein(s)/peptide(s).
3. Investigations on the pathophysiological significance of the isolated protein(s)/peptide(s).
4. Investigations on the possible biomedical application(s) of the isolated protein(s)/peptide(s).

CHAPTER II

REVIEW OF LITERATURE

2 REVIEW OF LITERATURE

2.1 Hemostatically active snake venom proteins

Over the past few decades, many snake venom proteins affecting the blood coagulation cascade and the process of fibrinolysis have been identified and their roles characterized. Most of the identified snake venom proteins that affect the haemostatic system can lead to either one or more effects such as,

(a) Reduced blood coagulability [58, 61, 85],

(b) Damaged blood vessels, resulting in bleeding [61, 241-242],

(c) Organ damage and hypovolaemic shock [61, 242], and/or

(d) Thrombosis [58, 61, 85].

Many of these snake venom proteins and peptides have been indispensable in illuminating the complex physiological mechanisms that govern the process of hemastasis. Moreover, many of these proteins and peptides show promising potential in the field of therapeutics and diagnostics [7, 9, 12, 14, 61, 202, 205, 243-244]. This chapter reviews the various aspects of hemostatically actives snake venom proteins with particular reference to RVV proteins and peptides.

2.2 RVV proteins affecting hemostasis

RVV is predominantly procoagulant in nature by virtue of the presence of several procoagulant proteins, all of which exhibit proteolytic effects on various factors of the coagulation cascade (Table 2.1). However, many anticoagulant proteins have also been reported from RVV (Table 2.1). In this chapter, the structure-functional properties, pathobiological effects and biomedical significance of the procoagulant and anticoagulant proteins from RVV has been emphasized.

2.2.1 Procoagulant proteins isolated from RVV: A brief review on their structure-functional characteristics and their mechanisms of action

2.2.1.1 Factor X activators from RVV

One of the major procoagulant metalloprotease identified from RVV is 92.8 kDa factor X activator named RVV-X [116-117]. RVV-X is a unique heterotrimeric glycoprotein containing a disintegrin and metalloprotease (ADAM)-like heavy chain (57.6 kDa) and two heterogenous C-type lectin-like light chains (19.4 kDa and 16.4 kDa), which are covalently held together by disulfide bonds [73, 116-117, 216]. RVV-X contains 13% carbohydrate by mass which includes 6.0% hexose, 1.7% N-acetyleneuraminic acid, and 5.3% galactosamine [116]. On the basis of its structural arrangements, RVV-X has been classified under P-IIId class of SVMPs [73].

Crystallographic studies of RVV-X indicate that RVV-X adopts a "hook-spanner-wrench"-like structure; the metalloprotease/disintegrin portion constitutes a mobile 'hook' whereas the lectin-like domains and the rest of the heavy chain, comprise the 'handle' [73]. The lectin-like domains form a tangled dimer with strong structural similarity to anticoagulant factor X-binding proteins [73]. Recently, the crystal structure of RVV-X has been elaborated by Takeda and his colleagues [73] (Fig. 2.1). It has been postulated that the structure of RVV-X displays a 6.5-nm separation between the catalytic zinc atom and a putative γ-carboxylglutamic acid (Gla) domain-binding exosite, which are involved in factor X activation by RVV-X [73].

Table 2.1 A list of procoagulant and anticoagulant proteins identified from RVV.

	Protein	Class	Molecular mass (kDa)	Mechanism of action	Other functions	Reference
Procoagulant	RVV-X	Metalloprotease	92.8	Factor X activator	FIX/Protein C activator	[117]
	FIa	Metalloprotease	34.4	Factor X activator	-	[119]
	RVBCMP	Metalloprotease	15.0	Unknown (but not thrombin-like mechanism)	α-fibrinogenase, tissue damage	[65]
	Russelobin	Serine protease	51.2	Thrombin-like enzyme	Defibrinogenation, *in vivo* anticoagulant	[81-82]
	RVV-V	Serine protease	29.0	Factor V activator	-	[116]
	RV-FVPα	Serine protease	32.9	Factor V activator	αβ-fibrinogenase	[83]
	RV-FVPβ	Serine protease	33.36	Factor V activator	αβ-fibrinogenase	[83]
	RV-FVPγ	Serine protease	33.35	Factor V activator	αβ-fibrinogenase	[83]
	RV-FVPδ	Serine protease	34.5	Factor V activator	αβ-fibrinogenase	[83]

	Protein	Class	Molecular mass (kDa)	Mechanism of action	Other functions	Reference
Anticoagulant	VRV-PL-VI	PLA_2	11.8	Phospholipids hydrolysis	Hemorrhagic	[99]
	VRV-PL-VIIIa	PLA_2	12.0	Phospholipids hydrolysis	Hemorrhagic and induces edema	[100]
	RVVA-PLA_2-I	PLA_2	58.0	Phospholipids hydrolysis and factor Xa inhibition	Membrane damage	[96]
	Neupholipase	PLA_2	13.0	Phospholipids hydrolysis	Membrane damage	[97]
	$RVAPLA_2$	PLA_2	13.8	Phospholipid hydrolysis and thrombin inhibition	-	[98]
	VRR–73	Metalloprotease	73.0	Fibrinogenase	Hemmorhagic	[64]
	Daborhagin-M	Metalloprotease	62.0	Fibrinogenase	Hemmorhagic	[66]
	Daborhagin-K	Metalloprotease	66.0	Fibrinogenase	Hemmorhagic	
	Rusvikunin	Kunitz-type protease inhibitor	6.9	Thrombin inhibitor	inhibits trypsin and plasmin	[106]
	Rusvikunin II	Kunitz-type protease inhibitor	7.1	Factor Xa inhibitor	inhibits trypsin and plasmin	[59]
	RVsnaclec	C-type lectin protein	66.3	Factor Xa inhibitor	Concentration dependent platelet aggregation and de-aggregation	[113]

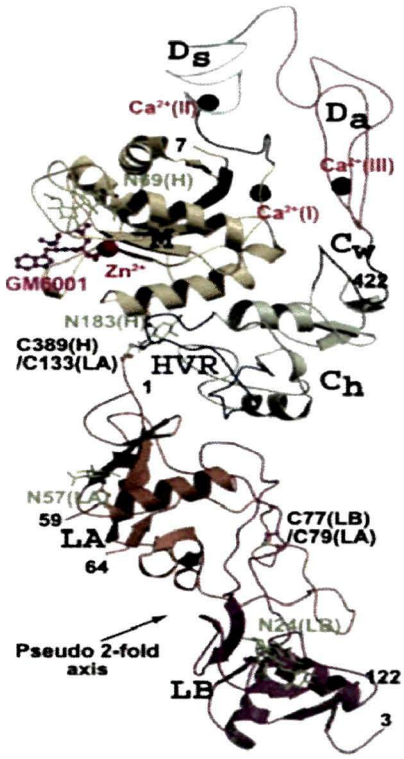


Fig. 2.1 Ribbon structure of RVV-X. Bound calcium and zinc ions are represented by black and red dots, respectively. The carbohydrate moieties (in green) linked to asparagine residues are shown in ball-and-stick representations [245].

RVV-X, a potent activator of factor X [246] specifically activates factor X via a single cleavage at the same internal Arg^{194}-Ile^{195} bond in factor X as done physiologically by factors IXa and VIIa during the process of blood coagulation [247]. It has been proposed that the two C-type lectin light chains of RVV-X serve as an exosite by which RVV-X familiarizes and binds to the Gla domain of factor X [147] (Fig. 2.2). In addition to the factor X activating property, RVV-X is also capable of factor IX and protein C activation [116, 248], since the Gla domains of

these proteins are very similar [249]. Moreover, activation of factor X by RVV-X was strictly dependent on the presence of Ca^{2+}-ions [116].

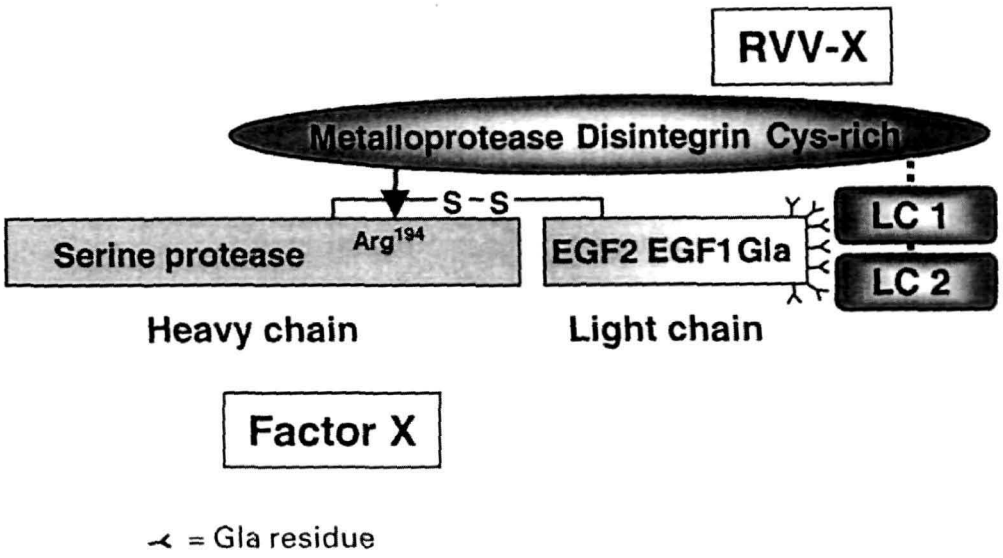


Fig. 2.2 Proposed mechanism of factor X activation by RVV-X [147]. The heavy chain of RVV-X contains the metalloproteinase domain responsible for the proteolytic cleavage of the target bond in factor X. The two C-type lectin light chains (LC 1 and LC 2) together form a secondary binding site specific for interaction with the Gla domain of factor X [144] (Tans and Rosing, 2001).

In 2007, Sun and his colleagues, reported the isolation and preliminary characterization of another factor X activator from RVV (*Daboia russelii siamensis*, Myanmar) named FIa [119]. The 34.5 kDa protease demonstrated strong procoagulant activity considering the fact that the hemostatic activity of 0.5 mg FIa was equal to that of 1.56 U thrombin [119]. FIa readily activated factor X, but had no influence on prothrombin and fibrinogen [119]. However, further structure-functional characterization and studies concerning its mechanism of factor X activation are still unreported.

2.2.1.2 Factor V activators from RVV

One of the earliest identified procoagulant serine protease from RVV is a thrombin-like factor V activating protease, RVV-V [125-126]. RVV-V was purified to homogeneity and is characterized to be a monomeric glycoprotein containing 6% carbohydrate by mass with an apparent molecular weight of 29.0 kDa [125, 250]. RVV-V is made up of 236 amino acids and is found to be abundant in RVV as three different isoforms designated as RVV-Vα, RVV-Vβ and RVV-Vγ [126]. The enzyme is composed of a single polypeptide chain having an amino-terminal sequence of Val-Val-Gly-Gly-Asp-Glu-Cys-Asn-Ile-Asn-Glu-His-Pro-Ile [126]. Tokunaga and his colleagues (1988) reported that the sequences of the RVV-V isoproteins exhibited 62% identity with that of batroxobin, a thrombin-like enzyme present in *Bothrops atrox* venom [251], and 33% identity with that of human thrombin B chain. The structural difference between RVV-V and other trypsin-like serine proteases was also elaborated [126]. It has been suggested that the conservative Ser^{214}-Trp^{215}-Gly^{216} sequence, which is considered as the site of antiparallel β-sheet formation between the protein substrate, in most of the serine proteases is replaced by the corresponding sequence Ala^{214}-Gly^{215}-Gly^{216} in RVV-V [126]. Recently, the structural basis of coagulation factor V recognition for cleavage by RVV-V was shown by Nakayama and colleagues [252] (Fig. 2.3).

Functionally, RVV-V activates factor V to the same extent as the physiological enzyme α-thrombin and mediates factor V activation by a single peptide bond cleavage at Arg^{1545}–Ser^{1546} bond in factor V, generating the heavy and light chains of factor Va [126]; unlike α-thrombin which results in the cleavage of three peptide linkages in factor V releasing a large connecting fragment [128, 253] (Fig. 2.4). RVV-V demonstrates a narrow range of specificity with no apparent proteolytic activity towards other coagulation factors such as fibrinogen, plasminogen, prothrombin, factor VIII and factor XIII [128].

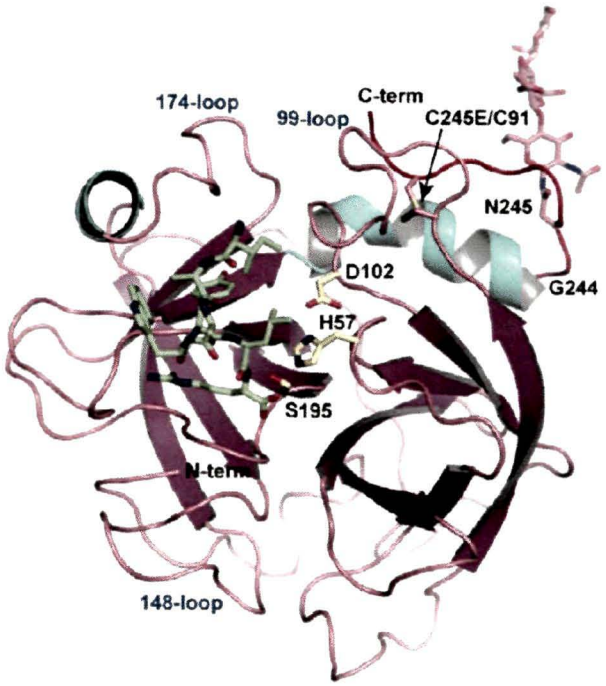


Fig. 2.3 Structure of the RVV-V/ factor V14 (residues 1533–1546 of human factor V) peptide complex. The factor V residues Ile^{1539}–Arg^{1545} and the side-chains of the active site residues in RVV-V are shown in green and yellow, respectively. The unique C-terminal nine residue extension of RVV-V (residues 244, 245, and 245A–245G, shown in red) contains an extra disulfide bond (Cys91–Cys245E) and an *N*-glycosylation site linked to Asn^{245}, which is not found in thrombin or other mammalian serine proteinases [252].

It is reported that RVV-V exhibits arginine esterase activity and is sensitive to the serine proteinase inhibitor diisopropyl fluorophosphates [126]. RVV-V is also inhibited by D-Phe-Pro-Arg-chloromethyl ketone (FPRCK) (97%), PMSF (76%), and benzamidine (50%) but remains unaffected by bovine pancreatic trypsin inhibitor (BPTI), antithrombin and EDTA [128].

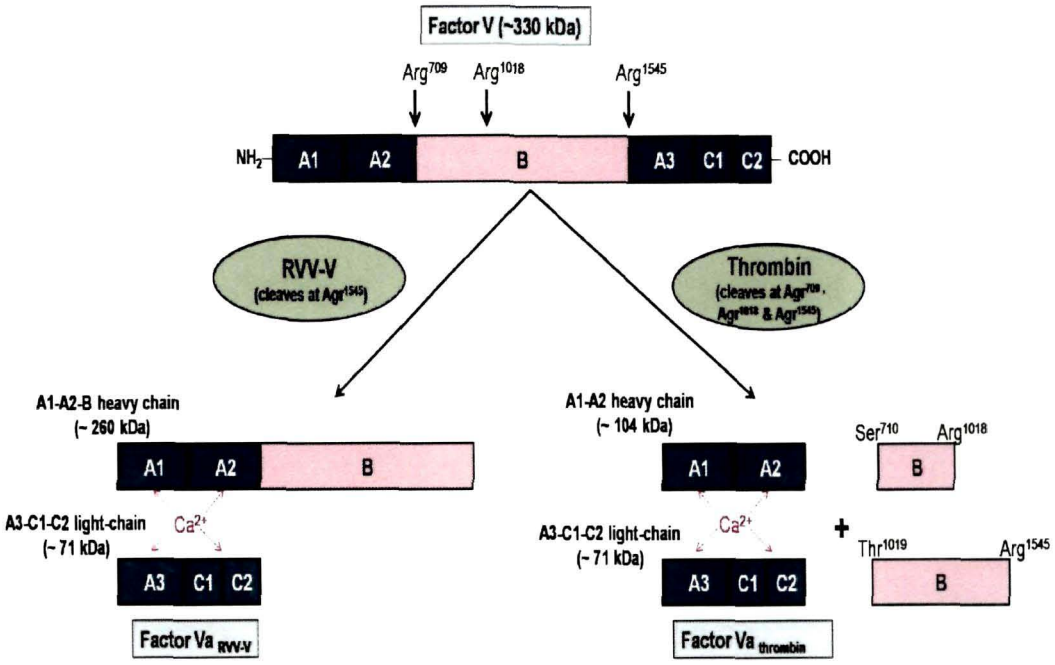


Fig. 2.4 Schematic representation of the activation of the single-chain factor V molecule by RVV-V. RVV-V cleaves factor V at Arg^{1545}. The ensuing factor Va molecule, consists of an A1-A2-B heavy chain (~260 kDa) which is non-covalently bound to the A3-C1-C2 light-chain (~71 kDa) of factor Va, through Ca^{2+}-dependent interactions (shown with red dashed arrows). α-thrombin, on the other hand, results in the cleavage of three peptide linkages in factor V releasing a large connecting fragment.

Of late, four new anticoagulant RVV serine protease isoenzymes namely RV-FVPα, RV-FVPβ, RV-FVPγ and RV-FVPδ of molecular masses corresponding to 32901.044 Da, 333631.179 Da, 333571.472 Da, and 34594.776 Da, respectively, have been purified and characterized [83]. The N-terminal residues of these isoenzymes (VIGGDECNIN for RV-FVPα, RV-FVPβ and RV-FVPδ; VIGGDEXNIN for RV-FVPγ) were shown to have significant sequence homology with the N-terminal sequences of SVTLEs and factor V-activating serine proteases [83].

These proteases have been reported to show significantly different fibrinogenolytic, BAEE-esterase and plasma clotting activities but no fibrinolytic, TAME-esterase or amidolytic activity against the chromogenic substrates for trypsin, thrombin, plasmin and factor Xa [83]. On the basis of the fibrinogen degradation pattern, these procoagulant isoenzymes have been classified as αβ-fibrinogenases [83]. These isoenzymes are extensively glycosylated with 42% to 44% of N-linked carbohydrates by mass [83]. *In vitro* studies suggested that the RV-FVP isoenzymes induced blood coagulation through factor V activation, whereas *in vivo* they provoked dose-dependent defibrinogenation and anticoagulant activity in mouse model [83].

2.2.1.3 Thrombin-like enzymes from RVV

Recently, a new thrombin-like serine protease named Russelobin has been identified from *Daboia russelii russselli* venom of Pakistan origin [81-82]. This 51.3 kDa glycosylated, monomeric protein is the first example of a thrombin-like enzyme purified from RVV which has a molecular mass greater than the typical masses of most of the thrombin-like serine proteases purified from snake venom [82]. Most of the SVTLEs have the molecular mass in the range of 25-35 kDa with few exceptions BJ-48 (52.0 kDa) [254] and BjussuSP-I (61.0 kDa) [255], including Russelobin (51.3 kDa) [82]. Studies conducted by Mukherjee and Mackessy suggested that Russelobin shares considerable sequence homology with a serine β-fibrinogenase-like protein precursor (28 kDa) from *D. r. siamensis* venom although Russelobin is predominantely an α-fibrinogenase [81-82]. Russelobin is reported to be extensively glycosylated and demonstrated a high degree of selectivity toward fibrinogen without detectable proteolysis of other plasma proteins, suggesting that fibrinogen is the primary physiological substrate for Russelobin [82].

Functionally, Russelobin hydrolyses both the Aα- and Bβ- chain of fibrinogen but does not release detectable quantities of either fibrinopeptide A or B at a

concentration of 40 nM [82]. However, at a concentration higher than this, Russelobin shows dose-dependent preferential release of fibrinopeptide A from fibrinogen (Fig. 2.5). The release of fibrinopeptide B by Russelobin was also reported nevertheless, to a much slower rate but the γ-chain of fibrinogen remained intact [82]. Russelobin is therefore found to be an αβ-fibrinogenase with high specificity toward fibrinogen, both *in vitro* and *in vivo* [82] (Mukherjee and Mackessy, 2013).

Russelobin, like other SVTLEs demonstrated *in vitro* procoagulant effects on citrated PPP by virtue of its effects on fibrinogen; however, *in vivo* it demonstrated defibrinogenating activity [82]. It has been suggested that, under controlled clinical setting, Russelobin could result in defibrinogenation which would be beneficial for treating thrombosis associated ailments such as cardiovascular disorders and even cancer progression [82].

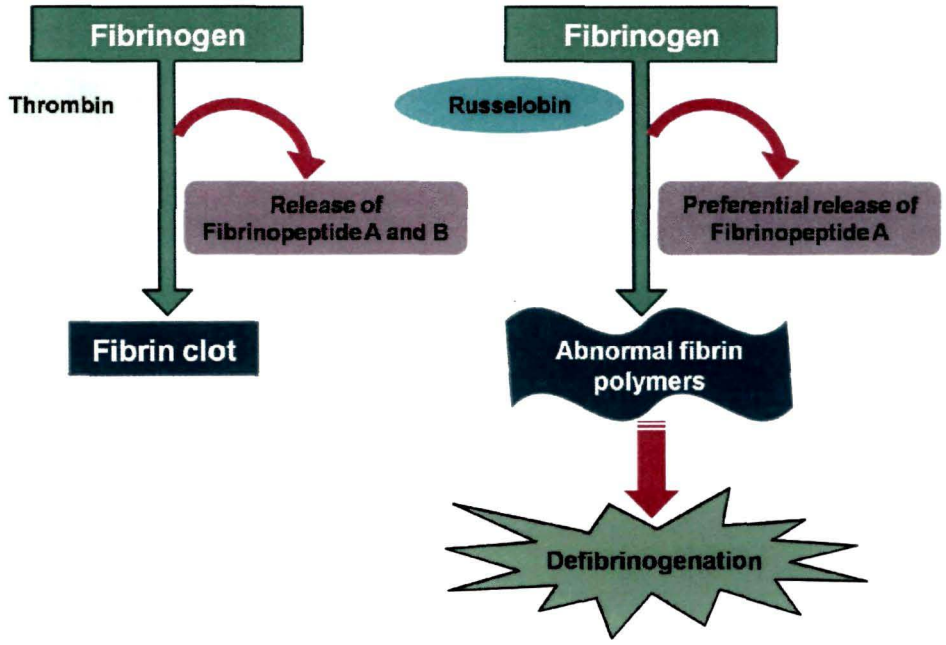


Fig. 2.5 Proposed mechanism of action of the thrombin-like enzyme-Russelobin purified from RVV. A comparison of its fibrinogenolytic activity with that of physiologically active enzyme thrombin.

2.2.1.4 Other procoagulant proteins from RVV

Studies on the biochemical composition of RVV from Eastern India origin has been carried out extensively by Mukherjee et al. [46]. A 15.0 kDa metalloprotease named RVBCMP (Russell's Viper basic coagulant metalloprotease) showing hemorrhagic and potent *in vitro* procoagulant activities was purified and characterized from RVV [65]. In fact, this is the smallest metalloprotease reported from RVV. Further, the carbohydrate content of RVBCMP was reported to be about 20% of total protein mass, which is much higher than the carbohydrate content of the many of the coagulant glycoproteins isolated from RVV [65]. Inhibition of procoagulant activity of RVBCMP post-treatment with EDTA resulted in more than 80% abolition of its protease (catalytic), coagulant as well as liver haemorrhagic activities indicating RVBCMP is a metalloprotease from RVV [65]. RVBCMP is reported to be an α-fibrinogenase as it specifically degraded the Aα-chain of fibrinogen in addition to showing tissue (liver) hemmorhagic activity [65]. Investigations such as neutralization studies with commercial antivenom revealed that RVBCMP contains distinct perhaps overlapping sites for rendering coagulant and tissue hemmorhagic effects [65]. However, the actual mechanism of its procoagulant activity remains unexplored.

In spite of the presence of several prothombin activators in the venom of snakes belonging to the Viperideae family (Table 2.2), there is no report on the presence of such proteins in RVV. The snake venom prothrombin activators reported till date, are classified into four different groups on the basis of the structural and functional requirements as described in section 1.9.1.1.3.

Table 2.2 Classification of snake venom prothrombin activators.

Group	Example	Molecular mass (kDa)	Type of protease	Source	Family	Reference
A	Ecarin	55.0	SVMP	*Echis carinatus*	Viperidae	[256]
	Bothrojaractivase	22.8	SVMP	*Bothrops jararaca*	Viperidae	[257]
	Basparin A,	70.0	SVMP	*Bothrops asper*	Viperidae	[258]
	Aharin	63.0	SVMP	*Agkistrodon halys pallas*	Viperidae	[259]
	Berythractivase	78.0	SVMP	*Bothrops erythromelas*	Viperidae	[260]
B	Carinactivase	62 and 25 subunit	SVMP	*Echis carinatus leucogaster*	Viperidae	[261]
	Multactivase	62 and 24 subunit	SVMP	*Echis multisquamatus*	Viperidae	[262]
	Cotiaractivase	22.9	SVMP	*Bothrops cotiara*	Viperidae	[263]
	Multisquamase	58 and 23 subunit	SVMP	*Echis multisquamatus*	Viperidae	[264]
	Mikarin	47.0	SVMP	*Micropechis ikaheka*	Elapidae	[265]

Group	Example	Molecular mass (kDa)	Type of protease	Source	Family	Reference
C	Oscutarin C	30.0	SVSP	*Oxyuranus s. scutellatus*	Elapidae	[266]
	Pseutarin C	25.0	SVSP	*Pseudonaja textilis*	Elapidae	[142]
	Textarin	50.0	SVSP	*Pseudonaja textilis*	Elapidae	[267]
D	Trocarin D	46.5	SVSP	*Tropidechis carinatus*	Elapidae	[143]
	Hopsarin D	46.1	SVSP	*Hoplocephalus stephensi*	Elapideae	[268]
	Notecarin D	46.5	SVSP	*Notechis scutatus scutatus*	Elapidae	[268]
	Notanarin D	46.0	SVSP	*Notechis ater niger*	Elapidae	[268]
Unclassified	Cotiarinase	29.0	SVSP	*Bothrops cotiara*	Viperidae	[269]

2.2.2 Anticoagulant proteins from RVV: A brief review on their structure-functional characteristics and their mechanisms of action

RVV proteins/peptides that prolong blood coagulation are either enzymatic or non-enzymatic proteins with molecular masses ranging from 6 kDa to 70 kDa [59, 64, 66, 96-100, 106, 113]. These proteins have been shown to exhibit unique structure-functional aspects and inhibit blood coagulation by different mechanisms. In the following sections, the structural and functional properties of various enzymatic and non-enzymatic anticoagulant proteins identified from RVV have been described.

2.2.2.1 Anticoagulant proteins from RVV with enzymatic properties

2.2.2.1.1 PLA_2 enzymes from RVV

The most prominent enzymatic anticoagulant proteins in RVV are the PLA_2 isoenzymes [96-100]. Strong anticoagulant PLA_2s such as VRV-PL-VI [99] and VRV-PL-VIIIa [100] which induce hemorrhage in the lung, liver, kidney and endocrine glands have been reported from RVV. Though, a detailed mechanism of their anticoagulant activities are not known, however, it is suggested that these PLA_2 enzymes show anticoagulant activity by virtue of their ability to hydrolyze and destroy the phospholipid surface [99-100].

Saikia et al. reported the presence of an acidic PLA_2 (RVVA-PLA_2-I) in RVV which demonstrated dose-dependent catalytic, mitochondrial and erythrocyte membrane damaging activities [96]. RVVA-PLA_2-I exerted its anticoagulant activity by enzymatic hydrolysis of plasma phospholipids and by non-enzymatic inhibition of factor Xa in a phospholipids/Ca^{2+} independent manner [96]. The non-enzymatic, Ca^{2+}-independent mechanism of inhibition of blood coagulation by RVVA-PLA_2-I was further evident from the fact that it retained about 38% of

anticoagulant activity although the catalytic activity was reduced to 80% of its original activity post addition of metal chelators such as EDTA [96].

A 13 kDa neutral anticoagulant PLA_2 (Neupholipase) showing *in vitro* and *in vivo* anticoagulant property was isolated from RVV by Saikia et al. [97]. The strong anticoagulant activity of Neupholipase was primarily attributed to the enzymatic hydrolysis of phospholipids (PS) and marginally to the binding of this PLA_2 to the phospholipids of plasma which are required in the coagulation process [97]. Neupholipase demonstrated preferential hydrolysis of PS, unlike most snake venom PLA_2s which show preference for the hydrolysis of PC rather than PS [97].

Recently, another RVV PLA_2 was identified by Mukherjee (2014b) [98]. The 13.84 kDa PLA_2 named $RVAPLA_2$ showed strong anticoagulant effect via thrombin inhibition with a K_i value of 380 nM as well as by binding to procoagulant phospholipids of plasma [98]. This is the first example of a thrombin inhibiting PLA_2 from RVV. Studies concerning this PLA_2 revealed that binding of $RVAPLA_2$ to thrombin was not a very rapid event and nanomolar inhibition constant (K_i) suggested tight binding of $RVAPLA_2$ with thrombin, via a non-covalent mechanism [98]. Moreover, the anticoagulant property of $RVAPLA_2$ was found to be enhanced in the presence of Ca^{2+} ions which favored its binding to free phospholipids [98].

2.2.2.1.2 Fibrino(geno)lytic proteases from RVV

Only a few fibrino(geno)lytic proteases have been identified from RVV. VRR–73, a hemmorhagic metalloprotease specific to skin and muscle was isolated from the venom of Russell's Viper of Eastern India origin by Chakraborty and his colleagues in 2000 [64]. VRR-73 contains Mg^{2+} in a mole per mole ratio in contrast to a large number of haemorrhagins where the associated metals are found to be Zn^{2+} or Ca^{2+} [54]. This 73 kDa metalloprotease was reported to exhibit strong fibrinolytic activity that was independent of its hemorrhagic activity [64].

Biochemical characterization revealed that VRR-73 did not cause any significant proteolysis of basement membrane proteins such as laminin and collagen, albeit it showed hydrolysis of Nα-Benzoyl-L-arginine 4-nitroanilide hydrochloride (BAPNA), a chromogenic substrate for trypsin, with a specific activity of 3000 U mg^{-1} [64].

The anticoagulant property of VRR-73 was attributed to its strong fibrin(ogeno)lytic activity [64]. VRR-73 also induced fibrinolysis by activating plasminogen to produce plasmin which in turn causes fibrinolysis [64]. A detailed study on its anticoagulant mechanism is warranted.

Two P-III class hemmorhagic metalloproteases were also identified from RVV are Daborhagin-M and Daborhagin-K [66]. Daborhagin-M was isolated from the RVV of Myanmar origin (*Daboia russelii siamensis*) whereas Daborhagin-K was isolated from RVV of Indian origin (*D. r. russelii*). Both Daborhagin-M and Daborhagin-K are glycosylated zinc metalloprotease with an apparent molecular mass of 62 and 66 kDa under non-reducing and reducing conditions, respectively [66]. Daborhagin-M hydrolyzed the β-chain of insulin at Ala^{14}-Leu^{15} and Tyr^{16}-Leu^{17} followed by His^{10}-Leu^{11} and Phe^{24}-Phe^{25} positions as well as type IV collagen *in vitro* [66]. Daborhagin-M was reported to bind to a single zinc and calcium ion per subunit [66]. Daborhagin-K on the other hand, binds to single zinc and two calcium ions per subunit [66].

The anticoagulant property of the daborhagins is attributed to its potent *in vitro* fibrinogenolytic activity [66]. At a low enzyme concentration of 75 nM, Daborhagin-M specifically cleaved the Aα-chain of human fibrinogen with no apparent effect on Bβ- and γ-chains [66]. Though the hemorrhagic properties of the daborhagins have been well characterized, however, a detailed study on its anticoagulant mechanism is unknown.

2.2.2.2 Non-enzymatic anticoagulant proteins/peptides from RVV

Venom of snakes belonging to the Viperideae family have been reported to possess anticoagulant peptides that are non-enzymatic in nature which are mostly C-type lectin proteins [112, 178-185, 187] or Kunitz-type serine protease inhibitor [107, 270]. Only a few non-enzymatic anticoagulant peptides have been identified from RVV which belong to the family of Kunitz-type serine protease inhibitors or C-type lectin proteins [59, 106, 113].

2.2.2.2.1 Kunitz-type serine protease inhibitor from RVV

A few Kunitz-type serine protease inhibitors with potent anticoagulant effects have recently been identified from RVV. A 6.9 kDa Kunitz-type serine protease inhibitor named Rusvikunin was reported from RVV [106]. This glycoprotein formed a complex with another 7.1 kDa basic Kunitz-type protease inhibitor named Rusvikunin-II to induce *in vitro* and *in vivo* anticoagulant action [59, 106]. Functionally, Rusvikunin-II was found to inhibit trypsin, plasmin and factor Xa whereas Rusvikunin inhibited the proteolytic activities of trypsin, thrombin and factor Xa [59, 106].

The anticoagulant property of Rusvikunin was attributed to its ability to inhibit the fibrinogen clotting activity but not the catalytic activity of thrombin in a manner similar to hirudin, which involves interaction of the peptide to the exosite–I of thrombin [106]. Interestingly, Rusvikunin could inhibit the fibrinogen clotting activity of thrombin but could not inhibit thrombin-like serine protease isolated from the same venom such as Russelobin [59, 106]. This suggested that snake venom Kunitz-type serine protease inhibitors such as Rusvikunin have probably undergone positive selection favoring the inhibition of protease(s) of the victim/prey, avoiding targeting of its own venom components [106].

Later, Mukherjee and his colleagues reported the isolation of another Kunitz-type protease inhibitor isoenzyme named Rusvikunin-II [59]. It was suggested that the interaction between Rusvikunin and Rusvikunin-II leads to the formation of a protein complex (Rusvikunin-complex) in RVV apparently augmenting the anticoagulant action and lethality of the individual components [59]. Both Rusvikunin-II and the Rusvikunin-complex dose-dependently inhibited the prothrombin activating property of factor Xa; with Rusvikunin-complex demonstrating better factor Xa inhibition property [59]. Rusvikunin alone was unable to inhibit factor Xa [106], therefore, it is suggested that the interaction between Rusvikunin and Rusvikunin-II, leading to the formation of a stable protein-complex, is responsible for the increase in biological potency and lethality [59].

2.2.2.2.2 C-type lectin proteins from RVV

An anticoagulant protein which belongs to the family of snake venom C-type lectin proteins (snaclec) named RVsnaclec has been recently identified from RVV [113]. RVsnaclec is a heterodimer of two subunits, α (15.1 kDa) and β (10 kDa) [113]. These subunits are covalently linked to form multimeric (αβ)2 and (αβ)4 structures [113]. LC-MS/MS analysis of RVsnaclec demonstrated its similarity to snaclecs purified from other viperid snake venoms with two tryptic peptide sequences exhibiting the putative conserved C-type lectin domains [113].

RVsnaclec dose-dependently increased the plasma recalcification time as well as prothrombin time of PPP; however, it did not affect the partial thromboplastin time (APTT) or thrombin time of PPP [113]. The *in vitro* and *in vivo* anticoagulant activity of RVsnaclec is correlated to its binding and subsequent uncompetitive inhibition of factor Xa (*Ki* =0.52 μmole) in a Ca^{2+}- independent manner; however, supplementation with 0.25 mM Ca^{2+} enhanced the Xa binding potency of RVsnaclec [113]. Monovalent or polyvalent antivenom failed to neutralize its anticoagulant potency, and RVsnaclec could not inhibit trypsin, chymotrypsin,

thrombin or plasmin [113]. RVsnaclec was devoid of hemolytic activity or cytotoxicity against several human cancer cell lines, demonstrated concentration independent aggregation and deaggregation of human platelets, and inhibited the ADP-induced aggregation of platelet [113].

However, till date, no anticoagulant protein showing dual inhibitory effects on thrombin and factor Xa has been identified from RVV. Also, there is a dearth of knowledge on the presence of three-finger toxin like molecules from RVV which induces anticoagulant and toxic effects in the presence of a fibrino(geno)lytic protease. Due to the low toxicity and potent *in vivo* anticoagulant effects of RVV proteins/peptides, most of these molecules show a high potential for development of antithrombotic and/or thrombolytic drugs [59, 98, 106, 113].

2.3 Pharmacological properties and toxicity of procoagulant and anticoagulant proteins/peptides purified from RVV

A number of RVV proteins/peptides have been found to be the key components responsible for hemostatic alterations in envenomed victim/prey. The various procoagulant and anticoagulant proteins/peptides identified from RVV are listed in Table 2.1. These proteins/peptides are shown to affect the mammalian coagulation cascade at different levels. On the basis of their effect on the coagulation pathway, RVV proteins/peptides are found either to prolong the clotting time of blood (procoagulant nature) or they exert anticoagulant activity by inhibition of blood coagulation (Table 2.1). Pharmacologically, RVV proteins such as Daborhagin is reported to affect the basement membrane proteins of different tissues, thereby leading to microvessel damage and symptoms such as tissue necrosis, internal tissue and dermal hemorrhage [66]. Both the Daborhagins- Daborhagin-M and Daborhagin-K are reported to display high levels of hemorrhagic activity with a MHD of 0.86 μg and 0.82 μg when subcutaneously injected into mice [66].

As reported by Suwansrinon and his colleagues, RVV fractions are known to affect the systemic and renal hemodynamics in dogs [271]. They suggested that the changes in hemodynamics were possibly caused by the presence of some proteolytic enzymes in the RVV although the RVV protease(s) actually responsible for this pharmacological effect was not identified by them. Later, it was reported that RVV-X is the key component that causes renal dysfunction and coagulopathy in experimental animals [272]. Changes in renal hemodynamics and renal functions such as decreased glomerular filtration rate (GFR), effective renal plasma flow (ERPF), effective renal blood flow (ERBF) and increased renal vascular resistance (RVR) suggested the formation of renal lesions due to RVV-X treatment in experimental animals [272]. In addition to renal dysfunctions, RVV-X treated animals also demonstrated intravascular clots and increased D-dimer level [272]. Also, a synergistic interaction between RVV-X and a trypsin inhibitor is reported to significantly increase the toxicity of RVV-X substantially [273].

Neutralization study of RVV-X with commercial antivenom against RVV revealed that the factor X activator activity of purified RVV-X towards its specific chromogenic substrate (S-2765) could be completely neutralized when taken in the ratio of 250:1 (w/w) [272]. Moreover, Russell's Viper antivenom could also effectively prevent the D-dimer formation caused by purified RVV-X, under *in vivo* conditions when injected intravenously prior to RVV-X administration [272]. Further, no apparent histopathological changes in kidneys and lungs were observed in rats receiving Russell's Viper antivenom along with purified RVV-X [272].

In addition to RVV-X, other metalloproteases from RVV are also reported to demonstrate appreciable tissue damage and induce hemorrhage in experimental animals [64, 66]. For example, VRR-73 showed a minimum hemorrhagic dose (MHD) of 0.5 mg in mouse skin as compared to 5.0 mg for whole RVV [64]. VRR-73 produced haemorrhagic spots in the abdominal muscle wall and adjacent skin near the site of intradermal introduction with very little haemorrhage in the

peritoneal cavity or in the intestine [64]. Systemic haemorrhage was not detected in animals injected with VRR-73 (both *i.p.* and *i.v.*) and the LD_{50} of VRR-73 was calculated to be 2.5 mg 20 g^{-1} in mouse [64].

Neutralization study of VRR-73 with commercial bivalent antiserum revealed that intradermal or intravenous administration of antiserum after 2 min of intravenous administration of VRR-73 to laboratory inbred rats did not show any visible sign of inhibition of VRR-73 induced haemorrhage [64]. However, when the antiserum was introduced simultaneously with VRR-73 a complete inhibition of haemorhagic activity of VRR-73 was observed. This suggested that binding of VRR-73 to its site of action was a very rapid event and it occurred virtually in an irreversible manner [64]. Though the detailed structural and functional analysis of VRR-73 is warranted, studies conducted by Chakraborty and his colleagues suggested that VRR-73 was likely to be consisted of multiple domains from where its activities were operated independently [64]. Therefore, by applying reversible thermal denaturation to VRR-73 it was possible to restore its fibrinolytic activity almost completely while losing the hemorrhagic activity permanently [64].

Pharmacological characterization of the procoagulant metalloprotease RVBCMP suggested that the identified protease lacked edema inducing property at the tested dose of 5.0 μg in experimental mice but demonstrated strong *in vitro* tissue damaging activity [65]. In view of the pharmacological properties of RVBCMP, it is suggested that RVBCMP contributes to consumption coagulopathy in treated mice and is likely to play a significant role in Russell's Viper envenomings.

None of the serine proteases identified from RVV are accounted to be lethal in nature. As reported by Nakayama et al., RVV-V undergoes substantial changes in the substrate-specificity pockets but only minimal changes both in the catalytic triad and in the oxyanion hole upon substrate/inhibitor binding [252]. Their results suggested that the catalytic activity of RVV-V is regulated by induced-fit substrate

binding mechanism which explains the narrow specificity of RVV-V towards its substrates [252].

Another non-lethal, factor V activating protease, RV-FVPα and its isoenzymes have been reported to demonstrate *in vitro* procoagulant effects but *in vivo* anticoagulant effects by virtue of its fibrinogenolytic and factor V activating properties [83]. None of the isoenzymes were found to be lethal in mice or house geckos, at the tested dose of 5 mg kg^{-1} [83]. The studies on RV-FVPα clearly suggested the suitability of these isoenzymes for clinical application as cardiovascular drugs [83].

The RVV thrombin-like enzyme, Russelobin on the other hand is an anticoagulant SVSP, *in vivo*, which does not show *in vivo* lethality and toxicity in NSA strain experimental mice or house geckos at a dose of 5.0 mg kg^{-1} [82]. The lack of histologically detectable changes to several tissues, lethal toxicity or cytotoxicity on mammalian cells underpins the non-toxic nature of Russelobin supporting its suitability for clinical applications [82]. Studies conducted by Mukherjee and Mackessy (2013) show that Russelobin forms a tenuous clot by preferentially releasing fibrinopeptide A and slowly releasing fibrinopeptide B from fibrinogen leading to formation of non-crossed linked, labile (soft) clot which is easily degraded by plasmin [82]. This indicates that Russelobin is a potent defibrinogenating agent in *in vitro* and *in vivo*, possibly contributing to uncontrolled bleeding seen after Russell's Viper envenomation [82].

The anticoagulant PLA_2s that have been identified from RVV are either highly toxic [99-100] or they are non-toxic in nature, in spite of their profound effects on the coagulation cascade [96-98]. It is interesting to note that though the anticoagulant PLA_2 (Neupholipase) was non-lethal to BALB/c mice (0.08 mg kg^{-1}), however; it showed detrimental effects on mice liver at the tested dose [97]. Pathological symptoms observed in Neupholipase-treated mice could be correlated with the actual clinical manifestations in Russell's Viper envenomed patients from

eastern India indicating contribution of this PLA_2 in RVV-induced toxicity and pathogenesis [97].

The non-enzymatic anticoagulant proteins such as Rusvikunin (4 mg kg^{-1}; *i.p.*), Rusvikunin-II (5 mg kg^{-1}; *i.p.*) and RVsnaclec (5 mg kg^{-1}; *i.p.*) are also reportedly non-cytotoxic and lacked *in vivo* toxicity in experimental mice [59, 106, 113]. However, these proteins demonstrated significant *in vivo* anticoagulant property by virtue of their ability to prolong the blood coagulation time in experimental animals [59, 106, 113].

2.4 Toxicity of snake venom complexes: A few examples

Snake venom contains a mixture of biologically active proteins which exhibit distinct pharmacological effects by virtue of their structural and functional specificities towards different molecular targets. Very often, snake venom proteins exist as monomers, but some of them form complexes in the venom [59, 274-278]. These complexes are shown to exhibit a higher level of pharmacological activity compared to the individual components and therefore, contribute to the pathophysiological effects during envenomation [59, 274-278].

The protein complexes are formed as a result of protein-protein interaction which happens via distinctive covalent and/or non-covalent interactions. In many cases, the subunits of the snake venom complexes are either identical (homodimers) or dissimilar (heterodimers; in some cases subunits) and might also belong to different families of proteins [59, 186, 275-278]. Formation of protein complexes eliminates the non-specific binding and enhances the binding of the complex to a specific target molecule, synergistically, making it a more effective molecule in comparison to the individual components [59, 186, 275-278].

Additionally, formation of protein complexes in snake venoms, via non-covalent protein-protein interaction also leads to the exhibition of distinct pharmacological

activities by the complexes in comparison to their individual components [59, 186, 275-278]. As such, it is important to characterize snake venom proteins which form venom complexes rather than individual components of the complex which might be responsible for inducing enhanced toxicity in victims/prey. Many snake venom proteins which form either covalent or non-covalent complex with other venom proteins have been studied and their pathobiological effects have been explored.

It has been reported that many snake venom PLA_2s participate in the formation of protein complexes via non-vovalent interactions with other PLA_2s, PLA_2 precursors or other venom proteins [275-278]. In 2010, Mukherjee reported the non-covalent interaction of a 14.3 kDa PLA_2 (NK-PLA_2) and two neurotoxins (KTX) named as kaouthiotoxin A (7.7 kDa) and kaouthiatoxin B (7.6 kDa), from venom of *Naja kaouthia* which exhibited marked synergism leading to enhanced cytotoxicity on virus infected Sf9 and normal Tn insect cells [277]. NK-PLA_2-KTX complex furnished an example of protein complementation for restoration of the biological activity by non-covalent interaction of two polypeptides of cobra venom which possibly plays an important role in the pathophysiology of cobra bite [277].

An anticoagulant protein complex consisting of a three finger toxin was first reported by Banerjee and his colleagues [186]. The complex, known as the Hemextin AB complex was isolated from *Hemachatus haemachatus* venom and consisted of two copies of Hemextin A and two copies of Hemextin B, which formed a 1:1 complex [186]. It inhibited the enzymatic activity of factor VIIa and factor VIIa/tissue factor complex. While hemextin A exhibited only mild anticoagulant activity and hemextin B was reportedly inactive, the latter synergistically enhanced the activity of Hemextin A in the complex [186]. Moreover, Hemextin AB complex is the first report of a natural inhibitor of factor VIIa that does not require either tissue factor or factor Xa scaffold to mediate its inhibitory activity [186].

RVV is also rich in protein components which form complexes with other venom proteins. In 1990, Jayanthi and Gowda, demonstrated a synergistic interaction between the protease, RVV-X and two trypsin inhibitors, TI-I and TI-II [273]. TI-I inhibited only the factor X activating property of RVV-X while TI-II inhibited the caseinolytic as well as factor X activating properties of RVV-X. Moreover, the edema inducing activity of RVV-X increased markedly in the presence of non-edema inducing doses of TI-I and TI-II [273]. Though RVV-X, TI-I and TI-II were non-lethal in mice, the combination of RVV-X and TI-II demonstrated enhanced toxicity [273].

In 2008, Kumar et al. reported the presence of a 44.6 kDa protein complex in RVV consisting of a 13 kDa PLA_2, 29.1 kDa protease, and a 6.5 kDa trypsin inhibitor. The complex showed an LD_{50} of 5.06 mg kg^{-1} body weight in mice and inhibited the frequency of spontaneous release of neurotransmitter in hippocampal neurons [275]. It also caused peritoneal bleeding, and edema in the mouse foot pads. Interestingly, the complex also caused degeneration of both the germ cells and the Leydig cells of mouse testis [275]. This was the first report demonstrating toxicity of a snake venom protein complex to the reproductive system [275]. It was also observed that the association of the three distinct proteins in Reprotoxin resulted in masking of the protease and trypsin inhibitory activities and enhanced the PLA_2 activity [275].

Another protein complex consisting of a PLA_2 and a neurotoxic peptide was reported from RVV by Venkatesh and Gowda [278]. The isolated complex named DR-HC-I was strongly hemmorhagic and consisted of a 13.4 kDa hemmorhagic PLA_2 and a 6.7 kDa non-enzymatic neurotoxic peptide (DNTx-II) [278]. DR–HC-I complex was lethal to mice with the LD_{50} dose of 0.7 mg kg^{-1} body weight with hemorrhagic and neurotoxic properties. It was reported that at 5:2 M ratio, the complex acted as a toxic molecule and induced hemorrhage at the site of injection [278]. On the other hand, neither the PLA_2 nor the DNTx-II induced hemorrhagic effects individually [278].

The first report of a protein complex comprising of two Kunitz-type protease inhibitors from RVV was reported by Mukherjee and coworkers [59]. Both the inhibitors, Rusvikunin (7.1 kDa) and Rusvikunin-II (6.9 kDa) demonstrated potent anticoagulant effects *in vivo* by virtue of their thrombin and factor Xa inhibiting properties, respectively [59, 106]. Rusvikunin-complex demonstrated better factor Xa inhibitory property in comparison to Rusvikunin-II but demonstrated a lower thrombin inhibitory property in comparison to Rusvikunin [59]. It is interesting to note that both Rusvikunin and Rusvikunin-II are individually non-toxic [59]. Conversely, Rusvikunin complex showed prey-specific toxicity towards NSA mice at a dose of 5 mg kg^{-1} (*i.p.*). It is suggested that these non-toxic, Kunitz-type inhibitors (Rusvikunin and Rusvikunin-II) interact synergistically thereby leading to increased toxicity which provides a definitive advantage for the Russell's Viper by inducing lethal toxicity in its victims/prey [59, 106].

2.5 Biomedical applications of RVV proteins/peptides affecting hemostasis

Not many RVV proteins have established themselves in the field of therapeutics and diagnostics (Table 1.2). RVV-X and RVV-V are two milestones isolated from RVV which have carved a niche for themselves in the field of diagnostic applications. Both RVV-V and RVV-X are marketed by Pentapharm Ltd (Basel, Switzerland) and are regularly used to diagnose factor deficient blood for routine assay of coagulation factors and as diagnostic reagents for studying both coagulation factors and hemostasis in general.

RVV-V (Pefakit® Pentapharm) is used in the laboratory for the estimation of level of factor V in human plasma [250]. In addition, RVV-V also serves as a useful tool in the study of factor V activation [12]. RVV-V is also used for studying the resistance of factor Va inactivation by activated Protein C caused by the Factor V Leiden mutation [12].

Like RVV-V, RVV-X is also used extensively in the diagnosis of quantitative conversion of the zymogen factor X into factor Xa [12]. The enzyme has become widely used for determination of factor X as well as for distinguishing between factor VII and factor X deficiency in plasma [203] and lupus anticoagulant assay [204]. The clotting time of plasma using RVV-X is known as the Stypven time^{TM}. A normal Stypven time used in conjunction with a prothrombin time (PT) suggests factor VII deficiency whereas a prolonged Stypven time indicates factor X deficiency in the plasma [203]. RVV-X enzyme is also used to assay the platelet factor 3 [279] and this may perhaps have some advantages over the routine coagulation tests in discriminating between thrombotic patients and healthy controls. In view of the exclusive characteristics, RVV enzymes and proteases have been used regularly for the improvement in the detection of Von-Willebrand disease [12].

Therapeutic defibrinogenation with SVTLEs and fibrinogenolytic proteases have been endeavored to treat peripheral arterial occlusion diseases in which surgical revascularization could not be satisfactorily achieved [280]. Potential non-toxic thrombin-like enzymes from RVV such as Russelobin and fibrinogenolytic enzymes such as RVFP isoenzymes, can be considered effective candidate for application as antithrombotic molecules [82-83]. Under controlled clinical setting, Russelobin and RVFP isoenzymes is likely to result in defibrinogenation and control hyperfibrinogenomia, which would benefit in the treatment of thrombus and cancer associated pathologies such as deep vein thrombosis, myocardial infarctions and other cardiovascular disorders. In view of the non-toxic nature of Russelobin and RVFP isoenzymes, the therapeutic application of these proteases is a welcome measure to deal with hyperfibrinogenomia-associated disorders and vascular ailments [82-83].

Unlike SVTLEs and fibrinogenolytic enymes, snake venom enzymes with specific activity towards fibrin lead to direct dissolution of blood clots without causing defibrinogenation; thereby suggesting potential application of fibrinolytic enzymes

for the treatment of cardiovascular ailments such as strokes, heart attacks and other diseases associated with the formation of thrombus. Fibrinolytic enzymes so far identified from RVV such as VRR-73 may serve as a potential candidate for the development of effective drugs for dissolving unwanted clot which results in fatal complicacies such as myocardial infarction, stroke and other cardiovascular disorders; provided its hemorrhagic property is neutralized and/or removed by thermal denaturation or other possible techniques [64].

Non-enzymatic inhibitors of thrombin and/or factor Xa isolated from RVV such as Rusvikunin, Rusvikunin-II and RVsnaclec, which are individually non-toxic, can serve as potential candidates for development of peptide-based antithrombotic agents [59, 106, 113].

2.6 Anticancer proteins/peptides from RVV

A few peptides such as *dr*CT-1 (7.2 kDa), RVV-7 (7.2 kDa) and Drs-PLA_2 (13.6 kDa) with anticancer potential has also been reported from RVV [92, 281-282]. All of these peptides are characterized as PLA_2 enzymes with cytotoxic and anticancer potential [92, 281-282].

The anticancer activity of drCT-I was evaluated against Ehrlich ascites carcinoma (EAC) cells *in vivo* and human leukemic cells (U937, K562) *in vitro* [281]. It was observed that under *in vivo* conditions, drCT-I (125 mg kg^{-1}, i.p day^{-1} for 10 days) significantly decreased EAC cell count, cell viability and increased the survival time of tumour bearing mice in comparison to untreated tumour bearing control [281]. Under *in vitro* conditions, drCT-I, produced dose and time-dependent inhibition of human leukemia cell lines, U937 and K562 cell growth with an IC_{50} of 8.9 and 6.7 mg ml^{-1} respectively [281]. drCT-I induced apoptosis in U937 and K562 through G1 phase arrest of the cell cycle [281].

The anti-tumor activity of a 7.2 kDa protein named RVV-7, towards *Mus musculus* skin melanoma (B16F10) cells has been attributed to the 'disintegrin' like function [92]. RVV-7 antagonizes platelet aggregation suppressing both metastasis and solid tumor growth [92, 226]. Pre-treatment of B16F10 cells with RVV-7 altered cell–cell adhesion property which in turn is suggested to affect the three-dimensional growth of tumor mass [92].

Khunsap et al. purified another PLA_2 (Drs-PLA_2) from RVV (*Daboia russelii siamensis* venom) which exhibited cytotoxic activities towards mammalian cancer cell lines [282]. Drs-PLA_2 decreased human skin melanoma (SK-MEL-28) cell viability in a dose-dependent manner [282]. Drs-PLA_2 also inhibited SK-MEL-28 cell migration with an IC_{50} of 25.6 nM [282]. Moreover, under *in vivo* conditions, Drs-PLA_2 inhibited the colonization of skin melanoma cells (B16F10) in BALB/c mice lungs by approximately 65% [282].

CHAPTER III

MATERIALS AND METHODS

3 MATERIALS AND METHODS

3.1 Materials

3.1.1 Venom and antivenom

Dry, pooled and desiccated crude Russell's Viper *(Daboia russelii russelii)* venom (RVV) of Eastern India origin was purchased from a licensed venom dealer, Calcutta Snake Park, Kolkata, India. Dried venom samples were stored in desiccators at room temperature (~23 °C) until further use. Polyvalent antivenom (PAV, effective against *Daboia russelii*, *Naja naja*, *Echis carinatus* and *Bungarus caeruleus* venoms) was purchased from Serum Institute of India Limited, Pune, India (Batch No. A2512012). Russell's Viper monovalent antivenom (MAV) was obtained from VINS Bioproduct Limited, India (Batch No. 30AS11001).

3.1.2 Chromatographic matrices and other fine chemicals

All chromatographic matrices such as Sephadex G-50, CM-Cellulose and DEAE-Cellulose were obtained from Pharmacia Fine Chemicals, Sweden. Prepacked chromatographic columns such as HiPrep CM FF 16/10 cation exchange column (16 mm x 100 mm) was purchased from GE Healthcare, Sweden. C_{18} Zip-tip, prestained protein ladder [Thermo-Pierce Fermentas, PageRuler prestained protein ladder, #26616 (10-170 kDa) or #26619 (10-250 kDa)] for SDS-PAGE, sinapinic acid matrix and α-cyano-4-hydroxycinnamic acid matrix for MALDI-TOF MS analysis was purchased from Sigma, USA.

Chromogenic substrates such as F3301, T1637, V0882, B2133, V3133 and I0898; extracellular matrix (ECM) proteins such as type-IV collagen, laminin and fibronectin; phospholipids such as phosphatidylcholine (PC) and phosphatidylserine (PS); chemical modifiers such as TPCK, TLCK and pBPB was also purchased from Sigma, USA. Chemical modifiers such as PMSF, IAA and DTT; metal chelator such as EDTA; reagents for buffer preparation such as Tris-

buffer, PBS etc; reagents for SDS-PAGE analysis; solvents such as acetonitrile and methanol; and all other chemicals and reagents were of analytical grade and purchased from Merck, Germany or HiMedia, Mumbai, India.

3.1.3 Coagulation proteins and kits for assessment of plasma/serum proteins and serum profile

Coagulation proteins such as α_2-macroglobulin, α_2-antiplasmin, streptokinase, heparin and human plasmin was purchased from Sigma, USA. Other coagulation proteins such as bovine fibrinogen, bovine thrombin, factor X, factor Xa, factor Va, prothrombin, plasminogen, antithrombin-III were purchased from either Calbiochem, Germany or HiMedia, Mumbai India.

Thromboplastin reagent for prothrombin time test (Liquiplastin®) and cephaloplastin reagent for activated partial prothrombin time test (aPTT) (Liquicelin-E) was obtained from Tulip Diagnostics, India.

All diagnostic kits used for plasma analysis such as total protein, glucose, uric acid, creatinine, cholesterol, urea, ALP, CPK, SGOT and SGPT were purchased from Crest Biosystem, Goa. All kits for measurement of serum/plasma fibrinogen (factor I), factor IX, factor X, PDGFR-α and level of pro- and anti-inflammatory cytokines (TNF-α, IL-1, IL-6 and IL-10) were purchased from USCNK Life Science Incorporation, Wuhan, China.

Fresh goat blood was collected from local slaughter houses.

3.1.4 Cell culture reagents

MTT(3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide), Sulpho-rhodamine B, Mitochondria Staining Kit for mitochondrial potential changes detection, dihydroethidium dye, growth mediums such as Eagle's Minimal Essential Medium (EMEM) and Dulbecco's Modified Eagle Medium (DMEM), fetal bovine serum, sodium bicarbonate, insulin, antibiotics such as antimycotic

solution containing streptomycin, amphotericin B and penicillin, Hoechst 33250, acridine orange (AO) and ethidium bromide (EB) were purchased from Sigma Aldrich, USA. Mouse monoclonal p53 and p21 antibody, and Annexin Apoptosis Kit was purchased from Santa Cruz Biotechnology, Santa Cruz, CA. Alexa 568-labeled antimouse IgG antibody was purchased from Invitrogen, California, USA.

3.2 Methods

3.2.1 Isolation and purification of procoagulant and anticoagulant proteins/peptides from RVV

3.2.1.1 Purification of a non-enzymatic anticoagulant peptide from RVV

25.0 mg of crude RVV was dissolved in 2.0 ml of 20 mM potassium phosphate buffer, pH 7.0 and kept overnight at 4 °C. The venom solution was gently vortexed and centrifuged at 10,000 rpm for 5 min at 4 °C in a cooling centrifuge (Heraeus Multifuge X1R Centrifuge, Thermo Scientific, USA). The clear supernatant was applied on a CM-cellulose (20 mm X 60 mm) column which was pre-equilibrated with 20 mM potassium phosphate buffer, pH 7.0. The proteins in the venom supernatant were eluted stepwise using potassium phosphate buffers of increasing molarities and pH at room temperature (~23 °C). The increasing molarities and pH values were (a) 20 mM potassium phosphate, pH 7.0; (b) 50 mM potassium phosphate, pH 7.0; (c) 50 mM potassium phosphate, pH 7.5; (d) 75 mM potassium phosphate, pH 7.5; (e) 100 mM potassium phosphate, pH 8.0; (f) 180 mM potassium phosphate, pH 8.0; (g) 230 mM potassium phosphate, pH 8.0; and (h) 260 mM potassium phosphate, pH 8.0; (i) 300 mM potassium phosphate, pH 8.0. The flow rate was adjusted to 24.0 ml h^{-1} and 1.0 ml fraction was collected. The protein content and plasma recalcification time of all the fractions were estimated as described in sections 3.2.3.1, and 3.2.4.1, respectively.

The fractions eluted with 300 mM potassium-phosphate buffer, pH 8.0 (CMAC-III) showing highest anticoagulant activity were pooled, lyophilized and applied to

Sephadex G-50 gel-filtration column (1 cm x 60 cm) previously equilibrated with 20 mM potassium phosphate buffer, pH 7.0. Elution was carried out with the same buffer at room temperature (~23 °C). The flow rate was adjusted to 24.0 ml h^{-1} and 1.0 ml fractions were collected. The protein content and plasma recalcification time of all the fractions were estimated as described in sections 3.2.3.1, and 3.2.4.1, respectively.

The gel-filtration fraction (GF-IV) showing anticoagulant activity was further subjected to a reversed phase (RP) C_{18}-μ-Nova pack column (4.0 μm, 3.9 mm x 150 mm) coupled to a HPLC system (Waters, Milenium-2000). Briefly, about 50.0 μg of GF-IV was dissolved in 20.0 μl of solvent A (0.1% v/v TFA in 5% v/v acetonitrile) and subjected to separation through a C_{18} RP-HPLC column, previously equilibrated with solvent A. Bound proteins were eluted using a gradient of 5-95 % solvent B (0.1% v/v TFA in 95% v/v acetonitrile) from 5 to 37 min at a flow rate of 1.0 ml min^{-1}, at room temperature (~23 °C). Protein elution was monitored at 280 nm and protein peaks were screened for anticoagulant activity on citrated PPP as described in section 3.2.4.1. The fraction showing anticoagulant activity was named Ruviprase.

3.2.1.2 Purification of a procoagulant metalloprotease from RVV

25.0 mg of crude RVV was dissolved in 1.0 ml of 20 mM potassium phosphate buffer, pH 7.0 and kept overnight at 4 °C. The venom solution was gently vortexed and centrifuged at 10,000 rpm for 5 min at 4 °C in a cooling centrifuge (Hereus Multifuge X1R Centrifuge, Thermo Scientific, USA). The clear supernatant was applied on Sephadex G-50 gel-filtration column (1 cm x 60 cm) pre-equilibrated with 20 mM potassium phosphate buffer; pH 7.0, and the proteins were eluted with the same buffer at a flow rate of 20.0 ml h^{-1}. 1.0 ml fractions were collected. The protein content, protease activity, fibrino(geno)lytic activity and plasma recalcification time of the fractions were screened by following the procedures described in sections 3.2.3.1, 3.2.3.2, 3.2.3.3.1 and 3.2.4.1, respectively.

The first gel-filtration peak (GF-I) (5.7 mg), showing fibrinogenolytic activity as well as strong procoagulant activity was subjected to HiPrep CM FF 16/10 cation exchange column (90 µm, 16 mm x 100 mm) coupled to an FPLC system (Akta Purifier 10, Wipro GE Healthcare). The column was pre-equilibrated with 20 mM potassium-phosphate buffer, pH 7.0 (1.5 column volume). After eluting the unbound and non-specifically adsorbed proteins with two volumes of equilibration buffer, the bound proteins were eluted with a linear gradient of 20 mM potassium-phosphate buffer, pH 7.0 containing 1 M NaCl. The flow rate was adjusted to 10.0 ml h^{-1} and 1.5 ml fractions were collected. Absorbance was measured at 280 nm. The protein content, protease activity, fibrino(geno)lytic activity and plasma recalcification time of all the fractions were estimated as described in sections 3.2.3.1, 3.2.3.2, 3.2.3.3.1 and 3.2.4.1 respectively. The fourth peak (GF-I-CM-IV) demonstrated strong fibrinogenolytic and procoagulant properties and was named Rusviprotease (Russell's Viper Protease).

3.2.1.3 Isolation of an anticoagulant toxic fraction (GF-VI DEAE-II) from RVV and purification of individual components of GF-VI DEAE-II

25.0 mg of crude RVV was dissolved in 1.0 ml of 20 mM potassium phosphate buffer, pH 7.0 and kept overnight at 4 °C. The venom solution was gently vortexed and centrifuged at 10,000 rpm for 5 min at 4 °C in a cooling centrifuge (Hereus Multifuge X1R Centrifuge, Thermo Scientific, USA). The clear supernatant was applied on Sephadex G-50 gel-filtration column (1 cm x 60 cm) pre-equilibrated with 20 mM potassium phosphate buffer; pH 7.0, and the proteins were eluted with the same buffer at a flow rate of 20.0 ml h^{-1}. 1.0 ml fractions were collected. The protein content, protease activity, fibrino(geno)lytic activity and plasma recalcification time of the fractions were screened by following the procedures described in sections 3.2.3.1, 3.2.3.2, 3.2.3.3.1 and 3.2.4.1 respectively.

The gel-filtration fraction showing strong proteolytic and fibrin(ogen)olytic activity (GF-VI), were pooled and further fractionated through DEAE cellulose

anion-exchange column (0.5 cm x 2.0 cm) pre-equilibrated with 20.0 mM potassium phosphate buffer, pH 7.0. After eluting the unbound and non-specifically adsorbed proteins with two volumes of equilibration buffer (20 mM potassium phosphate buffer, pH 7.0), the bound proteins were eluted by increasing the molarities (50 to 300 mM) of eluting phosphate buffers at room temperature (~23 °C). The increasing molarities were (a) 50 mM potassium phosphate, pH 7.0; (b) 75 mM potassium phosphate, pH 7.0; (c) 100 mM potassium phosphate, pH 7.0; (d) 150 mM potassium phosphate, pH 7.0; (e) 175 mM potassium phosphate, pH 7.0; (f) 200 mM potassium phosphate, pH 7.0; and (g) 300 mM potassium phosphate, pH 7.0. The flow rate was adjusted to 24.0 ml h^{-1} and 0.5 ml fraction was collected in each tube. The protein content, protease activity, fibrino(geno)lytic activity and plasma recalcification time of the fractions were screened by following the procedures described in sections 3.2.3.1, 3.2.3.2, 3.2.3.3.1 and 3.2.4.1 respectively. The fraction eluted with 50 mM potassium phosphate buffer, pH 7.0 (GF-VI DEAE-II) demonstrated highest fibrino(geno)lytic activity as well as anticoagulant property. The active fraction (GF-VI DEAE-II) was lyophilized and kept for further analysis.

40.0 µg of GF-VI DEAE-II was dissolved in 20.0 µl of solvent A (0.1% v/v TFA in H_2O) and subjected to fractionation through C_{18} RP-HPLC column (300 Å; 2.1mm x 150 mm) coupled to an UHPLC system (Ultimate 3000, Dionex, Germany). The column was equilibrated with solvent A (0.1% TFA (v/v) in H_2O) and eluted using a linear gradient of solvent B (80% acetonitrile containing 0.1% TFA (v/v) in water) at a flow rate of 0.35 ml min^{-1}. The protein elution was monitored at 215 nm and protein peaks were collected manually. The protein peaks were assessed for protein content, fibrin(ogen)olytic and anticoagulant activities as described in sections 3.2.3.1, 3.2.3.2, 3.2.3.3.1 and 3.2.4.1 respectively. The proteins (RP-I and RP-II/Rusvitoxin) thus obtained were lyophilized and stored at -20 °C until further use.

3.2.2 Biophysical characterizations

3.2.2.1 Purity of preparation and molecular mass determination

3.2.2.1.1 SDS-PAGE analysis

The homogeneity and molecular mass of the isolated venom protein/peptide (10.0-25.0 µg) was determined by SDS-PAGE analysis. Discontinuous polyacrylamide gels (12% or 15%) were freshly prepared according to Laemmli [283]. The composition of 10%, 12% and 15% resolving gels with their 4% stacking gel is given in the table below (Table 3.1). Freshly prepared running buffer (15.1 g l^{-1} Tris, 72 g l^{-1} glycine, 5 g l^{-1} SDS) was used for the SDS-PAGE analysis. Non reduced samples for SDS-PAGE were diluted appropriately and 5.0 µl of 3X loading dye (0.1% bromophenol blue (w/v), 10% SDS (w/v) and 10% glycerol in 0.05M Tris-HCl, pH 6.8) was added. For analyzing the samples under reducing conditions, 5.0 µl of the loading dye containing β-mercaptoethanol (30.0 µl ml^{-1} loading dye) and DTT (2.0 mM) was added and boiled at 100 °C for 5 min prior to loading in the gel. Electrophoresis was carried out at a constant current of 15 mA until the dye front reached the bottom of the gel.

Protein bands were visualized either by staining with Coomassie Brilliant Blue R-250 or by silver staining [284]. For staining with Coomassie Brilliant Blue R-250, gels were incubated (overnight) in Coomassie stain containing Coomassie Brilliant Blue R-250 (0.25% w/v), 40% methanol (v/v) and 10 % acetic acid (v/v) in distilled water. The gels were destained using destaining solution (distilled water containing 40% methanol (v/v) and 10% acetic acid (v/v)), until protein bands were clearly visible.

For silver staining, the gels were fixed for at least 30 min in fixing solution (40% methanol (v/v), 10% acetic acid (v/v) in MilliQ water). The gels were subjected to silver staining (solution containing 0.2% $AgNO_3$ (w/v) in MilliQ water) after sensitizing the gel using 0.02% $Na_2S_2O_3$ (w/v) in MilliQ water. After 25 min of

staining, the gels were developed (till bands were clearly visible), using the developing solution (3% Na_2CO_3 (w/v), 0.02% $Na_2S_2O_3$ (w/v), 0.02% 0.5 µl ml^{-1} formaldehyde in MilliQ water). Further development of the gel was stopped by adding 1.4% EDTA (w/v) solution.

For determination of molecular mass, the mobility of the isolated venom protein/peptide was compared with the protein molecular standards (see section 3.1.2) where a linear dependency of log molecular weight Vs. migration of protein bands (*Rf* value) was observed.

Table 3.1 Composition of Tris-glycine SDS-PAGE

Composition	Stacking gel (5 ml)	Resolving gel (10 ml)		
	4%	**10%**	**12%**	**15%**
30% Polyacrylamide (ml)	1.7	3.34	4.0	5.0
0.5 M Tris-HCl, pH 6.8 (ml)	1.25	-	-	-
1.5 M Tris-HCl, pH 8.8 (ml)	-	2.5	2.5	2.5
10% Ammonium persulphate (ml)	0.1	0.1	0.1	0.1
10% SDS (ml)	0.1	0.1	0.1	0.1
Water (ml)	6.8	4.0	3.3	2.3
TEMED (ml)	0.01	0.01	0.01	0.01

3.2.2.1.2 MALDI-TOF-TOF-MS analysis

The molecular mass of the isolated venom protein/peptide was determined by MALDI-TOF-TOF mass spectrometric analysis (MALDI ToF/ToF Analyzer, 4800 Plus MDS SCIEX, Applied Biosystems). Briefly, 0.5 µl of the purified venom protein/peptide (~1.0-5.0 µg) in 0.1% TFA was mixed with 0.5 µl of α-cyano-4-hydroxycinnamic acid matrix (10 mg ml^{-1}) or sinapinic acid matrix (10 mg ml^{-1}) dissolved in 50% (v/v) acetonitrile containing 0.1% (v/v) TFA. The mixtures were then spotted onto an Opti-TOF-384 plate (ABSciex), dried, and analyzed in

positive linear mode using an acceleration voltage of 25 kV and laser intensity of 3000. Spectra were acquired in low, mid and high molecular m/z range of 800-3,000 Da, 3,000-20,000 Da and 20,000-180,000 Da, respectively.

3.2.2.2 N-terminal protein sequencing

About 10.0 μg of the purified venom protein/peptide was dried or was subjected to 15% SDS-PAGE gel under reducing condition and electrotransferred to PVDF membrane (Millipore, USA). The purified venom protein/peptide was then subjected to Edman degradation using a fully automated Perkin-Elmer Applied Biosystems 492 pulsed-liquid phase protein sequencer (Procise) with an on-line 785A PTH-amino acid analyzer.

Protein homology searches were performed by using BLASTP (Basic Local Alignment Search Tool) program of the NCBI database (www.ncbi.nlm.nih.gov).

3.2.2.3 Peptide mass fingerprinting (PMF) analysis

For, MALDI-TOF-MS analysis, the protein bands of interest were excised, washed, reduced, alkylated and finally digested by trypsin [285]. Briefly, properly destained reduced protein bands were carefully excised from SDS-PAGE and digested with trypsin (100 ng). The samples were then rehydrated for 10 min in ice. The rehydrated gel pieces were overlaid with 25 mM NH_4HCO_3 and incubated at 37 °C for 16 h. The digested peptides were then extracted by adding 5% formic acid (v/v) in 50% acetonitrile (v/v) to the gel pieces. The peptide digest was then vortexed for 15 min at room temperature (~23 °C). Aliquots of trypsin-digested peptide fragments were withdrawn from the digest, desalted by C_{18} Zip-tip (using manufacturer's protocol) before subjecting to PMF analysis by MALDI-TOF-MS (Bruker Daltonik GmbH, Germany).

The PMF profiles were used to search for peptide matches from NCBI database following the MASCOT search (http://www.matrixscience.com). The sequences of

the peptides obtained were subjected for BLAST search in NCBI database of non-redundant protein sequences using BLASTP algorithm (http:// blast.ncbi.nlm.nih.gov/Blast.cgi). The sequences were finally aligned using CLUSTAL OMEGA program of EXPASY.

For LC MS/MS analysis, the purified venom protein/peptide in dried form was subjected to in-solution trypsin digestion [113]. Trypsin solution was added to the dried protein/peptide sample and kept overnight at 37 °C. The digested peptides were then extracted by adding 5% formic acid (v/v) in 50% acetonitrile (v/v) to the mixture. The peptide digest was then vortexed for 15 min at room temperature (~23 °C) and the volume was reduced by subjecting the mixture to SpeedVac [286]. The digested peptides were reconstituted in 15.0 µl of the 0.1% formic acid and were subjected to standard 70 min RP-HPLC-MS/MS analysis with collision-induced dissociation as the fragmentation method [113].

The generated data was searched following standard approach for the identity using MASCOT 2.4 as search engine on Proteome Discoverer 1.4. The data was searched against Uniprot Swiss-Prot database (non redundant database with reviewed proteins) from NCBI. The data was also searched against *D. r. russelii* database downloaded from NCBI using PEAKS 7.0 software [113]. Minimum of two high confident peptides were used as a prerequisite to identify the proteins. The matching peptides and proteins showing 10lgP value $\geq$ 15.4 and 20, respectively were only considered for the identification purpose and false discovery rate was kept very stringent (1%). The tryptic peptide sequences of individual proteins were subjected for BLAST search in NCBI database of non-redundant protein sequences using BLASTP algorithm (http:// blast.ncbi.nlm.nih.gov/Blast.cgi) [113].

3.2.2.4 Determination of secondary structure of protein/peptide

The secondary structure of the isolated venom protein/peptide was determined by circular dichroism (CD) analysis (Jasco J715 Spectropolarimeter, Japan). Briefly,

the far UV-CD spectra (190–250 nm) of native venom protein/peptide solution (0.3 mg ml^{-1} in 20 mM potassium phosphate buffer, pH 7.0) in a quartz cuvette with path length of 0.1 cm were recorded at room temperature (~23 °C). The bandwidth was set at 1 nm and response time was 2 s. The final spectrum was cumulative of five scans. The CD spectra were corrected by subtraction of buffer blank and expressed in molar ellipticity [θ] (degrees cm^2 $dmol^{-1}$), using 113 as mean residue molecular mass [287]. Yang's reference was set for the CD analysis. CDPRO CLUSTER software was used to determine the secondary structure of isolated venom protein/peptide [288].

3.2.3 Biochemical characterizations

3.2.3.1 Estimation of protein content

The protein content of crude RVV or chromatography fractions was estimated by measuring the absorbance at 280 nm or by Folin-Lowry method [289] using BSA (1.0 mg ml^{-1}) as a protein standard. The protein content of the unknown samples were calculated from a standard protein calibration curve established by plotting the absorbance at 660 nm vs. concentrations of BSA (10-100 μg ml^{-1}).

3.2.3.2 Casein/plasma protein hydrolysis activity

Digestion of casein/albumin/globulin was evaluated colorimetrically [290] with slight modifications [291]. Briefly, 1.0 ml of 1% (w/v) casein/albumin/globulin dissolved in 20 mM potassium phosphate buffer, pH 7.4 was incubated with specific amount of venom protein/peptide for 90 min at 37 °C. After the stipulated time interval, the reaction was stopped by addition of 0.5 ml of 10% ice-cold TCA. The reaction mixture was centrifuged (5000 rpm, 15 min) and 2.0 ml of 2% (w/v) Na_2CO_3 in 0.1 N NaOH was added to the mixture and incubated for 10 min at room temperature (~23 °C). This was followed by the addition of 0.5 ml of Folin Ciocalteu's reagent (1:2 dilutions). After 30 min, absorbance of the mixture was measured at 660 nm. One unit (U) of proteolytic activity has been defined as *n*

mole equivalent of tyrosine formed min^{-1} which is calculated from a standard tyrosine curve.

3.2.3.3 Fibrinolytic/Fibrinogenolytic activity

3.2.3.3.1 Fibrinolytic/fibrinogenolytic activity assay

The quantitative fibrinolytic assay was done following the protocol described by Mukherjee et al. [292]. Briefly, 40.0 µl of fibrinogen solution (final concentration 6.0 mg ml^{-1}) was clotted with human thrombin at room temperature (~23 °C) for 60 min. To the fibrin clot, specific amount of the venom protein/peptide was added and incubated at 37 °C for 30 min in a final reaction mixture of 90.0 µl adjusted with PBS (1X, pH 7.4). For quantitative fibrinogenolytic assay, instead of fibrin clot, 40.0 µl of fibrinogen solution (final concentration 2.5 mg ml^{-1}) was taken. The reaction was stopped by adding 10.0 µl of 10% ice-cold TCA and the supernatant obtained after centrifuging the reaction mixture at 3000 rpm for 10 min was used to determine the release of free amino acids (tyrosine) at 660 nm using Folin-Ciocalteus reagent. One unit (U) of fibrino(geno)lytic activity has been defined as *n* mole equivalent of tyrosine formed min^{-1}. Human plasmin (2.0 µg) assayed for fibrinolytic/fibrinogenolytic activity under similar conditions served as a positive control.

3.2.3.3.2 SDS-PAGE analysis of fibrin/fibrinogen degradation

To study the time course of fibrin/fibrinogen degradation, if any, a specific amount of the venom protein/peptide or human plasmin (2.0 µg) was incubated with fibrin/fibrinogen (see section 3.2.3.3.1) for different time intervals (0.5 h to 6 h) at 37 °C and the fibrin/fibrinogen degradation products were visualized by 12% SDS-PAGE analysis followed by staining with Coomassie Brilliant Blue R-250 as described in section 3.2.2.1.1. For control, 0.1 ml of PBS (1X, pH 7.4) was added instead of venom protein to the fibrin/fibrinogen solution and incubated under identical experimental conditions.

3.2.3.3.3 RP-HPLC analysis of fibrin(ogen) degradation products

The venom protein or human plasmin meditated degradation products of fibrin and fibrinogen obtained after 4 h of incubation at 37 °C, pH 7.4 were separated on a RP C_{18} (300 Å, 2.1 mm x 150 mm) column coupled to a UHPLC system (Ultimate 3000, Dionex, Germany). After washing the column with solvent A (5% acetonitrile with 0.1% TFA) for 5 min, the bound proteins were eluted with a linear gradient of solvent B (95% acetonitrile with 0.1% TFA) from 0% to 60% of B over 20 min at a flow rate of 1.0 ml min^{-1}. RP-HPLC elution profile of undigested fibrin/ fibrinogen served as a control.

3.2.3.3.4 Determination of kinetic constants for protease activity towards fibrinogen

A specific amount of venom protein (0.04 µmol l^{-1}) was incubated with graded concentrations of fibrinogen (0.6-6.0 µM) for 30 min at 37 °C, pH 7.4. Fibrinogenolytic activity of the isolated venom protein, if any, was assayed as described in section 3.2.3.3.1. The kinetic parameters (*Km* and *Vmax*) of fibrinogenolytic activity were determined by non-linear regression analysis using the GraphPad Prism 6.03 software.

3.2.3.4 Effect on extracellular matrix (ECM) proteins

ECM proteins such as laminin, type-IV collagen and fibronectin were incubated separately with the isolated venom protein in a substrate:enzyme ratio of 15:1 (w/w), in a total volume of 20.0 µl of 20 mM potassium phosphate buffer, pH 7.4 at 37 °C for 4 h. The reaction was stopped immediately after the stipulated time interval by chilling in ice and adding 5.0 µl denaturing buffer containing SDS and β-mercaptoethanol (see section 3.2.2.1.1). The degradation products were analysed on 10% SDS-PAGE after staining with 0.25% Coomassie Brilliant Blue R-250 as described in section 3.2.2.1.1.

3.2.3.5 Plasminogen and thrombin degradation assay

To study the effect of the isolated venom protein on coagulation factors such as plasminogen and thrombin, if any, plasminogen (20.0 µg) or thrombin (40.0 µg) was incubated separately with the venom protein (2.0 µg), in a total volume of 20.0 µl of 20 mM potassium phosphate buffer, pH 7.4 at 37 °C for different time intervals (0.5, 1, 2, 4, 6 and 8 h). The reaction was stopped immediately after the stipulated time interval by chilling in ice and adding 5.0 µl denaturing buffer containing SDS and β-mercaptoethanol (see section 3.2.2.1.1). The degradation products were analysed on 15% SDS-PAGE after staining with 0.25% Coomassie brilliant blue R-250 as described in section 3.2.2.1.1.

3.2.3.6 Amidolytic activity assay

Amidolytic assay was performed following the protocol described by Mukherjee et al. [292]. Briefly, the amidolytic activity of the isolated venom protein (0.04 µmole l^{-1} Rusviprotease or 1.0 µg ml^{-1} GF-VI DEAE-II) against various chromogenic substrates (0.2 mM) was measured at 37 °C in a final volume of 0.1 ml (50 mM Tris–HCl, pH 7.4). After incubation for 20 min at 37 °C, the amount of pNA released was determined by measuring the change in absorbance at 405 nm against reagent blank in a microplate reader (MultiSkan GO, Thermo Scientific, USA). One unit of amidolytic activity has been expressed as millimoles of p-nitroaniline released min^{-1} ml^{-1} of the enzyme. The chromogenic substrates used were B7632 (substrate for trypsin, thrombin and reptilase), B2133 (substrate for *Agkistrodon contortrix* thrombin like enzyme and plasma kallekrein), B3133 (substrate for trypsin and other proteolytic enzymes), V0882 (substrate for plasmin), V6258 (substrate for glandular kallekrein), I0898 (substrate for tissue plasminogen activator) and F3301 (substrate for factor Xa).

3.2.3.7 PLA_2 activity

PLA_2 activity of isolated venom protein/peptide was assayed by the turbidometric method of Joubert and Taljaard [293] with some modifications as described by Mukherjee [291]. One egg yolk was suspended in 250 ml of 0.9% (w/v) NaCl containing 0.02% (w/v) NaN_3. The egg yolk suspension was stored in 4 °C and used within 3 days of preparation. Before experiment, 1.0 ml of suspension was mixed with 10.0 ml of 0.1 M Tris-HCl, pH 8.0 and the absorbance of the resulting mixture was adjusted to 1.0 at 740 nm with the same buffer. For the PLA_2 assay, specific amount of the venom protein/peptide was mixed with 2.0 ml of the reaction mixture and the decrease in turbidity after 10 min was observed at 740 nm against reagent blank. One unit of PLA_2 activity has been defined as decrease in 0.01 absorbance in 10 min at 740 nm, under the assay conditions [287, 291].

3.2.3.8 Estimation of total carbohydrate content

Total carbohydrate was quantified by phenol-sulphuric acid method as described by Dubois et al. using D-glucose as a standard [294]. Briefly, 25.0 μg of the venom protein/peptide was mixed with 1.0 ml of phenol (5%, w/v). To the mixture, 5 ml of H_2SO_4 (96%, v/v) was added and kept at room temperature (~23 °C) for 30 min. Thereafter, the absorbance of the reaction mixture was measured at 490 nm against a reagent blank. The carbohydrate content of the unknown samples was calculated from a standard curve obtained by plotting absorbance Vs concentration of D-glucose (0.1 mg ml^{-1} in water).

3.2.3.9 Fibrinogen clotting time

Fibrinogen clotting activity was determined by the method described by Mukherjee with the following modifications [65]. Specific amount of the venom protein/peptide was added to 0.2 ml of 2.5 mg ml^{-1} bovine plasma fibrinogen solution (in 20 mM potassium phosphate buffer, pH 7.4). Reaction was incubated at 37 °C for 1.0 h and the formation of a fibrin clot, if any, was monitored by

visual inspection. The fibrin clot produced by thrombin (1.0 NIH U ml^{-1}, in 20 mM potassium phosphate buffer, pH 7.4) under identical experimental conditions served as positive control.

3.2.3.10 Serine protease inhibition

The isolated venom protein/peptide (2.4 µM) was incubated with either thrombin (0.16 µM), factor Xa (0.21 µM), trypsin (0.24 µM) or plasmin (0.12 µM) for 30 min at 37 °C. To the mixture, appropriate chromogenic substrate (T1637 for thrombin, F3301 for factor Xa, B3133 for trypsin or V0882 for plasmin) was added in a final concentration of 0.2 mM in a total reaction mixture of 100 µl adjusted with PBS (1X, pH 7.4). The change in absorbance as a result of release of pNA was measured at 405 nm after 10 min in a microplate reader (MultiSkan GO, Thermo Scientific, USA). The activity of the proteases towards their respective substrates in the absence of the venom protein/peptide was considered as 100% and considered for comparison.

3.2.3.11 Thrombin inhibition studies

3.2.3.11.1 Inhibition of thrombin amidolytic activity

Thrombin (3.0 µl, 10 NIH U ml^{-1}) was incubated with different concentrations of venom peptide (Ruviprase) (0-2.4 µM) for 30 min at 37 °C. After the stipulated time interval, 0.2 mM chromogenic substrate for thrombin (T1637) was added (0.2 mM) in a final reaction mixture of 100 µl adjusted with PBS (1X, pH 7.4). Substrate hydrolysis was measured by monitoring the change in absorbance at 405 nm for 20 min at 20 s intervals in a microplate reader (MultiSkan GO, Thermo Scientific, USA). Residual activity of thrombin due to the RVV peptide was calculated by considering the activity of thrombin (absorbance after 20 min of substrate hydrolysis) in the absence of inhibitor as 100%. Dose response curve for thrombin inhibition was fitted using GraphPad Prism 6.03 to calculate the Hill coefficient and IC_{50} value.

3.2.3.11.2 Determination of Inhibitory Constant (*Ki*) for thrombin inhibition

Thrombin (3.0 µl, 10 NIH U ml^{-1}) was pre-incubated with two different concentrations of the isolated venom peptide, Ruviprase (0.6 µM and 1.2 µM) at 37 °C for 30 min. Various concentrations (0.1-0.8 mM) of chromogenic substrate for thrombin (T1637) was added to the mixture in a final reaction volume of 100 µl adjusted with phosphate buffer saline (1X, pH 7.4). The release of pNA at 405 nM was determined after 10 min incubation at 37 °C in a microplate reader (MultiSkan GO, Thermo Scientific, USA). For kinetic analysis, the reaction rate (V) was plotted against substrate concentration (S) at each inhibitor concentration and the data was fit to a hyperbolic Michaelis-Menten model using GraphPad Prism 6.03 software.

The inhibitory constant (*Ki*) was determined using the uncompetitive model for enzyme inhibition using the same software. The inhibitory constant was determined by the software using the model given below,

$$VApp = \frac{Vmax}{1+\frac{I}{\alpha Ki}} \quad \text{-- [equation 1]}$$

$$KmApp = \frac{Km}{1+\frac{I}{\alpha Ki}} \quad \text{-- [equation 2]}$$

$$V = \frac{VApp \times S}{KmApp+S} \quad \text{-- [equation 3]}$$

where, αKi is the inhibition constant, expressed in the same units as I (inhibitor), *VApp* and *Vmax* denote the maximum velocity in the presence and absence of the inhibitor, and *KmApp* and *Km* denotes the Michaelis constants in presence and absence of inhibitor, respectively.

3.2.3.11.3 Effect on fibrinogen clotting activity of thrombin

The effect of venom peptide on the fibrinogen clotting activity of thrombin was performed following the protocol described by Ciprandi et al. [295]. Briefly, Thrombin (3.0 µl, 10 NIH U ml^{-1}) was incubated with different concentrations of the isolated venom peptide, Ruviprase (0-4.8 µM) for 30 min at 37 °C. To this mixture, 50.0 µg of fibrinogen was added in a final reaction volume of 100 µl adjusted with PBS (1X, pH 7.4). Fibrin clot formation was inspected by reading the absorbance at 605 nm at 30 s interval for 60 min in a microplate reader (MultiSkan GO, Thermo Scientific, USA). An increase in absorbance at 650 nm indicates fibrin clot formation by thrombin. Residual activity of thrombin was calculated by considering the absorbance after 60 min of fibrinogen clotting in the absence of Ruviprase as 100%. Dose response curves for inhibition of thrombin-induced fibrinogen clotting were fitted using GraphPad Prism 6.03 to calculate the Hill coefficient and IC_{50} value.

3.2.3.11.4 Effect on thrombin-induced platelet aggregation

The effect of isolated venom peptide (Ruviprase) on thrombin-induced platelet aggregation was evaluated using goat platelet following the protocol described by Horn et al. [296]. First, platelet rich plasma (PRP) was isolated from goat blood by centrifuging fresh blood at 250 x g for 10 min at 4 °C. The supernatant containing the PRP was carefully extracted and platelets were isolated. Briefly, PRP was centrifuged at 650 x g for 10.0 min and the pellet was washed twice in a tyrode buffer (5.0 mM Hepes, 13.0 mM NaCl, 2.7 mM KCl, 12.0 mM $NaHCO_3$, 0.42 mM Na_2HPO_4, 1.0 mM $MgCl_2$, 0.1% glucose and 0.25% BSA). The pellet was resuspended in the same buffer and the volume was adjusted with tyrode buffer to give an absorbance of 0.15 at 650 nm. Thrombin (0.2 µg, final concentration, 0.2 NIH U ml^{-1}) was incubated for 30 min at 37 °C with two different concentrations of Ruviprase (0.6 µM and 1.2 µM) in a final volume of 50.0 µl with tyrode buffer.

To the mixture, 100 µl of platelet solution was added and the decrease in absorbance was monitored for 20 min at 650 nm [296].

3.2.3.11.5 Effect on thrombin inhibition by AT-III in presence or absence of heparin

The effect of the isolated venom peptide, Ruviprase on thrombin inhibition by AT-III in presence or absence of heparin was performed as per the protocol described by Arocas et al. [297]. Briefly, thrombin equivalent to 0.16 µM (3 µl, 10 NIH U ml^{-1}) was pre-incubated with different concentrations of the venom peptide (0.6 µM, 1.2 µM and 2.4 µM) for 30 min at 37 °C in a final reaction mixture of 100 µl adjusted with PBS (1X, pH 7.4). Heparin free AT-III (100 nM) or 25 nM AT-III in the presence of 0.5 U ml^{-1} heparin, was added to the samples and the residual activity of thrombin was measured by T1637 hydrolysis after 10 min at 405 nm [297].

3.2.3.12 Factor Xa inhibition studies

3.2.3.12.1 Effect on the prothrombin activation property of factor Xa

To understand the effect of interaction of the isolated venom peptide (Ruviprase) with factor Xa, we studied the effect of the venom peptide on the physiological function (i.e. prothrombin activation) of factor Xa. Briefly, factor Xa (0.02 µM) was incubated with the venom peptide (0.6 µM) for 30 min at 37 °C. To the mixture prothrombin (1.0 µM) was added followed by the addition of 2.0 µl chromogenic substrate for thrombin (10 mM, T1637). The change in absorbance as a result of release of pNA was measured at 405 nm continuously for 15 min in a microplate reader (MultiSkan GO, Thermo Scientific, USA). Prothrombin incubated with PBS (1X, pH 7.4) and factor Xa (0.02 µM) under identical assay conditions served as negative and positive controls, respectively.

3.2.3.12.2 Determination of Inhibitory Constant (*Ki*) for factor Xa inhibition

Factor Xa (0.02 μM) was pre-incubated with two different concentrations of the isolated venom peptide, Ruviprase (0.6 μM and 1.2 μM) at 37 °C for 30 min. Various concentrations (1.0-5.0 μM) of prothrombin was added to the above mixture. The final reaction volume was adjusted to 100 μl with PBS (1X, pH 7.4) and the reaction was allowed to continue for 60 min at 37 °C. After the stipulated time interval, 2.0 μl of chromogenic substrate for thrombin (10 mM, T1637) was added to the mixture and the release of pNA was monitored at 405 nm after 10 min incubation at 37 °C. For kinetic analysis, the reaction rate (V) was expressed as nanomoles of thrombin generated min^{-1} μM^{-1} factor Xa and plotted against prothrombin concentration (S) at each inhibitor concentration and the data was fitted to a hyperbolic Michaelis-Menten model using GraphPad Prism 6.03 software. The inhibitory constant (*Ki*) was determined using the mixed-model for enzyme inhibition using the above software. The inhibitory constant was determined by the software using the model given below,

$$VApp = \frac{Vmax}{(1+I/(\alpha \times Ki))} \quad \text{-- [equation 4]}$$

$$KmApp = \frac{Kmax\,(1+I/Ki)}{(1+I/(\alpha \times Ki)} \quad \text{-- [equation 5]}$$

$$V = \frac{VApp \times S}{KmApp+S} \quad \text{-- [equation 6]}$$

where,

I denote the inhibitor concentration, *VApp* and *Vmax* denote the maximum velocity in the presence and absence of the inhibitor, and *KmApp* and *Km* denotes the Michaelis constants in presence and absence of inhibitor, respectively. Value of the constant α was set as 3.92.

3.2.3.13 Effect of pre-incubation on the serine protease inhibitor activities

To determine whether the isolated peptide (Ruviprase) showing inhibitory properties binds thrombin/ factor Xa with slow or fast kinetics, Ruviprase (0.6 μM) was pre-incubated with thrombin (3 μl, 10 NIH U ml^{-1})/ factor Xa (0.02 μM) for different time intervals (0-30 min) at 37 °C. Thereafter, the amidolytic activity of thrombin and prothrombin activation property of factor Xa was analyzed in the corresponding assay systems, as described in sections 3.2.3.11.1 and 3.2.3.14.1 respectively. Similarly, to study the effect of pre-incubation on the plasma recalcification time, Ruviprase (0.6 μM) was pre-incubated with pre-warmed PPP for different time intervals (0-30 min) at 37 °C and the clotting time was measured as described in section 3.2.4.1.

3.2.3.14 Prothrombin activation studies

3.2.3.14.1 Prothrombin activation assay

Prothrombin activation assay was performed according to Berger et al. with few modifications [257]. The concentrations of prothrombin and co-factors of prothrombinase complex were chosen to approximate physiological conditions [60, 298]. Briefly, the isolated procoagulant venom protease (0.04 μmol l^{-1}) or factor Xa (0.02 μmol l^{-1}) in 20 mM Tris-HCl buffer, pH 7.4 was incubated with 1.4 μmole l^{-1} of prothrombin either in the presence or absence of 2.5 mM Ca^{2+} or prothombinase complex [100 μM phospholipid vesicles (9:1 phosphatidylcholine (PC): phosphatidylserine (PS) and 3 nM factor Va] at 37 °C for 15 min. Thereafter, 0.2 mM chromogenic substrate for thrombin (T1637) was added and kinetics of substrate hydrolysis was monitored by measuring the initial rate of pNA released at 405 nm in a microplate reader (MultiSkan GO, Thermo Scientific, USA). The amount of thrombin generated was calculated from a calibration curve of standard thrombin. Prothrombin activation was expressed as nanomoles of thrombin generated min^{-1} μmol^{-1} of enzyme.

In a separate set of experiment, the fibrinogen clotting time of the thrombin (generated via prothrombin activation by the isolated procoagulant venom protease) was also determined. Briefly, 1.4 μmol l^{-1} of prothrombin was incubated with the venom protein (0.04 μmol l^{-1}) for 15 min at 37 °C either in presence or absence of Ca^{2+} (2.5 mM) followed by addition of 100 μl of fibrinogen solution (2.5 mg ml^{-1} in 1X PBS, pH 7.4). The time taken for fibrin clot formation was measured with a coagulometer (Coastat-1, Coagulation analyzer, Tulip Diagnostics, India). Prothrombin (1.4 μmol l^{-1}) incubated with 100 μl of fibrinogen solution (2.5 mg ml^{-1}) under identical conditions served as negative control and time of clot formation, if any, was recorded.

3.2.3.14.2 SDS-PAGE analysis of prothrombin activation

The kinetics of prothrombin activation by isolated procoagulant venom protease was also analysed by 12% SDS-PAGE under both non-reducing and reducing conditions. Briefly, the procoagulant venom protease (0.4 μmol l^{-1}) was incubated with prothrombin (14.0 μmol l^{-1}) for different time intervals (0.25 to 8 h) at 37 °C. The prothrombin activation products were separated on 12% SDS-PAGE and were visualized by staining the gels with Coomassie Brilliant Blue (0.25%) and destained with water, methanol and acetic acid (5:4:1, v/v) as described in section 3.2.2.1.1. To quantify the amount of thrombin formed, the band intensity of the generated thrombin separated on SDS-PAGE under non-reducing conditions was measured using ImageJ software.

3.2.3.14.3 PMF analysis of prothrombin activation products

The prothrombin cleavage products (corresponding to 36 kDa and 55 kDa bands in the gel) obtained after 6 h of enzyme treatment were subjected to in-gel tryptic digestion followed by PMF analysis (section 3.2.2.3). The PMF profiles were used to search for peptide matches from NCBI database following the MASCOT search. The tryptic sequences were subjected for BLAST search in NCBI database of non-redundant protein sequences using BLASTP algorithm.

3.2.3.14.4 Determination of kinetic parameters for prothrombin activation

The isolated procoagulant venom protease (0.04 μmol l^{-1}) which demonstrated prothrombin activation property was incubated with varying concentrations of prothrombin (0 to 10.0 μmol l^{-1}) for 15 min at 37 °C, in a final reaction mixture of 100 μl adjusted with 20 mM Tris-HCl, pH 7.4. Thereafter, 0.2 mM T1637 (substrate for thrombin) was added and thrombin generation was measured as described in section 3.2.3.14.1. The kinetic parameters (*Km* and *Vmax*) of prothrombin activation were determined by non-linear regression analysis using the GraphPad Prism 6.03 software.

3.2.3.15 Effect on factor Xa-prothrombinase complex

The kinetics of prothrombin activation by factor Xa-prothombinase complex in presence/absence of the procoagulant venom protease was also evaluated. Briefly, 0.04 μmol l^{-1} of the procoagulant venom protease was pre-incubated with 0.02 μmol l^{-1} factor Xa in presence of components of the prothrombinase complex [2.5 mM $CaCl_2$, 100 μM phospholipid vesicles (9:1 PC: PS) and 3 nM factor Va] for 30 min, at 37 °C. Thereafter, different concentrations of prothrombin (0 to 10.0 μM) were added to the mixture and thrombin generation after 15 min was measured using chromogenic substrate for thrombin as already described (section 3.2.3.14.1). The kinetic parameters *Km* and *Vmax* were obtained by non-linear regression analysis using GraphPad Prism 6.03. In another set of experiments, 0.04 μmol l^{-1} of the procoagulant venom protease was pre-incubated with 0.02 μmole l^{-1} factor Xa in the presence of prothrombinase complex [2.5 mM $CaCl_2$, 100 μM phospholipid vesicles (9:1 PC: PS) and 3 nM factor Va] at 37 °C for 30 min. Thereafter, 14.0 μM prothrombin was added to the mixture and incubated for 15 min at 37 °C and separated by 12% SDS-PAGE and stained with 0.25% Coomassie Brilliant Blue R-250 as described in section 3.2.2.1.1. The band intensity of the prothrombin activation products after separation by 12% SDS-PAGE under reducing conditions was quantified using ImageJ software.

3.2.3.16 Protein-protein interaction studies

3.2.3.16.1 Spectrofluorometric analysis

Interaction of venom protein/peptide with coagulation factors such as thrombin, prothrombin and factor Xa was measured by intrinsic fluorescence analysis using a fluorescence spectrometer (LS55, Perkin Elmer) following the protocol described by Saikia et al. [96]. Briefly, the coagulation proteins were mixed with the venom protein/peptide in a specific ratio either in the absence or presence of Ca^{2+} (5.0 mM) and the change in fluorescence intensity at an excitation of 280 nm was measured. The excitation and emission slits were set as 5 nm and emission spectra were recorded from 310-400 nm at room temperature (~23 °C). All the binding experiments were done in triplicates to ensure reproducibility. The fluorescence intensity of the coagulation protein or venom protein/peptide alone recorded under identical experimental conditions served as control.

3.2.3.16.2 Surface plasmon resonance (SPR) analysis of thrombin binding

Binding affinity was measured using Biacore X100 (GE Healthcare). Briefly, thrombin (0.4 mg ml^{-1}) dissolved in 10 mM sodium acetate buffer, pH 5.0 was immobilized on CM5 sensor chip using amine coupling following target level procedure for thrombin immobilization with a set target of 5000 RU [299]. The isolated venom protein/peptide under study (flow rate of 10 µl min^{-1}) was passed over thrombin bound chip in 20 mM potassium phosphate buffer, pH 7.4, at 25 °C. Each cycle consisted of (1) a 60 s buffer (20 mM potassium phosphate buffer pH 7,4) injection, (2) a 500 s injection where venom protein/peptide was present at 0.1, 0.2, 0.5, 1.0, 1.5, 2.0, 3.0 µM concentration in 20 mM potassium phosphate buffer pH 7.4. 0.25% Tween 20 was used in the buffer to reduce non-specific binding of the analyte (venom peptide) to the ligand (thrombin) (3) an undisturbed dissociation phase of 300 s, (4) a wash of the flow system (regeneration) using glycine pH 2.5.

The dissociation constant (K_D) from direct binding studies of the venom protein/peptide with thrombin was obtained by plotting the response in RU at steady state, Req, versus the concentration of compound, C, and by fitting these to the following equation

$$R_{eq} = C * R_{max}/(C + K_D) \quad \text{-- [equation 7]}$$

Where, R_{max} is the value in RU at complex saturation, and R_{eq} is the observed optimal change in RU at each given C.

3.2.3.17 Neutralization studies

3.2.3.17.1 Effect of PAV and MAV

In order to study the neutralizing ability of PAV/MAV on the biological functions of the isolated venom proteins/peptides, specific amount of the venom protein/peptide was pre-incubated with different amounts of antivenom (PAV or MAV) in an antigen: antivenom ratio of either 1:1 to 1:100 (w/w) for 30 min at room temperature (~23 °C) prior to the assessment of their respective activities such as fibrinolytic activity (section 3.2.3.3.1), thrombin/ factor Xa inhibiting activity (sections 3.2.3.13.1 and 3.2.3.14.1), prothrombin activating property (section 3.2.3.14.1) and their plasma-recalcification time (section 3.2.4.1) in the corresponding assay systems.

3.2.3.17.2 Effect of physiological inhibitors α_2-macroglobulin and α_2-antiplasmin

The neutralizing effect of α_2-macroglobulin or α_2-antiplasmin on the biological functions of the isolated venom proteins/peptides was assayed by pre-incubating specific amount of venom protein/peptide with 3.0 μM α_2-macroglobulin or α_2-antiplasmin for 30 min at 37 °C prior to the assessment of their respective functions such as fibrinolytic activity (section 3.2.3.3.1), thrombin/ factor Xa inhibition (sections 3.2.3.11.1 and 3.2.3.12.1), prothrombin activation (section

3.2.3.14.1) and plasma recalcification time (section 3.2.4.1) in the corresponding assay systems. The activity of venom protein/peptide in the absence of inhibitor was considered as 100% activity and all other values were compared with that.

3.2.3.18 Other biochemical characterizations

3.2.3.18.1 Chemical modification studies

The active sites of the isolated venom protein which demonstrated active proteolytic activity were chemically modified by the method of Mukherjee [65]. Different inhibitors viz. PMSF (serine protease inhibitor), EDTA (metal chelator), TPCK (chymotrypsin inhibitor), TLCK (trypsin inhibitor), DTT (reducing agent for disulphide bridge), IAA (cysteine modifier) and pBPB (histidine modifier) were tested for chemical modification of the active site of the isolated protein/peptide. Briefly, specific amount of the isolated venom protein (0.04 μmol l^{-1} Rusviprotease or 2.0 μg GF-VI DEAE-II) was incubated with different inhibitors (at 2.0 mM final concentration except EDTA which was tested at 5.0 mM final concentration) at room temperature (~23 °C) for 30 min. Thereafter, the respective enzymatic peoperties viz. fibrinolytic/fibrinogenolytic activity, prothrombin activating property and plasma recalcification time of the venom protein were analysed in their respective assay systems as described in sections 3.2.3.3.1, 3.2.3.14.1 and 3.2.4.1 respectively. The activity of venom protein in the absence of inhibitor was considered as 100% activity and all other values were compared with that.

3.2.3.18.2 Determination of optimum temperature

To study the effect of temperature on the catalytic properties of the isolated venom protein, 0.04 μmol l^{-1} Rusviprotease or 2.0 μg GF-VI DEAE-II was incubated with fibrinogen/fibrin for 30 min at different temperatures (25-75 °C). Thereafter, fibrinogenolytic/fibrinolytic activity of the venom proteins were analysed as described in section 3.2.3.3.1.

3.2.3.18.3 Determination of optimum pH

To determine the optimum pH for the functioning of the isolated venom protein, specific amount of the isolated venom protein (0.04 μmol l^{-1} Rusviprotease or 2.0 μg GF-VI DEAE-II) was incubated with fibrinogen/fibrin for 30 min at a 37 °C and different pH ranging from pH 5.0-9.0. The different buffers used for this study were 20 mM sodium acetate buffer, pH 5.0–6.5; 20 mM potassium phosphate buffer, pH 7.0-7.5 and 20 mM Tris-HCl buffer, pH 8.0-9.5. The fibrinogenolytic/fibrinolytic activity was analysed as described in section 3.2.3.3.1.

3.2.4 Pharmacological characterizations

3.2.4.1 Plasma recalcification time

The plasma clotting activity of the isolated proteins/peptides was measured by the method of Angulo et al. with some modifications as described by Mukherjee, et al [291, 300]. Briefly, fresh goat blood was collected in 3.8% trisodium citrate (9:1 ratio) and platelet-poor plasma (PPP) was prepared by centrifuging the citrated blood twice at 4300 rpm for 15 min at 4 °C. The PPP was used within 4 h of collection. For the assay of plasma recalcification time, specific amount of the isolated venom protein/peptide (in a final volume of 30.0 μl) was added to 300.0 μl of pre-warmed PPP at 37 °C and the mixture was incubated for 3.0 min at 37 °C. To the mixture, 40.0 μl of 250 mM $CaCl_2$ was added and the time taken for the appearance of first visible fibrin thread/clot formation was recorded with the help of a stop watch. As a control, plasma aliquots were incubated with 30.0 μl of 20 mM potassium phosphate buffer, pH 7.4, and the coagulation time was determined identically. One unit of procoagulant/anticoagulant activity of the isolated protein/peptide has been defined as 1.0 s decrease/increase in the clotting time of PPP as compared to the clotting time of normal PPP (control) under identical assay conditions [65, 291].

3.2.4.2 Prothrombin time test

The prothrombin time (PT) of goat PPP was tested by Quick's one-stage procedure following the protocol described by Mukherjee, using a commercial Liquiplastin kit [65]. Briefly, 0.1 ml of PPP was taken in a test-tube and placed in a waterbath for 3 to 5 min at 37 °C. Then specific amount of isolated venom protein/peptide was added to the PPP and the mixture was incubated in a water bath for 3 min at 37 °C. To the tube, 0.2 ml of liquiplastin reagent (pre-warmed at 37 °C for 3 min) was added and the tube was gently shaken to mix the contents. The first appearance of visible fibrin thread/clot formation was recorded with the help of a stop watch. As a control, plasma aliquots were incubated with 30.0 µl of 20 mM potassium phosphate buffer, pH 7.4, and the coagulation time was determined identically. One unit of procoagulant/anticoagulant activity has been defined as 1.0 s decrease/increase in the clotting time of PPP as compared to clotting time of normal PPP (control) under identical assay conditions.

3.2.4.3 Activated partial thromboplastin time test

The activated partial thromboplastin time (aPTT) of goat PPP was tested by following the principle described by Biggs and the protocol described by the manufacturer of commercial Liquicelin-E kit [301]. Briefly, 0.1 ml of PPP was taken in a test-tube and placed in a waterbath for 3 to 5 min at 37 °C. Then specific amount of isolated venom protein/peptide was added to the PPP and the mixture was incubated in a water bath for 3 min at 37 °C. To the tube, 0.1 ml of liquicelin reagent (pre-warmed at 37 °C for 3 min) was added and the mixture was again incubated in a water bath for 3 min at 37 °C. Following incubation period, 100 µl of pre-warmed $CaCl_2$ (25 mM) added to the plasma-Liquicelin mixture. The first appearance of visible fibrin thread/clot formation was recorded with the help of a stop watch. As a control, plasma aliquots were incubated with 30.0 µl of 20 mM potassium phosphate buffer, pH 7.4, and the coagulation time was determined identically. One unit of procoagulant/anticoagulant activity has been defined as 1.0

s decrease/increase in the clotting time of PPP as compared to clotting time of normal PPP (control) under identical assay conditions.

3.2.4.4 Platelet aggregation assay

The effect of isolated procoagulant protease (Rusviprotease) or anticoagulant fraction (GF-VI DEAE-II) on platelet aggregation was evaluated using goat platelet rich plasma (PRP) and/or fresh platelet suspension following the protocol described by Bednar et al. [302]. Platelet suspension was prepared as described in section 3.2.3.11.4. Briefly, the agonist, i.e. Rusviprotease (0.04-0.4 μmol l^{-1}) or GF-VI DEAE-II (1.0- 3.0 μg) was added to 100.0 μl of PRP or platelet suspension and the decrease in absorbance was continuously monitored for 300 s at 540 nm. Percent platelet aggregation after 300 s of incubation of platelets, with agonists at room temperature (~23 °C) was calculated by the following formula:

Percent aggregation= (A_{540} of PRP/platelet suspension before the addition of agonist – A_{540} of PRP/ platelet suspension after the addition of agonist) ÷ (A_{540} of PRP/ platelet suspension before the addition of agonist – A_{540} of PPP/tyrode buffer) x 100. The A_{540} denotes absorbance value at 540 nm [113].

3.2.4.5 Hemolytic activity

For the assay of hemolytic activity, 5% (v/v) washed erythrocyte suspension in 100 mM PBS, pH 7.4 was prepared from citrated goat blood by centrifuging at 4300 rpm for 15 min at 4 °C. For assay of direct hemolytic activity, specific amount of venom protein/peptide were incubated with the 5% (v/v) erythrocyte suspension (in a final volume of 3.0 ml) for 90 min at 37 °C. After the stipulated time interval, the reaction was stopped by chilling in ice. The reaction mixture was then centrifuged at 4300 rpm for 15 min at 4 °C and the released hemoglobin was measured spectrophotometrically at 540 nm [65].

For indirect hemolysis assay, 10.0 µl of egg yolk suspension was added to the reaction mixture prior to incubation. Total hemolysis (100%) was achieved by adding deionized water instead of venom protein to the erythrocytes suspension. The results were expressed as the percent of erythrocytes lysed by the isolated protein/peptide as compared to erythrocytes lysed by water (100% hemolysis) [96].

3.2.4.6 *In vitro* clot lysing activity

For the measurement of *in vitro* clot lysis, 1.0 ml citrated blood was taken in microfuge tube to which 100.0 µl $CaCl_2$ (250 mM) was added. The clot was allowed to form for 1 h at 37 °C. The dry weight of the clot was measured prior to addition of the protein. Then 1.0 µg of the isolated anticoagulant venom protease was added to the clot and incubated a 37 °C for 60 min. The supernatant was separated and the dry weight of the clot was measured. Clots incubated with buffer (20 mM potassium phosphate buffer, pH 7.4) served as negative control. Similarly, for measurement of amount of lysis of heat denatured clot, the blood clots were heated at 80 °C for 30 min prior to addition of the isolated anticoagulant venom protease. Clot lysing property of human plasmin (1.0 µg) was assayed under identical conditions to serve as positive control.

3.2.4.7 Cytotoxicity assay on tumour cells

3.2.4.7.1 MTT-based assay

The *in vitro* cytotoxicity of the isolated RVV protein/peptide was assessed on human glioblastoma-astrocytoma (U87MG), human colon adenocarcinoma (HT-29) and/or cervical cancer (HeLa) cell lines. The cells were cultured in Dulbecco's Modified Eagle Medium (DMEM) supplemented with 10% fetal bovine serum, penicillin (100 U ml^{-1}) and streptomycin (100 U ml^{-1}). Briefly, 1 x 10^5 cells were seeded in 96-well tissue culture plate and allowed to adhere by growing at 37 °C and 5% CO_2. After 24 h of seeding, the culture medium (DMEM) was replaced

with fresh medium containing different concentrations of the isolated protein/peptide (1.0-10.0 μg ml^{-1}). For control, cells were treated with 20 mM potassium phosphate buffer, pH 7.0. After 48 h of treatment, 100 μl of 1.0 mg ml^{-1} MTT was added to the cells and incubated for 3 h at 37 °C. The cells were then treated with 100 μl DMSO and incubated in dark under shaking conditions. The background absorbance of multi-well plates was measured at 690 nm and subtracted from the 550 nm measurement. The percent cell viability was calculated as the ratio of treated cells to the control cells [96].

3.2.4.7.2 Dual staining method

U87MG cells (1 x 10^5 cells ml^{-1}) were cultured as described in section 3.2.4.7.1. After 24 h of seeding, the culture medium (DMEM) was replaced with fresh medium containing different concentrations of crude RVV or isolated enzyme/protein (1.0-10.0 μg ml^{-1}). For control, cells were treated with 20 mM potassium phosphate buffer, pH 7.0. After 24 h of treatment, the culture media was carefully removed and the treated cells were then stained with 100 μl of AO/EB mix (10 mg mL^{-1} AO and EB in PBS). After 10 min of incubation, cells were washed with PBS (1X, pH 7.4) before visualizing under a fluorescence microscope (Nikon ECLIPSE, TS100, Tokyo) with an excitation filter of 480/30 nm.

3.2.4.7.3 Sulforhodamine B (SRB) assay for assessment of anti-proliferative activity against cancer cells

Human breast cancer (MCF-7) cells, EMT6/AR1 (multi drug resistant mouse mammary carcinoma) cells, and HeLa (cervical carcinoma) cells were cultured in Eagle's minimal essential medium (EMEM), supplemented with 10% fetal bovine serum, 2.2 g L^{-1} of sodium bicarbonate, 0.28 U mL^{-1} of insulin and 1% antibiotic antimycotic solution containing streptomycin, amphotericin B, and penicillin. Cells were maintained in a CO_2 incubator (Sanyo, San Diego, CA) at 37 °C in a humidified atmosphere having 5% CO_2 and 95% air. After 24 h of seeding of 1 × 10^5 cells ml^{-1} in a 96 well plate cells, the old medium was replaced with fresh

medium containing graded concentrations of isolated peptide (0-40 ng μl^{-1}) and cells were incubated further for 24 h in the above condition. A control was run in parallel where instead of purified peptide, cell culture medium was added. Thereafter, Sulforhodamine B (SRB) assay was performed following the procedure described by Vichai et al. using SRB assay kit [303]. The peptide-induced anti-proliferative activity (percent of survival cells) was determined from a standard curve of control cells.

3.2.4.7.4 Detection of apoptosis in MCF-7 cells

3.2.4.7.4.1 Annexin V/ propedium iodide (PI) staining

MCF-7 cells (0.5 x 10^5 cells ml^{-1}) were grown in 24 well cell culture plates. After 24 h of incubation at 37 °C/ 5% CO_2, old media were replaced with the fresh media containing the venom protein/peptide at its 2 IC_{50} value (8.0 ng ml^{-1}) and further incubated for 24 h in the above conditions. The trypsinized cells were collected by centrifugation at 3000 rpm for 10 min at 30 °C and washed twice with phosphate buffer saline. Annexin V/ PI staining was performed by using apoptosis kit and following the instructions of the manufacturer (Santa Cruz Biotechnology, Santa Cruz, CA). The coverslips were observed in a square cover glass dish under an Eclipse TE 2000U microscope (Nikon, Tokyo, Japan) at 40X magnification. The images were analyzed using Image-Pro Plus software (Media Cybernetics, Silver Spring, MD).

3.2.4.7.4.2 Flow cytometric analysis of apoptosis

Apoptotic populations were further quantified using flow cytometry (FACS Aria Beckton Dickenson). Briefly, 0.5 x 10^5 MCF-7 cells ml^{-1} were treated with the isolated peptide at a dose of 2 IC_{50} value (8.0 ng ml^{-1}) for 24 h at 37 °C/ 5% CO_2 and processed for apoptosis analysis using apoptosis kit protocol. The fluorescence intensity was determined by a flow cytometer with an excitation at 488 nm and

emission at 525 nm. Percentage of live, early apoptotic, late apoptotic and dead cells were analyzed using FACSDiva software.

3.2.4.7.4.3 Immunofluorescence microscopic analysis of induction of apoptosis

MCF-7 cells were grown on polylysine coated glass coverslips in 24 well plate with the density of 0.5 x 10^5 cells ml^{-1} per well for 24 h. Then the cells were incubated with the isolated protein/peptide at IC_{50} and 2 IC_{50} values (4.0 ng μl^{-1} and 8.0 ng μl^{-1}, respectively) for one cell cycle. For immune-staining trypsinized cells were collected by centrifugation, washed and fixed with 3.7% para-formaldehyde solution for 20 min. Cells were washed twice with PBS followed by membrane permeabilization with cold methanol at -35 °C for 30 min. Washing step was repeated to remove the methanol. To prevent the non specific interaction of antibodies, the cells were incubated with 2% w/v of BSA/PBS for 30 min at 37 °C. Subsequently cells were incubated with mouse monoclonal p53 antibody (1:300 dilutions in 2% BSA) or mouse monoclonal anti-p21 antibody (1:300 dilutions in 2% BSA) for 2 h at room temperature (~23 °C) in dark. The cells were then washed twice with PBS and incubated with Alexa 568-labeled anti-mouse IgG antibody (1:300 dilutions in 2% BSA) at room temperature (~23 °C) for 2 h. The coverslips were rinsed with PBS.

For DNA staining, the Ruviprase treated MCF-7 cells (as described in section 3.2.4.7.4.3) were fixed in 1% formaldehyde (in 1X PBS, pH 7.4) for 30 min at room temperature (~23 °C) and suspended in 100 µl EMEM. Thereafter, the cells were incubated with 5.0 µl Hoechst 33258 (10 mg ml^{-1}) for 30 min in the dark at room temperature (~23 °C). The coverslips were then washed twice with PBS and mounted in 80% glycerol in PBS containing 8 mg ml^{-1} of 1,4-diazabicyclo[2.2.2]octane. Sealed and dried coverslips were then observed using an Eclipse TE 2000U microscope (Nikon, Tokyo, Japan) at 40X magnification. The images were analyzed using Image-Pro Plus software (Media Cybernetics, Silver Spring, MD).

3.2.4.7.4.4 Detection of mitochondrial membrane potential and reactive oxygen species (ROS) activities

MCF-7 cells (0.5 x 10^5 cells ml^{-1}) were grown in 24 well cell culture plates on glass coverslips. After 24 h of incubation at 37 °C, old media were replaced with the fresh media containing protein solution (2 IC_{50}) and further incubated for 24 h. After incubation old media were replaced with fresh media containing dihydroethidium dye for ROS detection and JC1 dye for mitochondrial membrane potential activity detection [304-306]. The reaction was maintained in dark for 20 min and images were taken using fluorescence microscopy under 60X objective.

3.2.4.8 *In vivo* toxicity assessment

3.2.4.8.1 Ethical permissions

All experimenting protocols were performed as per the OECD/OCED guideline 425 and as approved by the Institutional Animal Ethical Committees of Tezpur University and Defense Research Laboratory, Tezpur, Assam (Approval nos. DORD-Pro/TUAEC/10-56/14/Res-07, DORD-Pro/TUAEC/10-56/14/Res-09 and DRL/02/Dec/2010-11/II).

3.2.4.8.2 Animals and housing conditions

The laboratory inbreed, pathogen free Balb/c mice (25-35 g) from the Central Facility of Animal House, Defence Research Laboratory, Tezpur, Assam were used for the experiments. The general conditions of captivity were maintained at 33-36 °C with relative humidity ≥ 75%. Animal food was obtained from Pranav Agrotech, Delhi, India. Animals were maintained in social groups before and after experimenting.

3.2.4.8.3 Determination of LD_{50}

For LD_{50} determination, individual groups of mice (n=6) were injected intraperitoneally (i.p.) with 0-4 mg kg^{-1} doses of the isolated venom protein dissolved in 0.25 ml of PBS (1X, pH 7.4). Groups of mice (n=6) injected with 0.25 ml PBS, pH 7.4 served as a control (placebo). The survival time for each animal was observed till 48 h. The LD_{50} value was calculated by probit analysis [307].

3.2.4.8.4 Assessment of acute *in vivo* toxicity

The *in vivo* toxicity of the venom protein/peptide was determined by analyzing the biochemical and hematological changes in mice plasma after its intraperitoneal (*i.p.*)/intravenous (*i.v.*) administration at 0.05-2.5 mg kg^{-1} body weight. Briefly, mice (n=6) were divided in two groups; the experimental group of mice were injected isolated venom protein/peptide (in a total volume of 0.25 ml 1X PBS pH 7.4) whereas the control (placebo) groups of mice (n=6) received injection of same volume of normal saline. The animals were observed at regular intervals up to 48 h post-injection for any behavioral changes viz. body weight, food and water intake, defecation and urination, grip strength, ear twist, rectal temperature, effect on circulatory system, and/or death.

Whole blood clotting time was measured at regular intervals by capillary glass method [308] or tail bleeding method [309]. Briefly, the tails of treated as well as control mice were transected at 2 mm from their tips after 6 h. Bleeding time was defined as the time elapsed until bleeding stopped [308]. Alternatively, blood from the transected tail was taken in capillary tube and the tubes were carefully broken at regular intervals to check the formation of fibrin clot [309].

The blood was collected immediately by cardiac puncture and all hematological parameters such as WBC, RBC, lymphocyte and monocyte count, Hb, MCV, MCH and MCHC were analyzed by Hematology auto analyzer MS4-S (Melet Schloesing Laboratories, France). The plasma obtained from the control as well as

treated groups of mice were analyzed for different biochemical parameters viz. total protein, glucose, uric acid, creatinine, cholesterol and urea levels as well as for different enzyme activity viz. alkaline phosphatase (ALP), creatine phosphokinase (CPK), plasma glutamic oxaloacetic transaminase (SGOT) and plasma glutamic pyruvic transaminase (SGPT) by using commercial diagnostic kits and following the instructions of the manufacturer. All biochemical parameters were analyzed on an autoanalyzer (Biochemical Systems International SRL Model 3000 evolution, Italy). The plasma levels of coagulation associated factors such as fibrinogen (factor I), factor IX, factor X, platelet derived growth factor receptor-α (PDGFR-α) and level of pro- and anti-inflammatory cytokines (TNF-α, IL-1, IL-6 and IL-10) were determined using commercially available kits and following the instructions of the manufacturers.

3.2.4.8.5 Histopathological studies

Groups of venom protein/peptide treated mice as well as the control group of mice were sacrificed with an overdose of sodium pentobarbital after 48 h of treatment. Various organs such as the liver, kidney and heart tissues were dissected out immediately, cut into 5 mm^3 sizes and washed with PBS (1X, pH 7.4) to remove the adhered blood. The tissues were fixed in 10.0% neutral buffered formalin and then processed by dehydrating with increasing grades of ethanol, and finally embedded in paraffin (ParaplastTM resin). Paraffin sections (5.0 μm) were processed for light microscopic observation (Leica DM 3000) by hematoxylin-eosin staining [310].

3.3 Statistical analysis

Data are expressed either as means or as mean ± standard deviation (S.D.) of three experiments or unless otherwise indicated. The statistical analysis of the data was done using the student's t-test. The value of $p \leq 0.05$ was considered as significant, unless otherwise stated. Statistical analysis and data fitting was performed using SigmaPlot 11.0 (Systat Software, San Jose, CA, USA).

CHAPTER IV

Purification and characterization of an anticoagulant peptide demonstrating anticancer activity from Indian Russell's Viper (Daboia russelii russelii) venom.

RESULTS AND DISCUSSION

4 Purification and characterization of an anticoagulant peptide demonstrating anticancer activity from Indian Russell's Viper (*Daboia russelii russelii*) venom

4.1 Brief introduction

The complex phenomenon of blood clotting requires the activation of zymogens of the coagulation cascade, in response to a vascular injury [195]. It is apprehended that thrombin and activated factor X are the key components of the blood coagulation cascade [195, 311]. As discussed in sections 1.7 and 1.9, factor Xa and thrombin are important pharmaceutical targets for the treatment and prevention of thrombotic-associated ailments [195-196]. Antithrombotic agents such as thrombin inhibitors are still under development for the treatment and prophylaxis of atrial fibrillation to avoid thromboembolism [197]. However, the research focus has recently shifted toward discovering dual inhibitors of thrombin and factor Xa that have a broader action mechanism. To date, most of these dual inhibitors have been chemically synthesized for better clinical efficacy and are still undergoing clinical trials [312-314]. Characterization of naturally occurring anticoagulant compounds possessing both thrombin and factor Xa inhibiting properties has been gaining focus for developing superior peptide-based therapeutic molecules to ameliorate cardiovascular disorders; nevertheless, discovering such potent molecules is still in its infancy worldwide.

It has been suggested that fibrin formation due to the unwanted action of thrombin, followed by deposition of excess fibrin surrounding the tumor cells plays an important role at different stages of tumor growth and dissemination [192]. Moreover, occurrence of venous thromboembolism is a common complication in cancer patients [192-194]. Although the degree of malignancy may not be correlated to the extent of fibrin deposition; nevertheless, the amount of fibrin deposited due to pathology of hemostatic system is a characteristic of a particular type of tumor [192]. Previous studies have suggested that the inhibition of thrombin expressed on the surface of tumor

thrombin expressed on the surface of tumor cells may help in controlling tumor progression [192]. Taken together, anticoagulants such as thrombin inhibiting peptides are important pharmaceutical targets not only for the treatment and/or prevention of thrombosis but also for preventing cancer progression [192]. Undoubtedly, the search for novel, non-toxic anticoagulant compounds with thrombin inhibiting properties has gained intense focus by researchers across the globe.

Snake venoms are reportedly rich in non-enzymatic serine protease inhibitors, many of which are functionally characterized as factor Xa or thrombin inhibitors [59, 94, 103, 106, 180]. A few serine protease inhibitors showing excellent thrombin and/or factor Xa inhibitory activity have been characterized from RVV that belong to the C-type lectin protein family (snaclec), the phospholipase A_2 (PLA_2) superfamily, or the Kunitz-type serine protease inhibitors [59, 98, 105-106, 113]. The goal of the present study is the pharmacological characterization and elucidation of the mechanism of anticoagulant action of a newly isolated, low molecular mass peptide purified from *D. r. russelii* venom. Furthermore, the physiological significance and therapeutic potential of the isolated peptide in terms of its anticoagulant effects and anticancer potential has also been explored.

4.2 Results

4.2.1 Purification of an anticoagulant peptide from RVV

4.2.1.1 Fractionation of crude RVV on CM-cellulose column

Fractionation of crude RVV (25.0 mg) on cation exchanger CM-Cellulose column (see section 3.2.1.1) resulted in the elution of three basic anticoagulant peaks (CMAC-I, CMAC-II and CMAC-III) (Fig. 4.1A).

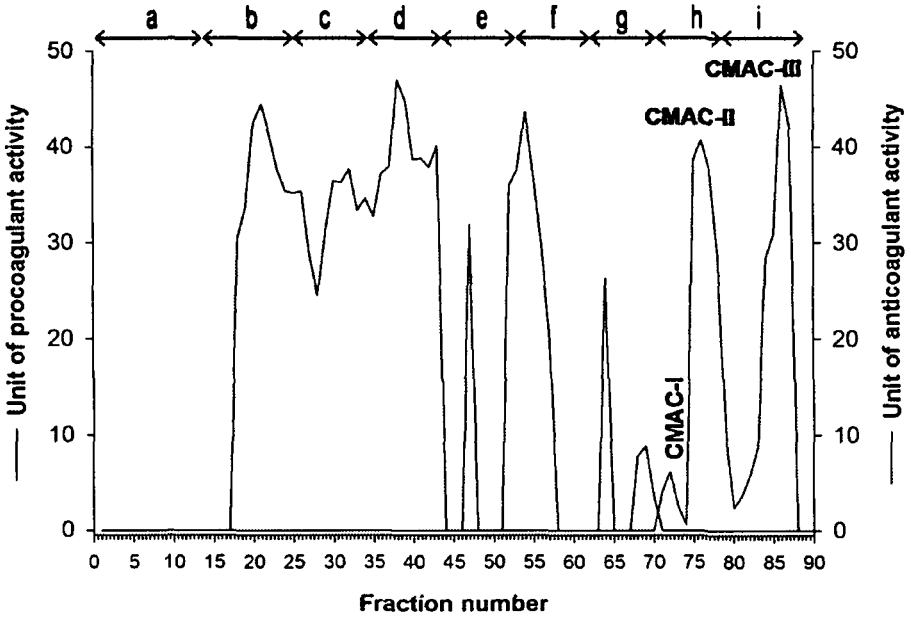


Fig. 4.1A CM-cellulose chromatography profile of crude RVV. The CM-cellulose column (20 mm x 60 mm) was pre-equilibrated with 20 mM potassium phosphate buffer, pH 7.0 and loaded with 25.0 mg crude RVV venom dissolved in 2.0 ml of the same buffer. The proteins in the venom supernatant were eluted stepwise using potassium phosphate buffers of increasing molarities and pH at room temperature (~23 °C). The increasing molarities and pH values were (a) 20 mM potassium phosphate, pH 7.0; (b) 50 mM potassium phosphate, pH 7.0; (c) 50 mM potassium phosphate, pH 7.5; (d) 75 mM potassium phosphate, pH 7.5; (e) 100 mM potassium phosphate, pH 8.0; (f) 180 mM potassium phosphate, pH 8.0; (g) 230 mM potassium phosphate, pH 8.0; and (h) 260 mM potassium phosphate, pH 8.0; (i) 300 mM potassium phosphate, pH 8.0. The flow rate was adjusted to 24.0 ml h^{-1} and 1.0 ml fraction was collected. The blue line represents protein peaks showing anticoagulant activity.

4.2.1.2 Fractionation of active CMAC-III on Sephadex G-50 column

Fractionation of RVV anticoagulant proteins (CMAC-III) eluted with 300 mM potassium phosphate buffer (pH 8.0) from a CM-cellulose column on a Sephadex

G-50 gel-filtration column resulted in separation of proteins in four peaks (GF-I, GF-II, GF-III, and GF-IV) (Fig. 4.1B). The fractions of both GF-III and GF-IV contained low molecular mass proteins and demonstrated anticoagulant property. However, GF-IV was selected for further studies as it demonstrated a better anticoagulant property as compared to GF-III.

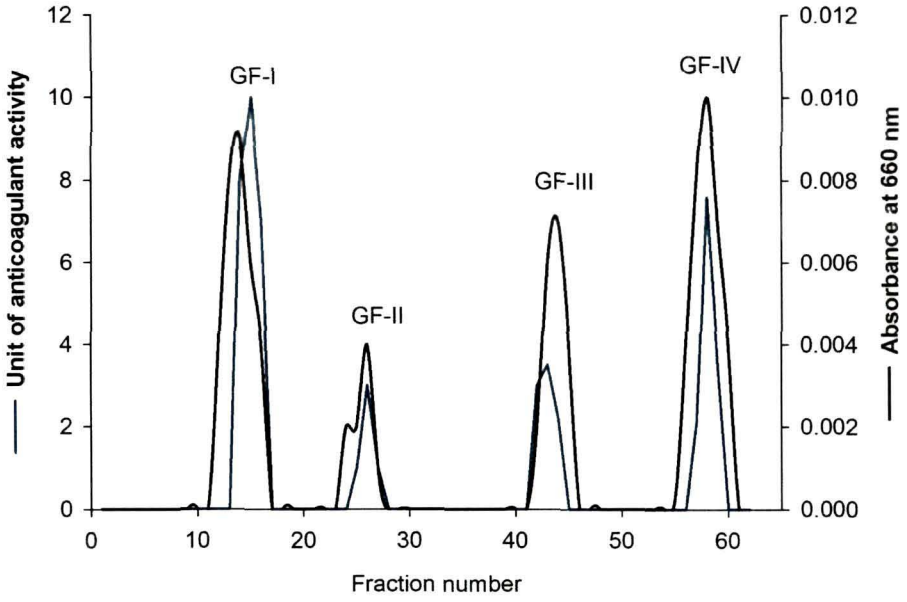


Fig. 4.1B Fractionation of CMAC-III through Sephadex G-50 gel-filtration column (1 cm x 60 cm). Gel-filtration chromatography of CMAC-III (1.0 mg) was performed as described in section 3.2.1.1. Each peak was screened for the protein content and anticoagulant property as described in sections 3.2.3.1 and 3.2.4.1, respectively.

4.2.1.3 RP-HPLC analysis of anticoagulant gel-filtration fraction GF-IV

The gel-filtration fraction, GF-IV, containing low molecular mass peptides and showing potent anticoagulant activity was subjected to fractionation through a C_{18} RP-HPLC column (4 μm, 3.9 mm x 150 mm), which resulted in the elution of

proteins in a single peak with a retention time of 15.8 min. This peak exhibited anticoagulant activity (Fig. 4.1C).

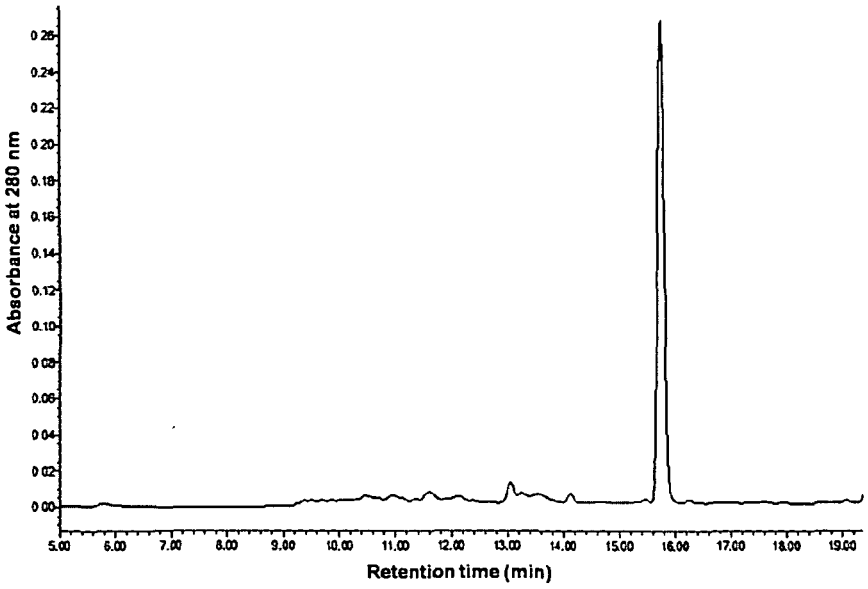


Fig. 4.1C RP-HPLC analysis of GF-IV on C_{18}-μ-Nova pack column (4 μm, 3.9 mm x 150 mm) coupled to HPLC system (Waters, Milenium-2000). 50.0 μg of GF-IV was dissolved in 20.0 μl of solvent A (0.1% v/v trifloro acetic acid (TFA) in 5% v/v acetonitrile) and subjected to RP-HPLC. Bound proteins were eluted using a gradient of 5-95% solvent B (0.1% v/v TFA in 95% v/v acetonitrile in H_2O) at a flow rate of 1.0 ml min^{-1} at room temperature (~23 °C). Detection was monitored at 280 nm and fraction collection was done manually. The peak eluted at 15.8 min was screened for protein content and anticoagulant activity.

The active anticoagulant fraction was named Ruviprase and it contributed to 0.1% of the total RVV proteins. A summary of purification of Ruviprase is shown in Table 4.1.

Ruviprase did not show detectable enzymatic activities viz. protease (caseinolytic) activity, fibrin(ogen)olytic activity and PLA_2 activity at a dose of 25.0 μg ml^{-1} (approximately 5.7 μM).

Table 4.1 Summary of purification of an anticoagulant peptide named Ruviprase from RVV.

Fraction	Total protein (mg)	Protein yield (%)	Anticoagulant activity		Purification fold
			Total activity (Unit*)	Specific activity (Unit mg^{-1})	
Crude RVV	25.0	100	ND	ND	0
CM-cellulose fraction (CMAC-III)	1.0	3.9	65.1	65.1	1
Gel-filtration fraction (GF-IV)	0.1	0.4	14.3	143.0	2.2
HPLC fraction (Ruviprase)	0.04	0.1	15.0	375	5.8

*Unit has been defined as 1 s increase in recalcification time of PPP in presence of crude RVV/venom fractions as compared to recalcification time of control PPP (108 ± 3.1 s, mean ± S. D., n=3).

The data represent a typical experiment.

4.2.2 Determination of molecular mass of Ruviprase

MALDI-TOF-MS analysis confirmed the homogeneity of Ruviprase preparation and determined its molecular mass at 4,423.6 Da (Fig. 4.2).

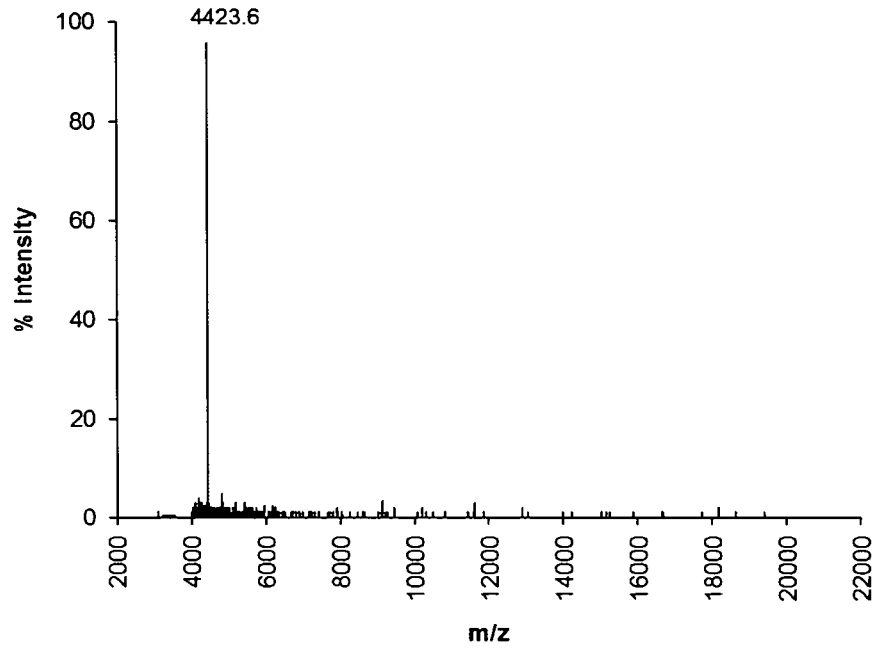


Fig. 4.2 Determination of molecular mass of Ruviprase by MALDI-TOF-MS analysis. Mixtures of 0.5 µl α-cyano-4-hydroxycinnamic acid matrix (10 mg ml^{-1}) and 0.5 µl of a Ruviprase solution (10.0 µg $µl^{-1}$, diluted in water containing 0.1% TFA) was spotted onto an Opti-TOF-384 plate (ABSciex), dried, and analyzed in positive linear mode on an AB 4800-Plus MALDI-TOF-TOF-MS instrument as described in section 3.2.2.1.2. Spectra were acquired in the m/z range of 2,000-22,000. The spectrum shows signal at m/z 4,423.6 corresponding to a single species with a molecular mass of 4.4 kDa.

4.2.3 N-terminal sequence analysis of Ruviprase

The partial N-terminal amino acid sequence of Ruviprase (E^1-V^2-X^3-W^4-W^5-W^6-A^7-Q^8-L^9-S^{10}) showed maximum sequence homology (80.0%) with the internal sequence of the apoptosis-stimulating factor of p53 protein 2 (Accession no: ETE62279.1) from *Ophiophagus hannah* and apoptosis-stimulating factor of p53 protein 2 isoform X2 and X1 (Accession no:

XP_007426348.1 and XP_007426347.1, respectively) from *Python bivittatus* (Table 4.2). A variation in the partial N-terminal sequence of Ruviprase was observed only at two positions; at position 2, 'I' of the apoptosis-stimulating factor of p53 proteins was replaced by 'E' in Ruviprase. Also, the amino acid residue at position 3 (E in similar peptides) could not be identified in Ruviprase and hence denoted as 'X'. Nevertheless, it is to be noted that the anticoagulant property of these apoptosis-stimulating factor of p53 proteins has never been investigated.

4.2.4 Determination of secondary structure of Ruviprase: Circular dichroism spectroscopy

Analysis of the CD spectrum of Ruviprase revealed that it was primarily composed of α-helices (61.9 %) with the characteristic minimum at 222 nm (Fig. 4.3). Random coils (38.1%) contributed to the rest of this protein structure (Fig. 4.3).

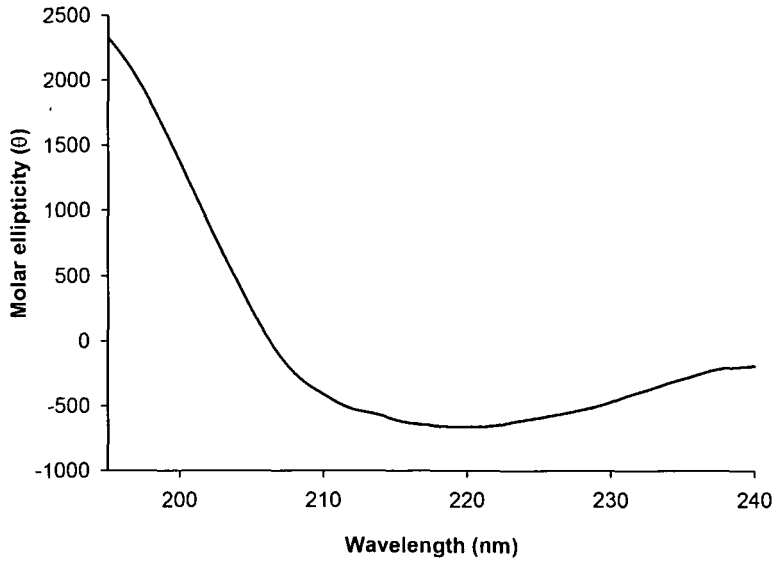


Fig. 4.3 Circular dichroism spectra of Ruviprase showing its secondary structure. Native Ruviprase (0.3 mg ml^{-1}) was dissolved in 20 mM potassium phosphate buffer pH 7.0 and the far UV-CD spectra was recorded at room temperature (~23 °C) between 195-250 nm against the appropriate buffer (blank) as described in section 3.2.2.4.

Table 4.2 N-terminal sequence alignment of Ruviprase with snake venom proteins deposited in NCBI database.

Accesion no.	Protein	Species	Sequence	Identity (%)
-	Ruviprase	*Daboia russelii*	E^{1} -V-X-W-W-W-A-Q-L-S^{10}	
ETE62279.1	Apoptosis-stimulating factor of p53 protein 2	*Ophiophagus hannah*	E^{981} -I- E-W-W-W-A-Q-L-S^{990}	80
XP_007426348.1	Apoptosis-stimulating factor of p53 protein 2 isoform X2	*Python bivittatus*	E^{1118}-I-E-W-W-W-A-Q-L-S^{1127}	80
XP_007426347.1	Apoptosis-stimulating factor of p53 protein 1 isoform X1	*Python bivittatus*	E^{1126}- I-E-W-W-W-A-Q-L-S^{1135}	80
	Apoptosis-stimulating of p53 protein 1	*Ophiophagus hannah*	E^{1046}-I-E-W-W-W-A-R-L-S^{1055}	70

4.2.5 Biochemical properties

4.2.5.1 Inhibition of serine proteases

The inhibitory effect of Ruviprase on the amidolytic activity of serine proteases such as thrombin, plasmin, factor Xa and trypsin has been shown in Fig. 4.4. Ruviprase inhibited the amidolytic activity of trypsin, plasmin and factor Xa to a minor extent but strongly inhibited the catalytic activity of thrombin towards its chromogenic substrate T1637 (Fig. 4.4).

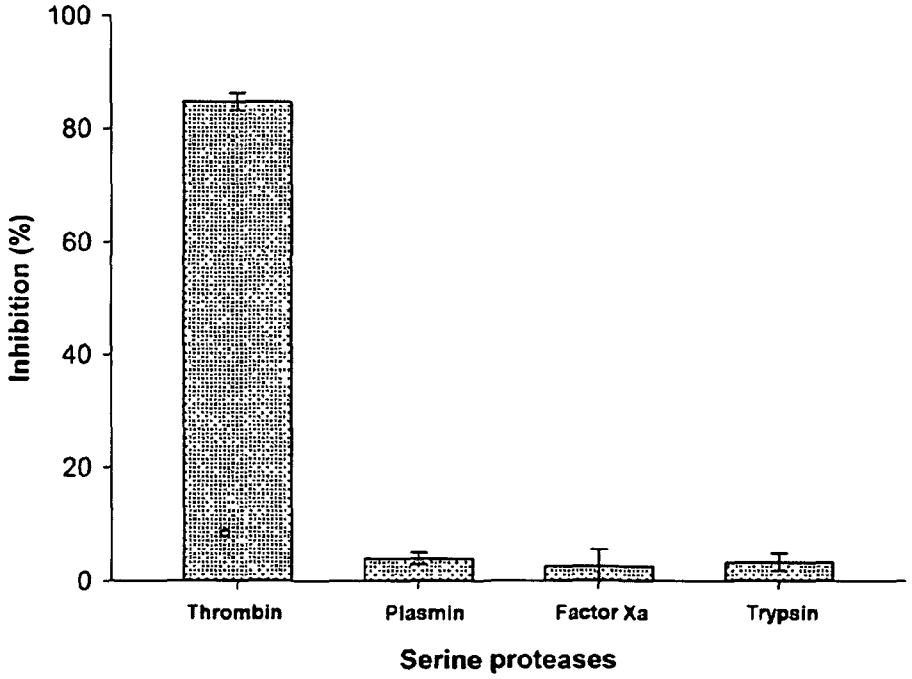


Fig. 4.4 Effect of Ruviprase on the amidolytic activity of different serine proteases. Ruviprase (2.4 μM) was incubated with either thrombin (0.16 μM), plasmin (0.12 μM), factor Xa (0.21 μM) or trypsin (0.24 μM) for 30 min at 37 °C. To the mixture, the appropriate chromogenic substrate (T1637 for thrombin, V0882 for plasmin, F3301 for factor Xa and B3133 for trypsin) was added (0.2 mM, final concentration), and amidolytic assay was performed as described in section 3.2.3.6. The activity of the above serine proteases towards their respective chromogenic substrates in the absence of Ruviprase was considered as 100% and other values were compared to this value. Values are shown as mean ± S.D. of triplicate values.

4.2.5.2 Thrombin inhibition studies

4.2.5.2.1 Inhibition of amidolytic activity of thrombin

Ruviprase inhibited the amidolytic activity of thrombin in a dose-dependent manner (Fig. 4.5). The IC_{50} value and Hill co-efficient of thrombin inhibition (amidolytic activity) were determined as 0.23 μM and 1.08, respectively.

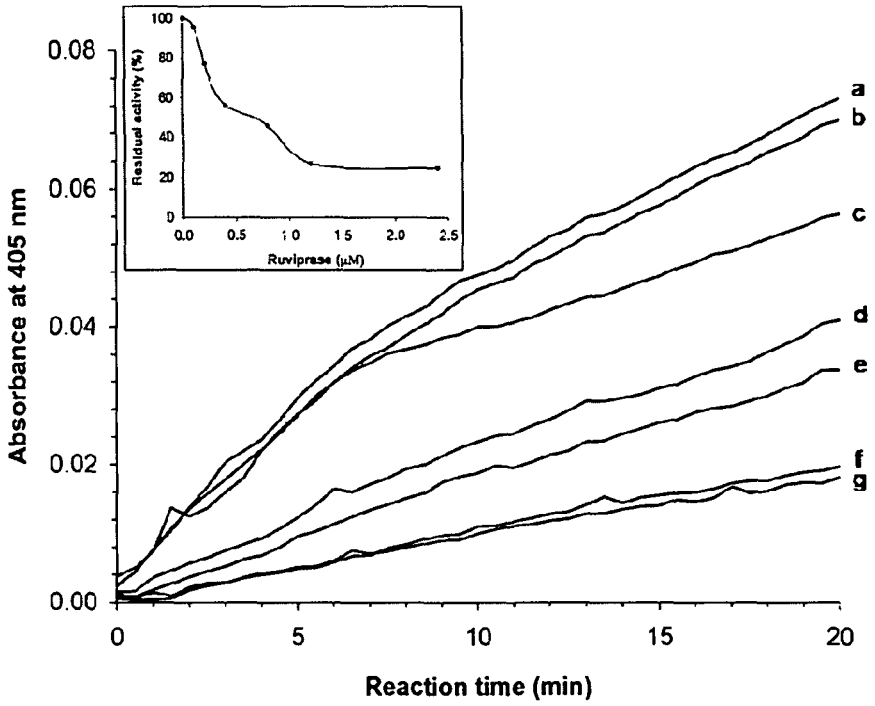


Fig. 4.5 Inhibition of the amidolytic activity of thrombin by different concentrations of Ruviprase. Thrombin (0.03 NIH ml^{-1}) was preincubated with (a) 0 μM (control); (b) 0.1 μM; (c) 0.2 μM; (d) 0.4 μM; (e) 0.8 μM; (f) 1.2 μM; or (g) 2.4 μM Ruviprase for 30 min at 37 °C. Chromogenic substrate for thrombin (T1637, 0.2 mM) was added and hydrolysis of the substrate was monitored for 20 min at 405 nm at 20 s intervals in a microplate reader. Inset: Dose response curve showing inhibition of the amidolytic activity of thrombin by Ruviprase. Residual activity of thrombin was calculated by considering the absorbance after 20 min of substrate hydrolysis by thrombin in the absence of inhibitor (Ruviprase) as 100%. Values are shown as mean of triplicate determinations.

4.2.5.2.2 Kinetics of thrombin inhibition

The Michaelis-Menten plot (Fig. 4.6) indicated a decrease in both the *Km* as well as *Vmax* values of thrombin towards its substrate (T1637) in presence of Ruviprase. The kinetics data analysis indicates that Ruviprase inhibits thrombin in an uncompetitive manner. The *Ki* value for the inhibition of thrombin by Ruviprase was determined as 0.42 μM.

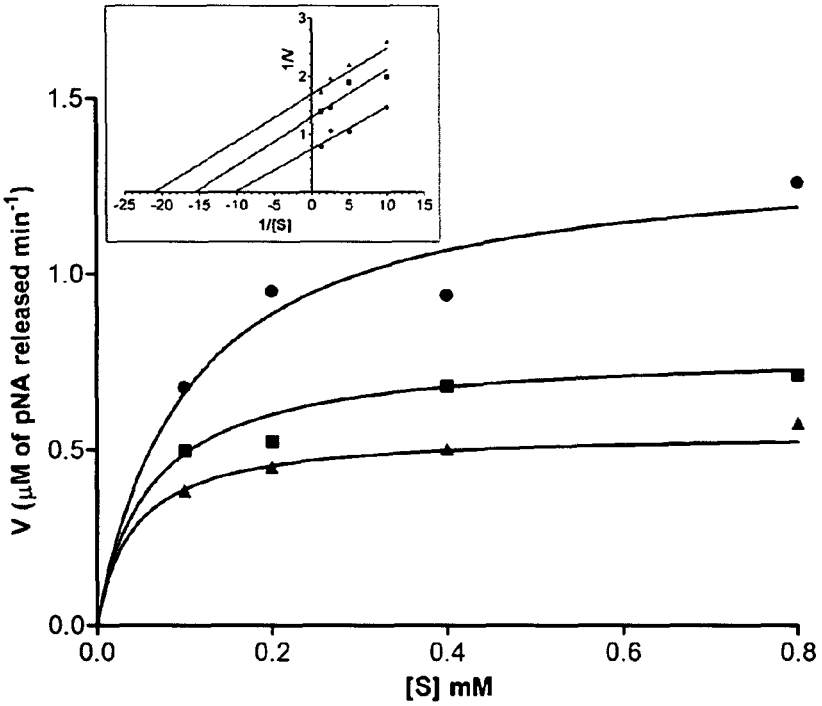


Fig. 4.6 Kinetics of inhibition of the amidolytic activity of thrombin by Ruviprase. Michaelis-Menten plots and Lineweaver-Burk plots (inset) showing inhibition of the amidolytic activity of thrombin towards T1637 in presence of (●) 0 μM (control); (■) 0.6 μM; or (▲) 1.2 μM Ruviprase. The experiment was performed at 37 °C, pH 7.4 (section 3.2.3.11.2). The plots are the means of 3 independent measurements.

4.2.5.2.3 Effect of Ruviprase on thrombin inhibition by AT-III in the presence or absence of heparin

Heparin-AT-III complex inhibited the amidolytic activity of thrombin towards its chromogenic substrate by 54.4%. It was observed that when thrombin was

pre-incubated with Ruviprase prior to addition of heparin-AT-III complex, the complex could display only 16.4% of its inhibitory activity (Table 4.3).

4.2.5.2.4 Effect of Ruviprase on the fibrinogen clotting activity of thrombin

The fibrinogen clotting activity of thrombin was also inhibited by Ruviprase in a dose-dependent manner (Fig. 4.7). The IC_{50} value and Hill co-efficient for the inhibition of fibrinogen clotting activity of thrombin by Ruviprase were determined as 1.52 μM and 5.21, respectively.

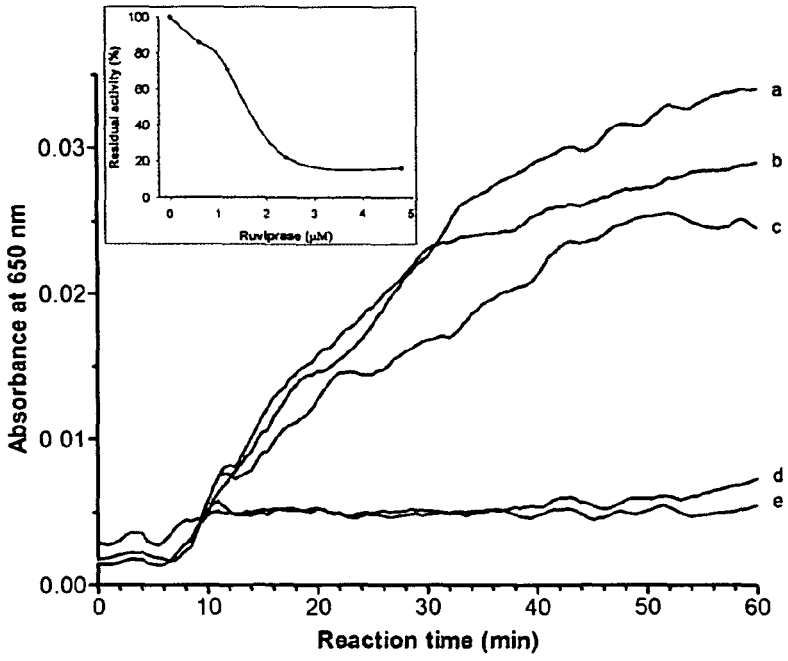


Fig. 4.7 Effect of Ruviprase on the fibrinogen clotting activity of thrombin. Thrombin (3.0 μl, 10 NIH U ml^{-1}) was pre-incubated with (a) 0 μM (control); (b) 0.6 μM; (c) 1.2 μM; (d) 2.4 μM; or (e) 4.8 μM of Ruviprase for 30 min at 37 °C prior to addition of fibrinogen (50.0 μg) in a final reaction mixture of 100 μl. Fibrin-clot formation was measured by monitoring the change in absorbance of the reaction mixture at 650 nm in a microplate reader (section 3.2.3.11.3). Inset: Dose-response curve showing inhibition of thrombin-induced fibrinogen clotting by Ruviprase. Residual activity of thrombin was calculated by considering the absorbance after 60 min of fibrinogen clotting in the absence of inhibitor (Ruviprase) as 100%. Values are shown as mean of triplicate determinations.

Table 4.3 Effect of Ruviprase on inhibition of amidolytic activity of thrombin by AT-III in presence or absence of heparin. Thrombin (3.0 µl, 10 NIH U ml^{-1}) was preincubated with 0.6 µM Ruviprase for 30 min at 37 °C. Heparin free AT-III (100 nM) or 25 nM AT-III in the presence of 0.5 U m^{-1} heparin, were added to the samples and the residual amidolytic activity of thrombin towards T1637 was measured as described in section 3.2.3.11.5. Percent thrombin activity was calculated by considering the absorbance after 10 min of T1637 hydrolysis in the absence of any inhibitor as 100% (i.e. 0% inhibition).

	% inhibition	
	Ruviprase	
Condition	**0 µM**	**0.6 µM**
Thrombin	0.0 ± 0^{a}	64.1 ± 3.2^{b}
Thrombin and AT-III	2.5 ± 1.2^{c}	67.9 ± 2.5^{d}
Thrombin, AT-III and heparin	54.4 ± 4.5^{e}	80.5 ± 1.6^{f}

Values are shown as mean ± S.D. of triplicate values. Values in the same row with different superscripts are statistically significant ($p<0.05$) from the control group (i.e. in absence of Ruviprase).

4.2.5.2.5 Effect of Ruviprase on platelet aggregation property of thrombin

Ruviprase inhibited thrombin-induced aggregation of washed platelet suspension at the tested doses (Fig. 4.8).

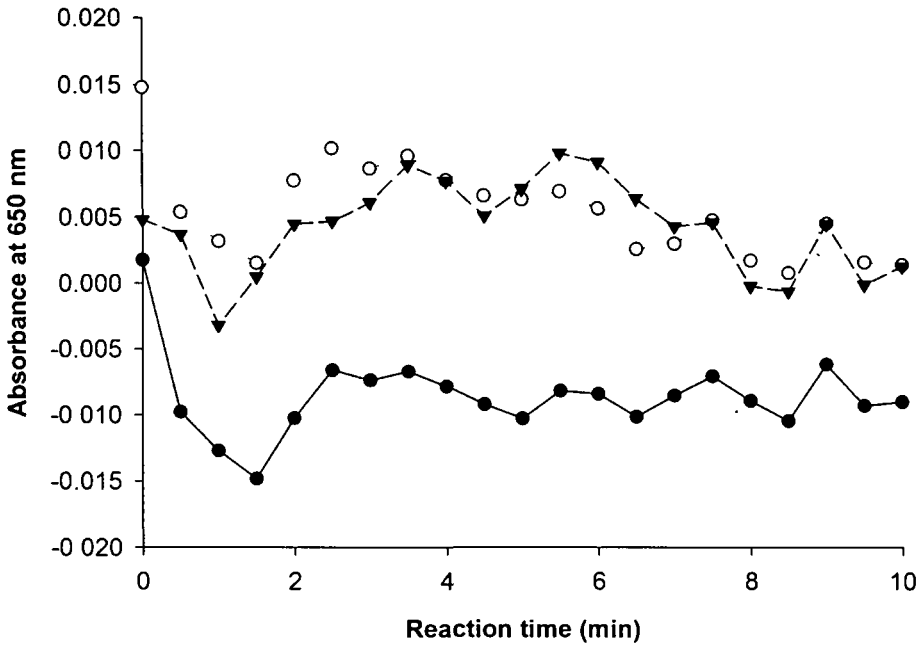


Fig. 4.8 Effect of Ruviprase on thrombin-induced platelet aggregation. Thrombin (0.2 NIH U ml^{-1}) was incubated with (●) 0 μM (control); (▼) 0.6 μM or (○) 1.2 μM Ruviprase, for 10 min at 37 °C in a microplate reader. Washed platelet suspension was prepared as described in section 3.2.3.11.4. The platelet suspension was added to the reaction mixture and the decrease in absorbance was monitored at 650 nm for 10 min in a microplate reader. Values are shown as mean of triplicate determinations.

4.2.5.3 Factor Xa inhibition studies

4.2.5.3.1 Effect of Ruviprase on the prothrombin activation property of factor Xa

Although Ruviprase failed to inhibit the amidolytic activity of factor Xa towards its chromogenic substrate F3301 (as shown in section 4.2.5.1);

however, it inhibited the physiological function of factor Xa (prothrombin activation) in the absence of components of the prothrombinase complex (Fig. 4.9).

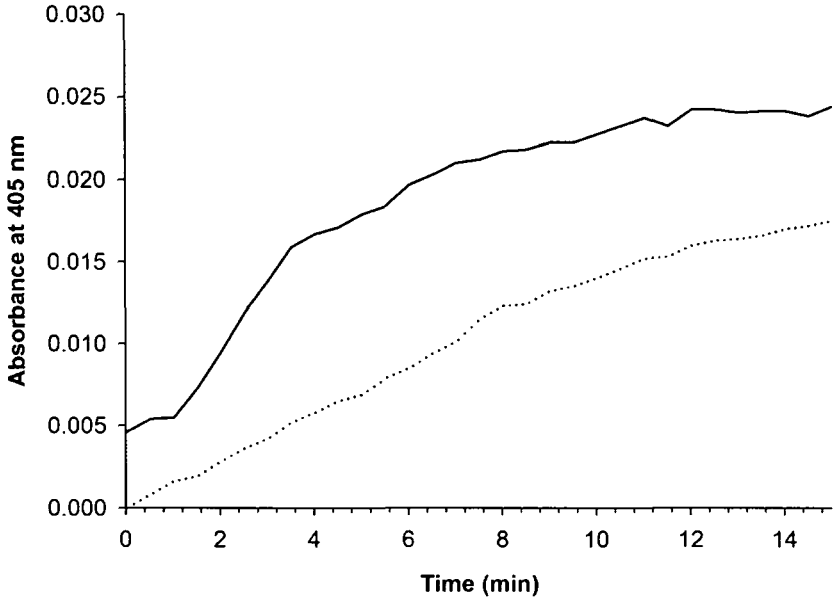


Fig. 4.9 Effect of Ruviprase on the prothrombin activation property of factor Xa. Factor Xa (0.02 μM) was preincubated with Ruviprase (0.6 μM) for 30 min at 37 °C. Prothrombin (1.0 μM) was added to the mixture followed by addition of 2.0 μl of chromogenic substrate for thrombin (10 mM, T1637) and the release of pNA was monitored at 405 nM at 37 °C for 15 min at 30 s intervals as described in section 3.2.3.11.1. Solid and dotted lines show prothrombin activation by factor Xa in the absence and presence of Ruviprase, respectively. Values are shown as mean of triplicate determinations.

4.2.5.3.2 Kinetics of factor Xa inhibition

The Michaelis-Menten plot showing inhibition of factor Xa (prothrombin activation) by Ruviprase indicated a mixed type of inhibition (Fig. 4.10). A decrease in the *Vmax* value with a subsequent increase in the *Km* value of factor Xa towards prothrombin in the presence of Ruviprase was observed. The *Ki* value for the inhibition of factor Xa (towards prothrombin activation) by

Ruviprase was determined as 0.16 μM. The IC_{50} value and Hill co-efficient for the inhibition of prothrombin activation by factor Xa were determined as 0.46 μM and 2.08, respectively.

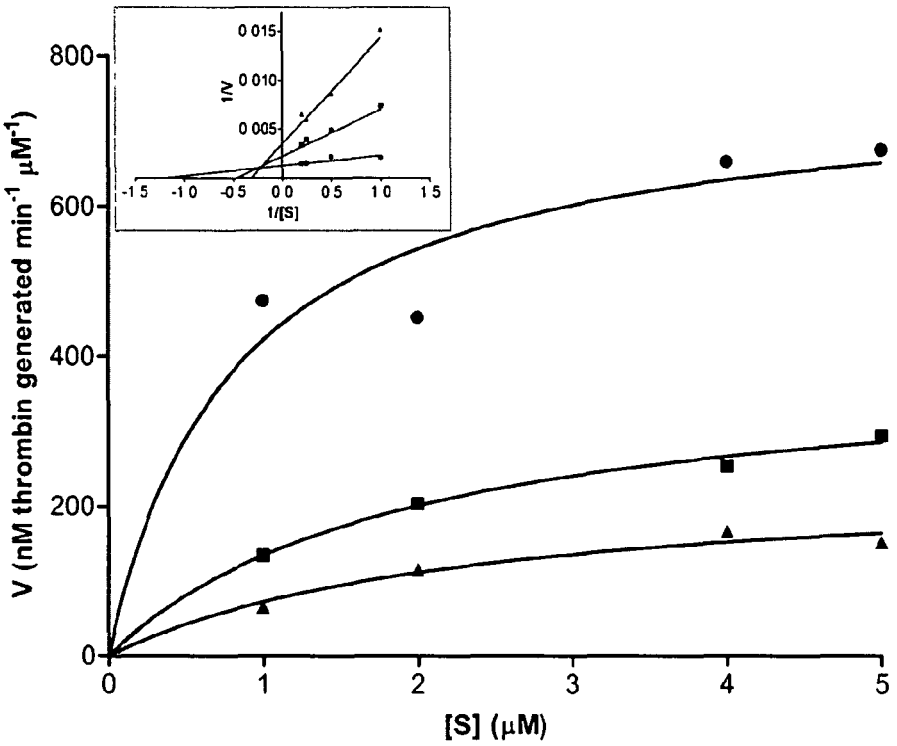


Fig. 4.10 Kinetic analysis of factor Xa inhibition by Ruviprase. Michaelis-Menten and Lineweaver-Burk plots (inset) showing inhibition of prothrombin activation property of factor Xa (0.02 μM) by (●) 0 μM; (■) 0.6 μM; or (▲) 1.2 μM Ruviprase. The experiment was performed as described in section 3.2.3.12.2, at 37 °C, pH 7.4. The plots are the means of 3 independent measurements.

4.2.5.4 Protein-protein interaction studies

4.2.5.4.1 Spectofluorimetric analysis

4.2.5.4.1.1 Interaction study of Ruviprase with thrombin

Excitation of Ruviprase/thrombin at 280 nm resulted in an emission maximum at 345 nm (Fig. 4.11A). The interaction between Ruviprase and thrombin resulted in an increase in the emission fluorescent signal even in the absence of

Ca^{2+} (Fig. 4.11A). Nevertheless, the presence of millimolar concentrations of Ca^{2+} increased the binding strength of Ruviprase with thrombin (Fig. 4.11A).

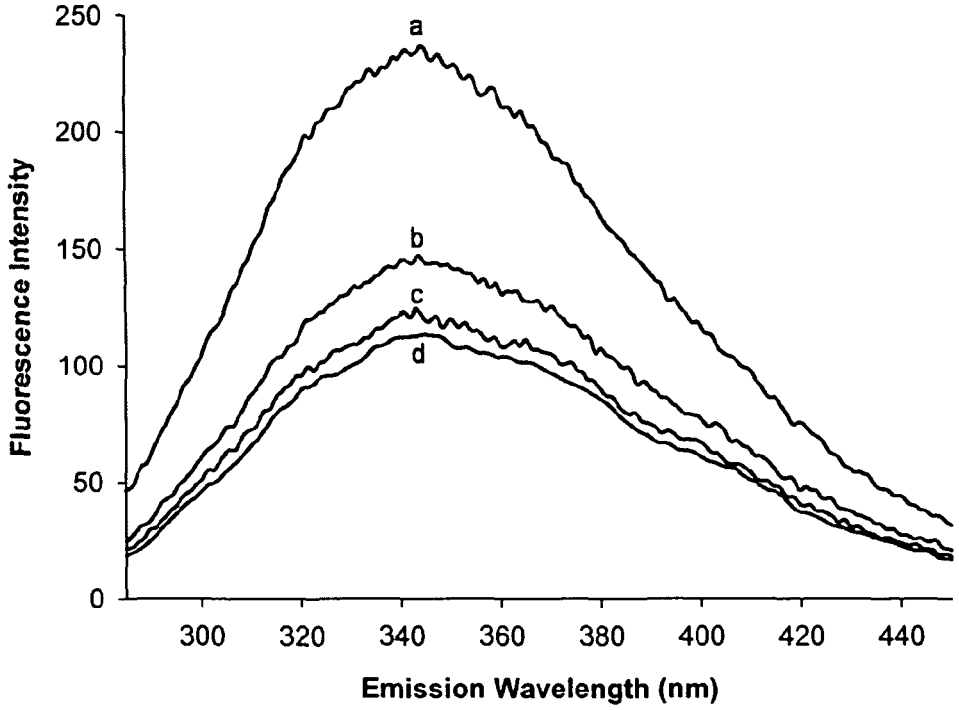


Fig. 4.11A Fluorescence spectra showing interaction of Ruviprase with thrombin. (a) thrombin (0.6 μM) in presence of Ruviprase (0.6 μM) and 5 mM Ca^{2+}, (b) thrombin (0.6 μM) and Ruviprase (0.6 μM) in absence of 5 mM Ca^{2+}, (c) thrombin (0.6 μM) and, (d) Ruviprase (0.6 μM). The experiment was performed as described in section 3.2.3.16.1, at room temperature (~23 °C), pH 7.4.

4.2.5.4.1.2 Interaction study of Ruviprase with factor Xa

The interaction between Ruviprase and factor Xa resulted in a slight increase in the emission fluorescent signal even in the absence of Ca^{2+} (Fig. 4.11B). The presence of millimolar concentrations of Ca^{2+} augmented the binding of Ruviprase with factor Xa (Fig. 4.11B).

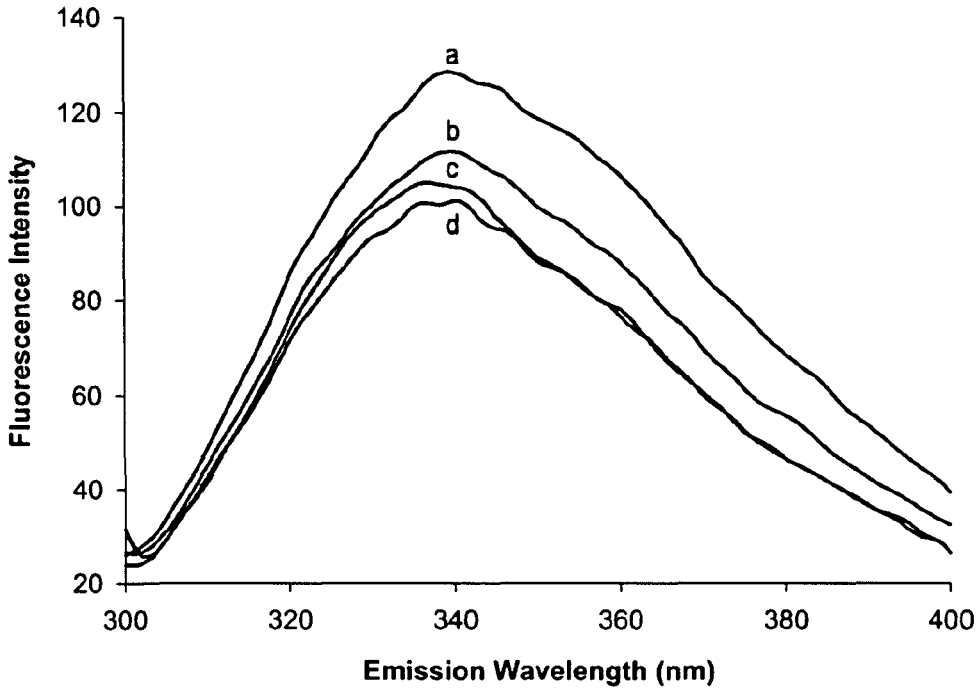


Fig. 4.11B Fluorescence spectra showing interaction of Ruviprase with factor Xa. (a) Factor Xa (0.2 μM) in presence of Ruviprase (0.6 μM) and 5 mM Ca^{2+}, (b) Factor Xa (0.2 μM) and Ruviprase (0.6 μM) in absence of 5 mM Ca^{2+}, (c) Factor Xa (0.2 μM) and, (d) Ruviprase (0.6 μM). The experiment was performed as described in section 3.2.3.16.1, at room temperature (~23 °C), pH 7.4.

4.2.5.4.2 Biosensor (SPR) analysis of thrombin binding

The binding of Ruviprase to thrombin was confirmed by SPR analysis. A multi-cycle binding study was performed, and the dissociation constant was determined using steady state data (Fig. 4.12). The dissociation constant (K_D) of Ruviprase for thrombin was found to be 0.46 μM.

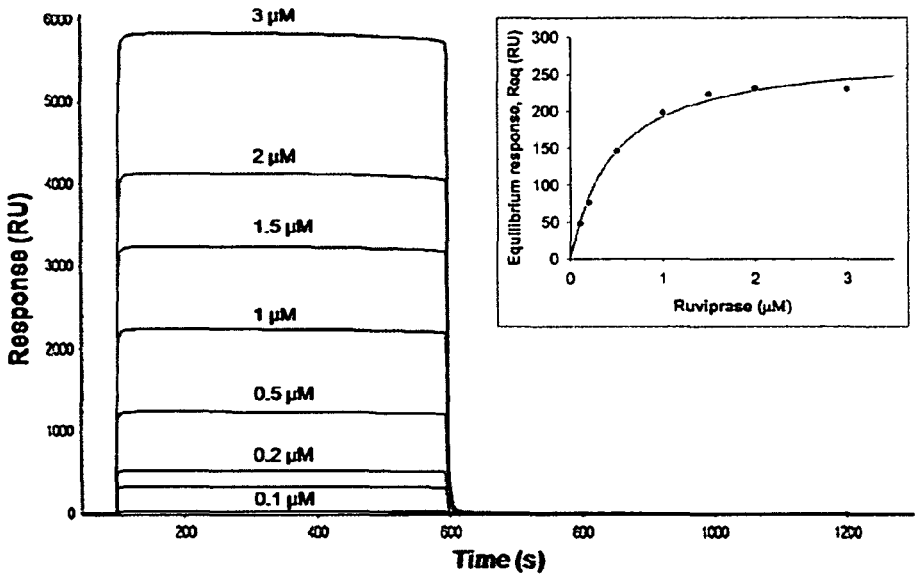


Fig. 4.12 Equilibrium binding assay of Ruviprase with thrombin by SPR. Sensorgram depicting interaction of Ruviprase with thrombin.Thrombin was immobilized at high density (~5000 RU) via amine coupling using a Biacore CM5 sensor chip as described in section 3.2.3.16.2. Inset: One site saturation binding curve of Ruviprase for thrombin. The concentrations of Ruviprase were: 0.1, 0.2, 0.5, 1.0, 1.5, 2.0 and 3.0 µM; the data is fitted to the equation, $R_{eq} = RU_{max}*C/(K_D + C)$. The K_D= 458.7 nM. R^2= 0.9890.

4.2.5.5 Total carbohydrate content

Ruviprase was devoid of carbohydrate content suggesting it is non-glycoprotein in nature.

4.2.6 Pharmacological properties of Ruviprase

4.2.6.1 Anticoagulant activity

4.2.6.1.1 Plasma recalcification time

Ruviprase prolonged the recalcification time of PPP in a dose-dependent manner (Fig. 4.13A). Moreover, Ruviprase demonstrated similar *in vitro*

anticoagulant potency as compared to that of commercial anticoagulants viz. heparin and warfarin (Fig. 4.13A).

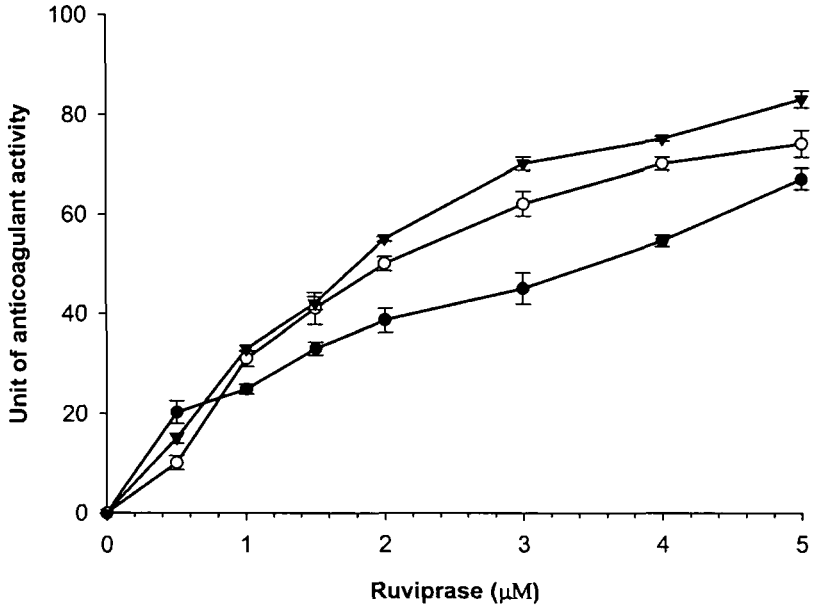


Fig. 4.13A Dose-dependent plasma recalcification activity of Ruviprase. Comparison of dose-dependent plasma recalcification activity of Ruviprase (▼) with warfarin (○) and Heparin (●). Different concentrations of Ruviprase/ Heparin/ Warfarin (0.5-5.0 μM) was incubated with 300 μl prewarmed PPP isolated from goat blood for 3 min at 37 °C. Then, 40 μl $CaCl_2$ (250 mM) was added to the mixture and clotting time was recorded as described in section 3.2.4.1. One unit of anticoagulant activity has been defined as increase in 1.0 s clotting of plasma compared to clotting time of normal citrated PPP. Values are mean ± S.D. of triplicate determinations.

4.2.6.1.2 Prothrombin time test

Ruviprase also prolonged the prothrombin time of PPP in a dose-dependent manner (Fig. 4.13B).

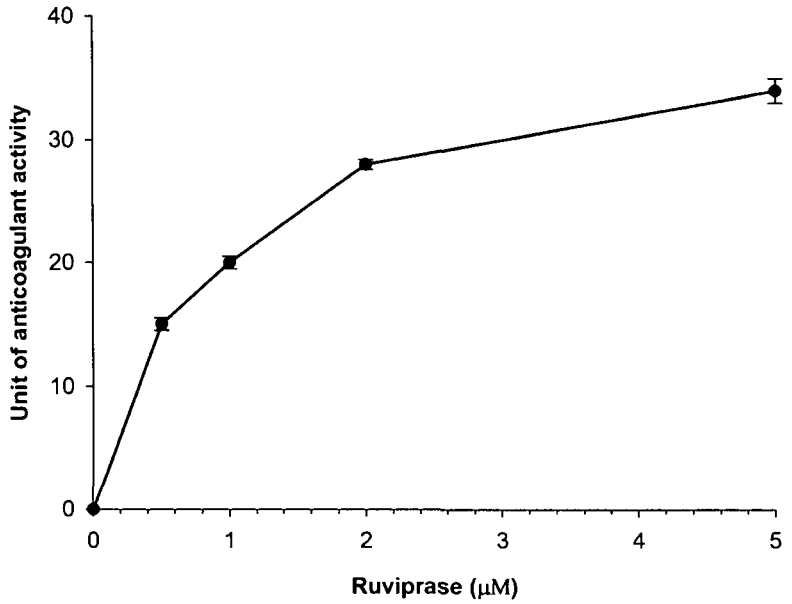


Fig. 4.13B Dose-dependent prothrombin time test of Ruviprase. PT assay was performed using commercially available kits (Liquiplastin) following the manufacturer's protocol as described in section 3.2.4.2. One unit of anticoagulant activity has been defined as 1.0 s increase in clotting time of citrated PPP as compared to clotting time of control (untreated) PPP under identical assay conditions. Values are mean ± S.D. of triplicate determinations.

4.2.6.1.3 Activated partial thromboplastin time test of Ruviprase

As shown in Fig. 4.13C, Ruviprase also prolonged the aPTT of citrated PPP obtained from goat blood in a dose-dependent manner (Fig. 4.13C).

4.2.6.2 Hemolytic activity

Ruviprase did not show *in vitro* direct or indirect hemolytic activity on mammalian erythrocytes at the tested dose of 10.0 μg ml^{-1} (2.5 μM).

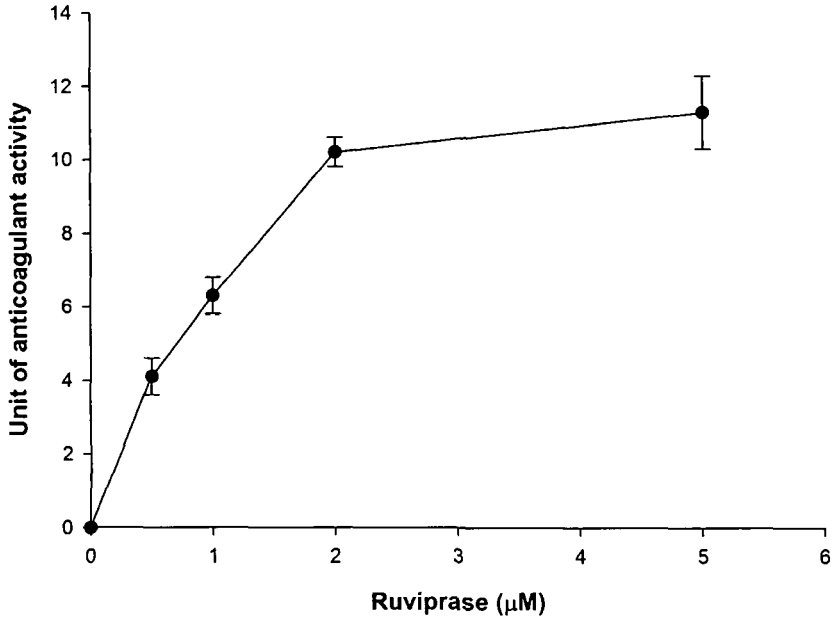


Fig. 4.13C Dose-dependent activated partial thromboplastin time test of Ruviprase. aPPT assay was performed using commercially available kits (Liquicelin) following the manufacturer's protocol as described in section 3.2.4.3. One unit of anticoagulant activity has been defined as 1.0 s increase in clotting time of citrated PPP as compared to clotting time of control (untreated) PPP under identical assay conditions. Values are mean ± S.D. of triplicate determinations.

4.2.6.3 Effect of pre-incubation on the anticoagulant property and thrombin/factor Xa inhibitory effects of Ruviprase

It was observed that with increased pre-incubation of Ruviprase with thrombin or factor Xa, the inhibitory effects of Ruviprase was also enhanced concomitantly up to 30 min of pre-incubation of Ruviprase with thrombin/ factor Xa (Fig. 4.14). Similarly, the anticoagulant activity (plasma recalcification activity) of Ruviprase was also observed to increase with subsequent increase in the pre-incubation time of Ruviprase with citrated PPP (Fig. 4.14).

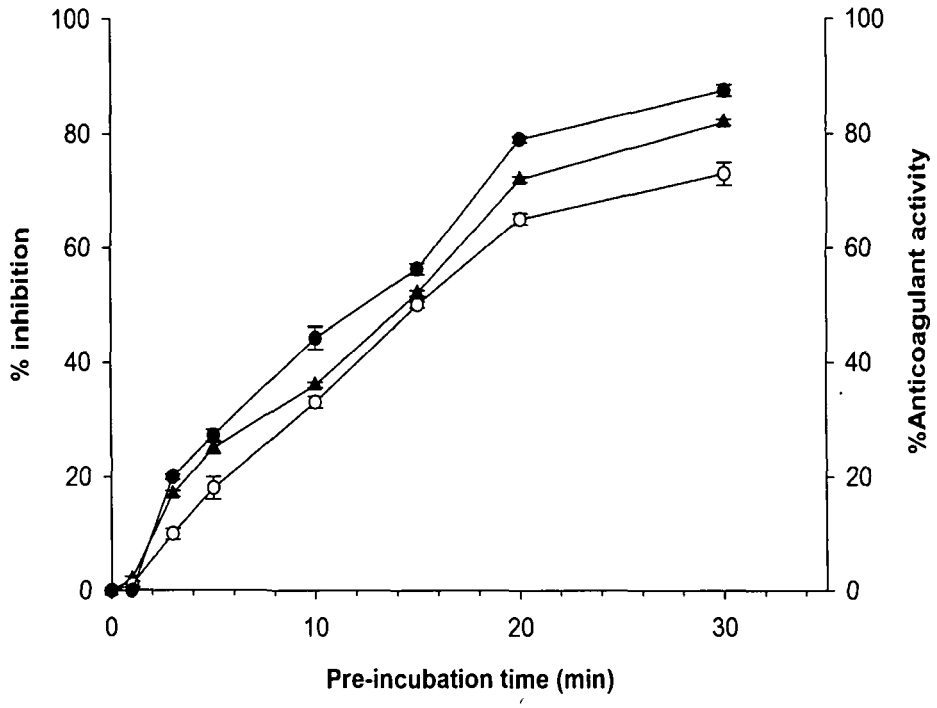


Fig. 4.14 Effect of pre-incubation on the anticoagulant property and thrombin/factor Xa inhibitory effects of Ruviprase. Ruviprase (2.4 μM) was pre-incubated with thrombin (3.0 μl, 10 NIH Unit ml^{-1}) or factor Xa (0.2 μM) for different time intervals (0-30 min) at 37 °C. Thereafter, the amidolytic activity of thrombin (▲) and prothrombin activation property of factor Xa (○) was analyzed as described in sections 3.2.3.11.1 and 3.2.3.12.1 respectively. Similarly, the plasma recalcification time (●) of Ruviprase was analyzed as described in section 3.2.4.1. Values are mean ± S.D. of triplicate determination.

4.2.6.4 Neutralization of Ruviprase by physiological inhibitors and commercial antivenom

Both polyvalent antivenom (PAV) and monovalent antivenom (MAV), at 1: 100 (Ruviprase: PAV/MAV, w/w) ratio, failed to neutralize the inhibitory effects of Ruviprase on the fibrinogen clotting activity of thrombin or the prothrombin activation property of factor Xa, as well as the anticoagulant property of Ruviprase (Fig. 4.15).

Physiological inhibitors such as α_2-macroglobulin and α_2-antiplasmin, at their physiological concentration of approximately 3 μM, failed to neutralize the anticoagulant property of Ruviprase as well as the inhibitory effects of Ruviprase on the fibrinogen clotting activity of thrombin or the prothrombin activation property of factor Xa (Fig. 4.15).

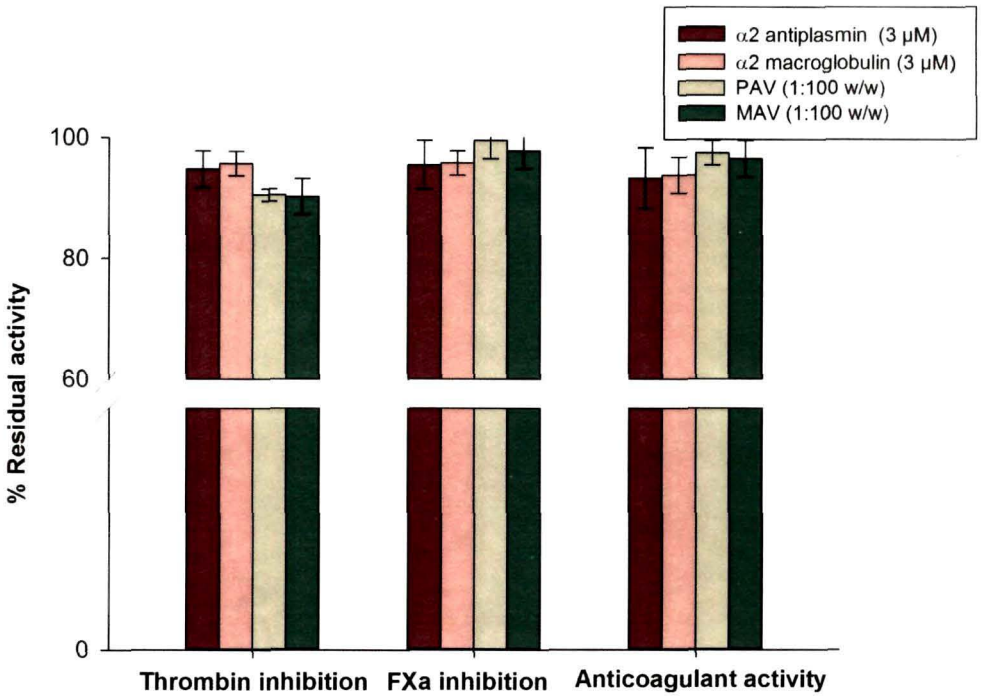


Fig. 4.15 Effect of physiological inhibitors and commercial antivenom on the different activities of Ruviprase. Ruviprase was pre-incubated with PAV or MAV in an antigen: antivenom ratio of 1:100 (w/w) or 3.0 μM inhibitor for 30 min at 37 °C. Thereafter, the amidolytic activity of thrombin, prothrombin activation property of factor Xa and plasma recalcification time of Ruviprase, was analyzed in the corresponding assay systems, as described in sections 3.2.3.11.1, 3.2.3.12.1 and 3.2.4.1, respectively. The inhibitory or anticoagulant activity of Ruviprase in the absence of antivenom/inhibitor was considered as 100% activity and other values were compared with that. Values are mean ± S.D. of triplicate determination.

4.2.6.5 Assessment of *in vitro* anticancer potential of Ruviprase

4.2.6.5.1 *In vitro* cytotoxicity assessment-dual staining method

Dual staining analysis of Ruviprase treated U87MG clearly depicted the cytotoxic property of this anticoagulant peptide (Fig. 4.16). The live cells were stained with AO and fluoresced green under green filter. The dead cells are stained with EB and fluoresced red under red filter. Clear pictures of the viable cells were visible by merging the pictures taken in the two filters. The results of dual staining depicted that most of the Ruviprase-treated cells were dead after 24 h as they stained with EB. Cells treated with 20 mM potassium phosphate buffer, pH 7.4 served as control.

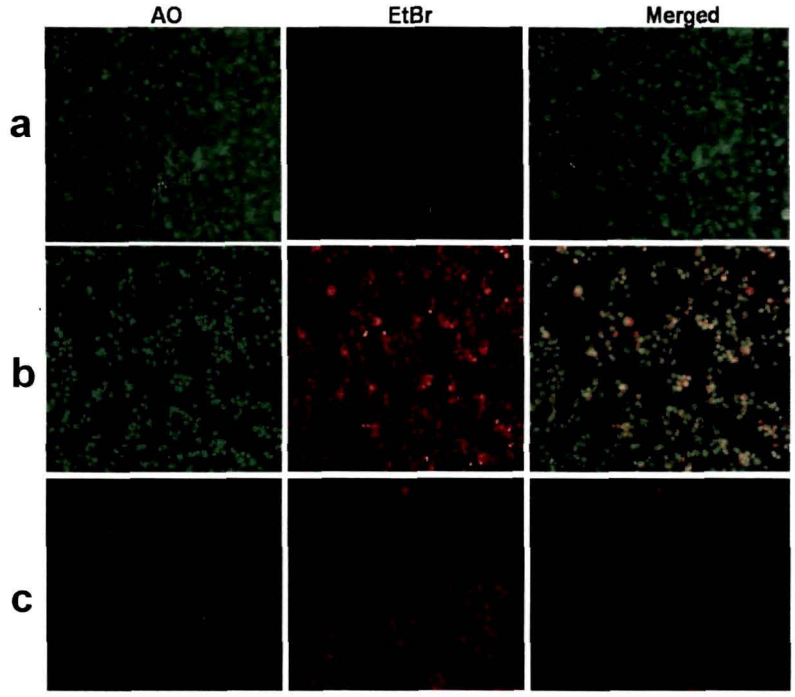
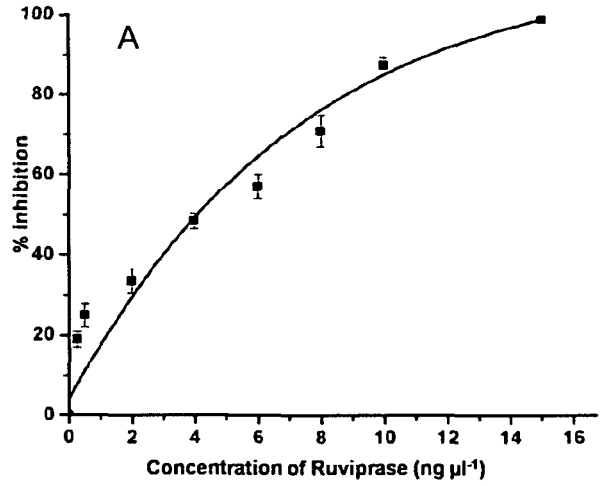


Fig. 4.16 *In vitro* toxicity assessment of Ruviprase on U87MG cells by dual staining method. (a) Untreated cells; (b) Cells treated with 20 mM potassium phosphate buffer, pH 7.4; (c) Ruviprase treated cells. U87MG cells (1 x 10^5 cells ml^{-1}) was treated with either 20 mM potassium phosphate buffer, pH 7.4 or Ruviprase (2.0 µM) for 24 h at 37 °C, 5% CO_2, before staining with 10 µl of AO/EB mix (10 mg mL^{-1} AO and EB in PBS) as described in section 3.2.4.7.2.

Cells were visualizing under a fluorescence microscope (Nikon ECLIPSE, TS100, Tokyo) with an excitation filter of 480/30 nm.

4.2.6.5.2 Assessment of anti-proliferative activity against cancer cells-Sulforhodamine B assay

The dose-dependent anti-proliferative activity of Ruviprase against MCF-7, EMT6/AR1, HeLa cells after 24 h of incubation at 37 °C, 5% CO_2 has been shown in Figs. 4.17A-C, respectively. The IC_{50} value of Ruviprase towards MCF-7, HeLa and EMT6/AR1 cells under identical experimental conditions was determined as 4 ± 0.3 ng μl^{-1} (Fig. 4.17A), 5.8 ± 0.7 ng μl^{-1} (Fig. 4.17B), and 23.0 ± 1.5 ng μl^{-1} (Fig. 4.17C), respectively. The results indicate that Ruviprase has highest cytotoxicity towards MCF-7 cells whereas it exhibits least inhibitory activity against EMT6/AR1 cells (Figs. 4.17A-C).



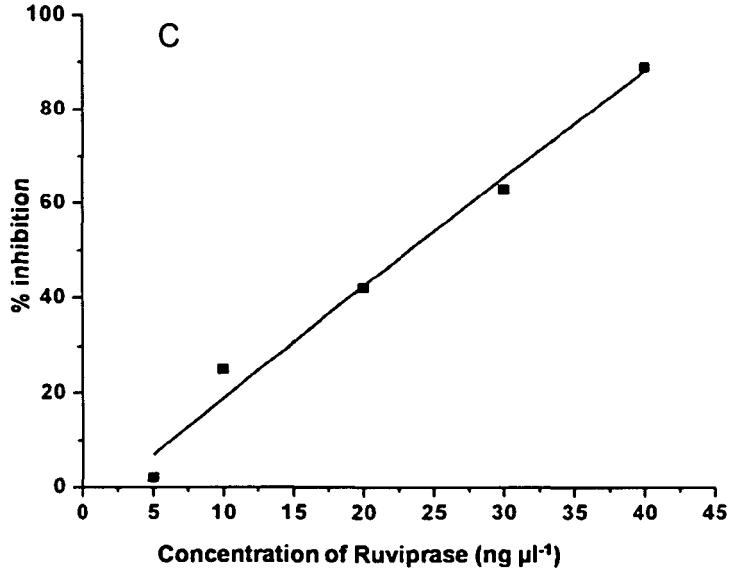


Fig. 4.17A-C Dose-dependent anti-proliferative activity of Ruviprase on (A) MCF-7, (B) HeLa, and (C) EMT6/AR1 cells. Cells were cultured as described in section 3.2.4.7.3. Cells were maintained in a CO_2 incubator at 37 °C and 5% CO_2. After 24 h of seeding of 1×10^5 cells ml^{-1} in a 96 well plate, the cells were treated with graded concentrations of Ruviprase (0-40 ng μl^{-1}) for 24 h in the above condition. A control was run in parallel where instead of purified peptide, medium was added. Thereafter, cells were fixed and Sulforhodamine B (SRB) assay was performed as described in section 3.2.4.7.3. Values represent mean of three independent experiments.

4.2.6.5.3 Detection of apoptosis in MCF-7 cells

4.2.6.5.3.1 Annexin V/ propedium iodide (PI) staining

Tthe apoptotic activity of Ruviprase towards MCF-7 cells was examined by PI/annexin V staining (Fig. 4.18). The figure indicates that Ruviprase damages the integrity of the plasma and nuclear membranes allowing PI to pass through the membranes which intercalate into nucleic acids and display red fluorescence.

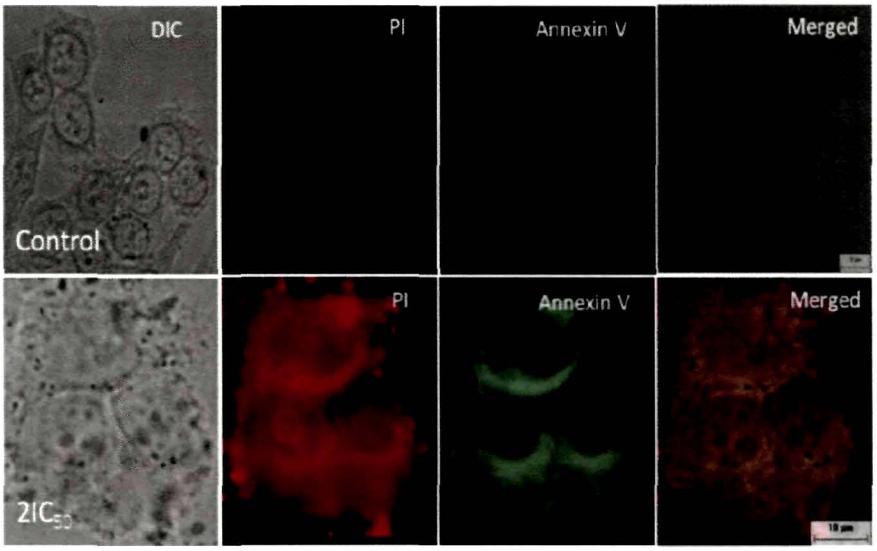


Fig. 4.18 Fluorescence microscopic detection of Ruviprase–induced apoptosis in MCF-7 cells by annexin V/propedium iodide (PI) staining. MCF-7 cells (0.5 x 10^5 cells ml^{-1}) were treated with Ruviprase (2 IC_{50} value) for 24 h at 37 °C, 5% CO_2. Annexin V/ PI staining was performed by using apoptosis kit and following the instructions of the manufacturer (Santa Cruz Biotechnology, Santa Cruz, CA). Cells were observed under an Eclipse TE 2000U microscope (Nikon, Tokyo, Japan) at 40X magnification. The images were analyzed using Image-Pro Plus software (Media Cybernetics, Silver Spring, MD).

4.2.6.5.3.2 Flow cytometric analysis of apoptosis

Ruviprase (2 IC_{50}) treated MCF-7cells were analyzed by flow cytometer to study the apoptotic status. Based on the flow cytometry analysis as shown in Fig. 4.19, it was revealed that about 48% of the Ruviprase-treated MCF-7 cells were apoptotic after 24 h of treatment at twice IC_{50} value dose (inset: Table 4.4).

Cell population that are not stained by either annexin V-FITC or PI (annexin V−/PI−) can be regarded as normal cells which are depicted in quadrant 3 (Q3), while those stained by annexin V-FITC but not by PI are in the early

apoptotic phase (annexin V+/PI−) and are depicted in quadrant 4 (Q4). Cells that are stained by both annexin V-FITC and PI (annexin V+/PI+) are the cells at late apoptosis or necrotic stage depicted in quadrant 2 (Q2). The cells that are stained with PI (PI+) are considered to be the dead cells and depicted in quadrant 1 (Q1).

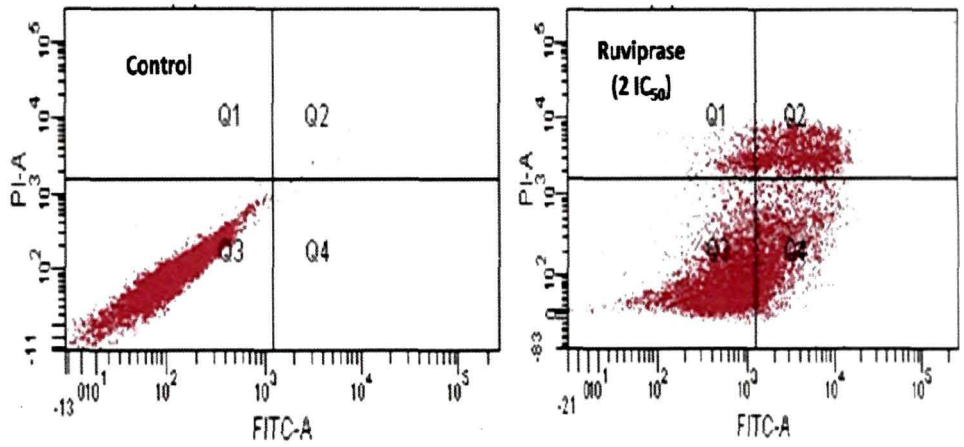


Fig. 4.19 Flow cytometry analysis of apoptosis induced by Ruviprase (2 IC_{50}) in MCF-7 cells after 24 h of treatment at 37 °C/ 5% CO_2. MCF-7 cells (0.5 x 10^5 cells ml^{-1}) were treated with Ruviprase (2 IC_{50}) for 24 h at 37 °C, 5% CO_2 and flow cytometric analysis was performed as described in section 3.2.4.7.4.2.

Table 4.4 Flow cytometry analysis of apoptosis induced by Ruviprase (2 IC_{50}) in MCF-7 cells**.** Flow-cytometric reading was taken using 488 nm excitation and band pass filters of 530/30 nm (for FITC detection) and 585/42 nm (for PI detection). Percentage of live, early apoptotic, late apoptotic and dead cells were analyzed using FACSDiva software.

Samples	**Live cells Q3**	**Early apoptotic cells Q4**	**Late apoptotic cells Q2**	**Dead cells Q1**
Control	99	0	1	0
Ruviprase (2 IC_{50})	49	25	23	3

As shown in Table 4.4, the flow-cytometric data revealed that after 24 h of Ruviprase-treatment, 25.0% and 23.0% of the MCF-7 cells were in the early and late apoptotic phase, respectively; whereas 3% cells were considered as dead cells.

4.2.6.5.3.3 Immunofluorescence microscopic analysis of induction of apoptosis

Elucidation of the mechanism of apoptosis induction in MCF-7 cells by Ruviprase shows involvement of p53 and p21 pathways (Figs. 4.20A-B). Activation of p53 produces downstream signal that leads to p21 expression and its nuclear accumulation. Ruviprase treatment strongly enhanced the nuclear accumulation and translocation of p53 (Fig. 4.20A) and p21 proteins (Fig. 4.20B) with a loss of DNA integrity of treated cells as compared to control cells (Figs. 4.20A-B).

Hoechst staining was performed in MCF-7 cells after 24 hr of Ruviprase-treatment (2 IC_{50} dose) to qualitatively analyze the effect of Ruviprase on mitotic progression. The fluorescence microscopic study suggested that Ruviprase treatment did not induce mitotic blockage or chromosomal aberration in cells; however, cell DNA appeared to be more condensed with respect to control which might be the prerequisite step for apoptosis initiation (Fig. 4.21).

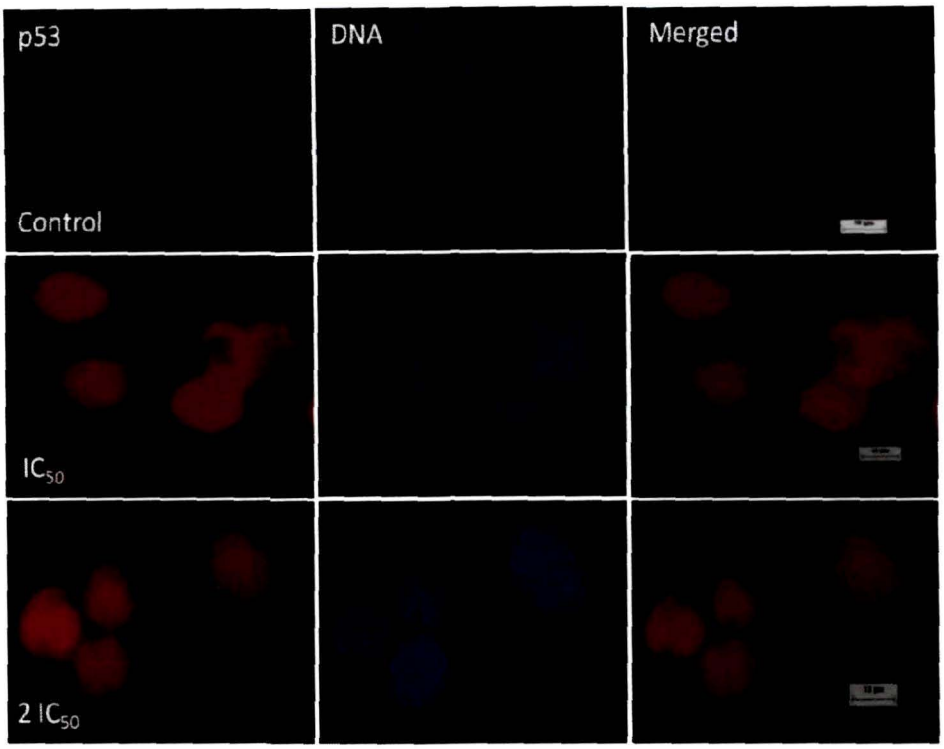


Fig. 4.20A Immunofluorescence microscopic analysis of Ruviprase-treated MCF-7 cells for assessment of nuclear accumulation of p53. Briefly, Ruviprase treated MCF-7 cells (0.5 x 10^5 cells ml^{-1}) were incubated with mouse monoclonal p53 antibody for 2 h at room temperature (~23 °C). Cells were washed and incubated with Alexa 568-labeled antimouse IgG antibody at room temperature (~23 °C) for 2 h (section 3.2.4.7.4.3). For DNA staining, cells were incubated with Hoechst 33250 for 15 min at room temperature (~23 °C) (section 3.2.4.7.4.3). The nuclear translocation and accumulation of p53 was determined by immunofluorescence confocal laser scanning microscopy at 40X magnification.

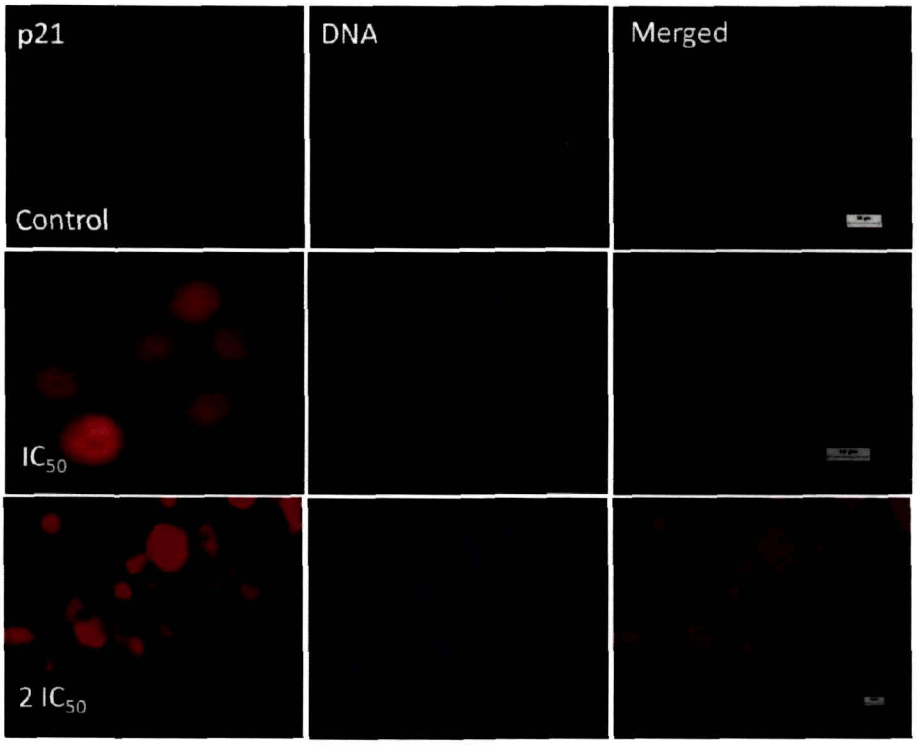


Fig. 4.20B Immunofluorescence microscopic analysis of Ruviprase-treated MCF-7 cells for assessment of nuclear accumulation of p21. Briefly, Ruviprase treated MCF-7 cells (0.5 x 10^5 cells ml^{-1}) were incubated with mouse monoclonal p21 antibody for 2 h at room temperature (~23 °C). Cells were washed and incubated with Alexa 568-labeled antimouse IgG antibody at room temperature (~23 °C) for 2 h (section 3.2.4.7.4.3). For DNA staining, cells were incubated with Hoechst 33250 for 15 min at room temperature (~23 °C) (section 3.2.4.7.4.3). The nuclear translocation and accumulation of p21 was determined by immunofluorescence confocal laser scanning microscopy at 40X magnification.

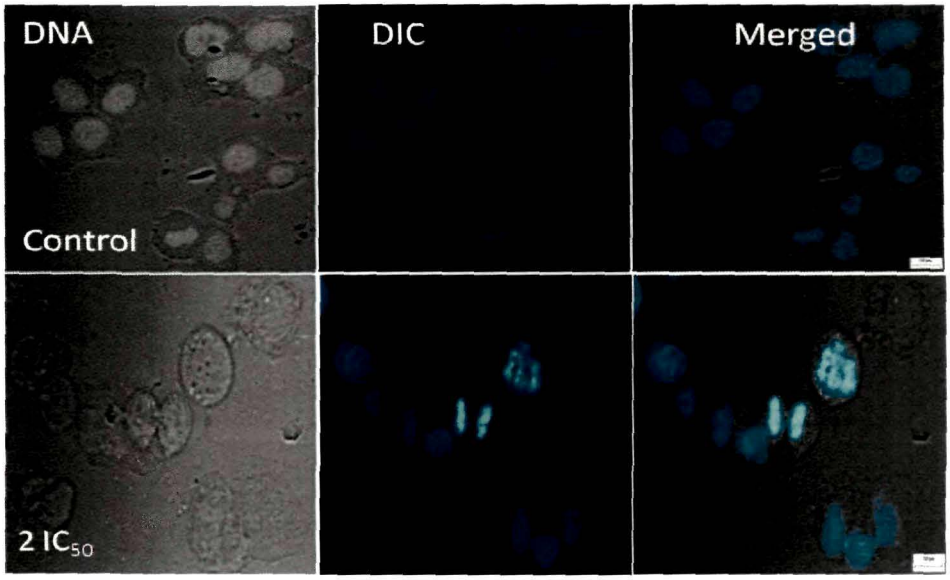


Fig. 4.21 Immunofluorescence microscopic analysis of Ruviprase-treated MCF-7 cells for assessment of nuclear accumulation of mitotic block. Ruviprase-treated MCF-7cells (0.5 x 10^5 cells ml^{-1}) were fixed in 1% formaldehyde for 30 min at room temperature (~23 °C) and suspended in 100 µl of EMEM. The cells were incubated with 5.0 µl of Hoeschst 33250 (10 mg ml^{-1}) for 30 min at 37 °C 5% CO_2 as described in section 3.2.4.7.4.3. Cells were observed under an immunofluorescence confocal laser scanning microscope at 40X magnification.

4.2.6.5.4 Determination of reactive oxygen species generation and mitochondrial transmembrane potential

Treatment of MCF-7 cells with Ruviprase (at 2 IC_{50} value) resulted in alteration of mitochondrial transmembrane potential (Fig. 4.22). As shown in the figure, mitochondrial depolarization is indicated by a decrease in the red/green fluorescence intensity ratio of the membrane-permeant JC-1 dye, in the Ruviprase-treated cells.

Ruviprase-treatment also induced formation of ROS (Fig. 4.23) in the treated cells as compared to control cells. As suggested in the figure, dihydroethidium dye is oxidized by the ROS produced in the treated cells leading to the release of ethidium which then fluoresces red when bound to cellular DNA.

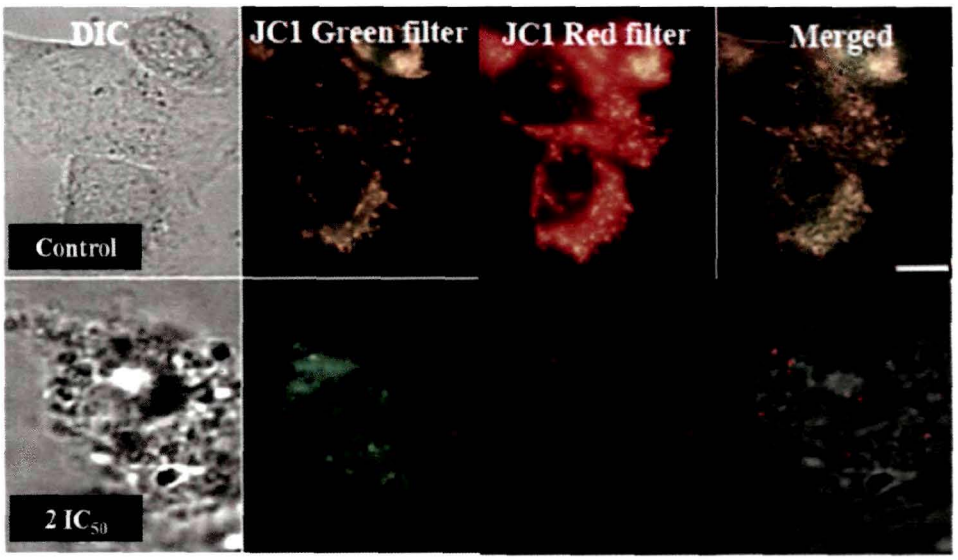


Fig. 4.22 Effect of Ruviprase treatment on the mitochondrial membrane potential of MCF-7 cells. MCF-7 cells (0.5 x 10^5 cells/ml) were treated with Ruviprase (2 IC_{50}) for 24 h at 37 °C and 5% CO_2. JC-1 dye was added for mitochondrial membrane potential activity detection as described in section 3.2.4.7.4.4. Images were taken using a fluorescence microscope under 60X objective.

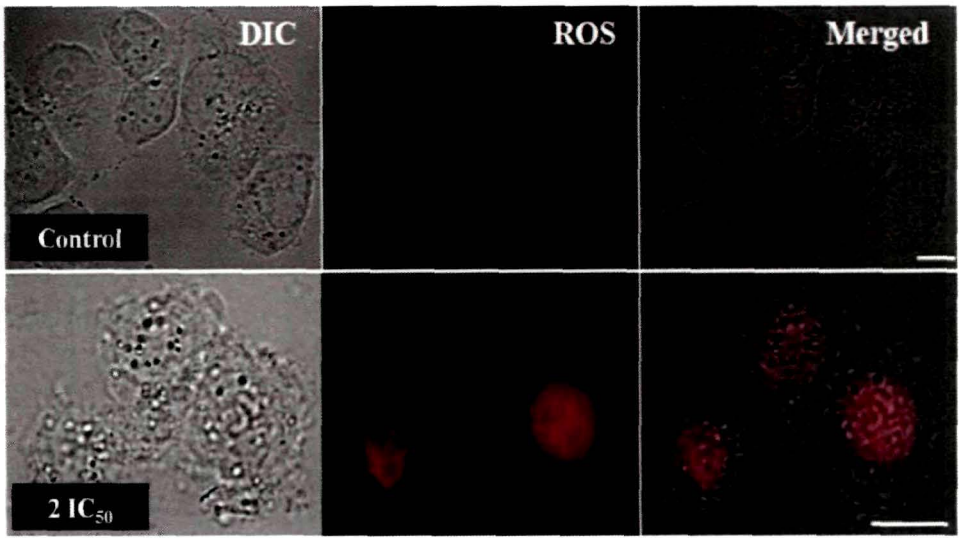


Fig. 4.23 Effect of Ruviprase treatment on the generation of ROS in MCF-7 cells. MCF-7 cells (0.5 x 10^5 cells/ml) were treated with Ruviprase (2 IC_{50}) and incubated for 24 h at 37 °C and 5% CO_2. Cells were treated with

dihydroethidium dye for ROS detection as described in section 3.2.4.7.4.4. Images were taken using a fluorescence microscope under 60X objective.

4.2.6.6 Acute *in vivo* toxicity assessment of Ruviprase on Balb/c mice

4.2.6.6.1 Lethality and *in vivo* toxicity in Balb/c mice

At a dose of 2 mg kg^{-1} body weight, Ruviprase was non-lethal to mice and the treated animals did not show any noticeable behavioral changes up to 48 h of *i.p.* treatment (Table 4.5).

4.2.6.6.2 *In vivo* anticoagulant effect

After 6 h of Ruviprase treatment, the whole blood clotting time of the treated group of mice (255.0 ± 8.0 sec) was found to be significantly higher ($p<0.05$) as compared to the whole blood clotting time of control group of mice (180.0 ± 9.0 sec) under identical experimental conditions (Table 4.5).

4.2.6.6.3 Effect on serum parameters

Biochemical analysis of serum from Ruviprase-treated mice (48 h post-treatment) showed that apart from alkaline phosphatase (ALP) and cholesterol, no significant changes were observed in the serum level of any of the other parameters tested as compared to their levels in the control group of mice (Table 4.6). However, the elevated levels of both ALP and cholesterol in Ruviprase-treated mice were found to be within the normal range (Table 4.6).

Table 4.5 Behavioral and physical changes in Ruviprase (2 mg kg^{-1}) treated mice (n=6) after 48 h of *i.p.* administration.

Parameters	**Control (Placebo)**	**Ruviprase-treated**
Survivality	No death	No death
Whole blood clotting time (s)	180.5 ± 21.2^{a}*	255.0 ± 8.0^{b}*
Grip strength (s)	30.18 ± 4.9^{c}	28.31 ± 4.2^{c}
Rectal temperature (°C)	35.0	35.0
Drooping of eyelids/ movement inability/ change in body weight	Not observed	Not observed
Food and water intake	Normal	Normal

*6 h of *i.p.* injection

Values are mean ± S.D. of triplicate determination. *Values in the same row with different superscripts are statistically significant ($p<0.05$)

Table 4.6 Biochemical parameters of serum of control and Ruviprase (2 mg kg^{-1}) treated mice (n=6) after 48 h of *i.p.* adminstration.

Serum parameters	**Control**	**Ruviprase-treated**
Glucose (mg dl^{-1})	237.25 ± 21.9	219.25 ± 14.5
SGOT (U L^{-1})	161.67 ± 8.5	150.25 ± 4.2
SGPT (U L^{-1})	53.0 ± 5.7	65.5 ± 6.3
CPK-MB (U L^{-1})	2.75 ± 0.4	2.83 ± 0.5
ALP (U L^{-1})	57.5 ± 14.2	72.75 ± 9.8*
Cholesterol (mg dL^{-1})	76.0 ± 3.8	100.2 ± 13.4*
Urea (mg dL^{-1})	33.0 ± 1.6	32.5 ± 1.2

All experimental values are represented as mean ± S.D.

The values marked with '*' were observed to be statistically significant with respect to control ($p<0.05$).

4.2.6.6.4 Effect on hematological parameters

Table 4.7 shows the *in vivo* effects of Ruviprase on the hematological parameters of blood. It was observed that none of the hematological parameters of mice was found to be significantly affected by Ruviprase treatment (Table 4.7).

Table 4.7 Hematological parameters of control and Ruviprase (2 mg kg^{-1}) treated mice (n=6) after 48 h of *i.p.* adminstration.

Hematological parameters	Control	Ruviprase-treated
WBC (M/mm^3)	6.2 ± 1.2	6.8 ± 1.3
Lymphocyte (%)	66.8 ± 3.6	67.0 ± 2.2
RBC (M/mm^3)	3.4 ± 0.2	4.4 ± 0.5
Mean Corpuscular Volume (MCV) (fl)	41.5 ± 5.7	43.3 ± 4.3
Mean Corpuscular Hb (MCH) (pg)	15.1 ± 1.5	14.7 ± 2.8
Mean Corpuscular Hb Conc (MCHC) (g dl^{-1})	36.5 ± 3.8	34.2 ± 4.9
Haemoglobin (Hb) (g dl^{-1})	5.2 ± 1.7	6.5 ± 1.5

All experimental values are represented as mean ± S.D.

4.2.6.6.5 Histopathological study

Histopathological study of liver, kidney, and heart tissues from Ruviprase-treated mice did not show any morphological alterations in these tissues as compared to tissues from the control group of mice (Fig. 4.24).

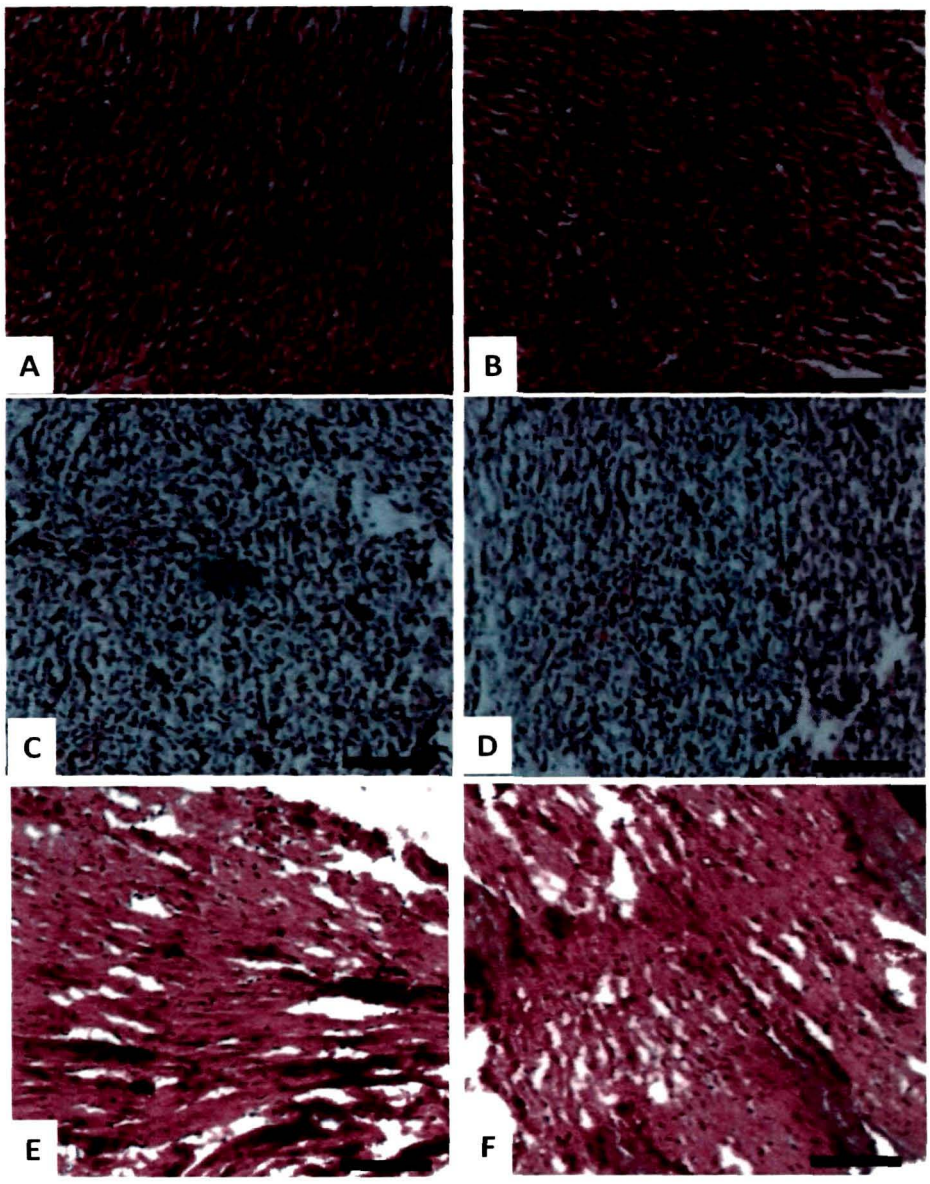


Fig. 4.24 Light microscopic observation of the effect of Ruviprase on liver, kidney, and heart tissues of Balb/c mice. The tissues were processed for microscopic observation by hematoxylene-eosin (H & E) staining as described in section 3.2.4.8.5. Tissues from mice treated with PBS served as respective controls. (A), Contol liver tissue, (B) Ruviprase-treated liver tissue, (C) Control kidney tissue, (D) Ruviprase-treated kidney tissue, (E) Control heart tissue, and (F) Ruviprase-treated heart tissue. Magnification 10X: Bar = 50 μm.

4.3 Discussion

The anticoagulant activity of a venom protein is not necessarily due its enzymatic activities [88]. There are several instances of venom proteins which render anticoagulant effects via non-enzymatic mechanisms, particularly due to interaction with blood coagulation proteins [59, 85, 98, 106, 113]. A number of potential inhibitors of thrombin and factor Xa have been reported from snake venom and from the salivary gland secretions of hematophagous insects [59, 94, 98, 106, 113, 180, 295-296, 315]. Venom of snakes belonging to the Viperidae family is reported to possess specific serine protease inhibitors [59, 98, 106, 113, 180] which belong to the C-type lectin protein family, the PLA_2 superfamily, or the Kunitz-type serine protease inhibitors. However, to date, there has been no report on the characterization of a dual inhibitor of thrombin and factor Xa from snake venom.

4.3.1 Ruviprase is a novel thrombin/factor Xa-inhibitory peptide isolated from RVV

Most of the serine protease inhibitors from snake venom have a molecular mass in the range of ~7 to 29 kDa [59, 98, 106, 113, 180]. However, low molecular mass thrombin and factor Xa inhibitors (<5 kDa) have been isolated from hematophagous insects such as tick [105, 240, 316]. Ruviprase, with a molecular mass of 4.4 kDa, is the smallest inhibitor of thrombin as well as factor Xa reported to date from snake venom. To ascertain whether Ruviprase is accidently produced during the extraction or drying of venom, we purified Ruviprase from several different batches of RVV that were from Eastern Indian origin. It is noteworthy that the results were identical in each case, suggesting that Ruviprase is produced and processed in the venom gland of Russell's Viper.

Partial N-terminal sequence analysis of Ruviprase suggests that, unlike most of the thrombin or factor Xa inhibitors isolated from snake venom, Ruviprase does not

belong to the C-type lectin protein family (snaclecs) or Kunitz-type serine protease inhibitors. Instead, the partial N-terminal sequence of Ruviprase displayed significant homology with the internal sequence of apoptosis-inducing peptides (10 kDa) identified by analysis of the venom-gland transcriptome of the *Ophiophagus hannah* and *Python bivittatus*, whose effect on the coagulation system is unknown. Secondary structure analysis showed that Ruviprase is primarily composed of α-helices and random coils, unlike some of the thrombin inhibitors isolated from tick such as Vareigin, which is a typical random coil protein [316], and snake venom Kunitz-type protease inhibitors such as Rusvikunin, which is primarily made of β-sheets and random coils [59, 106]. The secondary structure of Ruviprase showed partial similarity to snake venom thrombin inhibitors such as Bothrojaracin, a snaclec isolated from the venom of *Bothrops jararaca*, which typically consists of α-helices and β-sheets [317]. These data indicate that Ruviprase is a novel anticoagulant peptide isolated from RVV or any snake venom.

The catalytic activity of thrombin is regulated by the recognition domains – exosites-I and exosites-II, in addition to the active site [198-199]. Exosite-I of thrombin is the fibrinogen-binding site, whereas exosite-II is responsible for heparin binding [198-199]. Both sites are distant from the catalytic pocket but are involved in the specific binding of thrombin to several macromolecular substrates, inhibitors and modulators [198-199]. The binding of Ruviprase with thrombin or factor Xa was evident by fluorescence spectroscopy and SPR analyses. The binding was enhanced in the presence of Ca^{2+}, suggesting that divalent ions might contribute to increase in protein-protein interaction. However, Ruviprase demonstrated potent inhibitory effects on thrombin and factor Xa, even in the absence of divalent ions. Ruviprase binding resulted in the inhibition of the catalytic activity of thrombin towards its small chromogenic substrate, suggesting its interaction with the active site. The catalytic inhibition of thrombin by Ruviprase followed an uncompetitive model of inhibition similar to that shown by

rAaTI, a thrombin inhibitor isolated from the saliva of *Aedes aegypti* [318]. The mode of Ruviprase mediated thrombin inhibition was different from that of other potent snake venom thrombin inhibitors such as TI-Nh [94] and Rusvikunin [106], which follow a mixed model or non-competitive model of thrombin inhibition, respectively.

It has been well established that fibrinogen binding by thrombin is mediated through exosite-I, which is followed by the enzymatic cleavage of fibrinogen leading to its clotting [198]. Further, thrombin-induced aggregation of platelets is mediated by enzymatic cleavage of protease-activated receptors (PAR-1 and PAR-4 present on the surface of platelets) by thrombin [136]. As such, the enzymatic activity of thrombin is necessary for fibrinogen clotting and platelet aggregation [136]. Ruviprase inhibited the amidolytic activity, fibrinogen clotting, and platelet aggregation property of thrombin to an equal extent; therefore, it may be assumed that binding of Ruviprase to the catalytic site of thrombin results in inhibition of its enzymatic activity with a subsequent inhibition of its fibrinogen clotting or platelet aggregation property. Therefore, Ruviprase does not inhibit substrate binding (fibrinogen) to exosite-I of thrombin.

The heparin-AT-III complex strongly inhibits the catalytic properties of thrombin by binding to the heparin-binding site (exosite-II) of thrombin [198-199]. Pre-incubation of Ruviprase/thrombin prior to addition of heparin/AT-III lead to a partial decrease in the thrombin inhibitory property of the latter, suggesting that Ruviprase, in addition to binding to the active site, also binds to exosite-II (the heparin binding site) of thrombin. A probable mechanism of thrombin inhibition by Ruviprase is shown in Fig. 4.25. As shown in the figure, the mechanism of thrombin inhibition by Ruviprase is distinctly different from known snake venom thrombin inhibitors such as Bothrojaracin [180], TI-Nh [94], $RVAPLA_2$ [98] and Rusvikunin [106] as well as tick thrombin inhibitors such as Savignin [319] and Boophilin [320]. However, so far the data is not sufficient to conclude that

Ruviprase binds to exosite-II. Further structural analysis is essential to pinpoint the thrombin binding region of Ruviprase.

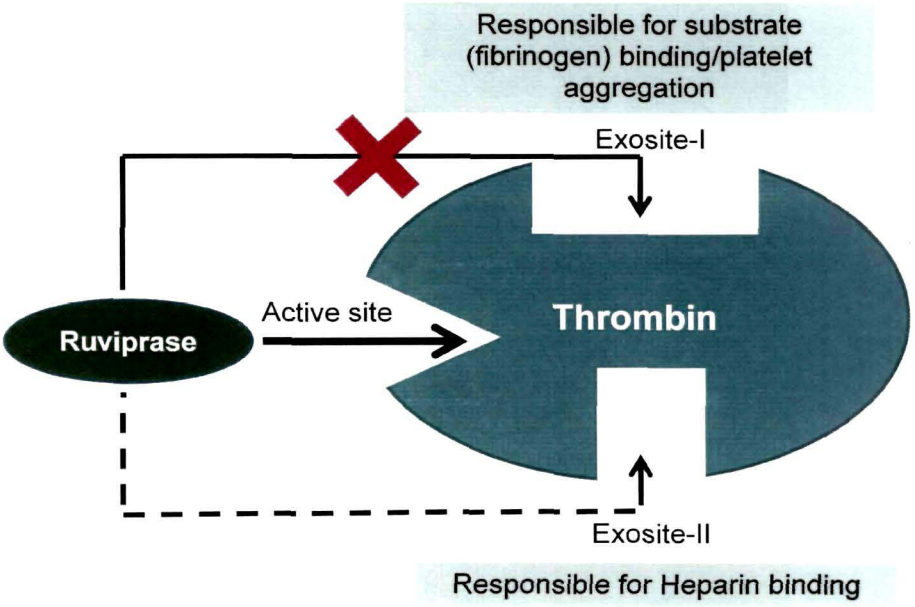


Fig. 4.25 A probable mechanism of thrombin inhibition by Ruviprase. Exosite-I of thrombin is the fibrinogen-binding site, whereas exosite-II is responsible for heparin binding. Active site is responsible for the catalytic property of thrombin. Ruviprase inhibited the amidolytic activity of thrombin suggesting its binding to the active site of thrombin. Heparin-AT-III complex demonstrated lesser thrombin inhibitory activity in presence of Ruviprase suggesting that Ruviprase also competed with heparin-AT-III complex for binding to Exosite-II. Ruviprase does not hamper substrate binding (fibrinogen) to exosite-I of thrombin.

Biosensor analysis suggested that the affinity of Ruviprase towards thrombin is higher than that of thrombin inhibitors isolated from hematophagus insects [321]. However, it showed lower affinity (K_D = 0.46 μM) in comparison to C-type lectins isolated from snake venom [180]. Ruviprase demonstrated a higher *Ki* value for thrombin inhibition compared with some other snake venom thrombin inhibitors such as Bothrojaracin (0.15 nM) [180], TI-Nh (73.0 nM) [93] and RVAPLA$_2$ (0.38

μM) [98]. Nevertheless, unlike these inhibitors, Ruviprase also demonstrated inhibition of the prothrombin-activating property of factor Xa; therefore, it turns out to be a superior anticoagulant agent compared to the snake venom thrombin inhibitors described to date.

The factor Xa inhibitors are basically classified under two groups- direct and indirect factor Xa inhibitors [198]. Direct factor Xa inhibitors are those which inhibit factor Xa directly without the requirement of any intermediate, whereas indirect factor Xa inhibitors demonstrate an AT-III mediated factor Xa inhibition [190]. To date there has been no report on the characterization of factor Xa inhibitors from snake venom. Ruviprase demonstrated direct factor Xa inhibition via a mixed model of inhibition, indicating that Ruviprase can inhibit free factor Xa or prothrombin (substrate) bound factor Xa. Besides, since Hill's co-efficient for binding between Ruviprase and thrombin (1.08) or factor Xa (5.21) was more than 1.0, a positive cooperative binding between the above coagulation factors and the inhibitor (Ruviprase) can be predicted. Additionally, binding of Ruviprase to thrombin or factor Xa follows slow binding kinetics because the binding strength of Ruviprase to thrombin/factor Xa increases with increased pre-incubation time. Taken together, the anticoagulant nature of Ruviprase is substantially attributed to its non-enzymatic mechanism of thrombin and factor Xa inhibition, and it furnishes an example of a component of snake venom exerting its biological activity through protein (venom)-protein (predator/victim) interaction [59, 94, 98, 106, 113].

Due to the low immunogenicity of low molecular mass snake venom proteins and peptides, neutralization of such venom components by commercial antivenoms is always a challenge [322]. The inability of both commercial polyvalent and monovalent antivenom to neutralize the physiological properties of Ruviprase clearly suggests the necessity of a better antivenom therapy for the efficient management of snakebite patients.

4.3.2 Ruviprase is a non-toxic anticoagulant peptide purified from RVV which might be suitable for development of peptide-based therapeutic drugs for treatment of cardiovascular ailments

Ruviprase, at the tested dose of 2 mg kg^{-1} (*i.p.*), did not induce mortality or toxicity in mice after 48 h of injection, suggesting that it is non-lethal. Ruviprase represents 0.1% (w/w) of total RVV protein and on average an adult Russell's viper may inject 225–250 mg of venom (the total amount in a bite) to its victim [96]. Therefore, after a full Russell's viper bite, 180-200 μg of Ruviprase gets injected, which corresponds to a nanomolar concentration (7.4 nM-8.2 nM) of Ruviprase in the blood of an adult victim. Therefore, it is unlikely that Ruviprase exerts anticoagulant activity in an adult human victim; however, its pathobiological effect in mice (concentration in blood ~1.6 μM) and rats (concentration in blood ~2.5 μM) cannot be ruled out. There are reports of several weak or non-toxic proteins/peptides of venom that exert a synergistic effect with other components of venom to enhance the lethality of venom [59, 186, 275-278]. Therefore, association of Ruviprase with other component(s) of the same venom to form a protein complex may contribute to the overall lethality of RVV envenomation (Mukherjee, A.K., Thakur, R, unpublished observation).

Thrombosis-associated disorders such as cardiovascular diseases are a leading cause of mortality in both developing and developed countries [206]. Unfortunately, many natural anticoagulants used as antithrombotic drugs are associated with adverse side effects. Heparin and warfarin are two of the potent commercial anticoagulants used for the treatment of thrombosis-associated ailments. Heparin does not possess anticoagulant activity by itself; however, it shows anticoagulant effect by binding with AT-III present in plasma leading to thrombin/ factor Xa inhibition [323]. Warfarin, on the other hand, inhibits the synthesis of biologically active forms of the calcium-dependent clotting factors II, VII, IX and X, as well as the regulatory factors such as protein C, protein S, and protein Z [324]. Both of these compounds demonstrate

severe adverse effects mostly accompanied with bleeding complications. As such, snake venom derived potent anticoagulants have been considered as an alternative for the development better antithrombotic agents [93, 98, 106, 113, 180]. Additionally, low toxicity anticoagulants are in great demand for therapeutic use in the prevention and treatment of occlusive thrombosis. The inability of physiological inhibitors in neutralizing the anticoagulant potency and inhibitory effects of Ruviprase suggests that it might be effective under *in vivo* conditions. Ruviprase at the therapeutic dose of plasmin (2 mg kg^{-1}) [292] does not show toxicity or adverse pharmacological effects but demonstrates strong anticoagulant activity. This suggests the suitability of Ruviprase for development of peptide-based therapeutic drugs for treatment of cardiovascular ailments [93, 98, 106, 113,180].

4.3.3 Ruviprase possesses potent anticancer activity against breast cancer cells

It has been estimated that breast cancer accounts for almost 28% of all cancers among women [325]. The chemotherapeutic agents such as paclitaxel and anthracycline used to treat all the stages of breast cancer have several side effects and cause drug resistance; therefore, there is an urgent need to search for safe and potent anticancer drugs [326]. It has also been established that thrombin mediated fibrin clot formation plays a vital role in camouflaging tumor cells in addition to promoting their attachment to the vascular endothelium, thus aiding cancer progression and metastasis [327]. Though, it has been postulated that thrombin inhibitors can be a potential anticancer agent nevertheless, none of the exogenous thrombin inhibitors identified from snake venom or hematophagous insects, except for Suramin [240], have been tested for anticancer property. Ruviprase by virtue of demonstrating potent cytotoxicity towards tested tumor cells and being particularly effective towards MCF-7 cells through induction of apoptosis by causing loss of membrane integrity and DNA damage in cancer cells suggests its anticancer potential specifically towards breast cancer cells. Though many snake venom

cytotoxins have been reported to demonstrate anticancer potential towards various cell line [328], the low IC_{50} value of Ruviprase towards the tested cells undoubtedly indicate its superiority to cytotoxins isolated from snake venom particularly from Russell's Viper venom, for example drCT-I [281].

Apoptosis is the process of programmed cell death and insufficient expression of various cellular proteins actively participate in apoptotic signaling may result in uncontrolled cell proliferation, such as cancer [329]. Therefore, cell cycle arrest along with DNA disorganization is regarded as an initial stage for apoptosis [329]. The p53 dependent apoptosis occurs by p53 protein activation and its nuclear localization which in turn leads to p21 up-regulation [330]. Furthermore, p53 is also involved in inhibiting cell cycle progression from the G2 phase to mitosis (the G2/M checkpoint). Therefore, Ruviprase treatment to MCF-7 cells results in a p53-dependent G2 arrest.

Production of ROS has been shown to be associated with inhibition of mitochondrial respiration chain resulting in rupture and change in membrane potential of mitochondria is a hallmark of apoptosis induction [306, 328]. Further, a close relationship between ROS generation and p53 activation has been shown in mitochondrial mediated apoptosis induction [331]. Venom cytotoxins exhibit anti-carcinogenic effects on different cancel cell-lines by causing DNA fragmentation and inducing apoptosis in a caspase-dependent mechanism by increasing the expression of p53 and moderating the expression of p21/p27 [321]. Certain toxins from snake venom decrease the mitochondrial transmembrane potential and increase the production of ROS leading to apoptosis in a manner similar to that of Ruviprase [328, 332]. Therefore, it may be anticipated that Ruviprase induces mitochondria dependent apoptosis via excess ROS generation which provokes oxidative modification of MCF-7 cells including inhibition of its protein function [331] as well as alteration of mitochondrial membrane potential [304-306, 328]. However, unlike Ruviprase, none of the snake venom cytotoxins with anticancer

potential have been reported to possess anticoagulant effects [328, 333] which may have additional advantage for prevention of cancer progression.

CHAPTER V

Characterization and elucidation of mechanism of action of a procoagulant metalloprotease purified from Indian Russell's Viper (Daboia russelii russelii) venom.

RESULTS AND DISCUSSION

5 Characterization and elucidation of mechanism of action of a procoagulant metalloprotease purified from Indian Russell's Viper (*Daboia russelii russelii*) venom

5.1 Brief introduction

Venoms from snakes belonging to the Viperideae family are generally procoagulant in nature. The procoagulant proteins characterized to date are either serine proteases or metalloproteases which possess the ability to activate the zymogen of specific coagulation factors in the coagulation cascade to induce or accelerate blood coagulation. Many of these procoagulant proteinases activate the coagulation factors such as factor VII [122], factor X [117, 144-146], and prothrombin [86, 139-143] or activate the protein cofactor, factor V [113, 125-128].

One of the central reactions of the coagulation cascade which is often a target of SVMPs is the process of prothrombin activation. Prothrombin activation is a vital reaction in the blood coagulation cascade resulting in the formation of thrombin which catalyzes the final step of the coagulation process [138]. Under physiological conditions, factor Xa catalyzed prothrombin activation is substantially enhanced (~300,000 fold) in the presence of the prothrombinase complex formed by the assemblage of Ca^{2+}, negatively charged phospholipids and factor Va [138]. On the basis of their structural properties and co-factor requirements during prothrombin activation, snake venom prothrombin activators have been classified into four groups as discussed in section 1.9.1.1.3 [86, 139-143].

Snake venom harboring procoagulant proteins such as prothrombin activators mainly provoke and promote coagulation in the victim/prey thereby leading to disseminated intravascular coagulopathy (DIC) and other symptoms [87]. In spite of the fact that Russell's Viper envenomation results in symptoms mainly

activating principle has been isolated from RVV till date. Therefore, the aim of the present study was to isolate, characterize and elucidate the mechanism of action of a procoagulant metalloprotease and to evaluate its *in vivo* pathophysiological affects in Balb/c mice.

5.2 Results

5.2.1 Purification of a procoagulant protease from RVV

5.2.1.1 Fractionation of crude RVV on Sephadex G-50 gel-filtration column

Fractionation of crude RVV (25.0 mg) on Sephadex G-50 gel-filtration column resolved the venom into 11 peaks (Fig. 5.1A). The peak GF-I was found to have proteolytic effect on fibrinogen and also demonstrated strong procoagulant effects on citrated PPP (Table 5.1).

5.2.1.2 Fractionation of GF-I through cation-exchanger HiPrep CM FF 16/10 FPLC column

The first gel-filtration peak (GF-I) (5.7 mg), showing fibrinogenolytic activity as well as strong procoagulant activity was subjected to HiPrep CM FF 16/10 cation-exchange column (16 mm x 100 mm) coupled to an FPLC system. This resulted in the separation of GF-I into 6 peaks (GF-I-CM-I, GF-I-CM-II, GF-I-CM-III, GF-I-CM-IV, GF-I-CM-V and GF-I-CM-VI) (Fig. 5.1B). The fourth peak, GF-I-CM-IV (named Rusviprotease) was found to be homogenous by SDS-PAGE analysis (described in section 5.2.2.1). This fraction demonstrated proteolytic activity as well as procoagulant property (Table 5.1) and was considered for further characterization.

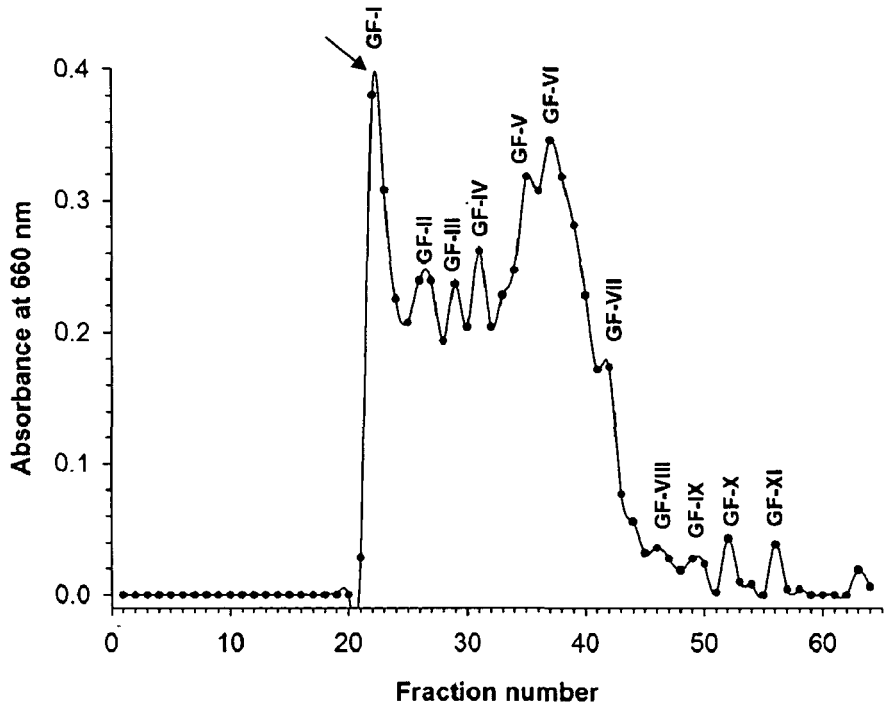


Fig. 5.1A Fractionation of crude RVV through Sepahdex G-50 gel-filtration column (1 cm x 60 cm). The column was equilibrated with 20 mM potassium phosphate buffer (pH 7.0) and was loaded with 25.0 mg of crude RVV dissolved in 1.0 ml of the same buffer. The elution of proteins was performed at a flow rate of 20 ml h^{-1} with the equilibration buffer at room temperature (~23 °C). 1.0 ml fractions were collected and the protein content of each fraction was measured at 660 nm as described in section 3.2.3.1. The fibrinogenolytic as well as plasma recalcification time of each protein peak was screened following the methods described in sections 3.2.3.3.1 and 3.2.4.1 respectively.

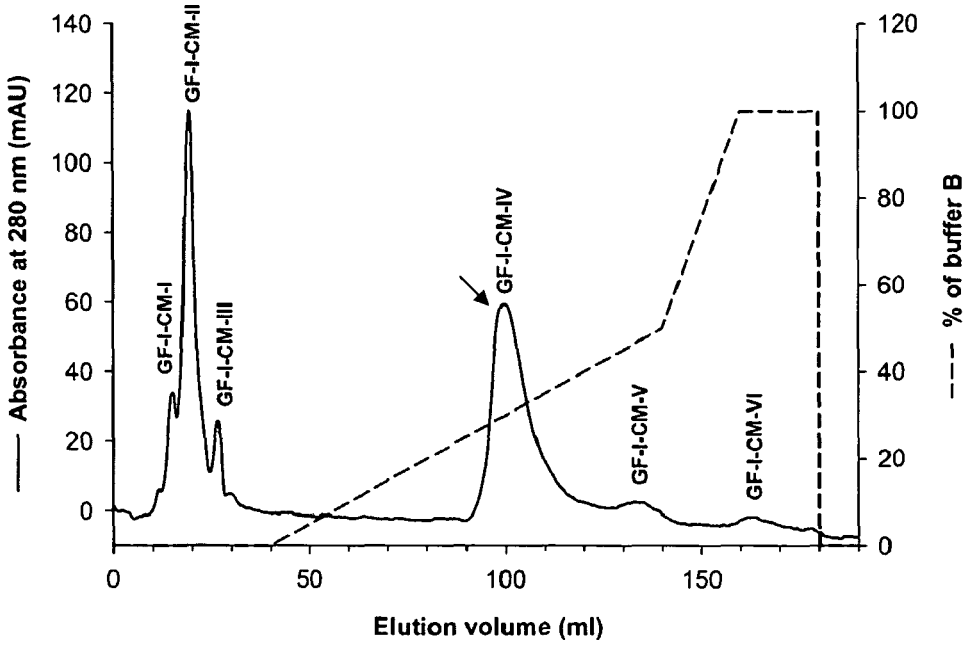


Fig. 5.1B Purification of Rusviprotease by FPLC on a HiPrep CM FF 16/10 column (20 mm x 100 mm). GF-I (5.7 mg) was applied to the cation-exchange (HiPrep CM FF 16/10) column, coupled to an FPLC system, and the elution was performed at a flow rate of 1.0 ml min^{-1} using 20 mM potassium phosphate buffer, pH 7.0 (solvent A) and a linear gradient of 20 mM potassium phosphate buffer, pH 7.0 containing 1.0 M NaCl (solvent B), at 4 °C, as described in section 3.2.1.2. The absorbance was monitored at 280 nm and 1.5 ml fractions were collected. The fibrinogenolytic as well as plasma recalcification activity of each protein peak was screened following the methods described in sections 3.2.3.3.1 and 3.2.4.1, respectively.

The yield of the isolated protein was 3.2% of the total crude venom load. A summary of purification of Rusviprotease is shown in Table 5.1.

At a tested dose of 25 μg ml^{-1} (1.0 μmol l^{-1}), Rusviprotease did not show any detectable PLA_2 activity.

Table 5.1 Summary of purification of a procoagulant protease (Rusviprotease) from RVV.

Fraction	Total protein (mg)	Protein yield (%)	Fibrinogenolytic activity		Coagulant activity		Purification fold	
			Total activity (Unit*)	Specific activity (Units mg^{-1})	Total activity (Unit#)	Specific activity (Units mg^{-1})	Fibrinogenolytic activity	Coagulant activity
Crude RVV	25.0	100	1.0×10^4	4.0×10^2	3.2×10^5	1.3×10^4	1	1
Gel-filtration fraction (GF-I)	5.7	22.8	2.1×10^4	3.7×10^3	2.1×10^5	3.7×10^4	9.25	2.8
Cation-exchange fraction (GF-I-CM-IV or Rusviprotease)	0.8	3.2	1.1×10^4	1.4×10^4	7.5×10^4	9.3×10^4	36.0	7.2

Values represent a typical experiment.

*One unit (U) of fibrinogenolytic activity has been defined as *n* mole equivalent of tyrosine formed min^{-1} [292].

#One unit of procoagulant activity has been defined as 1.0 s decrease in the clotting time of PPP as compared to the clotting time of normal PPP (control) under identical assay conditions [65, 291].

5.2.2 Molecular mass determination of Rusviprotease

5.2.2.1 SDS-PAGE analysis of Rusviprotease preparation

The molecular mass of Rusviprotease was analysed by 12% SDS-PAGE, under both reducing and non-reducing conditions (Fig. 5.2A). SDS-PAGE analysis of Rusviprotease under reducing condition revealed a single band with a molecular mass of 15.1 kDa (Fig. 5.2A). However, under non-reducing condition, Rusviprotease existed in the form of a probable dimer with a molecular mass of ~26 kDa (Fig. 5.2A). Further, Rusviprotease also existed naturally in other multimeric forms such a ~52 kDa tetramer and a ~104 kDa hexamer, which were visible under non-reducing SDS-PAGE conditions (Fig. 5.2A).

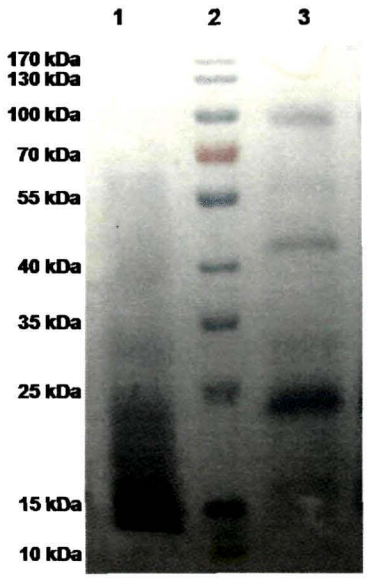


Fig. 5.2A SDS-PAGE analysis of Rusviprotease. Purified Rusviprotease was separated by 12% SDS-PAGE and the protein bands were visualized by staining with 0.25% Coomassie Brilliant Blue, under both reducing (R) and non-reducing (NR) conditions, as described in section 3.2.2.1.1. Lane 1, Rusviprotease (20 µg, R); lane 3, Rusviprotease (20 µg, NR), lane 2, prestained protein ladder (PageRuler prestained protein ladder #SM67172, Thermo Scientific, USA).

The molecular mass of the isolated protease was calculated on the basis of its *Rf* value in comparison with that of known protein standards, where a linear dependency of log Mw *Vs* migration of protein bands (*Rf* value) were observed (Fig. 5.2B).

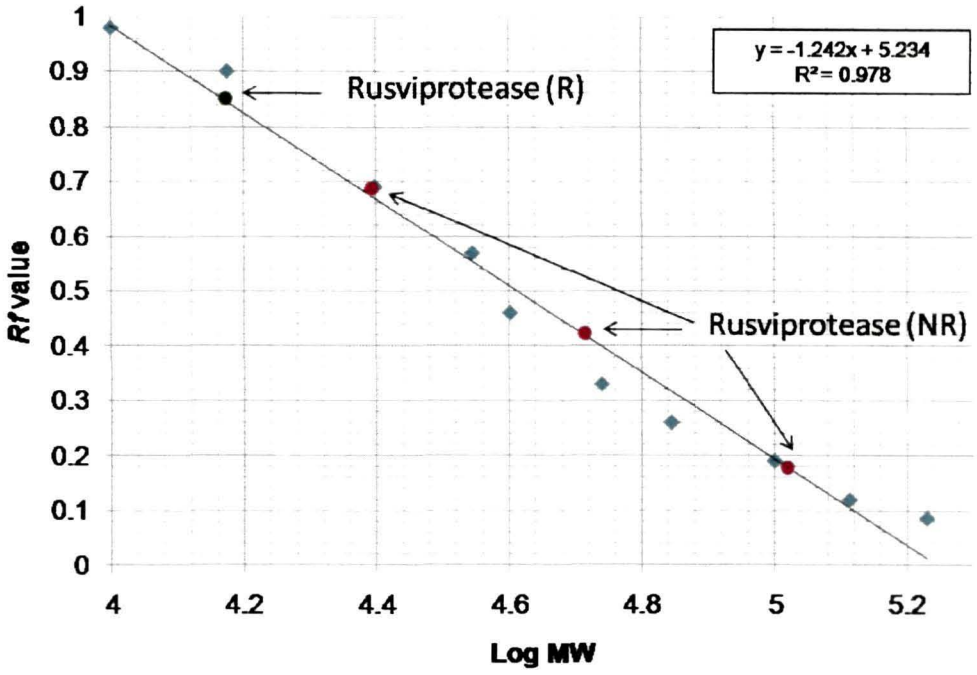


Fig. 5.2B Calibration curve of the protein ladder, #SM67172 (PageRuler prestained protein ladder) for molecular mass determination of Rusviprotease. The molecular mass of Rusviprotease under reducing (R) and non-reducing (NR) conditions was calculated by using the equation for straight line obtained in the above graph, where 'x' is the Log Mw and 'y' is the *Rf* value of Rusviprotease.

5.2.2.2 MALDI-TOF-MS analysis

MALDI-TOF-MS analysis of Rusviprotease revealed a major peak at 26.8 kDa suggesting the dimeric form of the protease with an additional peak at 53.8 kDa suggesting that Rusviprotease also existed in a tetrameric form (Fig. 5.3).

However, no peak was observed near 14-20 kDa range, suggesting that Rusviprotease did not exist in its monomeric form but existed in the form of a dimer and other multimeric forms in RVV (Fig. 5.3). Since Rusviprotease exists predominantly in a dimeric form, the molecular mass of 26.8 kDa was used for calculation of its molarity in the subsequent experiments.

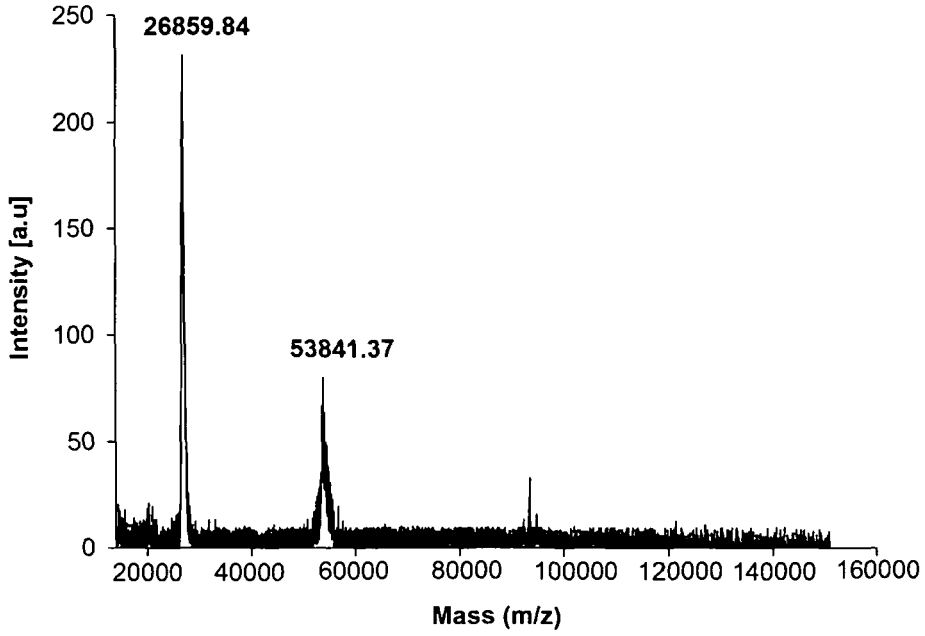


Fig. 5.3 MALDI-TOF-MS spectra of Rusviprotease. Mixtures of 0.5 µl sinapinic acid matrix (10 mg ml^{-1}) and 0.5 µl of a Rusviprotease solution (10.0 µg $µl^{-1}$, diluted in water containing 0.1% TFA) was spotted onto an Opti-TOF-384 plate (ABSciex), dried, and analyzed in positive linear mode using an acceleration voltage of 25 kV on an AB 4800-Plus MALDI-TOF-TOF instrument as described in section 3.2.2.1.2. Spectra were acquired in the m/z range of 20,000–150,000. The spectrum shows signal at m/z 26859.84 and 53841.37 corresponding to the molecular mass of 26.8 and 53.8 kDa, respectively.

5.2.3 Identification of Rusviprotease by LC MS/MS

LC MS/MS analysis of the tryptic digested peptides of Rusviprotease identified three peptides with greater than 62.1% probability (Table 5.2A). BLAST analysis of the tryptic fragments of Rusviprotease displayed 100% sequence similarity (100% sequence coverage) with snake venom metalloproteases (Table 5.2B). BLAST analysis of one of the tryptic fragments of Rusviprotease viz. R.NQCISLFGSR.A (1180.57 Da) displayed 100% sequence similarity (100% sequence coverage) with venom metalloproteases particularly factor X activators from *D. russelii* venom (accession numbers 2E3X_A, AAB22477.1 and ADW54332.1), fibrinogenases from *Echis ocellatus* venom (accession number Q2UXR0.1) as well as prothrombin activators such as Ecarin from *E. carinatus* venom (accession number Q90495.1), to name a few. The other tryptic peptide sequences, R.GYCYNGDCPIMR.N (1520.58 Da) and R.KIPCAPQDVK.C (1154.61 Da) of Rusviprotease also showed 100% sequence similarity with snake venom metalloproteases (78-100% sequence coverage) exhibiting factor X activating properties such as RVV-X from *D. russelii* venom (accession number 2E3X_A) and prothrombin activators such as EoMP06 isolated from venom *of E. ocellatus* (accession number Q90495.1) and Basparin A isolated from *Bothrops asper* (accession number P84035.1).

Table 5.2A Identification of Rusviprotease by LC MS/MS analysis. Tryptic peptides identified from Rusviprotease after LC MS/MS analysis.

Calculated mass (Da)	**m/z**	**z**	**Sequence**
1180.57	591.29	2	R.NQCISLFGSR.A
1520.58	761.30	2	R.GYCYNGDCPIMR.N
1154.61	383.87	3	R.KIPCAPQDVK.C

Table 5.2B Identification of the tryptic peptides of Rusviprotease (# According to NCBI database).

Peptide	Identified peptides	Identified protein	Species	Query coverage	Identity	Accession number#
1	R.NQCISLFGSR.A	RVV Metalloproteinase	*Daboia siamensis*	100	100	2E3X_A
	R.NQCISLFGSR.A	RVV-X-heavy chain	*Daboia siamensis*	100	100	AAB22477.1
	R.NQCISLFGSR.A	Group III SVMP	*Echis ocellatus*	100	100	ADW54332.1
	R.NQCISLFGSR.A	Eoc1	*Echis ocellatus*	100	100	Q2UXR0.1
	-. NQCI- LF ----. -	Ecarin	*Echis carinatus*	58	86	Q90495.1
2	R.GYCYNGDCPIMR.N	RVV Metalloproteinase	*Daboia siamensis*	100	100	2E3X.A
	R.GYCYNGDCPIMR.N	RVV-X-heavy chain	*Daboia siamensis*	100	100	AAB22477.1
	- .GYCYNGDCPIM-.-	EoMP06	*Echis ocellatus*	78	100	Q90495.1
3	R.KIPCAPQDVK.C	RVV Metalloproteinase	*Daboia siamensis*	100	100	2E3X_A
	R.KIPCAPQDVK.C	RVV-X-heavy chain	*Daboia siamensis*	100	100	AAB22477.1
	RKIPCAP-DVK	Basparin A	*Bothrops asper*	91	91	P84035.1

5.2.4 Determination of secondary structure of Rusviprotease: Circular dichroism spectroscopy

The secondary structure of Rusviprotease suggested that it was primarily composed of β-sheets (61.9%) and random coils (30.5%) (Fig. 5.4). The rest of the native structure of Rusviprotease was composed of α-helix (7.2%) and turns (0.3%) (Fig. 5.4).

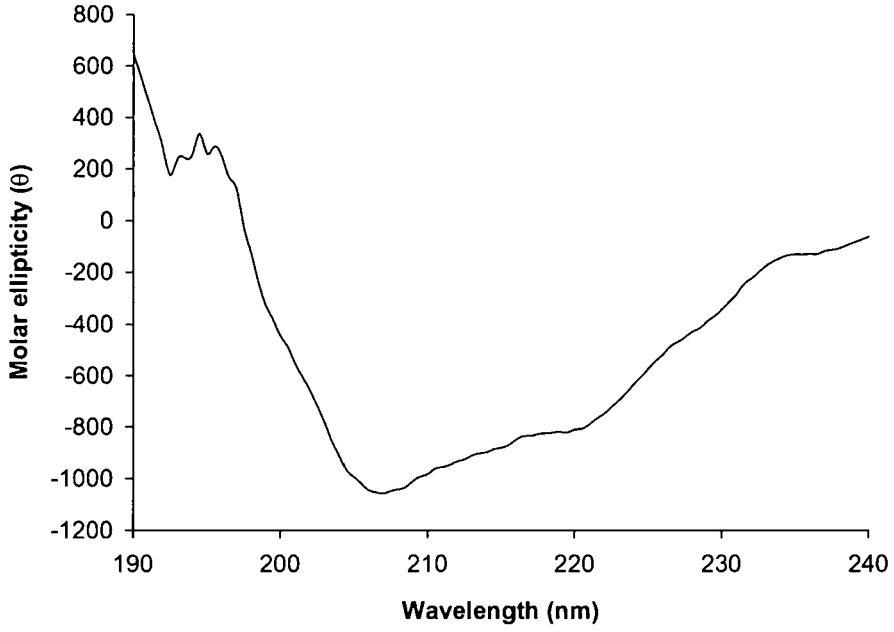


Fig. 5.4 Circular dichroism spectra of Rusviprotease showing its secondary structure. Native Rusviprotease (0.3 mg ml^{-1}) was dissolved in 20 mM potassium phosphate buffer pH 7.0 and the far UV-CD spectra was recorded at room temperature (~23 °C) between 190–240 nm against the appropriate buffer (blank) as described in section 3.2.2.4.

5.2.5 Biochemical characterizations of Rusviprotease

5.2.5.1 Protease activity

Protease activity of Rusviprotease towards the pseudo-substrate casein as well as different natural protein substrates is shown in Table 5.3. Rusviprotease demonstrated highest specificity towards bovine fibrinogen followed by casein and fibrin. Rusviprotease could not hydrolyse bovine albumin and globulin at the tested dose.

Table 5.3 Proteolytic activity of Rusviprotease towards protein substrates. Rusviprotease (1.0 μg ml^{-1} or 0.04 μmol l^{-1}) was incubated with the substrates for 90 min at 37 °C, pH 7.4 and the release of free tyrosine was calculated as described in section 3.2.3.2. One unit (U) of proteolytic activity has been defined as *n* mole equivalent of tyrosine formed min^{-1} which is calculated from a standard tyrosine curve.

Protease activity	Specific activity (Units mg^{-1} protein)
Bovine fibrinogen (0.25%)	$2.5 \times 10^4 \pm 115.0$
Casein (1.0%)	$1.4 \times 10^4 \pm 62.5$
Bovine fibrin (0.6%)	$2.9 \times 10^2 \pm 87.2$
Bovine albumin (1.0%)	Not detected
Bovine globulin (1.0%)	Not detected

Values are mean ± S.D. of triplicate experiments.

5.2.5.2 Fibrinogenolytic activity of Rusviprotease

5.2.5.2.1 Kinetics of fibrinogen hydrolysis

Because Rusviprotease demonstrated strongest proteolytic specificity towards bovine fibrinogen, therefore, the kinetics of fibrinogen hydrolysis by Rusviprotease was evaluated (Fig. 5.5). The *Km* and *Vmax* values of

fibrinogenolytic activity of Rusviprotease were calculated to be 3.14 μM and 78.7 nM min^{-1}, respectively.

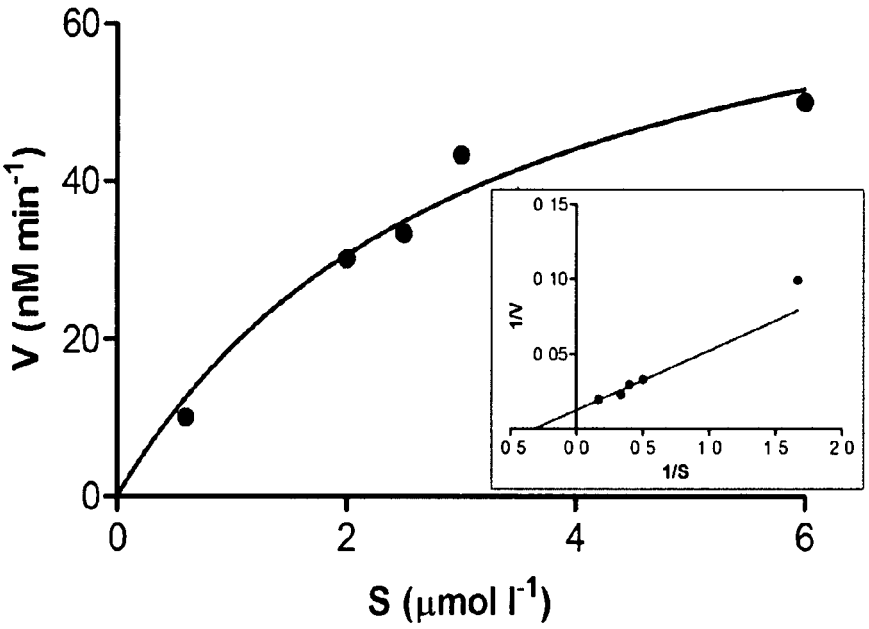


Fig. 5.5 Kinetics of fibrinogen hydrolysis by Rusviprotease. The Michaelis-Menten and Lineweaver-Burk double-reciprocal (inset) plots depicting fibrinogen hydrolysis (0.6-6.0 μmol l^{-1}) by Rusviprotease (0.04 μmol l^{-1}). The experiment was performed as described in section 3.2.3.3.4. The values are mean of triplicate determinations.

5.2.5.2.2 SDS-PAGE analysis of fibrinogenolytic activity

To elucidate the mechanism of fibrinogenolytic activity of Rusviprotease, the Rusviprotease-digested fibrinogen was analyzed by SDS-PAGE. Rusviprotease preferentially degraded the Aα-chain of fibrinogen (Fig. 5.6); however, the Bβ- and γ-chain of fibrinogen was not degraded even after 8 h of incubation (Fig. 5.6).

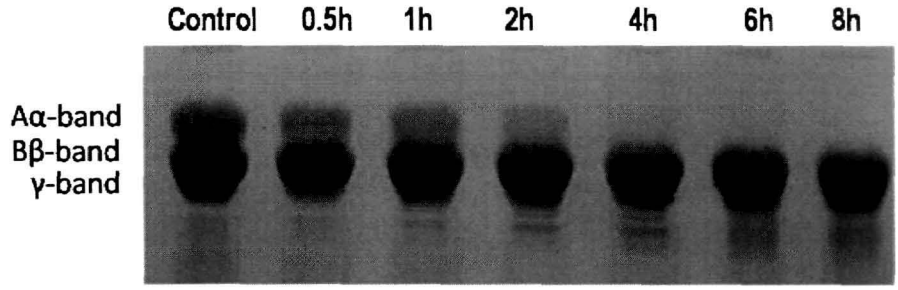


Fig. 5.6 SDS-PAGE analysis of fibrinogen degradation by Rusviprotease. Fibrinogen (final concentration of 2.5 mg ml^{-1}), was incubated with Rusviprotease (final concentration 0.4 µmol l^{-1}) and the fibrinogen degradation products were separated by 12% SDS-PAGE as described in section 3.2.3.3.2. Lane 1, control (fibrinogen without Rusviprotease; lanes 2-7, fibrinogen degradation pattern by Rusviprotease after 0.5, 1, 2, 4, 6 and 8 h of incubation; respectively at 37 °C, pH 7.4.

5.2.5.2.3 SDS-PAGE analysis of fibrinolytic activity

To elucidate the mode of action of Rusviprotease on fibrin, the protease-digested fragments of fibrin were analyzed using SDS-PAGE. Rusviprotease could not degrade fibrin even after 8 h of incubation (Fig. 5.7).

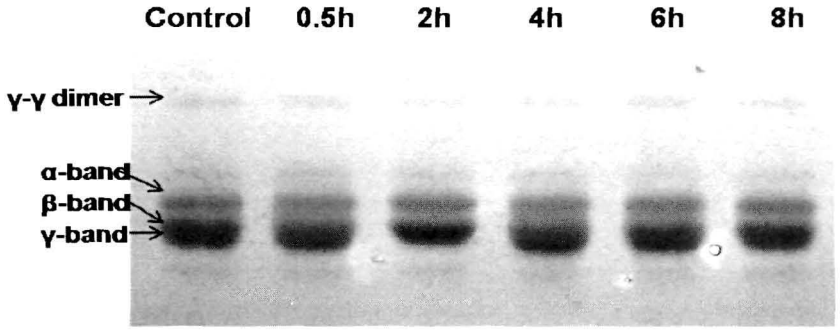


Fig. 5.7 SDS-PAGE analysis of fibrin degradation by Rusviprotease. Fibrin clot (40.0 µl of fibrinogen solution (6.0 mg ml^{-1}) clotted with 3.0 µl thrombin (0.2 NIH U ml^{-1}) at room temperature (~23 °C for 60 min)), was incubated with

Rusviprotease (final concentration of 0.4 μmol l^{-1}) and the fibrin degradation products were separated by 12% SDS-PAGE as described in section 3.2.3.3.2. Lane 1, control (fibrin without Rusviprotease); lanes 2-6, fibrin degradation pattern by Rusviprotease after 0.5, 2, 4, 6 and 8 h of incubation; respectively at 37 °C, pH 7.4

5.2.5.3 Optimum temperature for fibrinogenolytic activity

The optimum temperature for the protease (fibrinogenolytic) activity of Rusviprotease was assayed at a temperature range of 25-50 °C (Fig. 5.8). Rusviprotease showed optimum catalytic activity at 37 °C (Fig. 5.8).

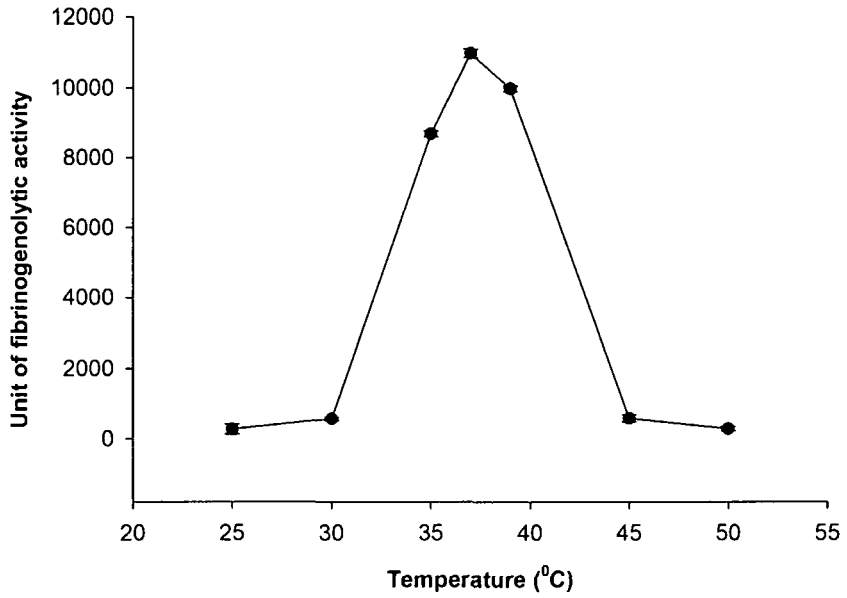


Fig. 5.8 Effect of temperature on the fibrinogenolytic activity of Rusviprotease. Fibrinogenolytic activity of Rusviprotease (0.04 μmol l^{-1}) was analysed at different temperature, after 30 min of incubation at pH 7.4. One unit (U) of fibrinogenotic activity has been defined as *n* mole equivalent of tyrosine formed min^{-1} which is calculated from a standard tyrosine curve. The experiment was performed as described in section 3.2.3.18.2. Values are mean ± S.D. of triplicate experiments.

5.2.5.4 Optimum pH for fibrinogenolytic activity

The optimum pH for the protease (fibrinogenolytic) activity of Rusviprotease was assayed at a pH range of 6.0-9.0 at a fixed temperature of 37 °C after 90 min of incubation (Fig. 5.9). Rusviprotease showed optimum catalytic activity at a pH range of 7.4-7.5 (Fig. 5.9).

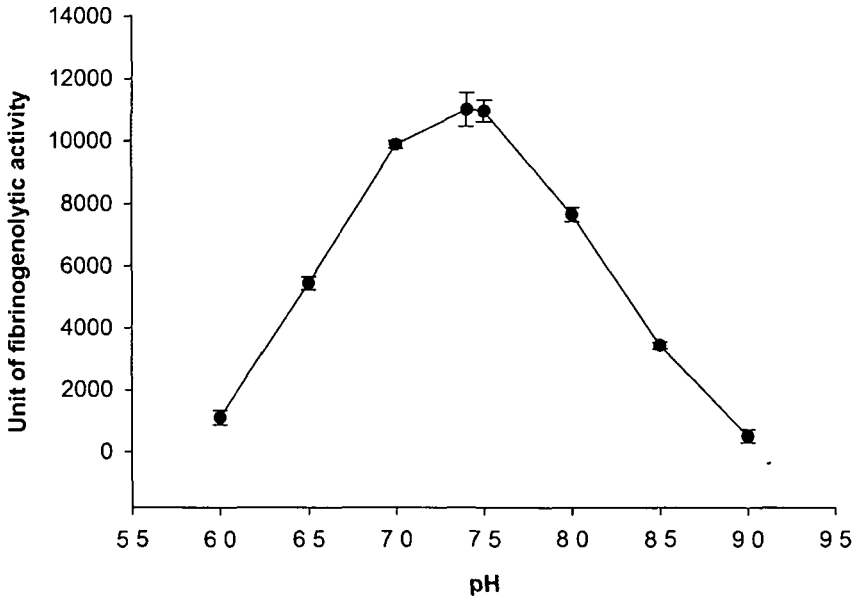


Fig. 5.9 Effect of pH on the catalytic activity of Rusviprotease. Fibrinogenolytic activity of Rusviprotease (0.04 μmol l^{-1}) was analysed at different pH, after 30 min of incubation at 37 °C. One unit (U) of fibrinogenolytic activity has been defined as *n* mole equivalent of tyrosine formed min^{-1} which is calculated from a standard tyrosine curve. The experiment was performed as described in section 3.2.3.18.3. Values are mean ± S.D. of triplicate experiments.

5.2.5.5. Amidolytic activity assay

Rusviprotease did not demonstrate detectable amidolytic activity for the tested chromogenic substrates for serine proteases such as F3301, I0898, V0882, T1637, B2133 and B3133.

5.2.5.6 Total carbohydrate content

The total neutral carbohydrate content of Rusviprotease was determined as 25.6 µg mg^{-1} of the protein.

5.2.6 Pharmacological characterizations of Rusviprotease

5.2.6.1 Procoagulant activity

5.2.6.1.1 Plasma recalcification time

The dose-dependent recalcification time of Rusviprotease on citrated PPP is shown in Fig. 5.10A. Rusviprotease shortened the recalcification time of citrated PPP even at a dose of 0.04 µmol l^{-1} (Fig. 5.10A). However, no further significant decrease in clotting time at the dose of 0.6-1.0 µmol l^{-1} Rusviprotease was observed (Fig. 5.10A).

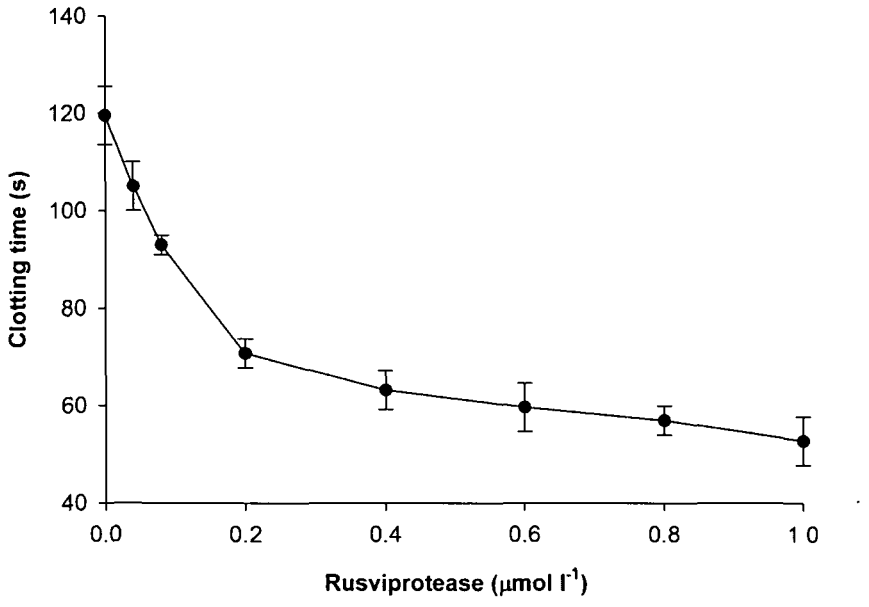


Fig. 5.10A Dose-dependent plasma recalcification activity of Rusviprotease. Different concentrations of Rusviprotease (0-1.0 µmoles l^{-1}) was added to 300.0 µl

of pre-warmed PPP and the mixture was incubated for 3.0 min at 37 °C. To this mixture, 40.0 µl of 250 mM $CaCl_2$ was added and the time taken for the appearance of first visible fibrin thread/clot formation was measured as described in section 3.2.4.1. The values are mean ± S.D. of triplicate determinations.

5.2.6.1.2 Prothrombin time test

Rusviprotease decreased the prothrombin time of PPP in a dose-dependent manner suggesting its probable effect on the extrinsic and/or common pathway(s) of the coagulation cascade (Fig. 5.10B).

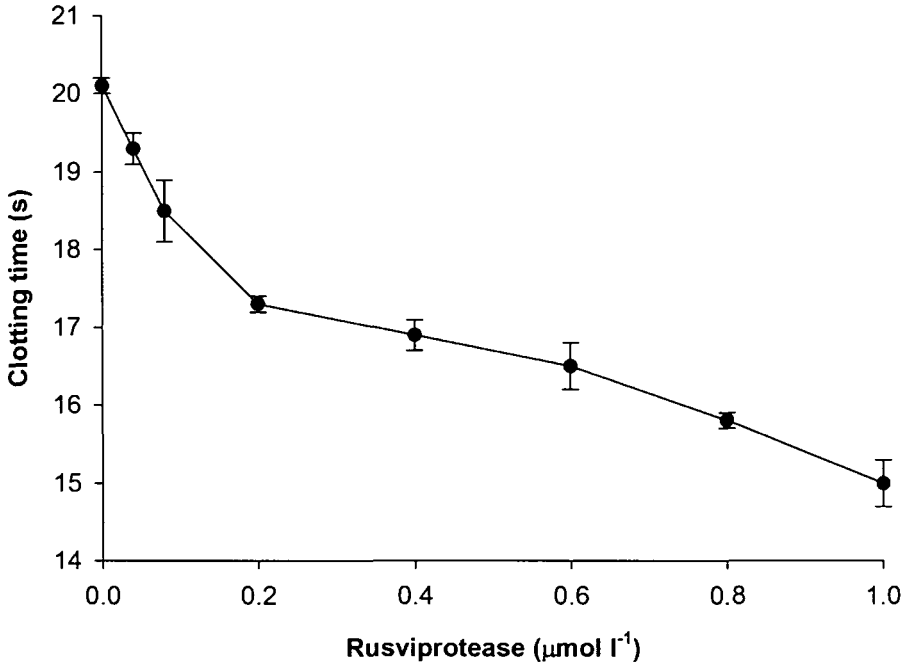


Fig. 5.10B Dose-dependent prothrombin time test of Rusviprotease. The prothrombin time (PT) of goat PPP was tested by Quick's one-stage procedure using a commercial Liquiplastin kit. Rusviprotease (0-1.0 µmol l^{-1}) was added to 0.1 ml of pre-warmed PPP and PT was determined as described in section 3.2.4.2. The values are mean ± S.D. of triplicate determinations.

5.2.6.1.3 Activated partial thromboplastin time (aPTT) test

Rusviprotease decreased the aPTT of PPP in a dose-dependent manner suggesting its probable effect on the common pathway of the coagulation cascade (Fig. 5.10C).

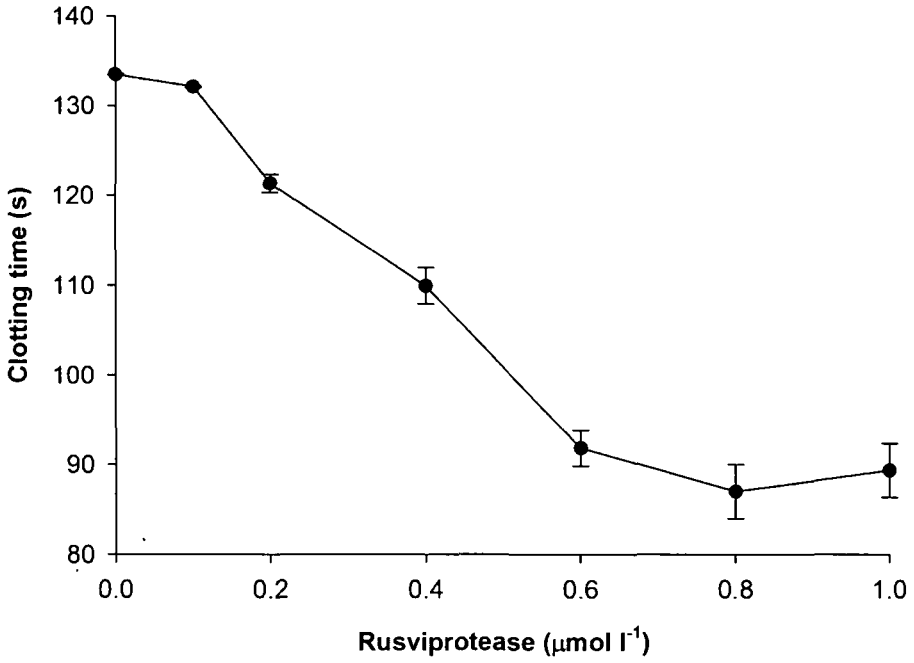


Fig. 5.10C Dose-dependent aPPT test of Rusviprotease. The aPTT of goat PPP was tested by Quick's one-stage procedure using a commercial Liquicelin kit. Rusviprotease (0-1.0 μmol l^{-1}) was added to 0.1 ml of pre-warmed PPP and aPPT was determined as described in section 3.2.4.3. The values are mean ± S.D. of triplicate determinations.

5.2.6.2 Fibrinogen clotting activity of Rusviprotease

Rusviprotease (0.4 μmol l^{-1}) was unable to clot fibrinogen (2.5 mg ml^{-1}) into fibrin clot as no visible fibrin thread formation was observed even after 4 h of incubation of fibrinogen with Rusviprotease at 37 °C, pH 7.4.

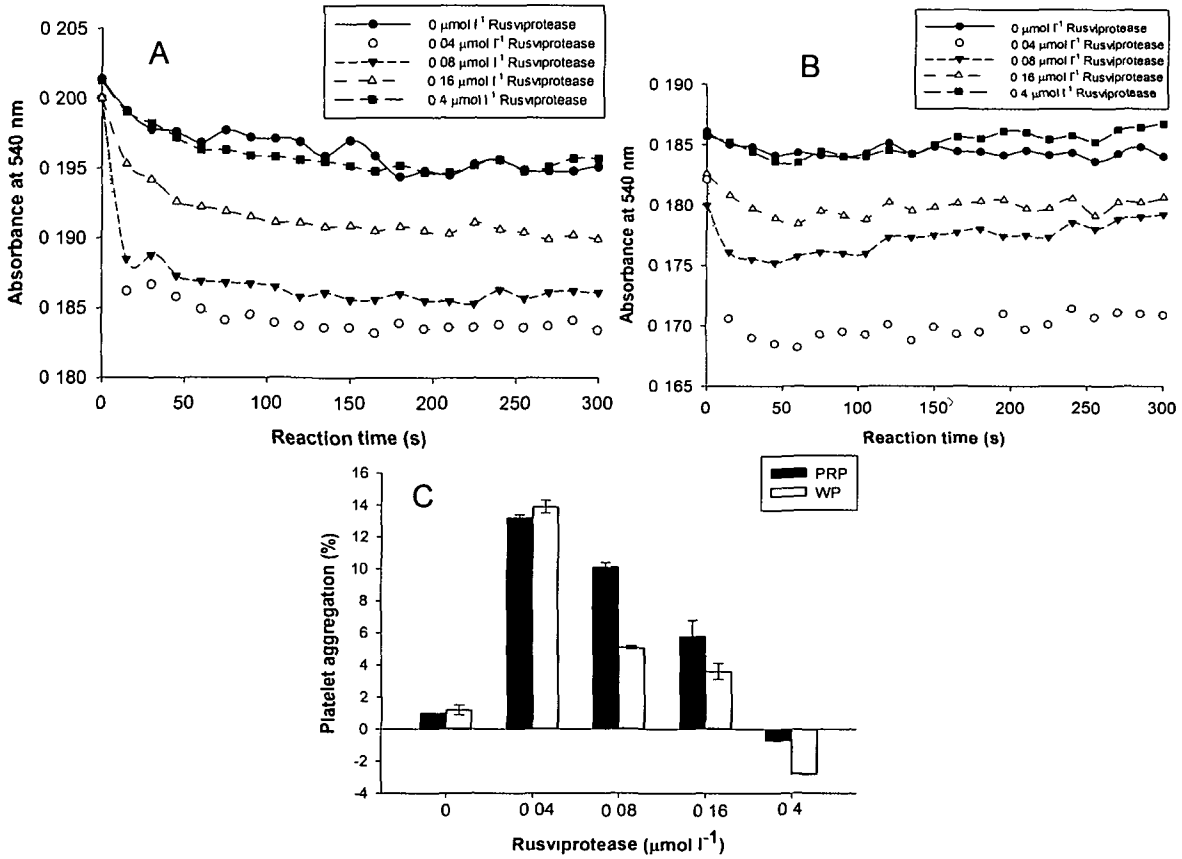
5.2.6.3 Platelet modulating activity of Rusviprotease

Rusviprotease displayed concentration-dependent platelet aggregation and disaggregation property (Figs. 5.11A-C). As shown in the figures, Rusviprotease-induced better platelet aggregation at lower doses (0.04-0.08 μmol l^{-1}) in both PRP (Fig. 5.11A) and washed platelet suspension (Fig. 5.11B). However, increasing concentration of Rusviprotease led to the reversal of the platelet aggregating effect, thereby leading to platelet disaggregation which was most prominent at the tested dose of 0.4 μmol l^{-1} (Figs. 5.11A-C).

A comparison of the platelet modulating property of Rusviprotease in PRP and washed platelets showed that Rusviprotease induced better platelet aggregating property in PRP than in washed platelet suspension (Fig. 11C).

5.2.6.4 Effect on extracellular matrix proteins

SDS-PAGE analysis of extracellular matrix protein degradation by Rysviprotease suggested that this metalloprotease was unable to degrade any of the tested extracellular matrix proteins namely Type-IV collagen, laminin and fibronectin even after 8 h of incubation, under physiological conditions (37 °C, pH 7.4) (Fig. 5.12).



Figs. 5.11A-C Effect of Rusviprotease on platelet aggregation. Rusviprotease (0.04–0.4 µmol l^{-1}) was added to 100 µl of (A) PRP or (B) washed platelet suspension (WP) and the decrease in absorbance was monitored for 300 s at 540 nm as described in section 3.2.4.4. (C) A comparison of the platelet modulating property of Rusviprotease in PRP and WP suspension. The values are mean ± S.D. of triplicate determinations.

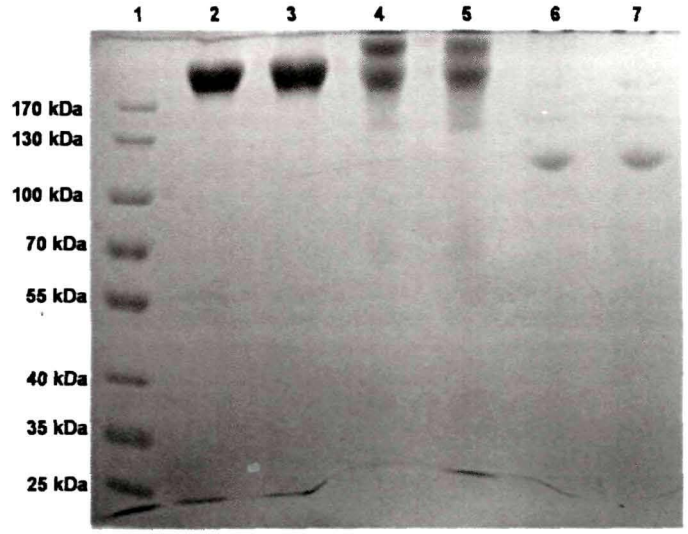


Fig. 5.12 Degradation of fibronectin, laminin and type-IV collagen by Rusviprotease. ECM proteins such as laminin, type-IV collagen and fibronectin were incubated with Rusviprotease in a substrate: enzyme ratio of 15:1 (w/w) at 37 °C, pH 7.4 for 4 h and the degradation products were analyzed on 10% SDS-PAGE after staining with 0.25% coomasie brilliant blue as described in section 3.2.3.4. Lane 1,molecular weight marker; lane 2, control fibrinonectin (without Rusviprotease); lane 3, Rusviprotease treated fibronectin; lane 4, control laminin (without Rusviprotease); lane 5, Rusviprotease treated laminin; lane 6 control type-IV collagen (without Rusviprotease); lane 7, Rusviprotease treated type-IV collagen.

5.2.6.5 Hemolytic activity

Rusviprotease lacked *in vitro* direct and indirect hemolytic activity on mammalian erythrocytes at the tested dose of 10.0 μg ml^{-1} (0.4 μmol l^{-1}) (Table 5.4).

5.7.2.6 Cell cytotoxicity assay

At the tested dose of 10.0 μg ml^{-1} (0.4 μmol l^{-1}), Rusviprotease did not exhibit cytotoxicity towards U87-MG, HeLa and HT-29 cells after 48 h of treatment at 37 °C (Table 5.4).

Table 5.4 Some pharmacological properties of Rusviprotease. Values are mean ± S.D. of triplicate determination.

Parameters	**Values**	
	Control	**Rusviprotease**
Direct hemolysis (% Hb released μg^{-1} protein, post 90 min of incubation at 37 °C)	0	ND
Indirect hemolysis (% Hb released μg^{-1} protein, post 90 min of incubation at 37 °C)	0	0.9 ± 0.1
In vitro cytotoxicity (% cell viability post 48 h of treatment and incubation at 37 °C, 5% CO_2, 98% humidity)		
*U87MG	100 ± 2.8	99.3 ± 1.5
*HeLa	100 ± 8.3	98.5 ± 1.5
*HT-29	100 ± 5.2	97.0 ± 1.6

ND= not detected.

*1 x 10^5 cells ml^{-1} were treated with 10 μg ml^{-1} (0.4 μmol l^{-1}) Rusviprotease for the cytotoxicity assays.

5.2.7 Mechanism of procoagulant activity of Rusviprotease

5.2.7.1 Effect of Rusviprotease on prothrombin and thrombin: Spectroflourimetric study

Rusviprotease exhibited positive interaction with prothrombin (Fig. 5.13A). A red shift along as well as quenching of fluorescence intensity of prothrombin was observed as a result of interaction between Rusviprotease and prothrombin (Fig. 5.13A). However, no interaction was observed between Rusviprotease and thrombin as the intensity of thrombin remained unchanged even in the presence of Rusviprotease (Fig. 5.13B).

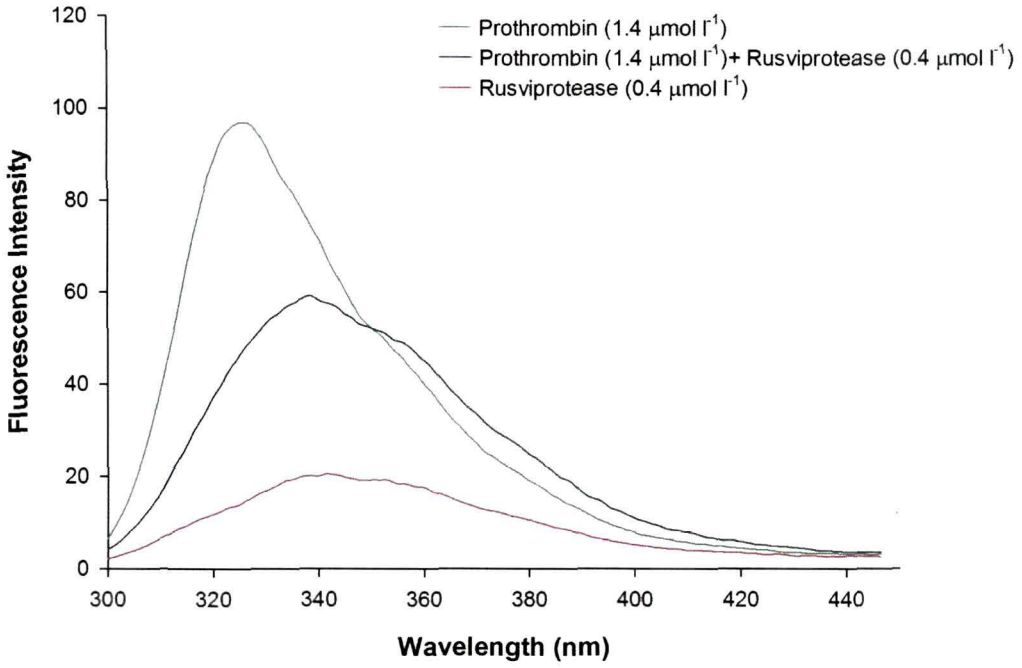


Fig. 5.13A Effect of interaction of Rusviprotease with prothrombin. Fluorescence spectra showing the interaction of Rusviprotease with prothrombin at room temperature (~23 °C), pH 7.4. The fluorescence intensity of a mixture of prothrombin (1.4 µmol l^{-1}) and Rusviprotease (0.4 µmol l^{-1}) was measured after excitation at 280 nm and emission spectra were recorded from 300-450 nm. The fluorescence intensity of prothrombin (1.4 µmoles l^{-1}) or Rusviprotease (0.4

μmoles l^{-1}) was also recorded under identical experimental conditions (section 3.2.3.16.1).

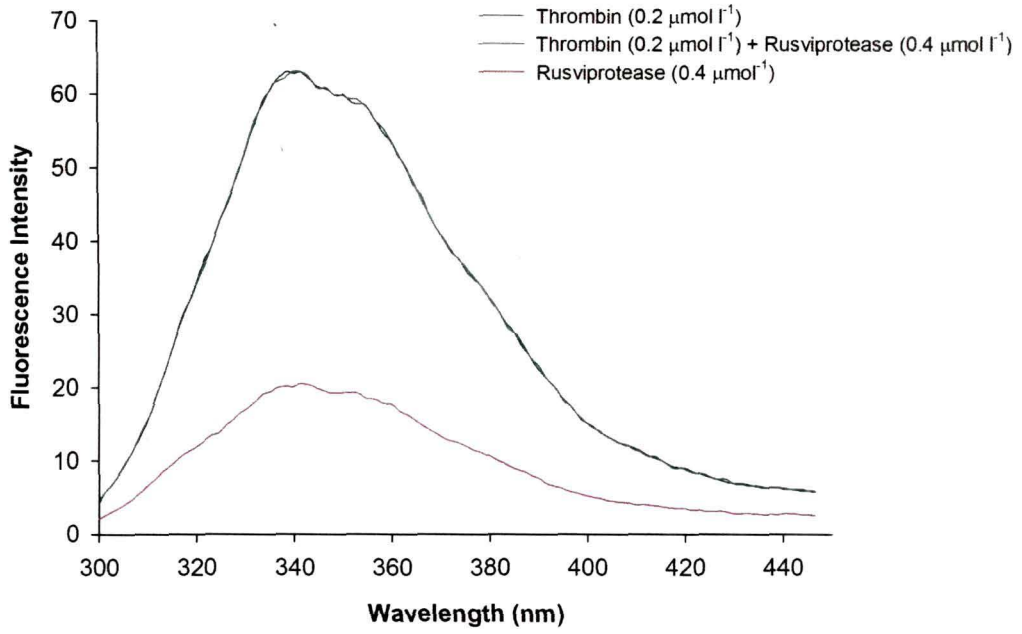


Fig. 5.13B Effect of interaction of Rusviprotease with thrombin. Fluorescence spectra showing the interaction of Rusviprotease with prothrombin at room temperature (~23 °C), pH 7.4. The fluorescence intensity of a mixture of thrombin (0.2 μmol l^{-1}) and Rusviprotease (0.4 μmol l^{-1}) was measured after excitation at 280 nm and emission spectra were recorded from 300-450 nm. The fluorescence intensity of thrombin (0.2 μmole l^{-1}) or Rusviprotease (0.4 μmol l^{-1}) was also recorded under identical experimental conditions (section 3.2.3.16.1).

5.2.7.2 Prothrombin activation studies

5.2.7.2.1 Prothrombin activation assay

Rusviprotease activated prothrombin to form thrombin, which subsequently hydrolysed the chromogenic substrate for thrombin (T1637) independent of the presence of Ca^{2+} (Fig. 5.14).

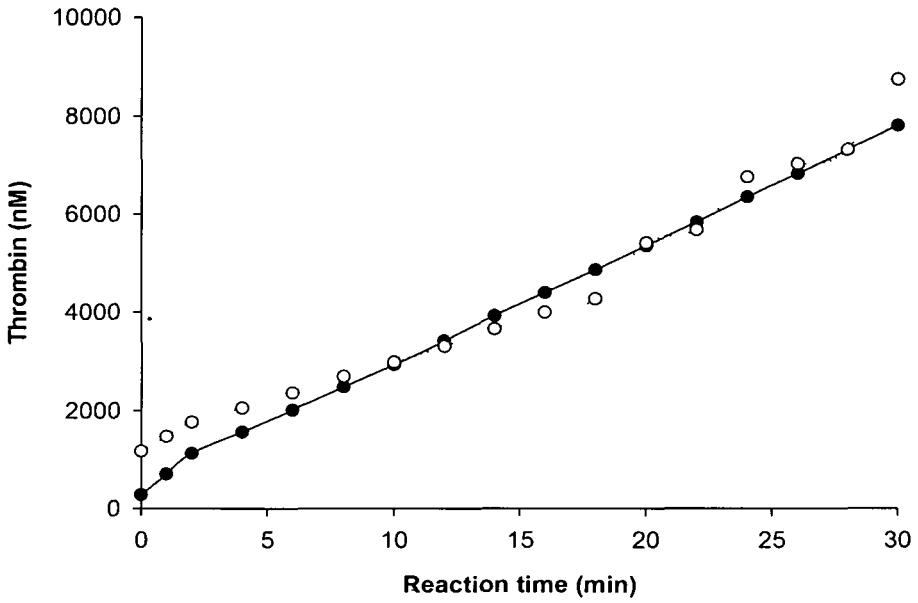


Fig 5.14 Prothrombin activation by Rusviprotease. Rusviprotease (0.4 μmol l^{-1}) was incubated with prothrombin (1.4 μmol l^{-1}) either in the presence (○) or absence (●) of 5 mM $CaCl_2$ for 10 min at 37 °C, pH 7.4. After 10 min, chromogenic substrate for thrombin (T1637, 0.2 mM final concentration) was added and the hydrolysis of T1637 was monitored by monitoring the release of pNA at 405 nm (section 3.2.2.14.1). Values are mean of three independent experiments.

Unlike factor Xa, the prothrombin activation by Rusviprotease was independent of the presence of the components of prothrombinase complex (Table 5.5). The prothrombin activating property of Rusviprotease, however, was found to be lower in comparison to that of factor Xa, under identical assay conditions (Table 5.5).

5.2.7.2.2 Kinetics of prothrombin activation

The *Km* and *Vmax* values of prothrombin activation by Rusviprotease in the absence of co-factors were calculated to be 1.73 μM and 153.5 nM thrombin generated min^{-1} μmol^{-1} activator, respectively (Fig. 5.15).

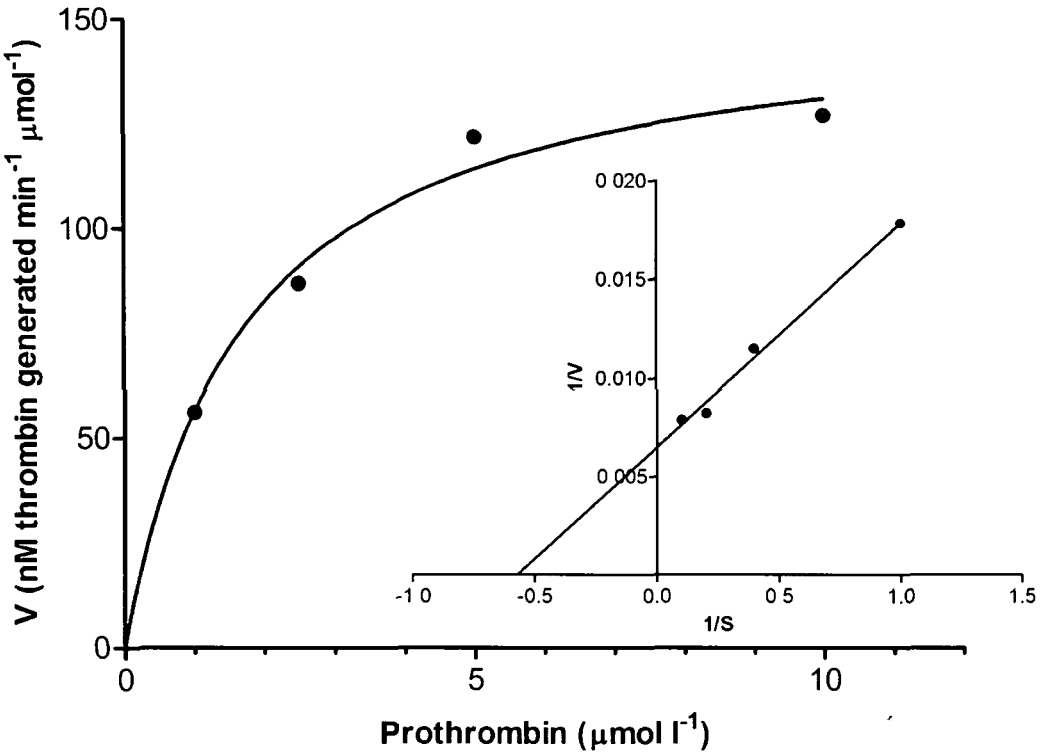


Fig. 5.15 Kinetics of prothrombin activation by Rusviprotease. The Michaelis-Menten and Lineweaver-Burk double-reciprocal (inset) plots depicting hydrolysis of prothrombin (0-10 µ µmol l^{-1}) by Rusviprotease (0.04 µmol l^{-1}). The kinetics of prothrombin activation were determined at 37 °C by measuring the pNA release from T1637 (substrate for thrombin) at 405 nm. The values are mean of triplicate determinations.

5.2.7.2.3 Fibrinogen clotting activity of Rusviprotease-mediated prothrombin activation products

Addition of Rusviprotease-activated prothrombin (incubated in absence or presence of Ca^{2+}) to fibrinogen resulted in fibrin clot formation within 162.0 ± 1.8 s and 158.5 ± 5.1 s (mean ± SD, n=3), respectively. This result indicates that Ca^{2+} did not influence ($p>0.05$) the prothrombin activating property of Rusviprotease.

Table 5.5. Prothrombin activation by Rusviprotease in presence or absence of co-factors. Rusviprotease (0.04 μmol l^{-1}) or factor Xa (0.02 μmol l^{-1}) was incubated with prothrombin (1.4 μmol l^{-1}) either in the presence or absence of 2.5 mM Ca^{2+} or prothombinase complex [100 μM phospholipid vesicles (9:1 phosphatidylcholine (PC): phosphatidylserine (PS) and 3 nM factor Va (FVa)] at 37 °C for 15 min. Thereafter, 0.2 mM chromogenic substrate for thrombin (T1637) was added and kinetics of substrate hydrolysis was monitored by measuring the initial rate of pNA released at 405 nm (section 3.2.3.14.1). The amount of thrombin generated was calculated from the action of graded concentrations of thrombin on T1637. Prothrombin activation was expressed as nanomoles of thrombin generated min^{-1} μmol^{-1} enzyme. Values are mean ± S.D. of triplicate data.

Conditions for prothrombin activation	**Prothrombin activation (nM of thrombin generated min^{-1} μmol^{-1} Rusviprotease)**
Rusviprotease (0.04 μmol/l)	157.2 ± 9.0
Rusviprotease (0.04 μmol/l), Ca^{2+} (2.5 mM)	153.5 ± 10.1
Rusviprotease (0.04 μmol/l), 100 μM phospholipids (9:1 PC: PS)	150.3 ±6.3
Rusviprotease (0.04 μmol/l), Ca^{2+} (2.5 mM), 100 μM phospholipids (9:1 PC: PS)	156.2 ± 5.4
Rusviprotease (0.04 μmol/l), FVa (3.0 nM), Ca^{2+} (2.5 mM), 100 μM phospholipids (9:1 PC: PS)	157.1 ± 3.0
FXa (0.02 μmol/l)	258.4 ± 2.8
FXa (0.02 μmol/l), FVa (3.0 nM), Ca^{2+} (2.5 mM), 100 μM phospholipids (9:1 PC: PS)	850.3 ± 5.1
FXa (0.02 μmol/l), Rusviprotease (0.04 μmol/l FVa (3.0 nM), Ca^{2+} (2.5 mM), 100 μM phospholipids (9:1 PC: PS)	1105.4 ± 7.3

5.2.7.2.4 SDS-PAGE analysis of prothrombin activation by Rusviprotease

Prothrombin (~72 kDa) activation by Rusviprotease, independent of any co-factor, was analysed by 12% SDS-PAGE under non-reducing and reducing sample conditions (Figs 5.16A-B). Formation of thrombin (~36 kDa) from prothrombin by Rusviprotease was observed within 15 min of incubation (Fig. 5.16A). Rusviprotease completely degraded prothrombin within 6 h of incubation. This resulted in the generation of meizothrombin(des)F1 (~55 kDa) and thrombin (~36 kDa) as major hydrolysis products (identified by PMF analysis, described in section 5.2.7.2.5) of prothrombin along with formation of minor hydrolysis products corresponding to fragment-1 (~27 kDa) and fragment-2 (~16 kDa) (Fig. 5.16A). The disassembly of meizothrombin(des)F1 due to the presence of β-mercaptoethanol as a reducing agent was also evident (Fig. 5.16B). The conversion of prethrombin-1 into thrombin B-chain and thrombin A chain-fragment 2 was also observed (Fig. 5.16B). Minor bands corresponding to fragment-1 and fragment-2 were also visualized by SDS-PAGE analysis under non-reducing conditions (Fig. 5.16A).

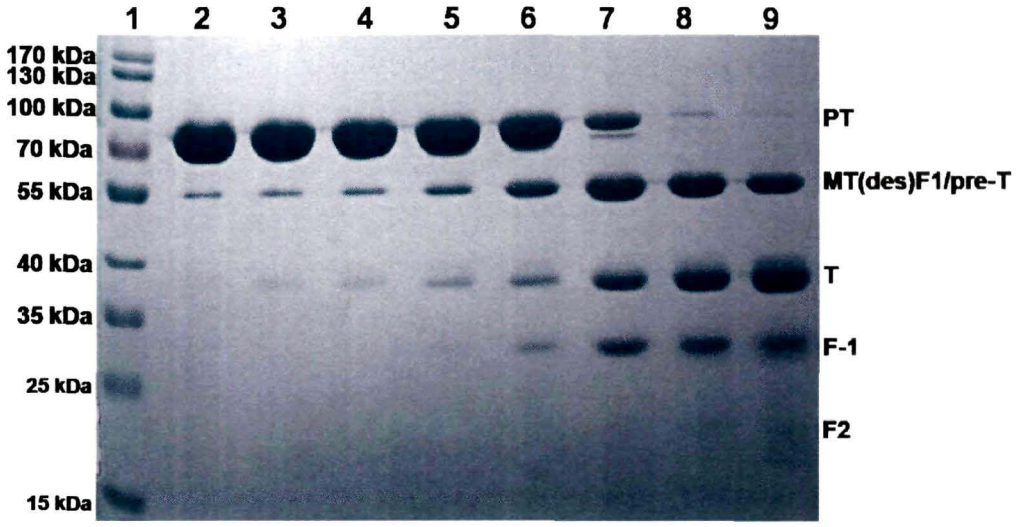


Fig. 5.16A Prothrombin activation by Rusviprotease analyzed by SDS-PAGE under non reducing conditions. Prothrombin (14.0 μmol l^{-1}) was incubated with

Rusviprotease (0.4 μmol l^{-1}) for the indicated time period at 37 °C, pH 7.4, and the prothrombin cleavage products were separated by 12% SDS-PAGE under non-reducing conditions. Lane 1, protein molecular markers; lane 2, prothrombin incubated with buffer for 8 h at 37 °C (control); lanes 3-9, products of prothrombin hydrolysis by Rusviprotease after 0.25, 0.5, 1, 2, 4, 6 and 8 h of incubation, respectively at 37 °C. The cleaved proteins are identified as follows: PT, prothrombin; MT, meizothrombin; MT(des)F1, meizothrombin(des)F1; T, thrombin; F1, fragment 1; F2, fragment 2; T-B, thrombin β-chain; PreT-1, prethrombin-1; F2+T-A, fragment 2 + thrombin α-chain.

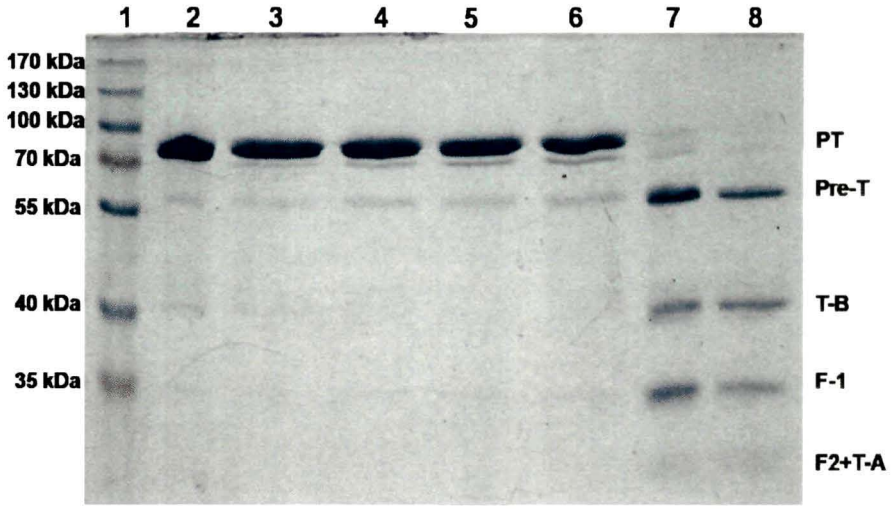


Fig. 5.16B Prothrombin activation by Rusviprotease analyzed by SDS-PAGE under reducing conditions. Prothrombin (14.0 μmol l^{-1}) was incubated with Rusviprotease (0.4 μmole l^{-1}) for the indicated time period at 37 °C and the prothrombin cleavage products were separated by 12% SDS-PAGE under reducing conditions. Lane 1, prothrombin incubated with buffer for 8 h at 37 °C (control); lanes 2-8, products of prothrombin hydrolysis by Rusviprotease after 0.5, 1, 2, 4, 6 and 8 h of incubation, respectively at 37 °C; lane 9, protein molecular markers. The cleaved proteins are identified as follows: PT, prothrombin; F1, fragment 1; T-B, thrombin β-chain; PreT-1, prethrombin-1; F2+T-A, fragment 2+thrombin α-chain.

5.2.7.2.5 PMF analysis of prothrombin activation products

PMF analysis of the three major prothrombin activation products of Rusviprotease (protein bands corresponding to 27 kDa, 36 kDa and 55 kDa in non-reducing SDS-PAGE) showed 100% identity with prothrombin (SWISS-PROT accession number Q69EZ8). The PMF analysis of the 36 kDa prothrombin activation product demonstrated putative conserved domains of thrombin light chain (Table 5.6). The 55 kDa fragment did not show putative conserved domain; however, BLAST analysis proved its identity (100%) with the light chain structure of human meizothrombin(des)F1 (Table 5.6). Similarly, the 27 kDa fragment also displayed no putative conserved domains but displayed identity (100%) with human prothrombin fragment 1 (Table 5.6).

5.2.7.3 Effect of Rusviprotease on factor Xa

5.2.7.3.1 Spectrofluorimetric study

The interaction of Rusviprotease with factor Xa resulted in a 4.5 fold increase in fluorescence intensity as compared to the fluorescence intensity of Rusviprotease or factor Xa alone suggesting the interaction between these two protein molecules (Fig. 5.17).

Table 5.6 Identification of the Rusviprotease generated prothrombin cleavage products by PMF analysis. The experiment was performed as described in section 3.2.2.3. The PMF profiles were used to search for peptide matches from NCBI database following the MASCOT search. The tryptic sequences were subjected for BLAST search in NCBI database of non-redundant protein sequences using BLASTp algorithm.

Band size	**Identified peptides**	**Delta mass**	**Query coverage**	**Maximum identity**	**Identified protein**	**Accession No#**
~27 kDa	SGIECQLWR	0.02	100	100	Prothrombin	Q69EZ8
	RQECSIPVCGQDQVTVAMTPR	0.04	100	100		
~36 kDa	TFGSGEADCGLRPLFEKK	0.08	100	100	Prothrombin (α-thrombin chain)	Q69EZ8
	TFGSGEADCGLRPLFEKKS	0.17	100	100		
~55 kDa	LAVTTHGLPCLAWASAQAK	0.12	100	100	Prothrombin	Q69EZ8
	TATSEYQTFFNPR	0.11	100	100		

#Values as provided by UniProtKB.

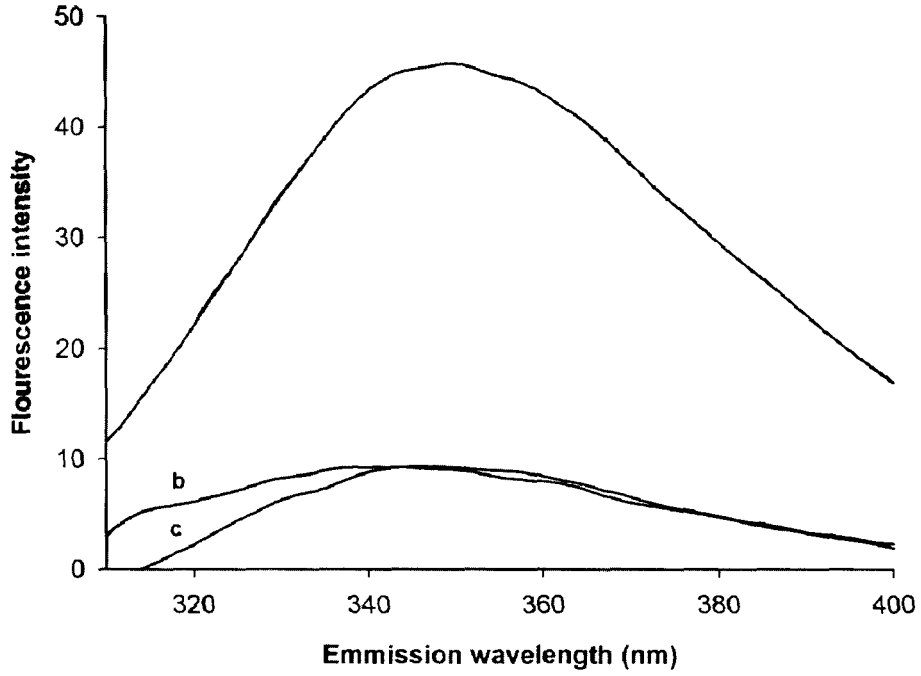


Fig. 5.17 Spectrofluorimetric study of interaction of Rusviprotease with factor Xa. Fluorescence spectra showing the interaction of Rusviprotease with factor Xa at room temperature (~23 °C), pH 7.4. The fluorescence intensity of a mixture of 0.02 μmol l^{-1} factor Xa (~1.0 μg) and 0.04 μmol l^{-1} Rusviprotease (~1.0 μg) was measured after excitation at 280 nm and emission spectra were recorded from 310-400 nm. The fluorescence intensity of factor Xa (0. 02 μmol l^{-1}) or Rusviprotease (0.04 μmol l^{-1}) was also recorded under identical experimental conditions. (a) factor Xa (0.02 μmol l^{-1}) in presence of Rusviprotease (0.04 μmol l^{-1}) (b) factor Xa (0.02 μmol l^{-1}), and (c) Rusviprotease (0.04 μmol l^{-1}).

5.2.7.3.2. Effect of Rusviprotease on the kinetics of prothrombin activation by factor Xa-prothrombinase complex

The kinetics (*Km* and *Vmax* values) of prothrombin activation by factor Xa-prothrombinase complex in the absence of Rusviprotease were found to be 0.036 ± 0.01 μM and 2350.9 ± 23.8 nM of thrombin generated min^{-1} μmol^{-1} of factor Xa,

respectively. However, in presence of Rusviprotease, the *Km* and *Vmax* values for prothrombin activation done by factor Xa-prothrombinase complex were calculated as 0.029 ± 0.02 μM and 3052.9 ± 20.5 nM of thrombin generated min^{-1} μmol^{-1} of factor Xa, respectively. The result shows a 1.3 fold increase in the maximal velocity (*Vmax*) of factor Xa catalyzed prothrombin activation in presence of Rusviprotease.

5.2.7.3.3. SDS-PAGE analysis of the effect of Rusviprotease on the prothrombin activation by factor Xa-prothrombinase complex

SDS-PAGE analysis of prothrombin activation by factor Xa-prothrombinase complex in presence of Rusviprotease clearly indicated more thrombin formation (Fig. 5.18).

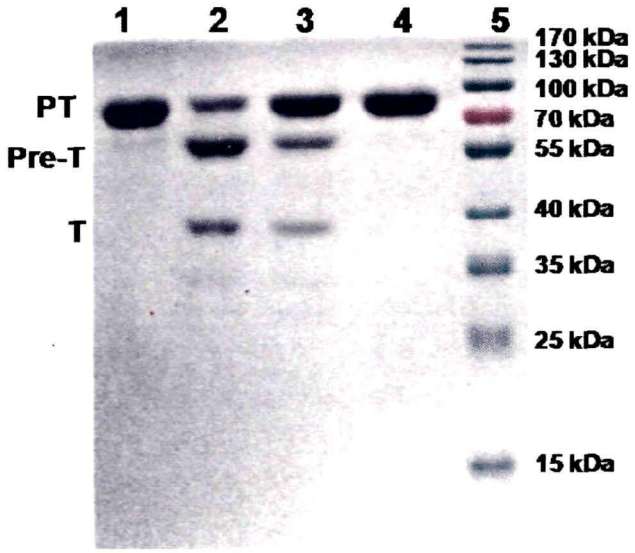


Fig. 5.18 SDS-PAGE analysis of the effect of Rusviprotease (0.04 μmol l^{-1}) on the prothrombin activation by factor Xa-prothrombinase (0.02 μM) complex. Rusviprotease was pre-incubated with factor Xa-prothrombinase complex for 30 min at 37 °C, pH 7.4, prior to addition of prothrombin (14.0 μmol l^{-1}). After 15 min, the prothrombin activation products were separated by 12% SDS-PAGE

under reducing conditions. Lane 1, prothrombin activation by Rusviprotease; lanes 2 and 3, prothrombin activation by factor Xa-prothrombinase complex in the presence or absence of Rusviprotease, respectively; lane 4, prothrombin; lane 5 prestained molecular weight marker. The cleaved proteins are identified as follows: PT, prothrombin; PreT-1, prethrombin-1 and T, thrombin β-chain.

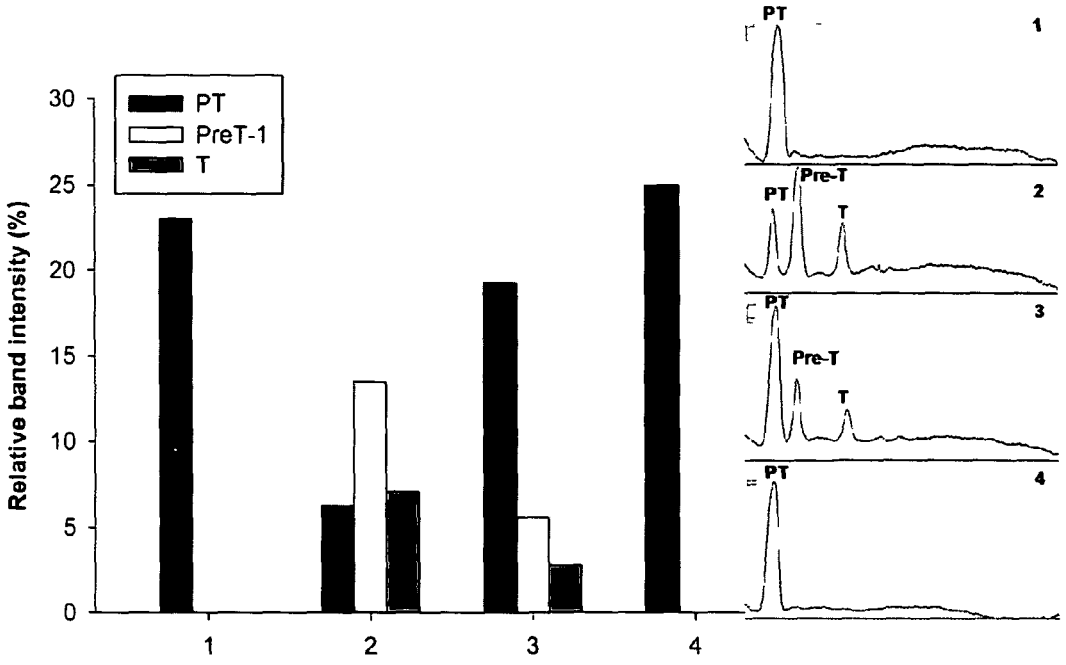


Fig. 5.19 Densitometric analysis of the effect of Rusviprotease on thrombin generation by factor Xa-prothrombinase complex. Graphical representation (relative band intensity, %) of the densitometric analysis (inset) of the prothrombin activation products prethrombin-1 (PreT-1) and thrombin (T) visualized in the SDS-PAGE data (Fig. 5.18) using ImageJ software.

Densitometric scanning of the gel revealed a 2.5 fold increase in the quantity of thrombin formation by factor Xa-prothrombinase-Rusviprotease complex as

compared with the quantity of thrombin generated by factor Xa-prothrombinase complex (Fig. 5.19).

5.2.7.3.4 Effect of Rusviprotease on the plasma recalcification time of factor Xa-prothrombinase complex

Factor-Xa-prothrombinase-Rusviprotease complex demonstrated better procoagulant activity ($p<0.05$) compared with the same activity displayed by factor Xa-prothrombinase complex under identical experimental conditions (Table 5.7).

5.2.8 Chemical modification and neutralization studies of Rusviprotease

5.2.8.1 Chemical modification studies

As shown in Table 5.8, the prothrombin activation property, fibrinogenolytic property and the overall procoagulant property of Rusviprotease was strongly inhibited by the metal chelator, EDTA. The catalytic as well as procoagulant activity of Rusviprotease was also strongly inhibited by disulphide bond breaker, DTT. Serine protease inhibitors such as PMSF, TPCK and TLCK as well as histidine modifier such as pBPB, did not affect the different activities of Rusviprotease (Table 5.8).

Table 5.7 Effect of Rusviprotease on the factor Xa-induced plasma recalcification time of citrated PPP. 7.2 nM Factor Xa (FXa) was pre-incubated with Rusviprotease (0.01 μmol l^{-1}) for 30 min at 37 °C in presence of the prothrombinase complex [2.5 mM $CaCl_2$, 100 μM phospholipid vesicles (9:1 PC: PS) and 3.0 nM factor Va (FVa)]. To this mixture, 300 μl PPP was added and the plasma recalcification time was determined as described in section 3.2.4.1. The plasma recalcification time of factor Xa-prothrombinase complex in presence or absence of Rusviprotease (0.01 μmol l^{-1}) was also determined under identical experimental conditions. At the above concentration, Rusviprotease did not show procoagulant effect ($p>0.05$).

Sample	**Clotting time (s)***
Control	120.0 ± 1.5^{a}
Rusviprotease (0.01 μmole l^{-1})	118.5 ± 1.5^{a}
FXa (7.2 nM) + FVa (3.0 nM) + Ca^{2+} (2.5 mM) +100 μM phospholipids (9:1 PC: PS)	110.6 ± 0.9^{b}
Rusviprotease (0.01 μmol l^{-1}) + FXa (7.2 nM) + FVa (3.0 nM) + Ca^{2+} (2.5 mM) +100 μM phospholipids (9:1 PC: PS)	93.1 ± 0.5^{c}

Values are mean ± S.D. of triplicate experiments.

Values in the same column with different superscripts are significantly different from control value ($p<0.05$).

Table 5.8 A comparison of neutralization of procoagulant and protease activity of Rusviprotease by chemical inhibitors. Rusviprotease (0.04 μmol l^{-1}) was pre-incubated with the inhibitors for 30 min at room temperature (~23 °C). Thereafter, prothrombin activation property, fibrinogenolytic activity and plasma recalcification time of Rusviprotease was analyzed in the corresponding assay systems, as described in sections 3.2.3.14.1, 3.2.3.3.1 and 3.2.4.1, respectively. The activity of Rusviprotease in absence of inhibitor was considered as 100% activity.

Inhibitor	**% Residual activity**		
	Prothrombin activation	**Fibrinogenolytic activity**	**Coagulant activity**
Control (Rusviprotease without inhibitor)	100	100	100
EDTA (5 mM)	0.0*	1.5 ± 0.2*	0.2 ± 0.04*
DTT (2 mM)	8.5 ± 0.7*	6.5 ± 1.5*	8.8 ± 0.5*
PMSF (2 mM)	97.0 ± 3.2	99.0 ± 0.2	96.0 ± 1.5
TLCK (2 mM)	ND	96.2 ± 3.3	98.3 ± 5.5
TPCK (2 mM)	ND	88.4 ± 9.2	92.6 ± 4.5
pBPB (2 mM)	ND	97.3 ± 5.8	98.0 ± 5.0

Values are mean ± S.D. of triplicate experimental values.
Values marked with '*' are statistically different from that of control ($p<0.05$).
ND, not determined.

5.2.8.2 Neutralization of Rusviprotease by physiological inhibitors and commercial antivenom

Both PAV as well as MAV inhibited the proteolytic and procoagulant properties of Rusviprotease in a dose-dependent manner (Table 5.9). Nevertheless, the neutralization of plasma clotting, prothrombin activating and fibrinogenolytic activities of Rusviprotease was found to be superior by MAV compared with PAV (Table 5.9). The physiological inhibitors, α_2-macroglobulin and α_2-antiplasmin

significantly inhibited the original catalytic and pharamcological potency of Rusviprotease (Table 5.9).

5.2.9 Acute *in vivo* toxicity assessment of Rusviprotease on Balb/c mice

5.2.9.1 Lethality and *in vivo* toxicity in Balb/c mice

At a dose of 0.07 mg kg^{-1} body weight, Rusviprotease was non-lethal to Balb/c mice and the treated animals did not show any noticeable behavioral changes up to 48 h of *i.p.* treatment (Table 5.10).

5.2.9.2 *In vivo* anticoagulant effects of Rusviprotease on mouse model

The tail-bleeding time of Rusviprotease-treated mice was found to be significantly higher ($p<0.01$) (185.5 ± 5.3 s, n=3) compared to the tail-bleeding time of control (52.7 ± 2.5 s, n=3) group of mice (Table 5.10). The plasma fibrinogen concentration in Rusviprotease-treated and control groups of mice was determined to be 0.38 ± 0.01 μg ml^{-1} and 0.46 ± 0.04 μg ml^{-1} (n=3), respectively ($p<0.05$) (Table 5.10) .

Table 5.9 A comparison of neutralization of procoagulant and protease activity of Rusviprotease by PAV and MAV. Rusviprotease was pre-incubated with PAV, MAV or physiological inhibitors (α_2-macroglobulin and α_2-antiplasmin) for 30 min at 37 °C. Thereafter, prothrombin activation property, fibrinogenolytic activity and plasma recalcification time of Rusviprotease was analyzed in the corresponding assay systems, as described in sections 3.2.3.14.1, 3.2.3.3.1 and 3.2.4.1, respectively. The activity of Rusviprotease in absence of inhibitor was considered as 100% activity.

Commerical Antivenom/Physiological inhibitor	**% Residual activity**		
	Prothrombin activation	**Fibrinogenolytic**	**Procoagulant**
Rusviprotease	100	100	100
Rusviprotease: PAV (w/w)			
1:1	66.1 ± 1.2^{a}	$74.7 \pm 2.7^{b\#}$	$72.0 \pm 1.3^{c\#}$
1:5	$64.5 \pm 1.5^{a\#}$	61.5 ± 3.2^{a}	$73.9 \pm 2.8^{b\#}$
1:10	$51.7 \pm 1.3^{a\#}$	$47.4 \pm 5.3^{a\#}$	$61.3 \pm 2.2^{c\#}$
Rusviprotease:MAV (w/w)			
1:1	67.3 ± 1.5^{a}	$80.2 \pm 2.1^{b\#}$	$78.2 \pm 0.3^{b\#}$
1:5	$57.7 \pm 2.4^{a\#}$	59.3 ± 6.8^{a}	$67.5 \pm 2.1^{b\#}$
1:10	$46.8 \pm 2.7^{a\#}$	$34.7 \pm 4.7^{b\#}$	$54.8 \pm 2.3^{c\#}$
Physiological inhibitor			
α_2-macroglobulin (3.0 μM)	44.1 ± 4.5^{a}	53.5 ± 1.5^{b}	64.5 ± 3.6^{c}
α_2.antiplasmin (3.0 μM)	75.6 ± 10.1^{a}	78.4 ± 14.3^{a}	76.8 ± 2.5^{a}

Values are mean ± S.D. of triplicate experimental values. Values in the same row with different superscripts are significantly different ($p<0.05$). '#' represents values that are significantly different from each other under similar experimental conditions.

Table 5.10 Behavioral and physical changes in Rusviprotease (0.07 mg kg^{-1}) treated mice (n=6) after 48 h of *i.v.* administration.

Parameters	**Control (Placebo)**	**Rusviprotease treated**
Survivality	No death	No death
Tail-bleeding time (s)	52.7 ± 2.5^{a}	185.5 ± 5.3^{b}*
Grip strength (s)	30.2 ± 4.9	28.31 ± 4.2
Rectal temperature (°C)	35.0	35.0
Drooping of eyelids/ movement inability/ change in body weight	Not observed	Not observed
Food and water intake	Normal	Normal
Plasma fibrinogen ($\mu g\ ml^{-1}$)	0.5 ± 0.04^{c}	0.4 ± 0.01^{d}*

*The test was performed after 6 h of *i.v.* injection of Rusviprotease.
Values are mean ± S.D. of triplicate determination.
Values in the same row with different superscripts are statistically significant ($p<0.05$).

5.3 Discussion

A number of RVV components influencing hemastasis are either procoagulants [83, 116-117, 119, 126] or anticoagulants [59, 82, 96-100; 106, 113]. Many of the procoagulant enzymes reported from RVV are glycoproteins which includes fibrin(ogen)olytic proteases, Factors X and V protease activators [65, 81-83, 117, 216]. However, there is no report on the presence of prothrombin activating principle from RVV. Therefore, the focus of the present study was to isolate and characterize as well as elucidate the procoagulant mechanism of a prothrombin activating metalloprotease (Rusviprotease) purified from RVV of Eastern India origin.

5.3.1 Rusviprotease is a basic oligomeric metalloprotease which is glycoprotein in nature

Rusviprotease is characterized to be a basic multimeric protein which predominantly exists naturally in the form of a 26.8 kDa dimer of 15.1 kDa monomer. Rusviprotease also exists in other multimeric forms (tertameric and hexameric forms) apart from its dimeric form. Elution of Rusviprotease in the void volume during the separation process using Sephadex G-50 matrix, also suggests that the isolated protease exists in RVV in a higher molecular mass form. It has been previously reported that protein oligomerization plays a key role in the regulation of the functional attributes of many enzymatic proteins [336]. Protein dimerization often results in the formation of hetero- and homodimers. Oligomeric proteins may be covalently linked to each other through disulfide bonds or through hydrophobic interactions and/or salt bridges for dimerization to occur [336]. There are few reports of oligomeric SVMPs that form homodimers through disulfide bonds [337-338]. In addition, procoagulant enzymes reported from venom of snakes belonging to the Viperideae family, more particularly from RVV, are acidic glycoproteins with molecular masses in the range of 35 to 145 kDa [83, 116-117, 216, 334]. Most of these proteases are composed of two to four polypeptides

chains [117, 216]. On the contrary, Rusviprotease is a basic glycoprotein, with carbohydrate content much less than previously reported proteases [65, 82-83, 117]. LC MS/MS analysis as well as carbohydrate content analysis unequivocally suggested the similarity of the isolated protease snake venom glycoproteins, such as RVV-X, which are also metalloproteases. It is interesting to note that RVV-X is also a multimeric protein which exists in heterotrimeric form [216]. However, Rusviprotease is distinctively different from RVV-X in terms of its functional properties. Further, the secondary structure analysis of Rusviprotease revealed partial similarity with snake venom fibrinogenolytic metalloproteases such as FIIa isolated from the venom of *Agkistrodon acutus* [339]. Taken together, Rusviprotease is a multimeric metalloprotease from RVV which is glycoprotein in nature and is distinctly different from previously reported RVV metalloproteases.

Inhibition of the procoagulant as well as proteolytic activities of Rusviprotease post-treatment with EDTA (a metal chelator) and DTT (which breaks disulphide bond) resulted in obliteration of protease as well as procoagulant properties, which clearly indicates that Rusviprotease is a metalloprotease whose catalytic and pharmacological properties rely on the oligomeric nature of the protein.

Although Rusviprotease and the previously reported basic metalloprotease, RVBCMP [65] purified from RVV appear analogous; however, Rusviprotease is distinctly different from RVBCMP in terms of its multimeric nature, carbohydrate content as well as lack of functional properties such as hemolytic and tissue damaging effects [65]. However, existence of several protease isoenzymes in snake venom is not uncommon [83] and therefore, isoenzyme nature of these two metalloproteases (RVBCMP and Rusviprotease) may not be ruled out. In fact, variation in the degree of glycosylation in the two proteins might be responsible for the difference in their overall pharmacological effects [83].

5.3.2 Rusviprotease exerts strong *in vitro* procoagulant effects by virtue of its prothrombin activating properties

Rusviprotease dose-dependently decreased the clotting time of PPP suggesting its strong procoagulant effect. Further, Rusviprotease also decreased the PT as well as aPTT of human PPP suggesting its involvement in the common pathway of the blood coagulation cascade [340]. However, we cannot ascertain the site of action of Rusviprotease on the coagulation cascade with these results. Rusviprotease demonstrated prothrombin activation property independent of the requirement of any co-factor. Like most of the prothrombin activators isolated from snake venoms, Rusviprotease activated prothrombin product(s) could induce fibrinogen clotting suggesting the formation of active thrombin [258]. On the contrary, Rusviprotease lacked fibrinogen clotting property thereby suggesting that unlike Russelobin [82], the procoagulant mechanism of Rusviprotease is not attributed to a thrombin-like mechanism. Since the prothrombin activation by Rusviprotease does not require any additional co-factor(s), therefore, Rusviprotease may be classified under "Group A" snake venom prothrombin activators [87]. The *Km* value of Rusviprotease for prothrombin activation was found to be higher as compared with that of previously reported prothrombin activators isolated from the venoms of *Bothrops neuweidi* and *Echis multisquamatus* [264, 341] as well as factor Xa-prothrombinase complex [342]. Besides, the α-fibrinogenolytic activity of Rusviprotease corroborates well with most of the prothrombin activating proteases isolated from snake venom [257, 260, 340]. Interestingly, Rusviprotease demonstrated concentration-dependent platelet aggregation and deaggregation unlike most of the snake venom metalloproteases and prothrombin activators. Similar observations have been reported for a C-type lectin protein, RVsnaclec, purified from RVV [113]. Moreover, the enhanced platelet aggregating property of Rusviprotease in PRP in comparison to washed platelets is probably due to the indirect effect of thrombin production by Rusviprotease. As such, further evaluation of the mechanism of platelet modulation by Rusviprotease is warranted.

These properties of Rusviprotease (procoagulant, fibrinogenolytic activity and platelet modulating property) might contribute to the overall pathophysiology of Russell's Viper bite [46] which is usually fatal but it may not lead to immediate death as it has been reported that signs of coagulopathy may persist for more than 48 h [46]. Therefore the contribution of Rusviprotease, which demonstrate enduring procoagulant effects on RVV-induced pathophysiology cannot be neglected.

In the present study, a perceptible scheme of prothrombin activation by Rusviprotease has also been proposed (Fig. 5.20). In a manner similar to the prothrombinase complex and Group A prothrombin activators [140, 241, 343], Rusviprotease first cleaves prothrombin (~72 kDa) generating meizothrombin, a prothrombin derivative with the same molecular mass as prothrombin [344]. Upon extended incubation, meizothrombin is cleaved to form meizothrombin(des)F1 (~55 kDa) and thrombin (~36 kDa) [344]. Conversely, in the absence of prothrombinase complex, factor Xa initially cleaves prothrombin giving rise to prothrombin fragment 1.2 and prethrombin-2 [344]. Prethrombin-2 is subsequently cleaved to generate active thrombin without the formation of meizothrombin intermediate [344]. As suggested by PMF analysis, Rusviprotease cleaves prothrombin yielding catalytically active α-thrombin which is composed of a light chain (A chain, ~6 kDA) and a heavy chain (B chain, ~31 kDa) linked together by a single disulfide bond, meizothrombin (des) F1 and fragment 1 [344]. Therefore, the mechanism of prothrombin activation by Rusviprotease is identical to the prothrombin activation by factor Xa-prothrombinase complex [342] rather than by factor Xa in absence of co-factors [344].

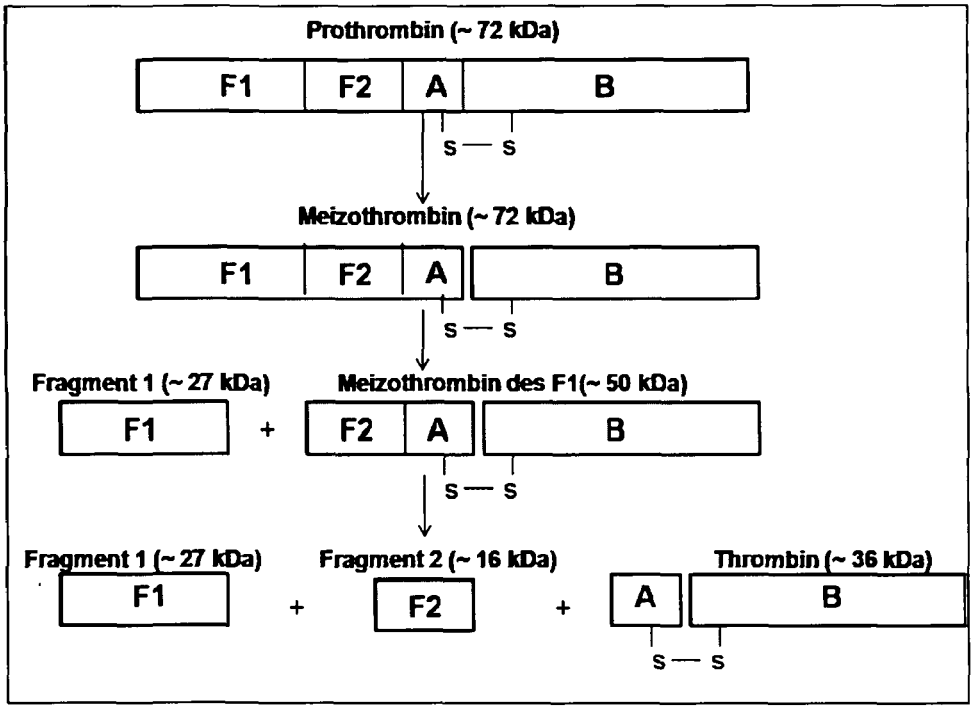


Fig. 5.20 A schematic representation of prothrombin activation by Rusviprotease. The procoagulant metalloprotease, Rusviprotease cleaves prothrombin to form meizothrombin and thrombin in a manner similar to group A prothrombin activators from snake venom. With an increase in incubation time, cleavage of meizothrombin by Rusviprotease results in the formation of meizothrombin(des)F1 and thrombin. The formation of latter demonstrates the overall procoagulant effect.

5.3.3 Rusviprotease enhances the catalytic property of factor Xa

Unlike any of the snake venom prothrombin activators, Rusviprotease significantly enhanced the thrombin formation (2.5 fold) by factor Xa-prothrombinase complex with a corresponding increase in *Vmax* value (1.3 fold) of Rusviprotease-factor Xa-prothrombinase complex, probably due to its additive effect on factor Xa-induced prothrombin activation. Although the prothrombin activating property of factor Xa is enhanced to more than 300,000 fold in the presence of factor Va [342]; however, even an additive 2.5 fold enhancement in the thrombin formation

by factor Xa in presence of Rusviprotease might play a significant role in RVV-induced consumption coagulopathy [15, 46].

5.3.4 Rusviprotease is partially neutralized by commercial antivenoms as well as physiological protease inhibitors

In vitro neutralization of many snake venom toxins by α_2-macroglobulin or α_2-antiplasmin may not be correlated with *in vivo* neutralization of the same toxins [345]. Therefore, it is not surprising that although prothrombin activation by Rusviprotease is significantly inhibited by α_2-macroglobulin under *in vitro* conditions, however, its pharmacological properties remain unaffected *in vivo*.

It has been reported that many snake venom proteins and peptides are difficult to neutralize by commercial polyvalent antivenom [322] owing to their small size or differences in the protein as a result of geographical variation. Further, PAV is the only choice for the treatment of snakebite victims in developing countries and therapeutic use of MAV is still in its infant stage [322]. The present study indicates that MAV moderately surpasses PAV in neutralizing the catalytic and pharmacological properties of Rusviprotease suggesting the therapeutic superiority of MAV over PAV in neutralizing Rusviprotease-induced coagulopathy.

5.3.5 Rusviprotease lacks *in vitro* toxicity and demonstrates *in vivo* anticoagulant effects in Balb/c mice

Unlike many of the SVMPs, Rusviprotease lacks *in vitro* basement membrane hydrolytic effects [340]. This clearly suggests the *in vitro* non-toxic nature of Rusviprotease. It is well known that haemostatic alteration is a common clinical manifestation after Russell's Viper envenomation mainly because of procoagulant components present in the crude RVV [15, 46]. Snake venom prothrombin activators are responsible for defibrinogenation leading to disseminated intravascular coagulopathy [87]. The role of Rusviprotease in the pathophysiology

associated with Russell's Viper envenomation is plausible since it constitutes 3.2% protein of RVV. In a full bite, Russell's Viper may inject 225-250 mg of venom to its victims [96]. Therefore, the concentration of Rusviprotease in the blood of an adult human would be approximately 0.1 mg kg^{-1} (~0.05 μmol^{-1}). But in the mouse, the concentration of the same injected dose will be ~142 μmol^{-1}. Because Rusviprotease shows procoagulant as well as fibrinogenolytic activity at a dose of ~0.04 μmol^{-1}; therefore, its role in pathogenesis of Russell's Viper envenomation is well anticipated. It is important to note that, RVV procoagulant proteases generally do not lead to thrombosis in humans unlike its affect in small animals, because the amount of venom injected in human is less compared to mouse [346]. Therefore, instead of thrombosis, RVV induces consumption coagulopathy in humans [346]. The *in vivo* anticoagulant effect of Rusviproteae on Balb/c mouse is probably due to its ability to exhaust vital coagulation factors such as prothrombin and fibrinogen rendering the blood incoagulable leading to hemostatic alterations via consumption coagulopathy [46]. These observations are similar to previously reported snake venom prothrombin activators such as Basparin A isolated from *Bothrops asper* venom [258].

CHAPTER VI

Isolation and characterization of a toxic, anticoagulant fraction and its cytotoxin-like component from Daboia russelii russelii venom.

RESULTS AND DISCUSSION

6 Isolation and characterization of a toxic, anticoagulant fraction and its cytotoxin-like component from *Daboia russelii russelii* venom

6.1 Brief introduction

Venom protein complexes or strongly associated venom proteins have been shown to demonstrate enhanced toxicity as compared to the individual components of the same complex [59, 274-278]. RVV is enriched with many low molecular mass (4-15 kDa) peptides [59, 98, 106, 288, 347-348] as well as proteases which bring about hemostatic alterations in the victims [64-66, 82-83, 125]. This study characterizes the *in vitro* biochemical and pharmacological properties of an anticoagulant venom fraction and provides insight into the contribution of the isolated anticoagulant fraction in immobilization of agile mammalian prey. Besides, the probable contribution of this anticoagulant fraction in the overall toxicity of Russell's Viper envenomation has also been explored.

6.2 Results

6.2.1 Purification of a lethal venom fraction from RVV

6.2.1.1 Fractionation of crude RVV on Sephadex G-50 column

Fractionation of crude RVV on Sephadex G-50 gel-filtration column resolved the venom into 11 peaks (Fig 6.1A). The peak GF-VI demonstrated fibrinolytic property and but did not show detectable anticoagulant activity on PPP (Table 6.1).

6.2.1.2 Fractionation of GF-VI through anion-exchanger DEAE-cellulose column

The gel-filtration peak (GF-VI), showing fibrinolytic activity was subjected to DEAE-cellulose column (0.5 cm x 2.0 cm). This resulted in the separation of GF-

VI into four peaks (GF-VI DEAE-I, GF-VI DEAE-II, GF-VI DEAE-III and GF-VI DEAE-IV (Fig. 6.1B). The anion-exchange fraction showing fibrinolytic activity (GF-VI DEAE-II) also prolonged the recalcification time of PPP (Table 6.1). This fraction was considered for further characterization.

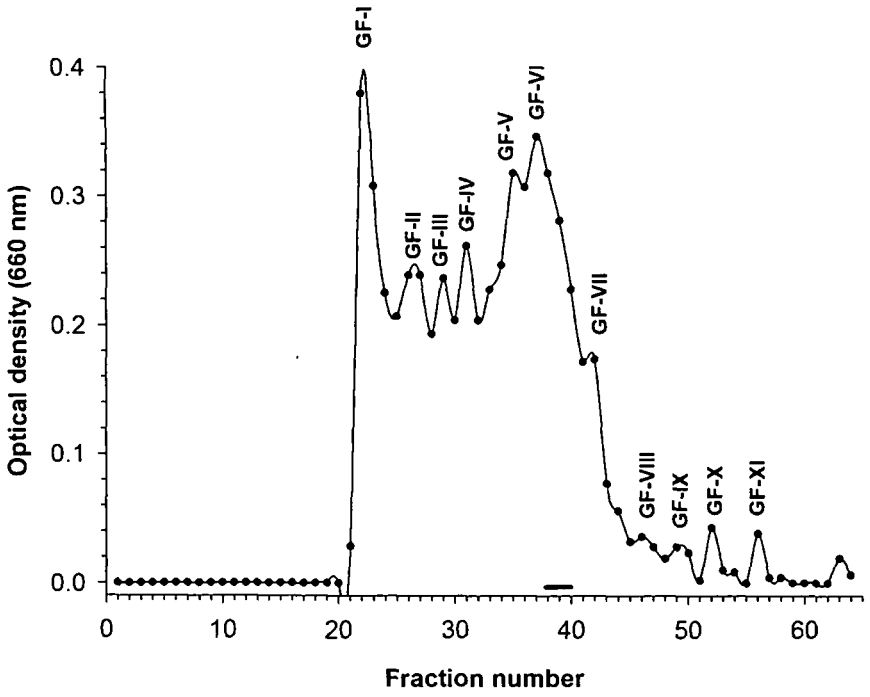


Fig. 6.1A Fractionation of crude RVV through Sepahdex G-50 gel-filtration column (1 cm x 60 cm). The column was equilibrated with 20 mM potassium phosphate buffer (pH 7.0) and was loaded with 25.0 mg of crude RVV dissolved in 1.0 ml of the same buffer. The elution of proteins was performed at a flow rate of 20 ml h^{-1} with the equilibration buffer at room temperature (~23 °C). 1.0 ml fractions were collected and the protein content of each fraction was measured at 660 nm as described in section 3.2.3.1. The proteolytic (fibrinolytic) activity as well as plasma recalcification time of protein peaks was also screened following the methods described in sections 3.2.3.3.1 and 3.2.4.1 respectively.

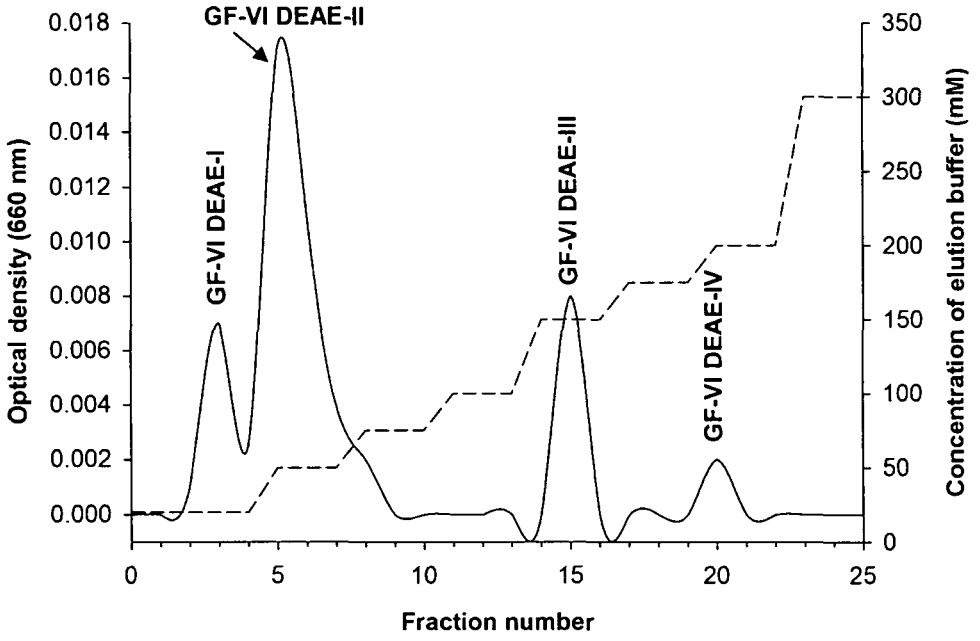


Fig 6.1B Fractionation of GF-VI through DEAE-Cellulose anion-exchange column (0.5 cm x 2.0 cm). GF-VI (2.3 mg) was loaded to a DEAE-Cellulose anion-exchange column and elution was carried out stepwise with potassium phosphate buffers, pH 7.0 of various molarities 20-300 mM potassium phosphate buffer at room temperature (~23 °C) as described in section 3.2.1.3.

6.2.1.3 Separation of the components of GF-VI DEAE II by RP-HPLC

RP-HPLC separation of the anion-exchange fraction (GF-VI DEAE II) demonstrating anticoagulant activity, yielded two protein peaks, a smaller (minor) peak RP-I (retention time 8.8 min) and a larger (major) peak RP-II (retention time 9.1 min) (Fig 6.1C).

The peak RP-I demonstrated weak protease activity (Table 6.1) whereas RP-II fraction did not show detectable protease or PLA_2 activity.

The yield of GF-VI DEAE-II was 1.6% of total RVV protein whereas RP-I and RP-II comprised 0.04% and 1.11% of the total RVV protein, respectively (Table

6.1). Because of the low protein yield as well as almost complete abolition of catalytic (fibrinolytic) activity, the RP-I fraction of GF-VI DEAE-II was not characterized further.

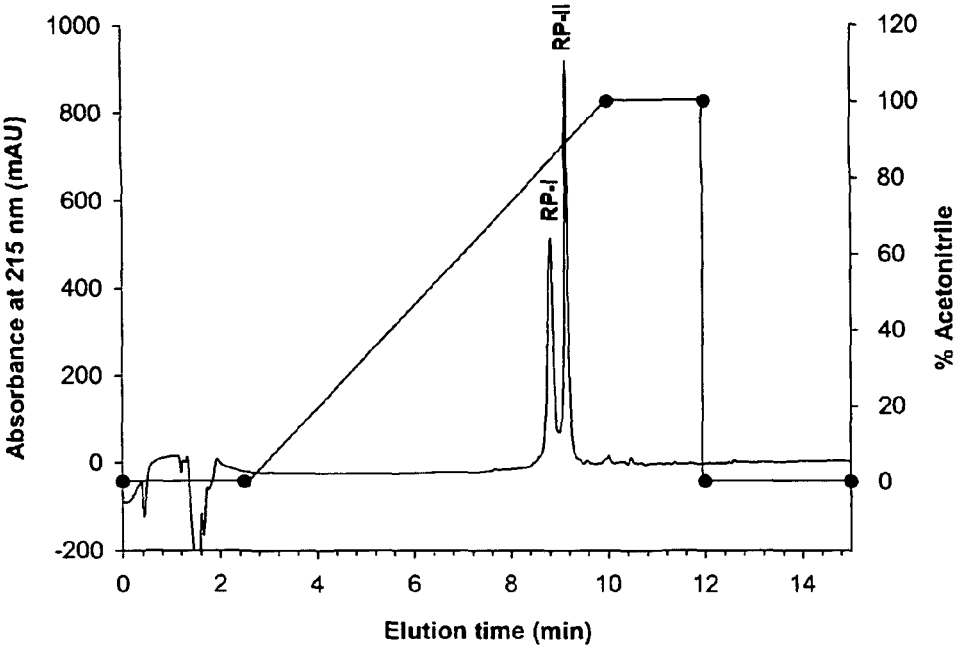


Fig. 6.1C RP-HPLC analysis of GF-VI DEAE-II on C_{18} RP-HPLC column (300 Å; 2.1 x 150 mm) coupled to an UHPLC system (Ultimate 3000, Dionex, Germany). 40.0 µg of GF-VI DEAE-II was dissolved in 20.0 µl of solvent A (0.1% v/v TFA in H_2O and subjected to fractionation through C_{18} RP-HPLC column (300 Å, 2.1 mm x 150 mm). Bound proteins were eluted using a gradient of 0-100% solvent B (0.1% v/v TFA in 80% v/v acetonitrile in H_2O) at a flow rate of 0.35 ml min^{-1}. Detection was monitored at 215 nm and the eluted fractions were collected manually. The peak was screened for protein content, protease and plasma recalcification time as discussed in sections 3.2.3.1, 3.2.3.3.1 and 3.2.4.1, respectively.

Table 6.1 Summary of isolation of the anticoagulant RVV fraction GF-VI DEAE-II and its individual components. Values in the table represent a typical experiment.

Fraction	Total protein (mg)	Yield (%)	Fibrinolytic activity		Anticoagulant activity	
			Total activity (Units)	Specific activity (Unit mg^{-1})	Total activity (Units)	Specific activity (Unit mg^{-1})
CRVV	25.0	100.0	12717.0	508.7	ND	ND
Gel-filtration (GF-VI)	2.3	9.2	2463.0	1070.0	ND	ND
Ion-exchange (GF-VI DEAE-II)	0.4	1.6	965.0	2412.5	14.3	35.8
RP-HPLC						
RP-I	0.01	0.04	3.9	392.0	1.2	30
RP-II	0. 30	1.11	ND	ND	ND	ND

ND= Not detected

6.2.2 Molecular mass determination

6.2.2.1 SDS-PAGE analysis

SDS-PAGE analysis of GF-VI DEAE-II under non-reducing conditions revealed a faint band at ~35 kDa and a smear between 12-22 kDa (Fig. 6.2). Under reducing conditions, a thin band was observed at ~35 kDa (RP-I) and distinct band was observed at ~9 kDa (RP-II). The ~9 kDa peptide was found to be the major component of the venom fraction whereas the higher molecular mass component (~35 kDa) was visible only when a higher amount of the fraction was subjected to SDS-PAGE analysis (Fig. 6.2).

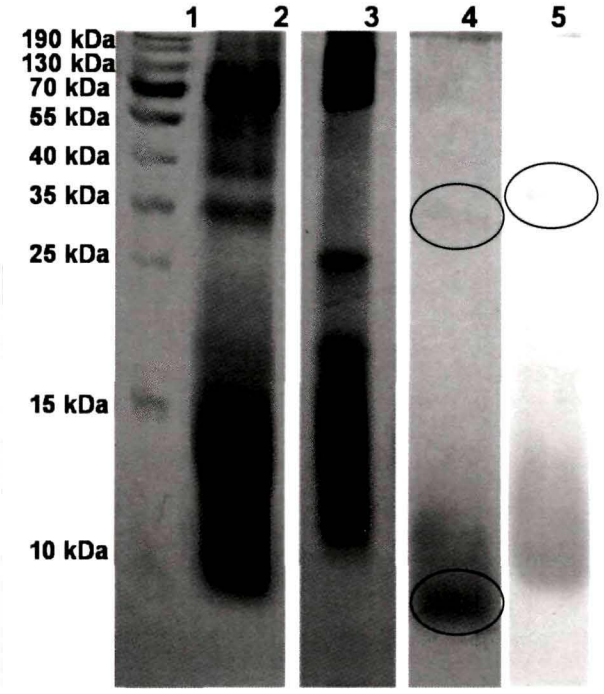


Fig. 6.2 SDS-PAGE (15%) analysis of GF-VI DEAE-II under non-reducing (NR) and reducing (R) sample conditions. Lane 1, Molecular mass marker; lane 2, crude RVV (R, 20.0 µg); lane 3, crude RVV (NR, 20.0 µg); lane 4, GF-VI DEAE-II (R, 20.0 µg); lane 5, GF-VI DEAE-II (NR, 20.0 µg). The experiment was done as described in section 3.2.2.1.1.

6.2.2.2 MALDI-TOF-MS analysis

MALDI-TOF-MS analysis of RP-I showed molecular masses in the range of 27.6-49.1 kDa (Fig. 6.3A), whereas the molecular mass of RP-II fractions was determined as 6.6 kDa by MALDI-TOF-MS analysis (Fig. 6.3B).

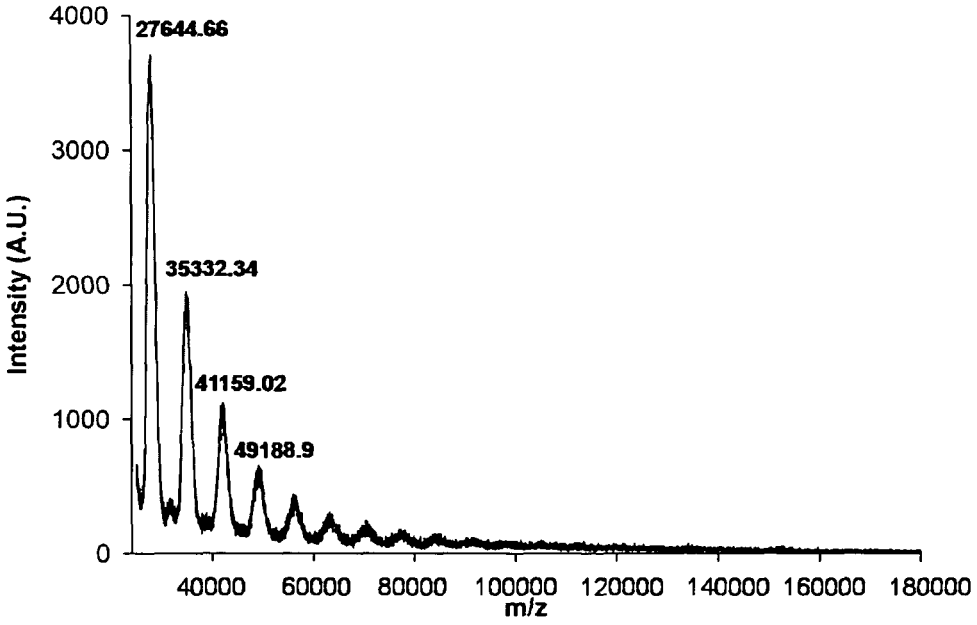


Fig 6.3A Determination of molecular mass of RP-I by MALDI-TOF-MS analysis. Mixtures of 0.5 µl α-cyano-4-hydroxycinnamic acid matrix (10 mg ml^{-1}) and 0.5 µL of a RP-I solution (10.0 µg $µl^{-1}$, diluted in water containing 0.1% TFA) was spotted onto an Opti-TOF-384 plate (ABSciex), dried, and analyzed in positive linear mode on an AB 4800-Plus MALDI-TOF-TOF instrument as described in section 3.2.2.1.2. Spectra were acquired in the m/z range of 20,000–180,000. The spectrum shows signal at m/z 27644.66, 35332.34, 41159.02 and 41612.19 corresponding to the molecular mass of 27.6 kDa, 35.3 kDa, 41.1 kDa and 41.6 kDa respectively.

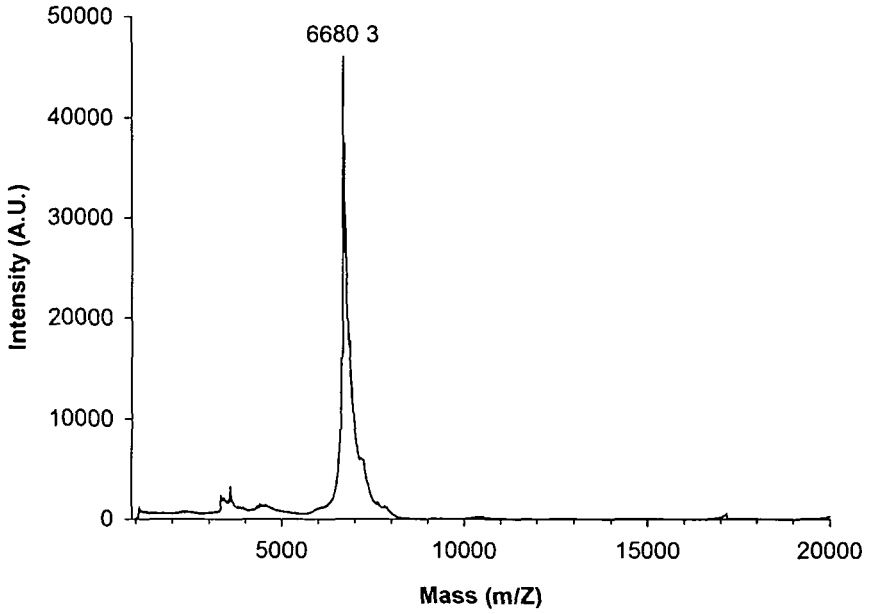


Fig 6.3B Determination of molecular mass of RP-II by MALDI-TOF-MS analysis. Mixtures of 0.5 µl α-cyano-4-hydroxycinnamic acid matrix (10 mg ml^{-1}) and 0.5 µL of a RP-II solution (10.0 µg $µl^{-1}$, diluted in water containing 0.1% TFA) was spotted onto an Opti-TOF-384 plate (ABSciex), dried, and analyzed in positive linear mode on an AB 4800-Plus MALDI-TOF-TOF instrument as described in section 3.2.2.1.2. Spectra were acquired in the m/z range of 2,000–20,000. The spectrum shows signal at m/z 6,680.3 corresponding to a single species with a molecular mass of 6.6 kDa.

6.2.3 Identification of the individual components (RP-I and RP-II) of the venom fraction

The LC MS/MS analysis of RP-I showed its significant identity with serine proteases isolated from snake venom, particularly from Viperidae venom. The RP-I showed highest identity with a 25.5 kDa serine protease fragment OS isolated from *Echis carinatus sochureki* (accession no. E9JG22_ECHCS, -10lgP: 74.91, 3% sequence coverage, 1 unique peptide). In addition, the RP-I fraction also demonstrated similarity with serine protease fragment OS from *Protobothrops*

flavoviridis (accession no. T2HP78_PROFL) and serine protease 10 from *Crotalus adamanteus* (accession no. J3S3W4_CROAD). BLASTp analysis of the unique peptide sequence, W.DNDIMLIK.L (m/z 481.25) against snake venom protein sequences deposited in the NCBI database showed significant sequence homology with previously reported trypsin-like serine proteases from snake venom (Table 6.2).

The PMF analysis of RP-II fraction showed its highest similarity (MASCOT score 147) to cytotoxin-10 (accession no. H3NJ0E) isolated from the venom of Egyptian cobra *Naja haje annulifera* (Table 6.3). By MS/MS sequence coverage of RP-II, three tryptic peptide sequences viz. GCIDVCPK (ion score 36), YVCCNTDR (ion score 38) and YVCCNTDRCN (ion score 43) were identified; although no putative conserved domains could be detected. The BLASTp analysis of these tryptic peptide sequences against the snake venom protein sequences deposited in the NCBI database showed significant sequence homology to cytotoxins, cardiotoxins and three finger toxin precursors from the elapid snake venom; besides, these sequences also revealed partial similarity to cytotoxins and neurotoxins reported from RVV (Fig. 6.4). Because of the similarity of RP-II with snake venom cytotoxins it has been named as Rusvitoxin (Russell's Viper cytotoxin).

Table 6.2 Identification of the peptides released from in solution trypsin digestion of RP-I. (# According to NR NCBI database).

Identified peptides	Identified protein	Species	Query coverage	Identity	Accession number#
WDNDIMLIKL	serine protease	*Echis carinatus sochureki*	100	100	ADI47573.1
	Beta-fibrinogenase brevinase	*Gloydius blomhoffii*	100	90	Q9PT51.1|VSPB_GLOBL
	serine protease	*Echis ocellatus*	100	90	ADI47564.1
	serine beta-fibrinogenase-like protein precursor	*Daboia siamensis*	100	90	ADP88560.1|
	Alpha-fibrinogenase	*Macrovipera lebetina*	100	90	Q8JH85.1|VSPA_MACLB

Table 6.3 Identification of RP-II component of GF-VI DEAE-II by peptide mass fingerprinting analysis.

Protein	Measured mass (Da)	Deviation from computed mass (Da)	Range	Ion score	Peptide Sequence
RP-II	947.42	0.006	37-44	36	R.GCIDVCPK.N
(Rusvitoxin)	1086.42	0.0108	51-58	38	K.YVCCNTDR.C
	1360.49	0.0056	51-60	43	K.YVCCNTDRCN.-

6.2.4 N-terminal sequence analysis of Rusvitoxin

The NH_2-terminal sequence of the first 10 residues of Rusvitoxin was determined as L-Q-C-N-I-L-V-P-I-A. A comparison of this sequence with other venom protein sequences deposited in NCBI database showed a significant sequence homology with the NH_2-terminal peptide sequence of cytotoxins from snake venoms particularly of *Naja* species (Table 6.4). The NH_2-terminal sequences of cytotoxins from snake venom were found to be highly conserved and variations were observed only at two positions; at position 2, K/E in similar peptides was replaced by Q in Rusvitoxin and at position 5, K/Q was replaced by I in RP-II (Table 6.4).

6.2.5 Secondary structure analysis of Rusvitoxin

The far UV spectrum of Rusvitoxin demonstrated minima at 208 nm and 220 nm suggesting the predominance of β-structures and random coils in its secondary structure (Fig. 6.4). Software analysis predicted that Rusvitoxin consisted of 29.5% α-helical content, 21.6% β pleated sheets, 15.8% turns and 33.1% random coils in its native structure (Fig. 6.4).

Table 6.4 Multiple sequence alignment of peptide fragments of Rusvitoxin generated after tryptic digestion with snake venom cytotoxins, cardiotoxins and three finger toxins. The N-terminal sequence of intact Rusvitoxin (RP-II) was obtained by automated Edman degradation. Conserved cysteine residues are shown in red color.

Accession no./Reference	1	10	20	30	40	50	60	Name	Mass (Da)	Species
P0CH80.1	LKCNKLVPLFYKTCPAGKNLCYKMFMVSNKTVPVKRGCIDVCPKNSLVLKYVCCNTDRCN-							CX1_NAJKA	6807	*Naja kaouthia*
P24780.1	LKCNKLIPLAYKTCPAGKNLCYKMFMVSNKTVPVKRGCIDVCPKNSLVLKYVCCNTDRCN-							CX3_NAJNA	6745	*N. naja*
P01445.1	LKCNKLIPLAYKTCPAGKNLCYKMFMVSNKTVPVKRGCIDVCPKNSLLVKYVCCNTDRCN-							CX2_NAJKA	6745	*N. kaouthia*
P86382.1	LKCNKLIPLAYKTCPAGKDLCYKMYMVSNKTVPVKRGCIDVCPKNSLLVKYECCNTDRCN-							CX7_NAJNA	6792	*N. naja*
P01447.1	LKCNKLIPLAYKTCPAGKNLCYKMYMVSNKTVPVKRGCIDVCPKNSLVLKYECCNTDRCN-							CX1_NAJNA	6791	*N. naja*
P01453.1	LECNQLIPIAHKTCPEGKNLCYKMFMVS--TSTVPVKRGCIDVCPKNSALVKYVCCNTDRCN-							CX10_NAJHA	6682	*N. haje Dabioa*
[347]	LECNKLQPIASK----------							DNTx-I	6675	*russelii russelii*
[348]	IRCFITPDUTSQACP----------							DNTx-III	6849	*D. r. russelii*
Present work	LQCNILVPIA----------RGCIDVCPKN---------KYVCCNTDRCN-							Rusvitoxin	6680	*D. r. russelii*

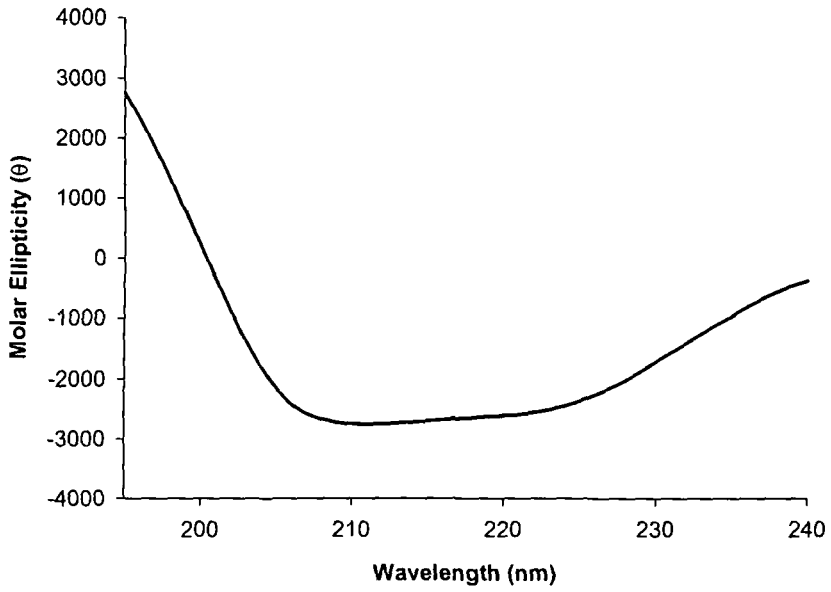


Fig 6.4 Circular dichroism spectra of Rusvitoxin showing its secondary structure. Native Rusvitoxin (0.3 mg ml^{-1}) was dissolved in 20 mM potassium phosphate buffer pH 7.0 and the far UV-CD spectra was recorded at room temperature (~23 °C) between 195–250 nm against the appropriate buffer (blank).

6.2.6 Biochemical characterizations of GF-VI DEAE-II

6.2.6.1 Protease activity

Protease activity of GF-VI DEAE-II towards the pseudo-substrate casein as well as different natural substrates was determined (Table 6.5). GF-VI DEAE-II demonstrated highest specificity for fibrin followed by casein and fibrinogen. GF-VI DEAE-II did not hydrolyze substrates such as albumin and globulin (Table 6.5).

The GF-VI DEAE-II fraction exhibited fibrinolytic property with a specific activity of 2412.5 ± 7.2 Units mg^{-1} protein. Conversely, the RP-I demonstrated fibrinolytic specific activity of 392.5 ± 5.6 Units mg^{-1} protein under identical experimental conditions. RP-II fraction did not show any detectable protease

activity. Human plasmin showed better fibrinolytic activity as compared to the GF-VI DEAE-II with a specific activity of 9056.8 ± 25.7 Units mg^{-1} protein.

Incubation of the GF-VI DEAE-II with 60% acetonitrile (the concentration at which RP-I was eluted from C_{18} column) for 10.0 min at room temperature (~23 °C) resulted in loss of more than 80% of the original fibrinolytic activity of the native (untreated) fraction.

Table 6.5 Proteolytic activity of GF-VI DEAE-II towards macromolecular substrates. GF-VI DEAE-II (2.0 μg) was incubated with protein substrates for 90 min at 37 °C and the release of free tyrosine was calculated as described in section 3.2.3.2.

Protease activity	**Specific activity (Units mg^{-1} protein)**
Bovine Fibrin (0.6%)	2412.5 ± 7.2
Casein (1.0%)	215.5 ± 5.8
Bovine fibrinogen (0.25%)	13.2 ± 3.4
Bovine albumin (1.0%)	Not detected
Bovine globulin (1.0%)	Not detected

Values are mean ± S.D. of triplicate experiments.

6.2.6.2 Fibrinolytic activity by SDS-PAGE analysis

GF-VI DEAE-II fraction could not hydrolyze fibrin up to 1 h of incubation at 37 °C, pH 7.4. However, beyond this period of incubation, the α-chain of fibrin was progressively hydrolyzed. On the contrary, the β and γ-chains of fibrin remained intact after 6 h of incubation at 37 °C (Fig. 6.5A).

SDS-PAGE analysis of fibrin degradation by the GF-VI DEAE-II showed that it degraded the α-chain of fibrin to release a single major peptide with an apparent molecular mass of 31.4 kDa (Fig. 6.5A) whereas human plasmin degraded fibrin differentially to release three major fragments of molecular mass of 41.1 kDa, 49.4 kDa and 56.4 kDa, respectively (Fig. 6.5B).

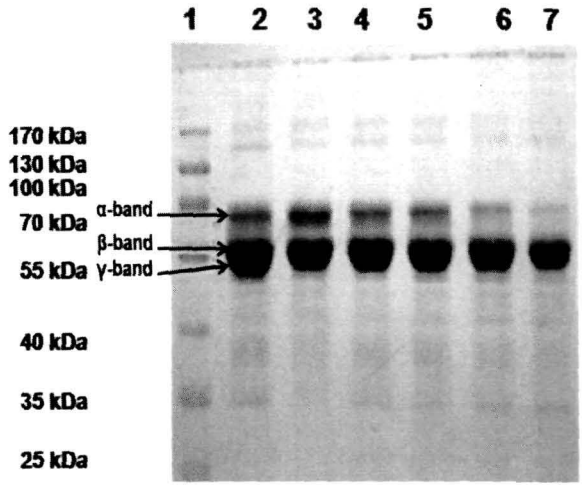


Fig. 6.5A Fibrin degradation by GF-VI DEAE-II. The fibrin degradation products were separated by 12% SDS-PAGE and stained with 0.25% Coomassie brilliant blue. Lane 1, molecular weight markers; lane 2, control (fibrin treated with buffer); lanes 3-7, fibrin degradation by 2.0 µg GF-VI DEAE-II after 0.5, 1, 2, 4 and 6 h of incubation, respectively at 37 °C, pH 7.4. The experiment was performed as described in section 3.2.3.3.2.

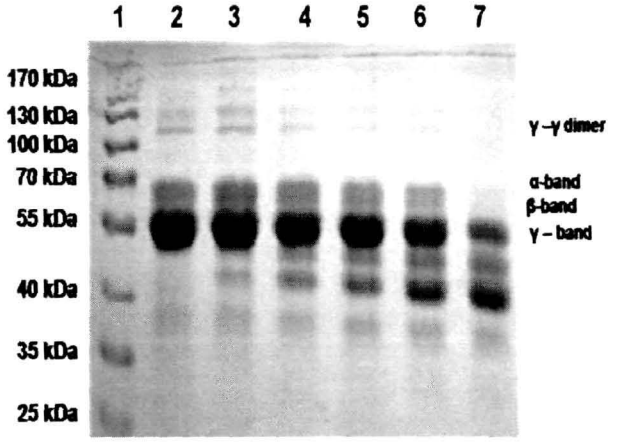


Fig. 6.5B Fibrin degradation by human plasmin. The fibrin degradation products were separated by 12% SDS-PAGE and stained with 0.25% Coomassie brilliant blue. Lane 1, molecular weight marker; lane 2, control (fibrin incubated with buffer); lanes 3-7, fibrin clot degradation by 2.0 µg human plasmin after 0.5,

1, 2, 4 and 6 h of incubation, respectively at 37 °C, pH 7.4. The experiment was performed as described in section 3.2.3.3.2.

6.2.6.3 Fibrinogenolytic activity by SDS-PAGE analysis

GF-VI DEAE-II could degrade fibrinogen only after 16 h of incubation at 37 °C suggesting its preference for fibrin over fibrinogen as a substrate (Fig. 6.6A). Human plasmin, on the other hand, degraded fibrinogen within 4 h of incubation (Fig. 6.6B).

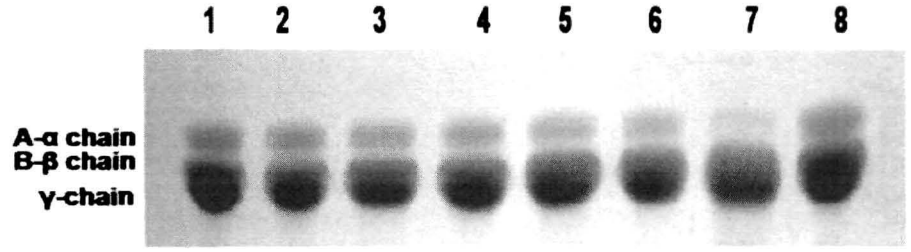


Fig. 6.6A Fibrinogen degradation by GF-VI DEAE-II. The fibrinogen degradation products were separated by 12% SDS-PAGE and stained with 0.25% Coomassie brilliant blue. Lanes 2-7, fibrinogen (2.5 mg ml^{-1}) degradation by 2.0 µg GF-VI DEAE-II after 2, 4, 6, 8, 12 and 16 h of incubation, respectively at 37 °C, pH 7.4. Lane 1 and 8, control (fibrinogen without enzyme). The experiment was performed as described in section 3.2.3.3.2.

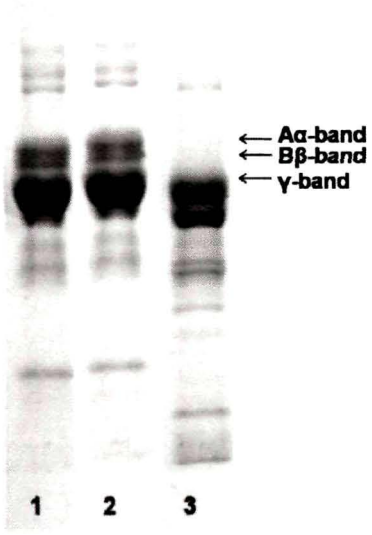


Fig. 6.6B Comparison of fibrinogen degradation by human plasmin and GF-VI DEAE-II. The fibrinogen degradation products were separated by 12% SDS-PAGE and stained with 0.25% Coomassie brilliant blue. Lane 1, Control (fibrinogen without any enzyme); lane 2, fibrinogen degradation by 2.0 µg GF-VI DEAE-II after 4 h at 37 °C, pH 7.4; lane 3, fibrinogen degradation by 2.0 µg human plasmin after 4 h of incubation at 37 °C, pH 7.4. The experiment was performed as described in section 3.2.3.3.2.

6.2.6.4 RP-HPLC analysis of fibrin/fibrinogen degradation

The elution profiles on UHPLC of the peptides released from fibrinogen and fibrin by GF-VI DEAE-II are shown in Fig. 6.7A and Fig. 6.7B, respectively.

As shown in Fig. 6.7A, human plasmin degraded fibrinogen to release peptides which had a retention time of 15.20, 15.75, 15.98, 16.16, 16.50, 16.93 and 17.19 min. However, none of these peptide peaks were visualized in the elution profile of peptides released by the action of GF-VI DEAE-II on fibrinogen after 4 h of incubation at 37 °C (Fig. 6.7A).

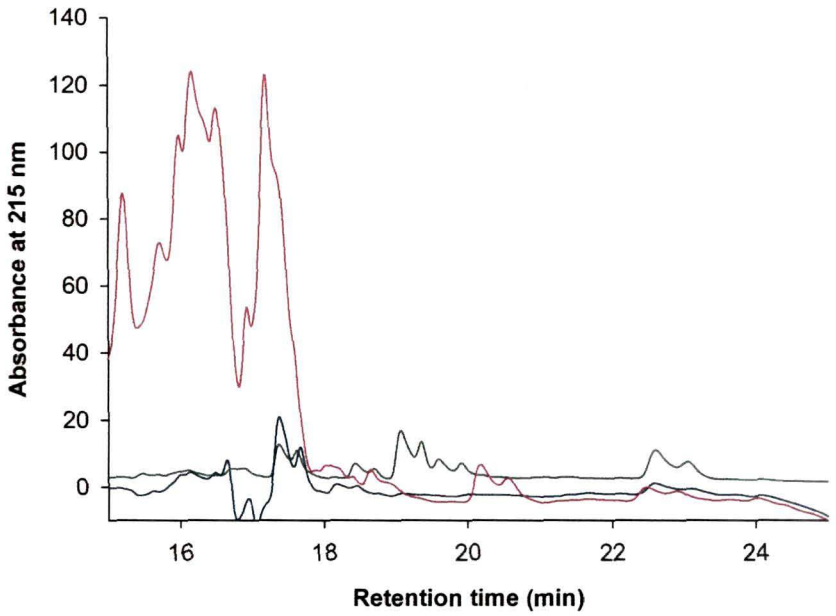


Fig. 6.7A A comparison of fibrinogen degradation between GF-VI DEAE-II and human plasmin by RP-HPLC analysis. Elution profiles on C_{18} RP-HPLC column (2.1 mm x 150 mm) coupled to a UHPLC system (Ultimate 3000, Dionex, Germany) of peptides released from fibrinogen incubated with 20 mM potassium phosphate buffer, pH 7.4 (shown in green), by the action of the GF-VI DEAE-II (2.0 µg) (shown in blue) and human plasmin (2.0 µg) (shown in red). The experimental conditions have been described in section 3.2.3.3.3.

Analysis of the elution profile of peptides released from fibrin by the action of GF-VI DEAE-II and human plasmin showed significant differences as well. As shown in Fig. 6.7B, GF-VI DEAE-II released peptides with a retention time of 16.79, 16.91, 17.46 and 17.84 min from fibrin similar to human plasmin though to a lesser extent. Additionally, unlike human plasmin, GF-VI DEAE-II could not release peptides with the retention time of 15.55, 15.89 and 16.20 min, respectively (Fig. 6.7B).

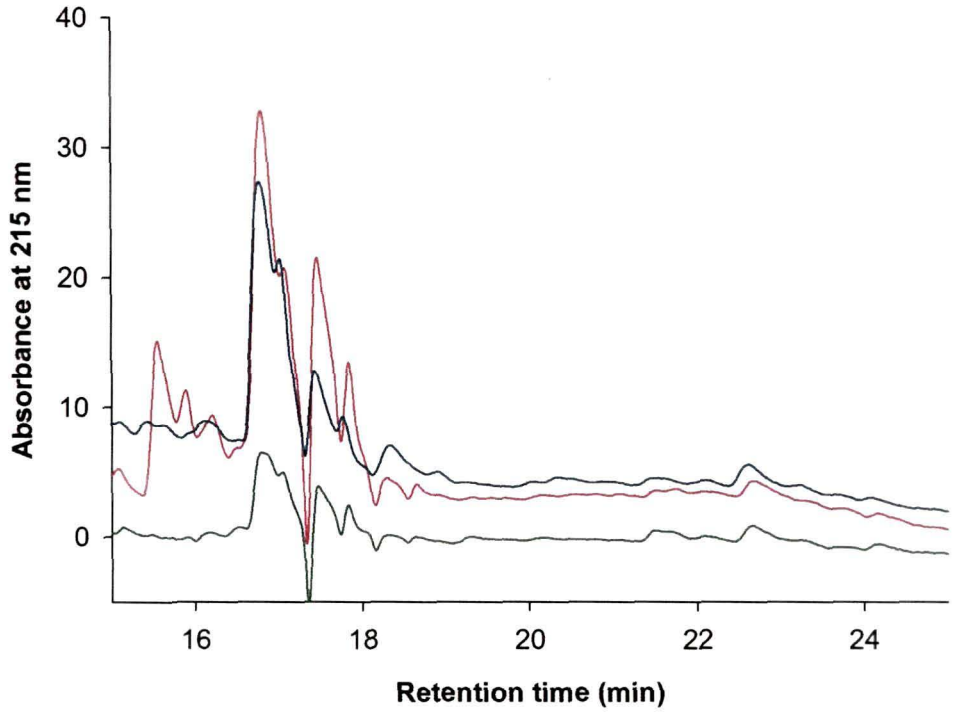


Fig 6.7B A comparison of fibrin degradation between GF-VI DEAE-II and human plasmin by RP-HPLC analysis. Elution profiles on C_{18} RP-HPLC column (2.1 mm x 150 mm) coupled to a UHPLC system (Ultimate 3000, Dionex, Germany) of peptides released from fibrin incubated with 20 mM potassium phosphate buffer, pH 7.4 (shown in green), by the action of the GF-VI DEAE-II (2.0 µg) (shown in blue) and human plasmin (2.0 µg) (shown in red). The experimental conditions have been described in section 3.2.3.3.3.

6.2.6.5 Amidolytic activity assay

The amidolytic activity of GF-VI DEAE-II was tested on different chromogenic substrates (Table 6.6). The GF-VI DEAE-II fraction exhibited highest specific activity towards chromogenic substrate for plasmin (V0882). Considering the activity of GF-VI DEAE-II towards V0882 as 100%, the relative activities towards other substrates were also calculated (Table 6.6).

Table 6.6 Amidolytic activity of GF-VI DEAE-II. The venom fraction (1.0 μg ml^{-1}) was incubated with the chromogenic substrates (0.2 mM) and the amount of pNA released was determined as described in section 3.2.3.6. One unit of amidolytic activity has been expressed as μmoles of pNA released min^{-1} ml^{-1}.

Chromogenic Substrate	Specific activity (Units mg^{-1} protein)	Relative activity (%)
V0882	$3.4 \times 10^{4} \pm 30.5$	100
T1637	$2.5 \times 10^{3} \pm 22.5$	7.5
B2133	ND	ND
F3301	ND	ND
I0898	ND	ND
B3133	ND	ND

ND- not detected

Values are mean ± S.D. of triplicate experiments.

6.2.6.6 Optimum temperature for fibrinolytic activity

The optimum temperature for the enzymatic activity of GF-VI DEAE-II towards fibrin was assayed at a temperature range of 25-50 °C (Fig. 6.8). GF-VI DEAE-II showed optimum catalytic activity at a temperature range of 35-37 °C (Fig. 6.8).

6.2.6.7 Optimum pH for fibrinolytic activity

The optimum pH for the enzymatic activity of GF-VI DEAE-II towards fibrin was assayed at a pH range of 6.0-8.0 at a fixed temperature of 37 °C after 90 min of incubation (Fig. 6.9). GF-VI DEAE-II showed optimum catalytic activity at pH 7.4 (Fig. 6.9).

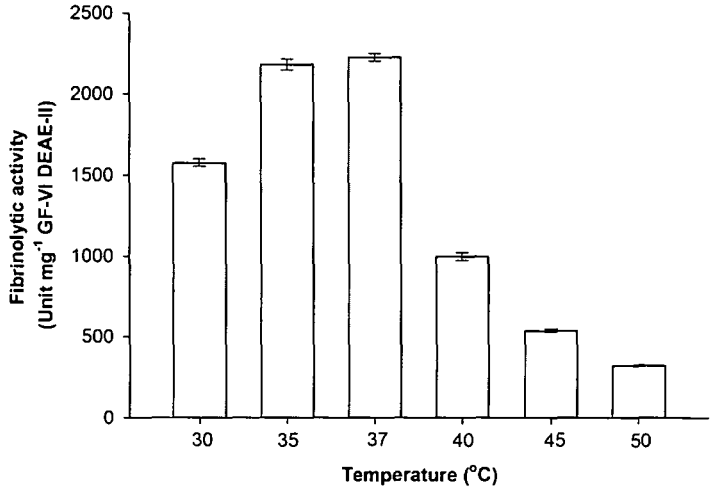


Fig. 6.8 Effect of temperature on the catalytic activity of GF-VI DEAE-II. Fibrinolytic activity of GF-VI DEAE-II (2.0 μg) was analyzed at different temperature after 90 min of incubation with fibrin at pH 7.4. One unit (U) of proteolytic activity is defined as *n* mole equivalent of tyrosine formed min^{-1}. The experiment was performed as described in section 3.2.3.18.1. Values are mean ± S.D. of triplicate experiments.

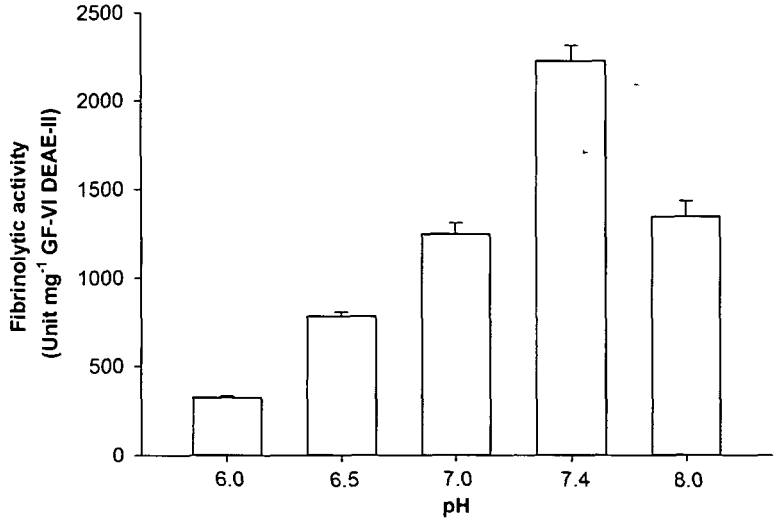


Fig. 6.9 Effect of pH on the catalytic activity of GF-VI DEAE-II. Fibrinolytic activity of GF-VI DEAE-II (2.0 μg) was analyzed at different pH after 90 min of incubation with fibrin at 37 °C. One unit (U) of proteolytic activity is defined as *n*

mole equivalent of tyrosine formed min^{-1}. The experiment was performed as described in section 3.2.3.18.2. Values are mean ± S.D. of triplicate experiments.

6.2.7 Pharmacological properties of GF-VI DEAE-II

6.2.7.1 Plasminogen activating property

SDS-PAGE analysis of plasminogen activation by GF-VI DEAE-II suggested that the isolated venom fraction could not activate plasminogen even after 12 h of incubation at 37 °C, pH 7.4 (Fig 6.10).

6.2.7.2 Thrombin-degrading property

Since SDS-PAGE analysis of thrombin incubated with GF-VI DEAE-II did not show any degradation products; therefore, it is suggested that GF-VI DEAE-II does not have thrombin-degrading property (Fig. 6.11)

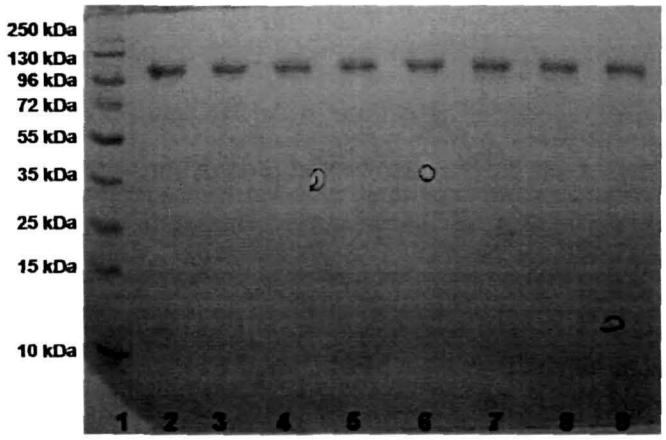


Fig. 6.10 SDS-PAGE analysis of plasminogen activation by GF-VI DEAE-II. Plasminogen activation was performed as described in section 3.2.3.5. The plasminogen activation products were separated by 15% SDS-PAGE and stained with 0.25% Coomassie brilliant blue. Lane 1, molecular weight marker; lane 2, control (plasminogen without GF-VI DEAE-II); lanes 3-8, plasminogen activation

by 2.0 μg GF-VI DEAE-II after 0.5, 1, 2, 4, 6, 8 and 12 h of incubation respectively at 37 °C, pH 7.4.

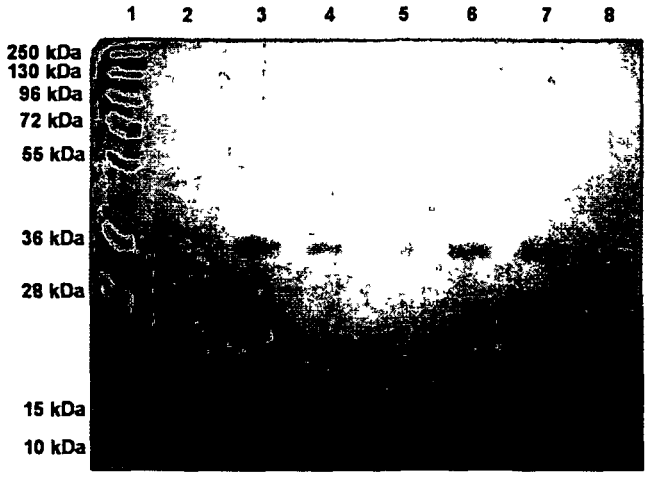


Fig. 6.11 SDS-PAGE analysis of thrombin degradation by GF-VI DEAE-II. Thrombin degradation was performed as described in section 3.2.3.5. The thrombin degradation products were separated by 15% SDS-PAGE and stained with 0.25% Coomassie brilliant blue. Lane 1, molecular weight marker; lane 2, control (thrombin without GF-VI DEAE-II), lanes 3-8, thrombin degradation by 2.0 μg GF-VI DEAE-II after 0.5, 1, 2, 4, 6 and 8 h of incubation, respectively at 37 °C, pH 7.4.

6.2.7.3 Fibrinogen clotting activity

GF-VI DEAE-II (10 μg ml^{-1}) was unable to clot fibrinogen (2.5 mg ml^{-1}) into fibrin as no visible fibrin thread formation was observed after 4 h of incubation of fibrinogen with GF-VI DEAE-II at 37 °C, pH 7.4.

6.2.7.4 Anticoagulant activity

6.2.7.4.1 Plasma recalcification time

In *in vitro* condition, the isolated fibrinolytic venom fraction (0.1 μg μl^{-1}) significantly ($p<0.001$) prolonged the re-calcification time of PPP (Table 6.7). The

anticoagulant activity of RP-I was found to be significantly lower ($p< 0.01$) compared to the same property displayed by GF-VI DEAE-II under identical experimental conditions (Table 6.7). Rusvitoxin did not show any anticoagulant activity on PPP.

Additionally, human plasmin (0.1 $\mu g\ \mu l^{-1}$) displayed superior anticoagulant activity compared to GF-VI DEAE-II (Table 6.7).

6.2.7.4.2 Prothrombin time test

In *in vitro* condition, GF-VI DEAE-II (0.1 $\mu g\ \mu l^{-1}$) significantly ($p<0.05$) prolonged the prothrombin time of PPP (Table 6.7).

6.2.7.4.3 Activated partial thromboplastin time test

GF-VI DEAE-II did not affect the aPTT of PPP at the tested dose of 0.1 $\mu g\ \mu l^{-1}$ (Table 6.7).

6.2.7.5 Hemolytic activity

GF-VI DEAE-II lacked *in vitro* hemolytic activity (direct and indirect) on mammalian goat erythrocytes at the tested dose of 10.0 $\mu g\ ml^{-1}$ (Table 6.7).

6.2.7.6 *In vitro* clot lysing (thrombolytic) activity

GF-VI DEAE-II showed significant *in vitro* clot lysing (thrombolytic) activity (Table 6.7). Human plasmin could lyse 29.8 ± 1.1 and 23.5 ± 2.6 mg of non heated and heated blood clot after 60 min at 37 °C, pH 7.4.

6.2.7.7 Platelet aggregation assay

As shown in Fig. 6.12, GF-VI DEAE-II did not induce platelet aggregation or disaggregation at the tested doses.

6.2.7.8 Effect on extracelullar matrix proteins

SDS-PAGE analysis of extracellular matrix proteins degradation by GF-VI DEAE-II suggested that the venom fraction was unable to degrade any of the tested extracellular matrix proteins namely type-IV collagen, laminin and fibronectin even after 8 h of incubation, at physiological conditions (37 °C, pH 7.4) (Fig. 6.13).

Table 6.7 Pharmacological properties of the GF-VI DEAE-II. Values are mean ± S.D. of triplicate determination.

Parameters	**Values**	
	Control	**GF-VI DEAE-II ($0.1\ \mu g\ \mu l^{-1}$)**
Plasma re-calcification time (s)[#]	162.1 ± 1.6	176.4 ± 2.8*
Prothrombin time (s)	18.6 ± 1.0	23.6 ± 1.5*
Activated prothrombin time (s)	13.5 ± 0.5	13.8 ± 0.6
Direct hemolysis (% Hb released μg^{-1} protein, post 90 min at 37 °C)	0	0
In vitro thrombolytic potency (mg of clot lysed μg^{-1} protein)		
Non heated blood clot	3.1 ± 0.6	26.78 ± 4.2*
Heated blood clot	2.1 ± 0.1	20.4 ± 5.5*

Plasma re-calcification time (s) (by 0.1 $\mu g\ \mu l^{-1}$ human plasmin) = 215.3 ± 6.0*

Plasma re-calcification time (s) (by 0.1 $\mu g\ \mu l^{-1}$ RP-I) = 165.2 ± 3.1

Plasma re-calcification time (s) (by 0.1 $\mu g\ \mu l^{-1}$ RP-II) = 162.5 ± 1.5

Values denoted with '*' are statistically different from the control value ($p<0.05$).

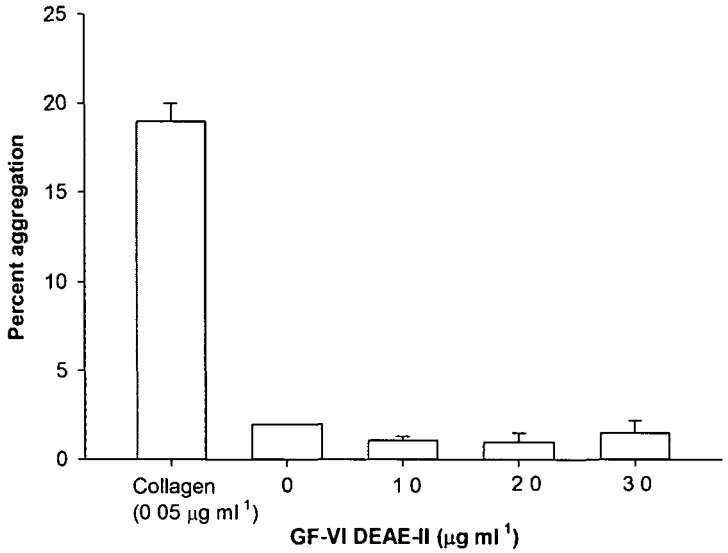


Fig. 6.12 Effect of GF-VI DEAE-II on platelet aggregation. GF-VI DEAE-II (1.0–3.0 µg ml^{-1}) was added to 100 µl of PRP and the decrease in absorbance was monitored for 300 s at 540 nm as described in section 3.2.4.4. Collagen (0.05 µg ml^{-1}) was taken as positive control. The values are mean ± S.D. of triplicate determinations.

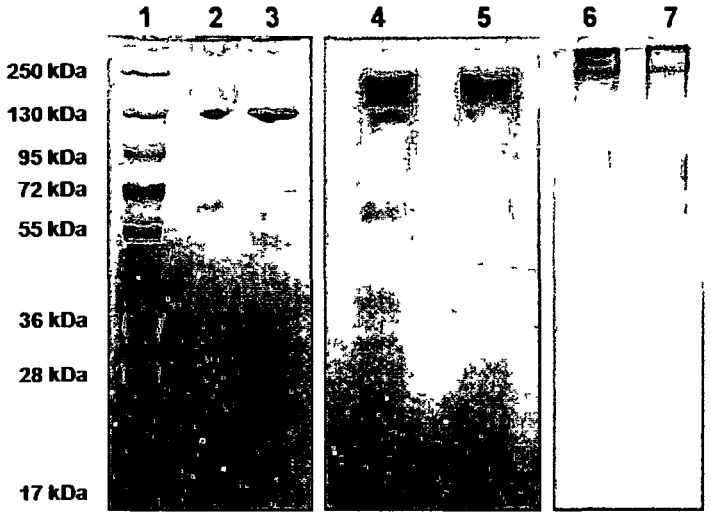


Fig. 6.13 Degradation of type-IV collagen, laminin and fibronectin by GF-VI DEAE-II. ECM proteins such as type-IV collagen, laminin and fibronectin were incubated with GF-VI DEAE-II in a substrate: enzyme ratio of 15:1, w/w for 4 h at

37 °C, pH 7.4. The degradation products were analyzed on 10% SDS-PAGE as described in section 3.2.3.4. Protein bands were visualized after staining with 0.25% Coomasie brilliant blue R-250. Lane 1,molecular weight marker; lane 2, control type-IV collagen (without GF-VI DEAE-II); lane 3, GF-VI DEAE-II treated type-IV collagen; lane 4, control laminin (without GF-VI DEAE-II); lane 5, GF-VI DEAE-II treated laminin; lane 6 control fibronectin (without GF-VI DEAE-II); lane 7, GF-VI DEAE-II treated fibronectin.

6.2.7.9 Cell cytotoxicity assay

6.2.7.9.1 *In vitro* cytotoxicity assessment by MTT based assay

Unlike crude RVV, GF-VI DEAE-II, up to a tested dose of 10.0 μg μl^{-1} did not exhibit cytotoxicity towards the tested cell lines- U87MG and HeLa cells after 48 h of treatment at 37 °C (Figs. 6.14A-B). Crude RVV on the other hand, showed better cytotoxicity towards U87MG cells compared to HeLa cells (Figs. 6.14A-B).

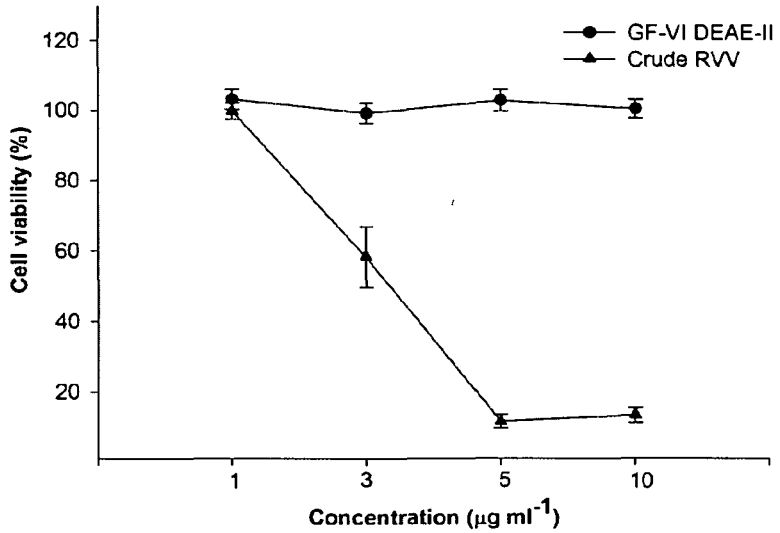


Fig. 6.14A A comparison of *in vitro* cell cytotoxicity between crude RVV and GF-VI DEAE-II towards U87MG cells. U87MG cells (1 x 10^5 cells ml^{-1}) were treated with different concentrations of crude RVV or GF-VI DEAE-II at 37 °C,

pH 7.4 and 5% CO_2 for 48 h. Experiment was performed as described in section 3.2.4.7.1. Values are mean ± S.D. of triplicate determinations.

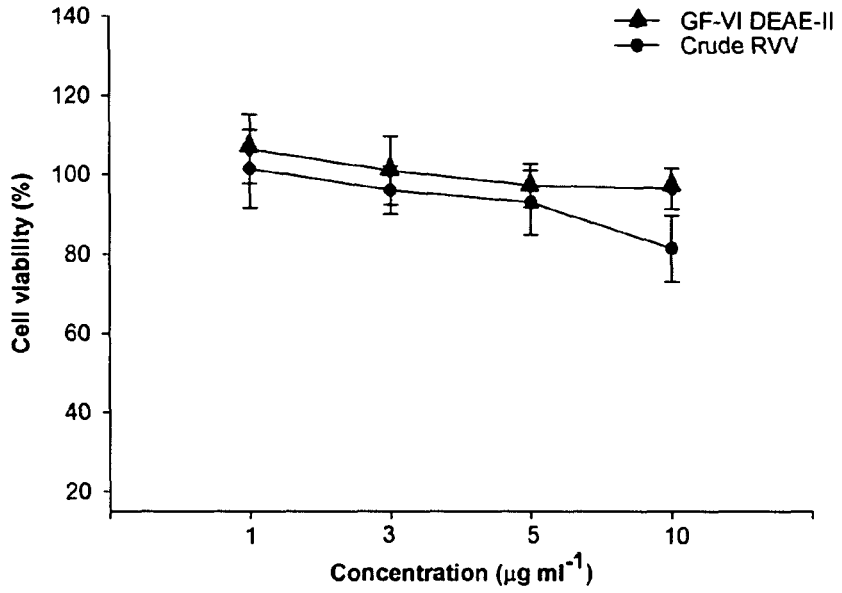


Fig. 6.14B A comparison of *in vitro* cell cytotoxicity between crude RVV and GF-VI DEAE-II towards HeLa cells. HeLa cells (1 x 10^5 cells ml^{-1}) were treated with different concentrations of crude RVV or GF-VI DEAE-II at 37 °C, pH 7.4 and 5% CO_2 for 48 h. Experiment was performed as described in section 3.2.4.7.1. Values are mean ± S.D. of triplicate determinations.

6.2.7.9.2 *In vitro* cytotoxicity assessment by dual staining method

Dual staining analysis of GF-VI DEAE-II treated U87MG cells clearly depicted the non-cytotoxic property of this venom fraction (Fig 6.15). The live cells were stained with AO and fluoresced green under green filter. The dead cells were stained with EB and fluoresced red under red filter. Clear pictures of the viable cells were visible by merging the pictures taken in the two filters. Cells treated with 20 mM potassium phosphate buffer, pH 7.4 and crude RVV (10 μg ml^{-1}) served as negative and positive control, respectively.

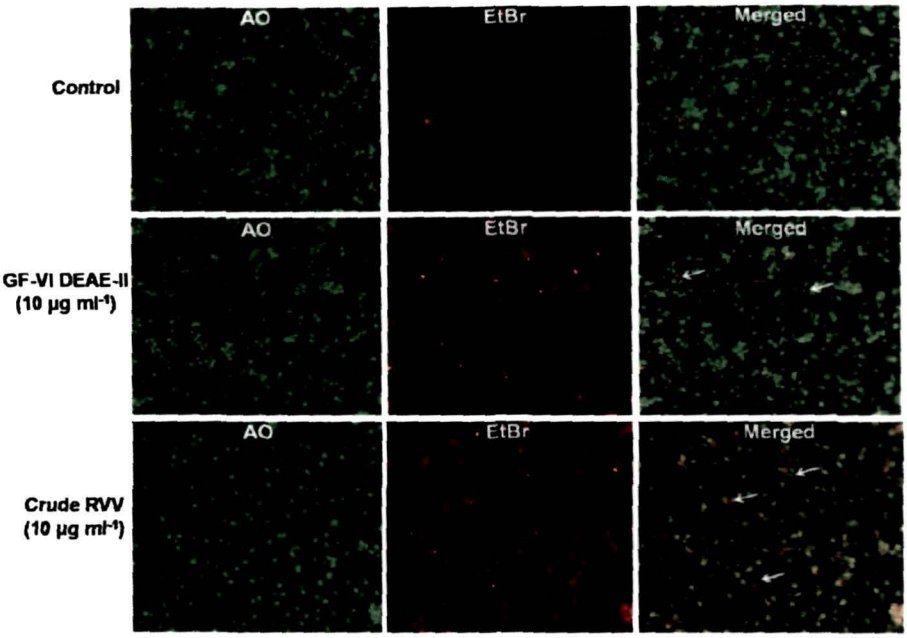


Fig. 6.15 Comparison of *in vitro* cell cytotoxicity between crude RVV and GF-VI DEAE-II towards U87MG cells by dual-staining method. Experiment was performed as described in section 3.2.4.7.2. U87MG cells (1 x 10^5 cells ml^{-1}) was treated with either crude RVV (10 µg ml^{-1}) or GF-VI DEAE-II (10 µg ml^{-1}) for 24 h at 37 °C, 5% CO_2, before staining with 10 µl of AO/EB mix (10 mg mL^{-1} AO and EB in PBS). Cells treated with 20 mM potassium phosphate buffer, pH 7.4 served as control. After 10 min of incubation, cells were washed with PBS (1X, pH 7.4) before visualizing under a fluorescence microscope (Nikon ECLIPSE, TS100, Tokyo) with an excitation filter of 480/30 nm.

6.2.8 Chemical modification and neutralization studies of GF-VI DEAE-II

6.2.8.1 Chemical modification studies

As shown in Table 6.8, the fibrinolytic and anticoagulant properties of GF-VI DEAE-II was strongly inhibited by serine protease inhibitor, PMSF; trypsin inhibitor, TLCK, and DTT which cleaves disulphide bonds. Metalloprotease inhibitors such as EDTA as well as histidine modifier such as pBPB could not

inhibit the catalytic as well as pharmacological properties of GF-VI DEAE-II (Table 6.8).

Table 6.8 A comparison of neutralization of anticoagulant and fibrinolytic activities of GF-VI DEAE-II by chemical inhibitors. GF-VI DEAE-II (2.0 μg) was pre-incubated with different inhibitors for 30 min at room temperature (~23 °C) as described in section 3.2.3.18.1. Thereafter fibrinolytic activity and plasma recalcification time of GF-VI DEAE-II was analyzed in the corresponding assay systems, as described in sections 3.2.3.3.1 and 3.2.4.1, respectively. The activity of GF-VI DEAE-II in absence of inhibitor was considered as 100 % activity.

Inhibitor	**% Residual activity**	
	Fibrinolytic activity	**Anticoagulant activity**
Control (GF-VI DEAE-II without inhibitor)	100	100
EDTA (5 mM)	98.9 ± 09.2	100.2 ± 5.4
DTT (2 mM)	19.5 ± 3.1*	28.5 ± 2.8*
PMSF (2 mM)	5.8 ± 1.5*	7.2 ± 1.3*
TLCK (2 mM)	1.5 ± 0.1*	2.8 ± 0.1*
TPCK (2 mM)	87.5 ± 7.5*	85.2 ± 9.5*
pBPB (2 mM)	95.3 ± 8.1	96.3 ± 5.4

Values are mean ± S.D. of triplicate experimental values.

Values marked with '*' are statistically significant ($p<0.05$).

6.2.8.2 Neutralization of GF-VI DEAE-II by physiological inhibitors

The indigenous protease inhibitors of blood viz. α_2-antiplasmin and α_2-macroglobulin neutralized the fibrinolytic as well as anticoagulant activity of GF-VI DEAE-II (Fig. 6.16). Both α_2-antiplasmin and α_2-macroglobulin showed slightly better inhibition of the anticoagulant property of GF-VI DEAE-II (Fig. 6.16).

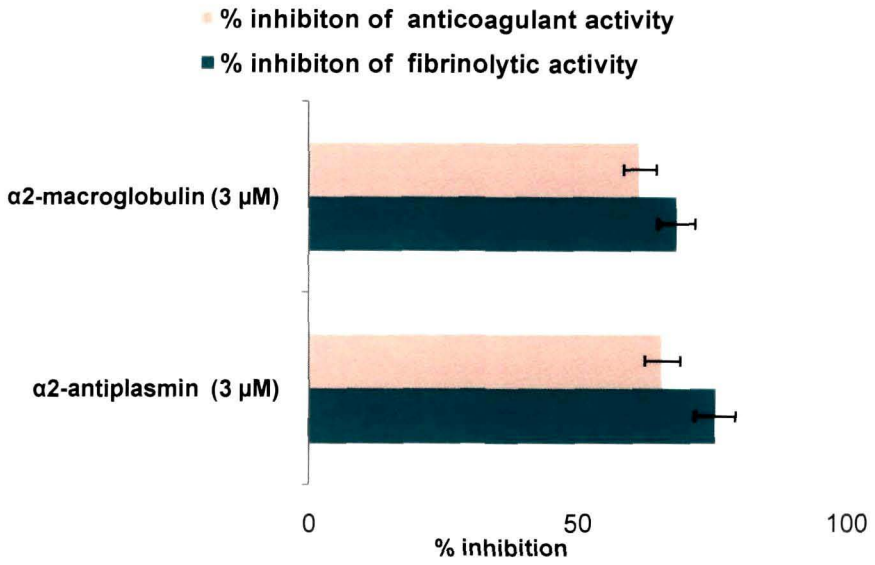


Fig. 6.16 Neutralization of GF-VI DEAE-II by physiological inhibitors. GF-VI DEAE-II (2.0 μg) was pre-incubated with 3.0 μM α_2-antiplasmin or α_2-macroglobulin for 30 min at 37 °C. Thereafter, fibrinolytic/anticoagulant activity was determined as described in sections 3.2.3.3.1 and 3.2.4.1, respectively. The fibrinolytic/anticoagulant activity in the absence of α_2-antiplasmin or α_2-macroglobulin was considered as 100% activity and other values were compared with that. Values are mean ± S.D. of triplicate experimental values.

6.2.8.3 Neutralization of GF-VI DEAE-II by commercial antivenom

The fibrinolytic (Fig. 6.17A) and anticoagulant property (Fig. 6.17B) of GF-VI DEAE-II was neutralized almost to an equal extent by both the PAV and MAV in a dose-dependent manner.

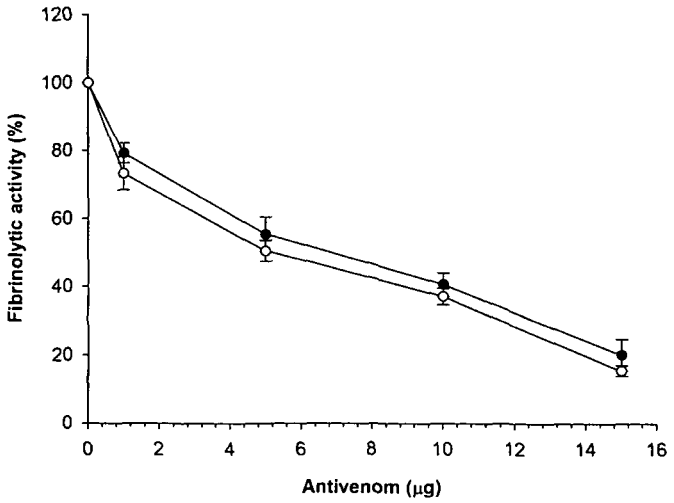


Fig. 6.17A A comparison of neutralization of the fibrinolytic property of GF-VI DEAE-II by PAV and MAV. GF-VI DEAE-II (2.0 µg) was pre-incubated with either PAV (•), MAV (○) for 30 min at 37 °C. The fibrinolytic activity of GF-VI DEAE-II was analyzed as described in section 3.2.3.17.1. The activity of GF-VI DEAE-II in absence of antivenom was considered as 100% activity and other values were compared with that. Values are mean ± S.D. of triplicate experimental values.

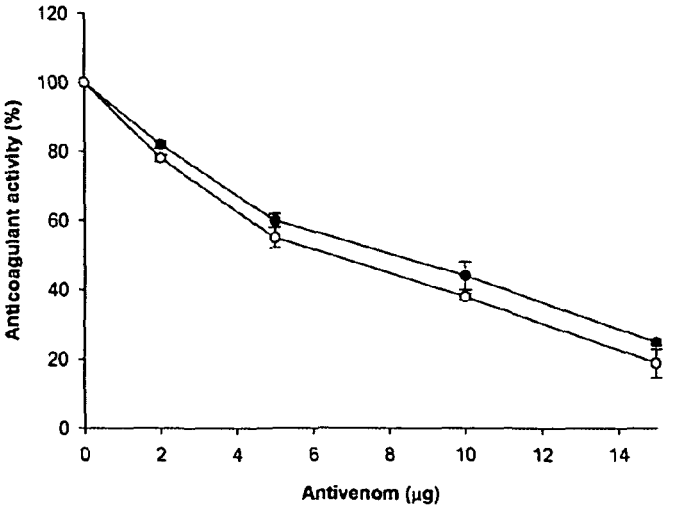


Fig. 6.17B A comparison of neutralization of the anticoagulant property of GF-VI DEAE-II by PAV and MAV. GF-VI DEAE-II (0.1 µg μl^{-1}) was pre-

incubated with either PAV (•), MAV (○) for 30 min at 37 °C. Thereafter, the plasma recalcification time of GF-VI DEAE-II was analyzed as described in section 3.2.4.1. The activity of GF-VI DEAE-II in absence of antivenom was considered as 100% activity and other values were compared with that. Values are mean ± SD of triplicate experimental values.

6.2.9 Acute *in vivo* toxicity assessment of GF-VI DEAE-II on Balb/c mice

6.2.9.1 Lethality and *in vivo* toxicity in Balb/c mice

The LD_{50} value of GF-VI DEAE-II was calculated to be 2.3 mg kg^{-1} for Balb/c mice. The animals treated with the LD_{50} dose of GF-VI DEAE-II (2.3 mg kg^{-1}) showed onset of stillness and ptosis, decreased grip strength and ear twitch, decreased appetite and polyuria within 10-20 min after injection (Table 6.9). However, with an increase in time (after 24 h of injection), the surviving animals were found to gradually recover from the above adverse effects which was evident from the recuperated grip strength, normal behaviour, as well as improved food and water intake (Table 6.9).

The treated group of mice also demonstrated hind leg paralysis in addition to more pronounced behavioural and physiological changes such as stillness, ptosis, decreased grip strength and polyuria after about 10 min of administration (Table 6.9). However, the surviving treated animals recovered from the above symptoms gradually.

Because the severity of any venom component can be partially judged on the basis of visible behavioural changes in experimental animal model, the above mentioned parameters were considered as the immediate *in vivo* toxic effect of the isolated lethal RVV fraction (GF-VI DEAE-II) in mice.

Animals treated with the cytotoxin fraction (Rusvitoxin) did not induce lethality or any visible behavioral changes in Balb/c mice at the dose of 2.3 mg kg^{-1}. However, due to very low yield, the lethality of the protease fraction (RP-I) could not be determined.

6.2.9.2 *In vivo* anticoagulant effect

After 10 min of GF-VI DEAE-II (LD_{50} value, 2.3 mg kg^{-1}) treatment, the whole blood clotting time of the treated group of mice was found to be significantly higher ($p<0.05$) as compared to the whole blood clotting time of control group of mice under identical experimental conditions (Table 6.9). However, the anticoagulant effect of GF-VI DEAE-II decreased gradually with increasing time after the treatment (Table 6.9).

6.2.9.3 Effect on serum parameters

It was observed that after 24 h of GF-VI DEAE-II (LD_{50} value, 2.3 mg kg^{-1}) treatment, a significant increase ($p<0.01$) in the serum levels of total protein, cholesterol, and uric acid as well as some organ specific marker enzymes such as SGOT, SGPT, ALP and CPK was observed in mice (Table 6.10). On the contrary, a significant decrease ($p<0.01$) in the levels of glucose, urea, and creatinine in blood of treated mice as compared to control mice was observed (Table 6.10). Nevertheless, it is to be noted that with an increase in time interval (24-48 h) the deviated parameters of plasma in the treated mice rebounded to their normal level (Table 6.10).

6.2.9.4 Effect on hematological parameters

The *in vivo* effect of GF-VI DEAE-II (LD_{50} value, 2.3 mg kg^{-1}) on several hematological parameters of blood is shown in table 6.11. The *i.p.* injection of GF-VI DEAE-II at the LD_{50} dose did not alter most of the tested hematological parameters (WBC, RBC, Hb, Pct and Ht) of the treated mice (Table 6.11).

Table 6.9 Behavioural and physical changes in GF-VI DEAE-II (LD_{50} value, 2.3 mg kg^{-1}) treated mice.

Parameter	Control mice	Treated mice (2.3 mg kg^{-1})			
		10 min	24 h	36 h	48 h
Initial Body weight (gm)	32.44 ± 2.6	ND	ND	ND	28.54* ± 4.4
Final Body weight (gm)	35.96 ± 7.6	ND	ND	ND	26.31* ± 3.2
Food intake (gm)	76.03 ± 13.35	ND	ND	ND	47.53* ± 10.3
Water intake (ml)	57 ± 5.4	ND	ND	ND	23* ± 3.2
Grip strength (s)	30.25 ± 12.4	8.5* ± 1.2	9.3* ± 2.2	14.6* ± 3.4	18.59* ± 9.1
Ear twitch (s)	7.14 ± 1.3	12.4* ± 2.9	14.5* ± 6.4	13.4* ± 3.4	15.98* ± 6.4
Rectal temperature (°C)	35.0	35.0	35.0	35.0	35.0
Drooping of eyelids	-	+	+	-	-
Movement inability	-	+	-	-	-
Stillness	-	+	-	-	-
Hind leg paralysis	-	+	-	-	-
Clotting time (min)	3.1 ± 2.3	10.1 ± 0.3	7.4 ± 0.4	5.91 ± 0.5	5.32 ± 0.7

Values are mean ± SD of three animals. The significance of differences between the means of control and treated groups was determined by student's t-test. *$p<0.05$ was considered significant compared to control group of mice. - = Not observed. + = Observed. ND= Not determined.

Table 6.10 Biochemical estimation of plasma parameters of GF-VI DEAE-II (2.3 mg kg^{-1}) treated and control mice.

Parameter	Control	Treated (2.3 mg kg^{-1})		
		24 h	36 h	48 h
Glucose (g dl^{-1})	322 ± 22.65	186.9 ± 12.2*	191.3 ± 6.9#	193.6 ± 9.1*
SGPT (U L^{-1})	66.6 ± 9.5	125.2 ± 2.3*	119.4 ± 5.3#	116.2 ± 16.0#
SGOT (U L^{-1})	87.5 ± 20.7	128.3 ± 14.2#	141.5 ± 10.1#	154.6 ± 20.5#
ALP (U L^{-1})	93.5 ± 2.1	154.4 ± 3.5*	103.2 ± 5.1#	96.5 ± 12.0
CK-MB (U L^{-1})	47.0 ± 7.9	112.4 ±6.3*	72.9 ± 5.4*	67.0 ± 9.6#
Cholesterol (mg dl^{-1})	66.6 ± 4.7	108.7 ± 10.8#	118.3 ± 4.9#	119.2 ± 7.0*
Uric acid (mg dl^{-1})	4.15 ± 0.77	6.3 ± 0.4#	5.9 ± 0.8*	5.13 ± 0.05
Urea (mg dl^{-1})	108.9 ± 12.0	67.6 ± 6#	75.4 ± 5.5#	80.6 ± 6.9#
Creatinine (mg dl^{-1})	61.1 ± 1.7	25.4 ± 0.8#	32.2 ± 3.4#	39.2 ± 1.3*

Values are mean ± SD (n=3). The significance of differences between the means of control and treated groups was determined by student's t-test. $p<0.05$ was considered significant.*$p<0.001$; #$p<0.05$.

Table 6.11 Changes in the hematological parameters of the GF-VI DEAE-II (2.3 mg kg^{-1}) treated and control (placebo) mice.

	WBC (M/mm^3)	**Lymphocyte (%)**	**Monocyte (%)**	**RBC (M/mm^3)**	**HB (g dl^{-1})**
Control group	4.85* ± 1.3	52.5* ± 9.8	20.8* ± 5.9	5.19* ± 0.9	5.5* ± 1.1
Treated (2.3 mg kg^{-1})#	6.49* ± 4.03	60.9* ± 3.3	19.4* ± 3.0	6.06* ± 1.6	6.4* ± 1.9

Values are mean ± SD (n=3). The significance of differences between the means of control and treated groups was determined by student's t-test. $p<0.05$ was considered significant.*$p>0.05$; #The values were observed in animals treated for 24 h.

6.2.9.5 Histopathological study

Histological analysis of heart, liver and kidney tissues from GF-VI DEAE-II treated mice revealed some gross morphological alterations (Figs. 6.18A-C).

Histopathological study of the liver tissues obtained from the animals after 24 h of GF-VI DEAE-II treatment showed signs of karyolysis (L) and karyopyknosis (P) in hepatocytes which was not observed in the control group of mice (Fig. 6.18A). Besides, liver tissues from the treated mice also depicted sinusoidal congestions (S) (Fig. 6.18A).

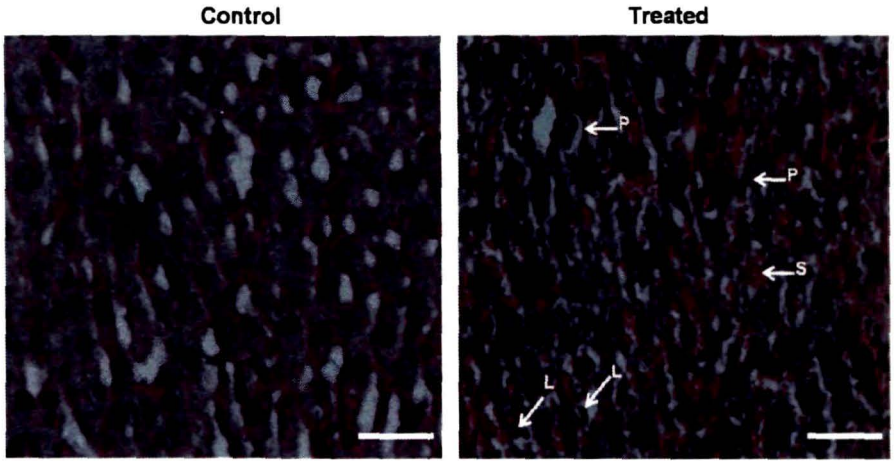


Fig. 6.18A Light microscopic observation of the effect of GF-VI DEAE-II (2.3 mg kg^{-1}) on liver tissues of Balb/c mice. Tissues from mice treated with PBS served as controls. Hematoxylene-eosin (H & E) staining was performed as described in section 3.2.4.8.5; Magnification 10X: Bar=50 µm. Hepatocytes with karyolysis are marked with 'L'. Hepatocytes demonstrating karyopyknosis are shown with arrow and denoted with 'P'. Liver section showing sinusoidal congestion is shown with an arrow with the notation 'S'.

Light microscopic study of sections of myocardial tissues obtained from mice that died due to treatment of GF-VI DEAE-II (2.3 mg kg^{-1}) demonstrated substantial

degeneration along with loss of striations in addition to massive degeneration and deleterious damage (Fig. 6.18B).

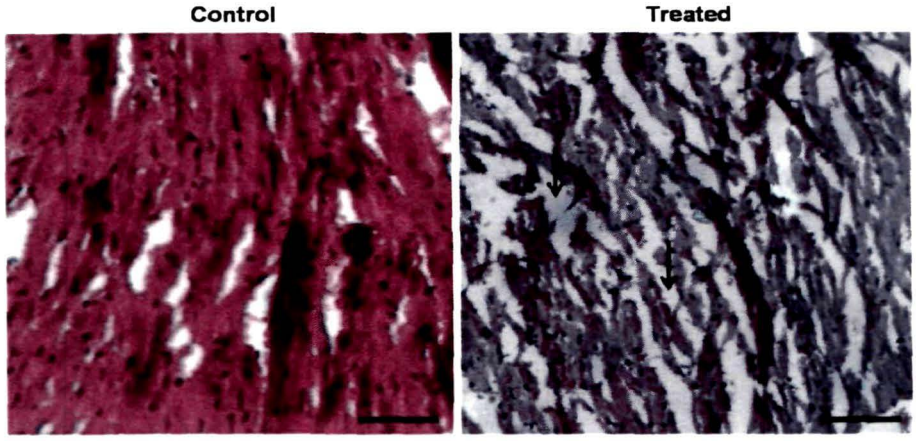


Fig. 6.18B Light microscopic observation of the effect of GF-VI DEAE-II (2.3 mg kg^{-1}) on heart tissues of Balb/c mice. Tissues from mice treated with PBS served as controls. Hematoxylene-eosin (H & E) staining was performed as described in section 3.2.4.8.5; Magnification 10X: Bar=50 µm. Degenerating muscle fibers with loss of striations are depicted in the figure with an arrow.

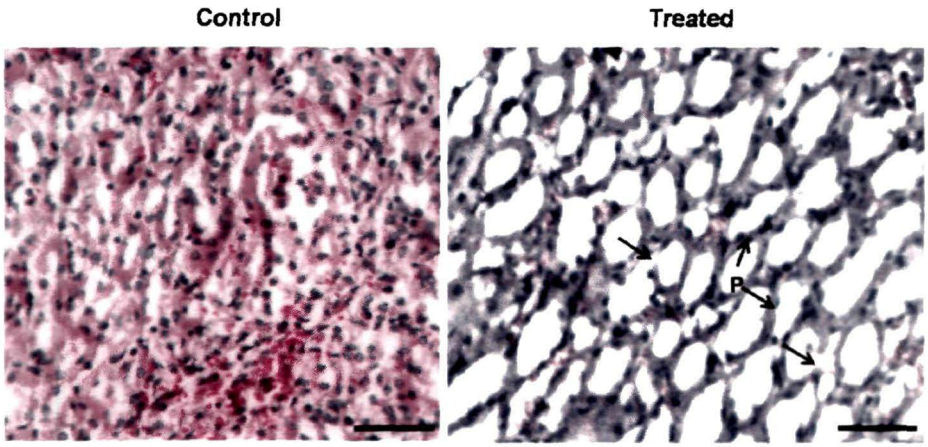


Fig. 6.18C Light microscopic observation of the effect of GF-VI DEAE-II (2.3 mg kg^{-1}) on kidney tissues of Balb/c mice. Tissues from mice treated with PBS served as controls. Hematoxylene-eosin (H & E) staining was performed as

described in section 3.2.4.8.5; Magnification 10X: Bar=50 μm. Pyknotic nuclei are shown with arrows with the notation 'P'. Damage to kidney epithelial lining is shown with arrow.

The renal tissues from the treated mice showed structural alterations viz. damage to the epithelial cells of the convoluted tubules and the presence of pycnotic nuclei (P) (Fig. 6.18C).

6.2.9.6 Effect of GF-VI DEAE-II on total protein and blood coagulation factors (factor IX, factor X, fibrinogen and PDGFR-α)

Administration of GF-VI DEAE-II to mice at its LD_{50} dose (2.3 mg kg^{-1}) resulted in the significant increase ($p<0.01$) in the level of plasma fibrinogen (Fig. 6.19) and factor X (Fig. 6.20) 24 h after treatment as compared to the plasma concentration of the above coagulation factors in the control group of mice.

Factor IX level was also found to be enhanced significantly in GF-VI DEAE-II treated animals after 48 h of treatment (Fig. 6.20). Plasma total protein level was however found to be insignificantly affected by GF-VI DEAE-II treatment (Fig. 6.19).

Furthermore, plasma concentration of PDGFR-α was also found to enhance significantly ($p<0.001$) in GF-VI DEAE-II treated mice (152.0 ± 1.4 ng ml^{-1}) as compared to the control group of mice (116.0 ± 1.4 ng ml^{-1}) after 24 h of treatment.

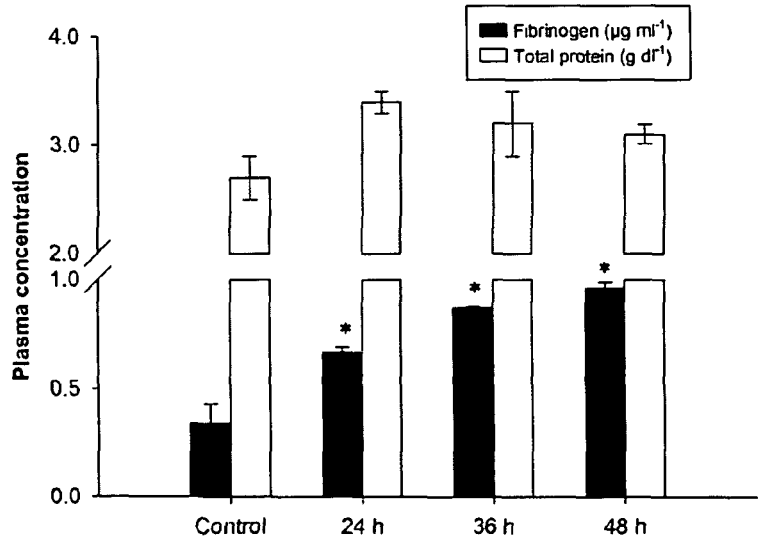


Fig. 6.19 Plasma profile of total protein and fibrinogen of GF-VI DEAE-II treated mice. A comparative analysis of total protein (g dl^{-1}) and fibrinogen concentrations (µg ml^{-1}) in plasma from control and GF-VI DEAE-II (2.3 mg kg^{-1}) treated mice. Values are mean ± S.D. of triplicate determinations. '*' represent values that are significantly different from that of control ($p<0.01$).

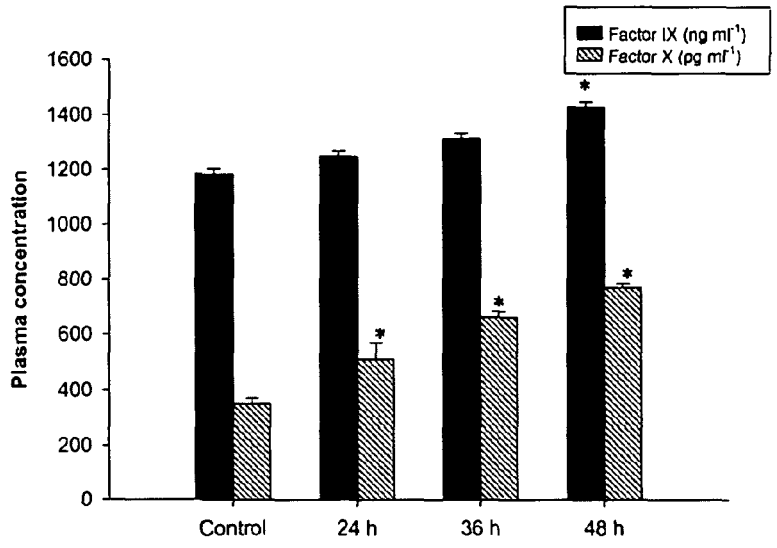


Fig. 6.20 Plasma profile of factor IX and factor X of GF-VI DEAE-II treated mice. A comparative analysis of Factor IX (ng ml^{-1}) and Factor X (pg ml^{-1})

concentrations in plasma of control and GF-VI DEAE-II (2.3 mg kg^{-1}) treated mice. Values are mean ± S.D. of triplicate determinations. '*' represent values that are significantly different from that of control ($p<0.01$).

6.2.9.7 Effect of GF-VI DEAE-II on inflammatory cytokines IL-1, IL-6, IL-10 and TNF-α

Plasma concentrations of the tested cytokines viz. IL-6, IL-10 and TNF-α in the GF-VI DEAE-II treated group of mice were found to be altered significantly compared to their plasma level in control group of mice (Fig. 6.21).

Pro-inflammatory cytokines such as TNF-α concentration increased significantly in treated animals after 24 h of *i.p.* injection of GF-VI DEAE-II (2.3 mg kg^{-1}). It was observed that TNF-α level continued to increase upto 48 h of GF-VI DEAE-II treatment.

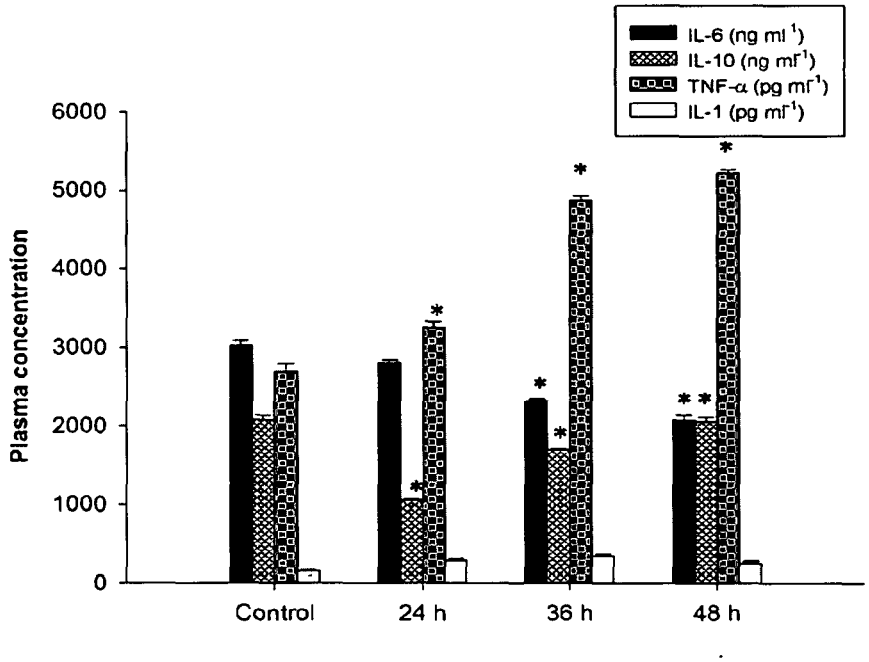


Fig. 6.21 Effect of GF-VI DEAE-II on inflammatory cytokines. A comparative analysis of plasma cytokines (IL-1, IL-6, IL-10 and TNF- α) level in control and

GF-VI DEAE-II (2.3 mg kg^{-1}) treated mice. Values are mean ± SD (n=3) of triplicate determinations. '*' represent values that are significantly different from that of control ($p<0.01$).

In contrast, plasma concentration of IL-1 remained unaffected due to GF-VI DEAE-II treatment (Fig. 6.21). A decrease in anti-inflammatory cytokine IL-10 was observed 24 h after injection of GF-VI DEAE-II. However, after 24 h the plasma level of IL-10 seemed to recover to its normal level (Fig. 6.21). The plasma concentration of IL-6 was observed to decrease in GF-VI DEAE-II treated mice compared to control mice (Fig. 6.21).

6.3 Discussion

6.3.1 The isolated toxic fraction (GF-VI DEAE-II) from RVV is composed of a fibrinolytic serine protease and a three-finger cytotoxin-like peptide

It is well established that snake venom comprises of numerous enzymatic and non-enzymatic components which often form complexes or co-elute in a single fraction [274, 277]. Co-elution of venom proteins is often attributed to the strong non-covalent interaction(s) between the protein components present in a particular venom [59, 274, 277]. It is important to study such co-eluting proteins and protein complexes since the biological effects of the individual components are significantly enhanced when present together [59, 75, 77-78]. Analysis of the GF-VI DEAE-II fraction by SDS-PAGE under non-reducing conditions indicated that the proteins present in this fraction shows a pattern of polydisperse molecules similar to that observed by Calvete and his colleagues (2007) in case of PLA_2s isolated from *Bitis caudilis* [349]. Similar observations were also observed for a Kunitz-type serine protease inhibitor (Rusvikunin-II) by Mukherjee and Mackessy [59]. The fact that the fraction separated into two distinct protein components

under reducing conditions, a ~35 kDa protein (RP-I) and a ~9 kDa peptide (Rusvitoxin), unequivocally indicated that these proteins co-eluted in a single fraction. Since the components of the toxic fraction could not be separated under various non-denaturing chromatographic conditions except RP-HPLC, it is suggested that the two molecules remain in close association with each other. The co-elution of a high molecular mass enzyme (~35 kDa) along with a low molecular mass peptide (~9 kDa) upon size exclusion chromatography advocates the above explanation. Moreover, the protein with the higher molecular mass of ~35 kDa protein was visible in SDS-PAGE, only when a higher amount of GF-VI DEAE-II was loaded in SDS-PAGE, suggesting that this protein is a minor component of the isolated venom fraction.

It is well established that the snake venom serine proteases (SVSPs) belong to the trypsin family S1 of clan SA [79]. The high degree of sequence homology displayed by RP-I towards trypsin-like snake venom serine proteases indicates that RP-I is a trypsin-like serine protease of RVV. Since RP-I did not display significant homology with previously reported fibrino(geno)lytic proteases and thrombin-like enzymes from RVV [64-65, 83-84]. Therefore, it may be anticipated that RP-I is a previously uncharacterized fibrino(geno)lytic protease from RVV. Chemical modification of the catalytic and pharmacological activities of GF-VI DEAE-II suggested the presence of a serine protease in the toxic venom fraction. Our study suggested that one of the active component of the isolated toxic fraction is a trypsin-like, fibrinolytic serine protease (RP-I) which is, in fact, a minor component of the venom fraction, GF-VI DEAE-II. However, the molecular mass of RP-I is comparable to the reported snake venom fibrinolytic proteases [75]. Although crude RVV exhibits potent fibrinolytic activity [15, 64]; however, apart from VRR-73 [64] no other proteases showing fibrinolytic activity has been identified from *D. r. russelii* venom.

Characterization of the fibrinolytic property of GF-VI DEAE-II revealed that the enzymatic component of the venom fraction showed maximum catalytic efficiency

at physiological conditions. GF-VI DEAE-II fraction specifically degraded the α-chain of fibrin similar to many of the snake venom fibrinolytic proteases [75]. Present study also suggests that GF-VI DEAE-II preferentially degrades fibrin over fibrinogen unlike most of the previously identified snake venom fibrinolytic enzymes [75, 350-351]. The preferred fibrin specificity of the protease component of GF-VI DEAE-II is probably due to its better binding ability with the cross-linked fibrin rather than fibrinogen [352]. Furthermore, a comparison of the fibrino(geno)lytic activity of human plasmin with GF-VI DEAE-II also suggests that these two direct-acting fibrinolytic proteases have different specificities and cleavage preference for fibrin(ogen); where the former is proven to be responsible for maintaining cellular homeostasis, whereas the later may contribute to the toxicity of RVV by disturbing normal homeostasis system in victim.

In vitro neutralization of many snake venom toxins by endogenous inhibitors such as α_2-macroglobulin and α_2-antiplasmin may not always be correlated with the *in vivo* neutralization of the same toxins [345]. Therefore, it may be anticipated that although GF-VI DEAE-II could be neutralized in *in vitro* conditions, however, its pharmacological properties could remain unaffected *in vivo.* The *in vitro* proteolytic (fibrinolytic) and anticoagulant properties of GF-VI DEAE-II could be partially neutralized by both PAV and MAV suggesting that these antivenoms could possibly neutralize the effects of this fraction in the whole RVV as well.

Most of the snake venom fibrinolytic proteases exhibit tissue necrotic effects [64] owing to their ability to degrade the extracellular membrane proteins such as fibronectin, laminin and type-IV collagen lining the tissues. Additionally, unlike most of the reported snake venom toxins, GF-VI DEAE-II did not display cytotoxic effects on the tested carcinoma cells-U87MG and HeLa cells, possibly due to the absence of specific-U87MG or HeLa cell membrane target molecules in GF-VI DEAE-II [353]. The lack of cell cytotoxicity of GF-VI DEAE-II may also be attributed to its lack of hydrolytic effects on the above membrane proteins.

The non-enzymatic component of the GF-VI DEAE-II (Rusvitoxin) demonstrated presence of conserved cysteine residues which are characteristic features of three-finger toxins [166-167]. The gene structure of three-finger toxins from Viperidae family of snakes such as *Sistrurus catenatus edwardsii* has been described [91] and existence of such proteins in RVV has been suggested [347-348]. However, Rusvitoxin is significantly different from the previously isolated short chain neurotoxins from RVV such as DNTx-I [347] and DNTx-III [348] in various fronts, particularly in terms of its primary sequence, pathological properties as well as LD_{50} value. Because the primary structure of Rusvitoxin does not share substantial sequence homology with any of the cytotoxins or neurotoxins isolated from RVV [347-348] and exhibits the presence of conserved cysteine residues of three finger toxins; it may be suggested that Rusvitoxin is a newly isolated, three-finger cytotoxin like molecule from RVV.

6.3.2 GF-VI DEAE-II aids in prey immobilizing and subsequent killing by Russell's Viper envenomation

On an average, an adult Russell's Viper may deliver approximately 225–250 mg of venom (total amount in a bite) to its prey/victim [96]. This value corresponds to approximately 3.6 to 4.0 mg of GF-VI DEAE-II being injected into its target prey or victim. The relatively strong LD_{50} value of the venom fraction (2.3 mg kg^{-1}) in mice reflects the contribution of this fraction in crude RVV in prey immobilization followed by its killing before being swallowed. Most of the non-conventional toxins as well as their protein complexes such as Reprotoxin are found to display an LD_{50} value in the range of 5-80 mg kg^{-1}in mice [275, 354]. Therefore, the contribution of GF-VI DEAE-II fraction in RVV-induced toxicity cannot be negated. Although RVV from eastern India is reported to demonstrate hind limb paralysis on experimental animals [48]; nevertheless, this phenomenon has never been observed in Russell's Viper envenomed patients from eastern part of India [291]. This suggests that GF-VI DEAE-II may presumably induce discrete effects in humans and rodents. Alternatively, the quantity of GF-VI DEAE-II injected in

an adult human (i.e. approximately 0.7 mg l^{-1}) may not be sufficient to produce the identical symptoms as shown in mice.

6.3.3 GF-VI DEAE-II induces toxicity by hemostatic disturbances and multiple organ dysfunctions

The *in vivo* pharmacological studies of GF-VI DEAE-II on Balb/c mice also throw some light on its other clinical attributes. Every living cell contains different intracellular enzymes many of which are released into the circulation after the breakdown/impairment of cell membranes [291]. Therefore, release of organ specific marker enzymes into circulation suggests damage to those organs by the isolated venom fraction. Previous reports have clearly shed light on the detrimental effects of crude RVV on myocardial and muscle tissues [15, 64-65, 291]. A significant rise in the plasma level of CPK and SGOT are important markers for muscle damage. It has been reported that RVV elevates CPK level in experimental animals as well as in envenomed patients [15, 65]. Therefore, a significant rise in CPK level in plasma of GF-VI DEAE-II treated mice as compared to control group of mice suggested muscle damage by the venom fraction.

It is notable to mention that renal failure is accountable for maximum number of deaths in Russell's Viper envenomed patients all throughout the South East Asian countries [15, 65, 356]. A significant elevation of plasma uric acid level depicts acute renal damage [357]. Elevated uric acid level was observed in GF-VI DEAE-II treated mice suggesting it induces nephrotoxicity in mice. However, the observed significant decrease in plasma urea and creatinine levels in treated mice may be attributed either by the venom fraction-induced renal damage or due to reduced diet consumed by the experimental groups of mice as compared to control mice.

In Russell's Viper envenomed patients, liver is reported as a primary target of action for various venom components that leads to hepatocellular damage. In the

present study, increase in plasma concentration of liver marker enzymes (SGOT, SGPT) and cholesterol following GF-VI DEAE-II treatment provided evidence of liver damage [291]. Further, amplified level of the intracellular enzyme, ALP suggests loss in the integrity of the hepatobilliary system [358]. Therefore, it may be well understood that GF-VI DEAE-II fraction has detrimental effects on liver tissues which contributes to its overall toxicity. It is to be noted that crude RVV also exhibits signs of hepatic dysfunction effects in experimental animal models [15] as well as in RV envenomed patients [359-360]. These observations suggest that the isolated venom fraction may have detrimental effects at the organ-system level leading to combined dysfunction of vital organs such as the heart, kidney and liver in its victim/prey.

The above detrimental effects were supported by histological observations of different tissues from the GF-VI DEAE-II treated mice. Nuclear alterations and sinusoidal condensations in the liver tissues are clear signs of increased cellular activity associated with the process of detoxification [361]. The possibility of intravascular tension and hemorrhage is also suggested in the treated animals due to damage in myocardial muscles.

It is interesting to note that despite of lacking *in vitro* cytoxicity, GF-VI DEAE-II demonstrated strong *in vivo* toxic effects on Balb/c mice suggesting that *in vitro* cytotoxicity cannot always be co-related with *in vivo* toxicity and vice-versa [353].

Unlike most of the three-finger cytotoxins, Rusvitoxin at a dose of 5.0 mg kg^{-1} lacks *in vivo* toxicity or lethality in mice [347-348]. Moreover, most of the snake venom components including serine proteases generally affect the coagulation cascade but do not demonstrate pathological damage to tissues or show toxicity in mice at a dose of 5 mg kg^{-1} [45, 82-83, 362]. However, many of the relatively non-toxic components of snake venom exert potent toxicity in experimental animals when present in the form of a protein complex [59, 275, 277-278]. Therefore, it

may be anticipated that the *in vivo* toxic effects of GF-VI DEAE-II may be due to the synergistic action of the protease and cytotoxin components.

6.3.4 GF-VI DEAE-II induces hyperfibrinogenomia and enhances the level of coagulation factors IX and X

Serum levels of acute phase proteins such as fibrinogen are important markers for understanding the condition of organs such as liver and the myocardium [363]. Studies have suggested that a 2-3 fold rise in plasma fibrinogen level (hyperfibrinogenemia) is the most common phenomenon after myocardial infarction [364]. Additionally, hyperfibrinogenemia may also be contributed by liver necrosis or toxic hepatitis. Interestingly, although GF-VI DEAE-II failed to hydrolyze plasminogen and thrombin (*in vitro*); however, it demonstrated *in vivo* anticoagulant activity probably due to the enhanced production of coagulation factors such as fibrinogen, factors IX and X resulting in fibrin formation which serves as a substrate for the venom fraction. In Russell's Viper bite patients, the clotting time is unanimously reported to be increased due to defibrination [15, 365]. Therefore, a possible explanation of the observed effect might be that GF-VI DEAE-II enhances the production of fibrinogen by damaging the liver tissues [366] which builds a better platform for the RVV proteases to act upon resulting in overall anticoagulant effect [83, 277]. This results in systematic consumption coagulopathy, which is a common clinical symptom of Russell's Viper bite [15, 46]. It is interesting to note that, like human plasmin, GF-VI DEAE-II also demonstrates anticoagulant property in *in vitro* conditions. Because the non-enzymatic component (Rusvitoxin) did not affect the *in vitro* clotting time of platelet-poor plasma, therefore, it may be anticipated that the protease component of GF-VI DEAE-II contributes to the overall anticoagulant effect shown by this fraction of RVV.

PDGFR-α is reported to play an important role in mediating inflammation and epithelial hyperproliferation [367]. However, the role of PDGFR-α in RV

envenomed patients has not been evaluated and henceforth its contribution in RVV-induced inflammation cannot be assessed in the present study. A concomitant increase in the level of PDGFR-α in mouse after treatment with GF-VI DEAE-II lead us to conclude that the venom fraction under study might play an important role in mediating inflammatory reactions.

6.3.5 GF-VI DEAE-II enhances the production of pro-inflammatory cytokines with a corresponding decrease in anti-inflammatory cytokines

The change in the balance between anti-inflammatory (IL-10) and pro-inflammatory (TNF-α) cytokines may provide insight into the progression of cardiovascular diseases [368]. The significant decrease in IL-10 level as well as in the ratio of IL-10 to TNF-α in the GF-VI DEAE-II treated mice, correlates with the altered cardiac function due to *in vivo* toxic effects of the venom fraction. Increased TNF-α level leads to tissue injury and necrosis [368] which is evident from the microscopic observation of the tissues obtained from the GF-VI DEAE-II treated mice. Decreased IL-10 level may also be due to reduced cell proliferation and immune-suppression that eventually leads to immunological imbalance [369]. However, further studies are necessary to correlate the immunological imbalance observed in GF-VI DEAE-II treated animals and its subsequent role in Russell's Viper envenomed patients. Though elevated IL-6 is also reported to be associated with increased risk of cardiovascular ailments [370]; however, for an unexplained reason, the plasma level of IL-6 decreased significantly ($p<0.05$) in Rusvitoxin treated mice.

Vascular cells are major sites of cytokines production in the blood circulation [371]. It may be well anticipated that some of the components of RVV in blood circulation may activate the production of pro-inflammatory cytokines with simultaneous deactivation and/or decrease in quantity of anti-inflammatory cytokines from these cells to inflict toxicity. It has been reported that the level of pro-inflammatory cytokines is considerably increased within 30 min of RV

envenomation [372]. The GF-VI DEAE-II fraction therefore may be considered as one such components of RVV which exerts its toxicity by local tissue damage and inflammation.

CHAPTER VII

CONCLUSIONS AND FUTURE PERSPECTIVES

7 Conclusion and future perspectives

In the present investigation, some RVV proteins affecting the process of blood coagulation and fibrinolysis have been isolated, characterized and their pathophysiological significance have been evaluated. The possible therapeutic applications of the isolated protein/peptide have also been explored, provided the isolated protein is non-toxic in nature.

The first major finding of the study is the purification, characterization and elucidation of the anticoagulant and pro-apoptotic mechanisms of a 4.4 kDa peptide named Ruviprase. Ruviprase demonstrated strong *in vitro* as well as *in vivo* anticoagulant properties by virtue of its ability to inhibit coagulation factors, thrombin and factor Xa. Moreover, the isolated peptide was found to be non toxic to Balb/c, at the therapeutic dose of 2 mg kg^{-1} suggesting its probable application as an antithrombotic agent. Moreover, this is the first report of a natural inhibitor isolated from snake venom, with the dual ability of inhibiting thrombin and factor Xa. This property makes it an interesting molecule with broader specificity. This new basic knowledge about Ruviprase permits us to foresee the development of this peptide as a therapeutic agent to prevent and manage unwanted blood clot formation (thrombosis) and fight cancer. In addition to its potent anticoagulant properties, Ruviprase was also found to demonstrate pro-apoptotic properties on breast cancer cell lines such as MCF-7. Ruviprase caused p53-dependent cell cycle arrest in MCF-7 cells and also induced mitochondria dependent apoptosis via excess ROS generation as well as alteration of mitochondrial membrane potential.

Future studies on the structural aspects of Ruviprase designed to understand the mechanisms involved in its binding to thrombin and factor Xa as well as its therapeutic potential in terms of its inhibitory effects on thrombin and factor Xa is warranted. A more advanced research focusing on the pro-apoptotic nature of Ruviprase and application of the peptide as a peptide-based anticancer agent holds promising ventures.

The second important finding of the present study is the purification, characterization and elucidation of procoagulant mechanism of a oligomeric metalloprotease named Rusviprotease which predominantly exists in a 26.8 kDa dimeric form. The present study reinforces that Rusviprotease is the first group A prothrombin activator that has been isolated from RVV. Further, Rusviprotease is the first report of a basic and oligomeric prothrombin activator isolated from snake venom. Interestingly, Rusviprotease lacked *in vitro* toxic effects and amplified the prothrombin activating property of factor Xa-prothrombinase complex suggesting its probable contribution to Russell's Viper-induced consumption coagulopathy. Further, neutralization study suggests that MAV is significantly better than PAV in neutralizing the procoagulant and proteolytic properties of Rusviprotease at higher doses (1:10 w/w, Rusviprotease:antivenom ratio). Besides, increased tail bleeding time and *in vivo* defibrinogenation in mice undoubtedly suggested the pathophysiological significance of Rusviprotease and its contribution in Russell's viper envenomations.

Future studies on the structural aspects of Rusviprotease as well as its therapeutic potential in terms of its *in vivo* defibrinogenation property are to be explored.

The third major finding of the present study is the isolation and characterization of a lethal RVV-fraction named GF-VI DEAE-II. The RVV fraction under study is composed of a 27.6 kDa fibrinolytic protease and a 6.6 kDa three finger cytotoxin-like peptide which are always co-eluted in a single fraction suggesting their close association. Furthermore, it may be well anticipated that the components of this toxic venom fraction (GF-VI DEAE-II) contributes to the overall toxicity of RVV in its prey by multiple organ-system damage. The isolated venom fraction increases the production of acute phase proteins viz. fibrinogen as well as alters the cytokine levels in mice leading to hemostatic and immunological imbalances.

Future studies on the molecular characterization of the individual components of this fraction as well as their structure-function analysis are warranted.

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PUBLICATIONS

A. Publications in peer-reviewed journals/Revised manuscripts communicated

1. **Thakur, R.**, Kumar, A., Bose, B., Panda, D., Saikia, D., Chattopadhyay, P., Mukherjee, A. K.. A new peptide (Ruviprase) purified from the venom of *Daboia russelii russelii* shows potent anticoagulant activity via non-enzymatic inhibition of thrombin and factor Xa, *Biochimie*, **105**, 149-158, 2014.
2. **Thakur, R.**, Chattopadhyay, P., Mukherjee, A.K. Biochemical and Pharmacological Characterization of a Toxic Fraction and its Cytotoxin-like Component Isolated from Russell's Viper (*Daboia russelii russelii*) Venom, *Comp Biochem Physiol-Part C*, in press, accepted manuscript.
3. **Thakur, R.**, & Mukherjee, A.K. A Brief Appraisal on Russell's Viper Venom (*Daboia russelii*) Proteinases, in *Handbook of Toxinology, Volume: Snake Venom*, Gopalakrishnakone, P., & Mukherjee, A.K., eds., Springer, revised manuscript communicated (invited article).
4. **Thakur, R.**, Chattopadhyay, P., Ghosh, S.S., Mukherjee, A.K. Elucidation of procoagulant mechanism and pathophysiological significance of a new prothrombin activating metalloprotease purified from *Daboia russelii russelii* venom, *Toxicon*, revised manuscript communicated.
5. Mukherjee, A. K., Saikia, D., **Thakur, R.** Medical and diagnostic application of snake venom proteomes, *J Proteins Proteomics,* **2**, 31-40, 2011.

B. Publications in National Conferences/General Publications

1. **Thakur, R.**, Kumar, A., Bose, B., Panda, D, Saikia, D., Chattopadhyay, P., Mukherjee, A.K. Anticoagulant mechanism of a peptide purified from *Daboia russelii russelii* (Russell's viper) venom. Proceeding of '83rd

Annual Meeting of Society of Biological Chemists (India), Evolution: Molecules to Life', 17th-21st December, 2014, KIIT University, Bhubeneshwar, Odisha.

2. **Thakur, R.**, Kumar, A., Bose, B., Panda, D, Saikia, D., Chattopadhyay, P., Mukherjee, A.K. Characterization of an anticoagulant peptide with pro-apoptotic properties purified from *Daboia russelii russelii* (Russell's Viper) venom. Proceeding of 'National Seminar on Recent Advances in Biotechnological Research in North East India: Challenges and Prospects', 27th-29th November, 2014, Tezpur University, Assam.
3. **Thakur, R.** & Mukherjee, A.K. Biochemical and pharmacological properties of a low molecular mass cytotoxin-like molecule purified from Indian Russell's viper (*Daboia russelii russelii*) venom. Proceeding of 'National Conference on Snakebite Management', 10th-12th December, 2012, University of Mysore and Karnataka State Open University, Karnataka.
4. **Thakur, R.**, Saikia, D., Mukherjee, A.K. Insight into the anticoagulant mechanism of a low molecular weight protein isolated from Indian Russell's Viper venom. Proceeding of '1st National Conference on Animal, Microbial, Plant Toxins & Snakebite Management: Biotoxins in Health & Disease', 11th-12th December, 2010, IICB and KPC Medical College & Hospital, Kolkata, West Bengal.
5. **Thakur, R.**, & Mukherjee, A.K. Screening for fibrinolytic isoenzymes from venom of Indian Russell's viper (*Daboia russelii russelii*). Proceeding of 'Two-Day National Symposium on Recent Advances in Snake Venom Research and Snakebite Therapy: National and International Perspectives', 18th-19th December, 2009, Tezpur University, Assam.

C. General publications

1. **Thakur, R.**, & Mukherjee, A. K. Venom Therapeutics: A breakthrough in the field of medicine, *TSI Newsletter*, **2**, 8-10, 2011.

2. **Thakur, R.**, & Mukherjee, A. K. Focusing on an emergent class of venom component-Thrombin inhibitors: a potential lead compound for cardiovascular drug development, *TSI Newsletter*, **2** (1), 22-24, 2012.
3. **Thakur, R.**, & Mukherjee, A. K. Snake venom prothrombin activators: their importance and pathophysiological significance, *TSI Newsletter*, **2** (2), 35-38, 2012.

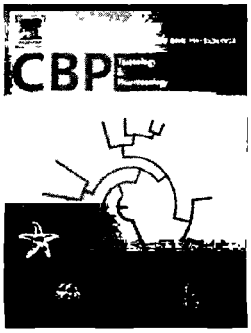
D. Other publications related to snake venom research

1. Mukherjee, A.K., Kalita, B., **Thakur, R.** Two acidic, anticoagulant PLA_2 isoenzymes purified from the venom of monocled cobra *Naja kaouthia* exhibit different potency to inhibit thrombin and factor Xa via phospholipids independent, non-enzymatic mechanism, *PLOS One*, **9**(8), e101334, 2014.
2. Mukherjee A.K., Chatterjee, S., Majumdar, S., Saikia, D., **Thakur, R.**, Chatterjee, A. Characterization of a pro-angiogenic, novel peptide from Russell's viper (*Daboia russelii russelii*) venom, *Toxicon*, **77**, 26-31, 2014.
3. Saikia D., **Thakur, R.**, Mukherjee, A. K. An acidic phospholipase A_2 (RVVA-PLA_2-I) purified from *Daboia russelli* venom exerts its anticoagulant activity by enzymatic hydrolysis of plasma phospholipids and by non-enzymatic inhibition of factor Xa in a phospholipids/Ca^{2+} independent manner, *Toxicon*, **57**(6), 841-850, 2011.

AWARDS RECEIVED IN NATIONAL CONFERENCES AND SEMINARS/OTHER AWARDS

1. **First Prize of ONGC-CPBT Young Scientist Award** recipient in the 'National Seminar on Recent Advances in Biotechnological Research in North East India: Challenges and Prospects', 27th-29th November, 2014, Tezpur University, Assam.
2. **Second Best Paper Award for Oral Presentation** recipient in the 'National Conference on Snakebite Management', 10th-12th December, 2012, University of Mysore and Karnataka State Open University, Karnataka.
3. **Young Scientist Award** recipient in the 'Two-Day National Symposium on Recent Advances in Snake Venom Research and Snakebite Therapy: National and International Perspectives', 18th-19th December, 2009, Tezpur University, Assam.
4. Recipient of the prestigious **INSPIRE fellowship for University first rank holders** at post graduate-level, from the Department of Science and Technology, Government of India, **for pursuing full time PhD** since 2010-till date.

Accepted Manuscript



Biochemical and Pharmacological Characterization of a Toxic Fraction and its Cytotoxin-like Component Isolated from Russell's Viper (*Daboia russelii russelii*) Venom

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MEDICAL AND DIAGNOSTIC APPLICATIONS OF SNAKE VENOM PROTEOMES

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Abstract: Snake venom is a highly toxic secretion produced and stored in specialized salivary glands of snakes which constitutes a vast array of biologically-active compounds, such as enzymes, proteins, peptides and low molecular weight compounds. These substances target an immense number of receptors and membrane proteins as well as coagulation proteins with high affinity, selectivity and potency, and can serve as potential drugs or scaffolds for drug design. During the recent years, much attention has been given to understand the mechanism of action of complex venom proteins for the development of novel drugs and therapeutic agents to treat life-threatening diseases such as cardiovascular diseases, cancer, thrombosis, arthritis, microbial infections and hypertension etc. Further, snake venom components have found uses in the diagnosis of haemostatic disorders. This paper reviews the various biomedical applications of snake venom proteins in terms of therapeutic and diagnostic values.

Key words: anticancer drug; antimicrobial agent; diagnostic reagents from snake venom; medical application of venom toxins; thrombolytic agents.

Introduction

The Indian subcontinent apart from being a rich source of flora and fauna is also home to a rich diversity of ophidian species. Only 242 species of snakes have been identified in India so far which includes 57 poisonous or harmful species. Snake venom is an evolutionary adaptation to immobilize, incapacitate as well as to digest their prey, which is also used as a defense mechanism by the snakes when encountered by enemies. In the recent years, snake venom and its components has gathered enormous interest by researchers across the globe. Scientists over the past few decades have postulated that despite of the harmful and life-threatening affects of snake venom, its component may provide highly specific research tools for the development of novel life-saving medicines and drugs against some common and life threatening diseases (Stocker, 1990) besides providing insight to basic coagulation process.

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A Brief Account of Snake Venom Composition

Over 90% of the solid snake venom components are pharmacologically active proteins and polypeptides, responsible for exerting pharmacological effects in victims (Sarkar and Devi, 1968; Stocker, 1990). Non-protein ingredients of venom include carbohydrate and metals (often in the form of glycoprotein and metalloprotein enzymes), lipids, free amino acids, nucleotides and biogenic amines.

The proteins/ polypeptides constituting snake venom can further be divided into enzymatic and non-enzymatic toxins, which in turn may be coagulant, anticoagulant or fibrinolytic in nature. Non enzymatic toxins are found to be predominant in cobra venom, whereas viper venom is found to be composed mainly of enzymatic proteins (Sarkar and Devi, 1968; Stocker, 1990). Snake venom toxins may have more than one specific activity, and therefore they may play multiple roles in the overall effect of envenoming. Considering this fact, the isolation and characterization of individual venom

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Research paper

A new peptide (Ruviprase) purified from the venom of *Daboia russelii russelii* shows potent anticoagulant activity via non-enzymatic inhibition of thrombin and factor Xa

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ABSTRACT

Compounds showing dual inhibition of thrombin and factor Xa (FXa) are the subject of great interest owing to their broader specificity for effective anticoagulation therapy against cardiovascular disorders. This is the first report on the functional characterization and assessment of therapeutic potential of a 4423.6 Da inhibitory peptide (Ruviprase) purified from *Daboia russelii russelii* venom. The secondary structure of Ruviprase is composed of α-helices (61.9%) and random coils (38.1%). The partial N-terminal sequence (E^1-V^2-X^3-W^4-W^5-W^6-A^7-Q^8-L^9-S^{10}) of Ruviprase demonstrated significant similarity (80.0%) with an internal sequence of apoptosis-stimulating protein reported from the venom of *Ophiophagus hannah* and *Python bivittatus*; albeit Ruviprase did not show sequence similarity with existing thrombin/FXa inhibitors, suggesting its uniqueness. Ruviprase demonstrated a potent *in vitro* anticoagulant property and inhibited both thrombin and FXa following slow binding kinetics. Ruviprase inhibited thrombin by binding to its active site via an uncompetitive mechanism with a *Ki* value and dissociation constant (K_D) of 0.42 μM and 0.46 μM, respectively. Conversely, Ruviprase demonstrated mixed inhibition (Ki = 0.16 μM) of FXa towards its physiological substrate prothrombin. Furthermore, the biological properties of Ruviprase could not be neutralized by commercial polyvalent or monovalent antivenom. Ruviprase at a dose of 2.0 mg/kg was non-toxic and showed potent *in vivo* anticoagulant activity after 6 h of intraperitoneal treatment in mice. Because of the potent anticoagulant property as well as non-toxic nature of Ruviprase, the possible application of the peptide as an antithrombotic agent for combating thrombosis-associated ailments appears promising.

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1. Introduction

In response to a vascular injury, the complex phenomenon of blood clotting requires the activation of zymogens of the coagulation cascade [1]. Thrombin and activated factor X (FXa) are the key components of blood coagulation [1,2]. FXa is the major component of the prothrombinase complex, which also comprises factor Va, negatively charged phospholipids, and calcium ions [2]. The prothrombinase complex eventually converts prothrombin to thrombin in the liver, which catalyzes the final stage of the blood coagulation cascade by cleaving fibrinogen to fibrin [1]. Besides, thrombin also activates procoagulant factors V, VIII, and XIII, in addition to contributing to the generation of a platelet plug [3].

Abbreviations: ALP, alkaline phosphatase; AT-III, antithrombin-III; BLASTP, basic local alignment search tool; CD, circular dichroism; CK-MB, creatine kinase; FXa, factor Xa; MAV, Russell's viper monovalent antivenom; OD, optical density; PBS, phosphate buffered saline; PLA_2, phospholipase A_2; PAV, polyvalent antivenom; pNA, p-nitroanilide; PPP, platelet poor plasma; RVV, Russell's viper venom; RU, response unit; SGOT, serum glutamic oxaloacetic transaminase; SGPT, serum glutamic pyruvic transaminase; TFA, trifloro acetic acid.

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